Springer Series on Cultural Computing

Vladimir Geroimenko *Editor*

Human-Computer Creativity

Generative AI in Education, Art, and Healthcare

Springer

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In celebration of the 20th anniversary of the British University in Egypt, and in memory of its founder, Mr. Mohamed Farid Khamis (1940–2020)

Preface

The transformative era of generative Artificial Intelligence (AI) has arrived, redefining the relationship between humans and technology in profound and promising ways. This book, *Human-Computer Creativity: Generative AI in Education, Art, and Healthcare*, seeks to explore the evolving interplay of creativity and computation across disciplines. Far from serving merely as tools, generative AI systems have emerged as creative collaborators, opening pathways to innovation and expression that were once the realm of imagination alone.

With contributions from 53 experts spanning 21 countries, this research monograph embodies a truly global perspective on the revolutionary impact of generative AI. Organised into four distinct parts, the chapters offer a comprehensive journey through conceptual frameworks, applications in education and art and groundbreaking innovations in healthcare. Each section reflects the diversity of thought and interdisciplinary insights that have shaped this pioneering effort, making it a vital resource for academics, practitioners and anyone curious about the future of human-computer collaboration.

Through this volume, we invite readers to engage with a new vision of creativity. In this vision, human ingenuity and artificial intelligence intersect to solve problems, inspire artistry and transform industries. This book represents not only a snapshot of our current understanding but also a beacon for what lies ahead.

The monograph consists of 19 chapters, which can be read in sequence or randomly. These chapters are arranged in four parts as follows:

Part I, titled *Conceptual Facets of Generative AI and Human-Computer Creativity*, includes four chapters (Chaps. 1–4).

Chapter 1, "Generative AI: From Human–Computer Interaction to Human–Computer Creativity", explores the transformative shift in human-computer relationships driven by generative artificial intelligence. Moving beyond traditional Human-Computer Interaction (HCI) frameworks focused on task completion and efficiency, this chapter delves into Human-Computer Creativity (HCC), where AI systems act as co-creators rather than passive tools. Through advancements in generative models, AI enables a collaborative environment that fosters creativity in diverse fields, including education, art and healthcare. By examining the historical progression of HCI, the capabilities of generative AI and its applications, the chapter illuminates the paradigm shift from interaction to co-creation. Key themes include the evolving roles of humans and AI, ethical considerations and the comparative analysis of interaction-based and creativity-based human-computer models. This shift highlights the potential of generative AI to reshape our understanding of creativity, challenging traditional notions of authorship and originality. In sum, the chapter provides a comprehensive analysis of how generative AI enhances and redefines human creativity, offering insights into the future of human-computer collaboration.

Chapter 2, "Emerging Paradigms in Human–AI Creativity: From Visualisation to Biofeedback in Adaptive Interfaces", posits that the effectiveness of artificial intelligence for creative purposes hinges on the development of appropriate interfaces. Such interfaces must be dynamic and adaptable, capable of responding to humans holistically rather than relying solely on language-based prompts. This chapter explores emerging trends in the landscape of human-computer interaction and creativity, underscored by cutting-edge advancements in generative AI revealing the dynamic interfaces reshaping user experiences. The chapter studies how creatives are using new modes of engaging with AI to shape new forms of creative expertise. The author surveys several artists using novel modes of AI interaction to facilitate the creative process. In particular, the artists featured here are making strides towards incorporating biological data to redefine the paradigms of creative human-computer interaction through visualisation and technologies such as EEG and BCIs. Culminating in a forward-looking perspective, the chapter ends by envisioning the future design of adaptive and biologically informed AI systems for creative settings.

Chapter 3, "Creativity in the Age of AI: Comparing Human and Machine Performance Using Standardised Tests", discusses AI potential in creative fields and how humans and generative AI creativity have recently been compared in the scientific literature. Generative AI models like ChatGPT have recently demonstrated remarkable capabilities in producing art, music and literature. Research comparing human and AI creativity reveals that AI can match or surpass human performance in specific tasks. Some studies have found that the best of humans can still outperform existing AI models regarding creative behaviours. Studies employing standardised creative tests have often shown that generative AI excels in tasks measuring fluency and originality, yet some aspects of creativity still see humans prevailing. The fact that AI excels and surpasses human performance in specific, experimentally controlled tasks does not automatically mean it will replace humans in creative jobs, which often require complex skills. Furthermore, AI's ability to produce innovative products, such as artworks, does not diminish the subjective experience of human artists or the natural human tendency to communicate through art. However, AI-generated high-quality art that is fast and cheap to produce may pose economic challenges for human artists. Future research should focus on human–AI collaboration, exploring how AI can augment human creativity while addressing ethical considerations.

Chapter 4, "The Role of AI in Transforming Educational Models: A Critical Analysis of Opportunities and Challenges", reflects on the impact of artificial intelligence on the educational sector by considering several implications for creating personalised learning and the infrastructure of academic institutions. Hence, this chapter aims to understand the role of artificial intelligence in helping and transforming the educational framework in a more creative environment. The use of artificial intelligence in educational fields facilitates the creation of new innovative processes that are useful for improving creativity among students and educators. In fact, by exploiting artificial intelligence, it is possible to help students develop their ideas and help educators create study plans based on innovative projects enriched by immersive experiences. This chapter, based on an overview of artificial intelligence in education, aims to identify the main elements of the transition from the traditional education system to a new educational model based on immersive technologies. In particular, the chapter seeks to specify how immersive technologies that exploit artificial intelligence can improve students' learning abilities and generate a dynamic environment for both students and educators. Finally, the chapter explores the challenges and opportunities accompanying AI integration into higher education, such as data privacy, the digital divide and concerns about the authenticity of AI-generated content.

Part II, titled *Generative AI and Creativity in Education*, consists of five chapters (Chaps. 5–9).

Chapter 5, "Transforming Education: The Potential of Generative AI Tools for Improving Learning Experience", presents how Generative Artificial Intelligence (GAI) tools are revolutionising the existing educational praxis. Multimodal GAI, such as text, images, video and games, are improving interaction skills in classrooms, promoting interculturality and creating a more playful and engaging educational environment. Qualitative analyses have emphasised its capacity to stimulate learning processes and have proposed innovative applications in the field of teacher education, showing significant improvements also in students' learning outcomes in cognitive and affective domains, enhancing flow experiences. However, it is essential to acknowledge that GAI applications, despite offering personalised tutoring, automated essay grading, language translation and interactive and adaptive learning, have certain limitations. These include a lack of human interaction, limited understanding and creativity and biases in the training data. Therefore, careful implementation and oversight are necessary to address these drawbacks effectively. In this sense, game-based learning platforms with generative AI are designed to improve digital literacy by providing engaging and interactive experiences. Although these initiatives hold potential, they necessitate additional refinement and assessment before they can be practically implemented. Although AI presents remarkable prospects, it necessitates diligent supervision to guarantee its responsible and efficient incorporation into educational frameworks.

Chapter 6, "Enhancing Digital Pedagogy and Creativity: Generative AI, Video Avatars, and Personalized Learning in Online Education", explores the transformative potential of generative artificial intelligence in online education, particularly within Learning Management Systems (LMS). By leveraging AI to create dynamic instructional content (including videos, images and texts), this chapter highlights how these tools can enhance creativity and engagement in digital learning environments. A focal point is the innovative use of video avatars, which utilise generative AI to produce interactive and personalised teaching personas. These avatars offer a scalable and individualised alternative to conventional pedagogical methods. Additionally, the chapter examines AI's role in designing assignments, rubrics and interactive simulations aimed at improving student engagement and learning outcomes. AI-assisted grading is presented to reduce instructor workload while providing personalised feedback. The ability of AI to autonomously analyse course objectives and suggest relevant assignments and evaluation metrics is also investigated. The integration of generative AI technologies, particularly video avatars, presents a significant opportunity to revolutionise the delivery and effectiveness of online education. The chapter posits that such advancements not only create a more engaging learning environment but also substantially elevate the quality of instruction and educational outcomes.

Chapter 7, "Creative Learning Using Generative AI: Empowering Problem-Solving and Programming Skills", discusses the different opinions about integrating generative artificial intelligence (GenAI) technologies into education systems in general and programming education in particular. Computer programming with writing code scripts is a way to empower learners' creativity. Through coding, they can create projects about things they care for and learn strategies for design and problem-solving. Traditional approaches to education computer programming are thriving through the GenAI era because our digital world still needs enormous contributions. However, the rocketing development of GenAI technologies is now raising the bars of education standards and revealing the widening gap between traditional education systems and the labour market requirements for skills graduates need. This rapid development was met with a move by the concerned educational authorities to explore ways to benefit from these advanced technologies to improve the quality of education and qualify students for the new jobs that will be created based on it. This chapter summarises several studies conducted to test the integration of GenAI tools. It offers best practices and recommendations reached by researchers to maximise the benefit of these technologies and mitigate their adverse effects.

Chapter 8, "Generative AI Applications in Education: A Low-Code/No-Code Approach", demonstrates how a Low-Code/No-Code (LC/NC) approach with generative artificial intelligence can be applied to adult education. Generative AI has rapidly changed the educational landscape by offering educators and students various tools and applications and adding new potential for creativity and personalisation. Most educators perceive AI as an opportunity; however, they face challenges of their own, mainly due to the increasing complexity of the applications and their limited knowledge and skills in AI-related topics. At the same time, there is a rising trend internationally regarding 'Citizen Developers', that is, power users who can create their software solutions with minimal or no programming skills, using user-friendly LC/NC frameworks and platforms in Artificial Intelligence as a Service (AIaaS) offering. The described AiHub platform prototype uses open-source frameworks and tools to simplify the Generative AI development process, allowing users without programming skills to design and develop their solutions and build applications with increasing complexity, from novice to expert. Finally, a case study on adult education is presented to demonstrate the power of development that can be given to educators to act as Citizen Developers and contribute to the AI revolution of our time.

Chapter 9, "Generative AI and Digital Twins: Fostering Creativity in Learning Environments", explores the intersection of creativity in education with digital twins and Generative AI (GenAI), highlighting their potential to transform teaching and learning. Building on theories from influential educators, such as Vygotsky and Gardner, the authors discuss how advances in digital technology enhance human-computer collaboration and position creativity as a sociocultural act shaped by interactions between humans and artificial agents. As part of the digital revolution, the narrative of education emphasises essential skills for successful citizenship, known as twenty-first-century skills, which include creativity, communication, collaboration and critical thinking. This chapter argues that creativity is not just a personal trait but a cultivable competency fostered through engagement with digital environments. The authors examine how GenAI and digital twins can improve creative learning experiences, enabling personalised growth and collaborative creativity with AI as a partner. Strategies for nurturing a culture of creativity in education are discussed, along with the challenges and opportunities these technologies present. Finally, the authors suggest future trends in creativity, digital twins and GenAI, redefining the creative potential of learners and educators.

Part III, titled *Generative AI and Creativity in Art*, comprises five chapters (Chaps. 10–14).

Chapter 10, "Generative AI and the Evolution of Artistic Creativity", explores the transformative impact of generative artificial intelligence on the evolution of artistic creativity, examining how advanced models like DALL-E, Midjourney and GPT are reshaping the artistic landscape by enabling unprecedented collaboration between human artists and machines. It provides an in-depth overview of key generative AI models and their capabilities across various domains, such as visual art, music generation and creative writing. It delves into the evolution of the creative process by comparing traditional artistic methods with modern practices enhanced by AI. The chapter addresses philosophical and ethical considerations and investigates the emergence of new artistic movements enabled by AI-driven aesthetics, showcasing innovative styles and reinterpretations of historical art forms. Introducing a comprehensive framework for assessing AI systems' aesthetic and creative capabilities, it proposes criteria for evaluating their contributions to art while critically examining ethical challenges related to authorship, bias, the impact on human artists, misinformation and accessibility. Concluding with a reflection on the future of human-AI collaborative creativity, the chapter envisions a landscape where human intuition and machine precision intertwine, potentially leading to a renaissance in creative expression and prompting a re-evaluation of creativity.

Chapter 11, "The Next-GenAI-Rtists: A New Generation of Artists with Generative Tools", explores the emergence and influence of Generative Artificial Intelligence (GenAI) within the realm of artistic creation, highlighting its transformative potential in shaping a new generation of creators known as 'Next-GenAI-rtists'. These artists redefine traditional practices by integrating GenAI as both a tool and a collaborative partner, fostering innovative co-creative processes. Employing the Hybrid Intelligence Technology Acceptance Model (HI-TAM), the chapter examines the psychological, technical and ethical dimensions of this integration, emphasising iterative learning, shared conceptual frameworks and the pivotal role of psychological safety. By juxtaposing GenAI's impact with historical shifts (such as photography's influence on art), the text situates the current revolution within a broader technological and cultural context. Practical examples of AI-generated art across various media underscore the democratisation and expanded accessibility GenAI offers, alongside its challenges to authorship and aesthetic norms. The chapter also delves into critical concerns regarding bias, transparency and intellectual property, advocating for responsible and ethical GenAI implementation. Concluding with an exploration of the evolving skill sets and interdisciplinary collaboration central to Next-GenAI-rtists, the chapter reimagines creativity as a dynamic interplay between human ingenuity and computational innovation in the digital age.

Chapter 12, "Augmented Creativity: Augmenting the Archaeological Imagination with Generative AI Tools", addresses the complexity of visual rendering of the past with software tools using Generative AI to fill in a large volume of data not found in the archaeological record, data that may come from ethnographic studies or archaeological experiments, as a result of a collaborative human-computer process using prompts. The interior of a Chalcolithic house (5th millennium BC) in Southeast Europe was reconstructed with traditional 3D modelling methods for the understanding of prehistoric daily life. In the virtual reconstructions, a series of objects generated by Generative AI tools based on publicly available ethnographic and experimental archaeological data (texts, images) were positioned as augmentations of virtuality. During this research, two commercial Generative AI tools, one of general purpose and one dedicated to image generation, were used for experimentation and evaluation. The results of the present research dedicated to the Augmented Creativity concept are shared through this chapter with the purpose of proposing the augmentation of the archaeological imagination and creativity with the use of Generative AI tools.

In Chap. 13, "Artist-Computer Collaboration and the Treachery of AI Images: This Pipe Does not Exist", the author embarks on a reflective journey through the evolving landscape of human-computer creativity, drawing on over 25 years of experience in digital arts. This essay explores the author's nuanced collaboration with artificial intelligence as a concept and technology, challenging the traditional view of computers as mere tools and highlighting the dynamic, symbiotic relationship that now exists. Through personal anecdotes, including the development of multimodal projects augmented by Generative Adversarial Networks (GANs) and Large Language Model (LLM) algorithms and through speculative collaborations with her avatar C.A.R.L.A. G.A.N. (Crossplatform Avatar for Recursive Life Action Generative Adversarial Network), the author examines the implications of AI in redefining creativity and the potential for a future where art transcends human and machine dichotomies. Addressing the intersection of AI with art, education and mixed reality, the essay advocates for a critical, intentional approach to AI collaboration, where the unique 'special sauce' of individual perspective and lived experience remains central to the creative process.

Chapter 14, "Disrupting Scholarly Writing Creatively Using Large Language Models", has been written to include generated commentaries from a large language model based on ideas that the authors present. It is written in this way to reveal three main features of large language models that also serve as creative methods through which to approach scholarly writing. The first is a response to the critique of many large language models that the content they generate is homogeneous or uniform. Identifying patterns of homogeneity in the content that large language models create might also help us identify similar patterns of uniformity within our own writing practices. The second section of the chapter explores the ability of a large language model to offer different viewpoints on the same subject, which reflects a key practice of many scholarly traditions that employ argumentation as a rhetorical tool. Finally, the way in which large language models generate content based on a corpus of data, which is composed of a multitude of sources written by many different authors, is similar to a long-established perspective that challenges the idea of a singular author when it comes to the composition and interpretation of any scholarly text.

Part IV, titled *Generative AI and Creativity in Healthcare*, includes five chapters (Chaps. 15–19).

Chapter 15, "Human Doctors Versus AI Models: Advice Perceptions in Serbia, Kazakhstan, and Kyrgyzstan", examines perceptions among individuals from Serbia, Kazakhstan and Kyrgyzstan regarding the trustworthiness, usefulness and empathy of medical advice provided by AI models compared to that offered by human doctors. A mixed-methods approach was employed, involving online surveys where 442 participants rated medical advice from three AI models (GPT-4o, Llama 3.1, Gemini Pro 1.5) and a real doctor across three medical scenarios. Participants evaluated each piece of advice on a 7-point Likert scale for trustworthiness, usefulness and empathy. They indicated whether they believed the advice was generated by AI or provided by a real doctor. Results showed that participants across all three countries rated advice from human doctors significantly higher in trustworthiness, usefulness and empathy than AI-generated advice. Familiarity with AI technology and trust in AI-generated information was highest in Kazakhstan, correlating with greater acceptance and usage of AI for health queries. Participants exhibited moderate ability to distinguish between AI and human advice, suggesting that AI models are advancing in producing human-like communication. The study concludes that although AI models show promise in generating medical advice, the human elements of trust and empathy remain indispensable in healthcare.

Chapter 16, "AI Technology in Psychological Education and Treatment: An Overview of Practical and Creative Applications", explores innovative applications of artificial intelligence technology in the education of psychology undergraduates as well as in psychological treatment. It begins by examining how generative AI can assist with specialised training and supervision for undergraduate students. The chapter discusses how practising interview techniques with AI-generated patients can help psychologists develop their professional communication skills. Additionally, it highlights the creative uses of AI in clinical practice, such as its potential to support diagnostic processes and enhance prevention efforts by identifying individuals who need mental health support in a timely manner. The chapter also provides an overview of the practical applications of AI in psychotherapy, including 'chatbot therapists' and text-to-image generation in art therapy. Finally, it addresses ethical considerations, risks and limitations associated with these technologies. Overall, this chapter focuses on the creative possibilities of generative AI in the fields of psychological education and treatment.

Chapter 17, "Generative AI Hallucinations in Healthcare: A Challenge for Prompt Engineering and Creativity", delves into the implications of generative AI hallucinations, particularly focusing on how they challenge the principles of prompt engineering and creative applications within the healthcare field. These hallucinations occur when AI systems generate information that appears plausible but is factually incorrect or unsupported by data. In healthcare, where accuracy and reliability are paramount, AI hallucinations can have severe consequences. Prompt engineering, the practice of designing effective prompts for AI models, plays a crucial role in mitigating the risks caused by AI hallucinations. The chapter surveys the best practices of prompt engineering in the context of healthcare (precision and clarity; ethical considerations and bias mitigation; iterative refinement and user feedback; incorporating multimodal inputs; human oversight), exploring its key principles and strategies to minimise AI hallucinations (incorporating constraints and boundaries; incorporating structured formats; providing contextual specificity; keeping a balance between innovation and reliability). Afterwards, the chapter explores a controversial relationship between AI hallucinations and creativity. It shows that AI hallucinations have been primarily associated with negative consequences. However, their potentially positive role deserves exploration in several healthcare areas, such as drug design, innovative treatment approaches, creative thinking stimulation, medical training and education and patient engagement.

Chapter 18, "Dentistry in the Digital Age: Exploring Human–Computer Creativity in Dental Education and Practice", explores the intersection of dentistry and educational technologies, focusing on the integration of generative artificial intelligence into dental practice and pedagogy. With the advent of innovative technologies, dentistry has undergone transformative advancements, leading to improved patient care and enhanced learning experiences for dental students. Through the lens of human-computer creativity, the chapter delves into the use of AI-powered tools for tasks ranging from diagnostics and treatment planning to the simulation of dental procedures in virtual environments. Furthermore, it examines the role of artificial intelligence in revolutionising dental education, offering personalised learning pathways, virtual patient encounters and immersive simulations that complement traditional teaching methodologies. By embracing the synergy between human expertise and computational capabilities, dentistry is entering a new era of innovation, where creativity thrives in a symbiotic relationship between humans and intelligent machines.

Chapter 19, "Generative AI with Quantum Computing for Visualizing Proteins in Mixed Reality", explores the transformative role of cutting-edge technologies (quantum computing, generative AI and mixed reality) in overcoming the challenges of simulating protein folding. Protein folding, a fundamental process in systems biology, is central to understanding cellular mechanisms, drug discovery and disease treatment. However, simulating protein folding at an atomic level remains computationally challenging due to the complexity of protein structures and the vastness of their conformational space. This chapter aims to provide a comprehensive overview of how integrating quantum computing and mixed reality reshapes protein folding research, opening new avenues for scientific exploration and breakthroughs in the life sciences.

Last but not least, we hope that readers will not judge the book's editor and contributors too harshly. We have accepted the challenge of being the first, and we have done our best to bring this pioneering work on human-computer creativity in the era of generative AI. Just go ahead and read this book. We sincerely hope that you will enjoy it.

Cairo, Egypt Vladimir Geroimenko

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Part I
Conceptual Facets of Generative AI and Human-Computer Creativity

Chapter 1
Generative AI: From Human–Computer Interaction to Human–Computer Creativity

Vladimir Geroimenko

Abstract This chapter explores the transformative shift in human–computer relationships driven by generative artificial intelligence (AI). Moving beyond traditional Human–Computer Interaction (HCI) frameworks focused on task completion and efficiency, this chapter delves into Human–Computer Creativity (HCC), where AI systems act as co-creators rather than passive tools. Through advancements in generative models, AI enables a collaborative environment that fosters creativity in diverse fields, including education, art, and healthcare. By examining the historical progression of HCI, the capabilities of generative AI, and its applications, the chapter illuminates the paradigm shift from interaction to co-creation. Key themes include the evolving roles of humans and AI, ethical considerations, and the comparative analysis of interaction-based and creativity-based human–computer models. This shift highlights the potential of generative AI to reshape our understanding of creativity, challenging traditional notions of authorship and originality. In sum, the chapter provides a comprehensive analysis of how generative AI enhances and redefines human creativity, offering insights into the future of human–computer collaboration.

Keywords Human-computer creativity · Human-computer interaction · Generative AI in education · Generative AI in art · Generative AI in healthcare

1.1 Introduction: The Dawn of Human–Computer Creativity

In the early years of computing, the relationship between humans and machines was largely transactional. Human–Computer Interaction (HCI) was focused on creating intuitive, efficient systems to allow users to accomplish specific tasks, from word processing to data analysis (Rogers et al. 2023; Lazar et al. 2017; MacKenzie 2012; Dix et al. 2003). This interaction was bound by structured commands and a clear

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divide between human intention and machine execution. But as technology evolved, so did the boundaries of this relationship. The arrival of artificial intelligence (AI) and, more recently, generative AI has introduced a new era that transcends interaction and opens a frontier of shared creativity between humans and machines.

Generative AI is a transformative leap in AI capabilities, shifting from merely responding to user commands to producing original content and ideas (Ishwarya et al. 2024; Ivcevic and Grandinetti 2024; Rafner et al. 2023). Unlike traditional AI, which focuses on automation and predictive tasks, generative AI actively engages with the user to co-create, envision, and expand upon human input (Vinchon et al. 2023). This transition from interaction to creative collaboration represents a radical departure from conventional HCI paradigms, embedding creativity directly into the human–computer relationship (Mackay 2023).

Creativity is at the heart of human endeavour, shaping fields as varied as education, art, and healthcare (Esling and Devis 2020; Farina et al. 2024; Tredinnick and Laybats 2023). In education, generative AI has become a tool for interactive learning experiences, helping students explore complex ideas through personalised narratives and adaptive content. In the arts, it acts as both muse and medium, facilitating new forms of expression that blend human ingenuity with algorithmic power. In healthcare, generative models transform diagnostics, patient communication, and therapeutic interventions, offering novel ways to understand and support the human experience.

This chapter explores the evolution from traditional human–computer interaction to this burgeoning field of human–computer creativity. By examining the capabilities and implications of generative AI, we uncover a paradigm shift that redefines what it means for humans and machines to work together—not merely as users and tools but as partners in creation. The chapter delves into the transformative impact of generative AI on the landscape of human–computer interaction. We will explore the evolution of HCI, the emergence of generative AI, and the significance of creativity in various domains. Our primary focus will be transitioning from traditional human–computer interaction to the emerging paradigm of human–computer creativity. By examining this paradigm shift's theoretical underpinnings, practical applications, and ethical considerations, we aim to shed light on the future of human–computer collaboration and its potential to reshape our world.

1.2 Human–Computer Interaction: Key Concepts, Principles, and Milestones

Human–computer interaction has bridged the cognitive, sensory, and operational gaps between humans and machines. This area of study emerged in the mid-twentieth century with the birth of computer science, but its philosophical roots trace back to early contemplations about mechanised intelligence. Its practical roots align with developments in usability during World War II. Each milestone has brought HCI closer to a seamless, intuitive human–machine connection, from command-line interfaces to graphical user interfaces (GUIs) and now to voice and gesture controls.

1.2.1 Historical Context of HCI: From Tools to Interfaces

The 1960s marked the earliest formal studies in HCI, propelled by pioneers like Douglas Engelbart, whose mouse development and early GUI concepts offered a glimpse into the potential of accessible computing. By the 1980s, HCI had grown as an interdisciplinary field, incorporating psychology, design, and engineering principles to make computers more accessible to the general public. This era introduced structured methods in usability testing and emphasised user-centred design, setting the stage for the following HCI revolution.

Key milestones punctuated this trajectory. The introduction of the Xerox Star workstation in 1981 showcased the first commercial GUI, changing the perception of computers from specialised tools to devices for everyday interaction. Apple's Macintosh launched in 1984, and Windows' ascent in the early 1990s further popularised intuitive, visual interfaces, solidifying HCI as critical to technology adoption and success.

1.2.2 Key Concepts, Principles, and Methodologies of HCI

HCI hinges on several foundational principles: usability, accessibility, efficiency, and satisfaction. Usability prioritises ease of learning and operation, while accessibility emphasises inclusivity for diverse user abilities. Efficiency seeks to reduce cognitive load, enabling users to achieve tasks quickly, and satisfaction relates to the emotional experience, turning interaction into an engaging activity rather than a burdensome task.

Methodologies in HCI evolve from these principles. User-centred design (UCD) has remained central, advocating iterative development with continuous feedback loops from users. Task analysis and usability testing are vital methodologies, dissecting user needs, and refining interfaces to maximise productivity and reduce error.

1.2.3 Issues in Traditional HCI

While traditional HCI frameworks advanced the accessibility and usability of computing devices, they also encountered limitations. The rigid models often required users to adapt to fixed workflows and input methods, which limited spontaneous creativity. Moreover, as HCI focused on task efficiency and minimal friction, it left little room for subjective or creative experiences. These challenges highlighted a gap for more adaptive, intelligent, and personalised systems that could better anticipate diverse user needs and even augment human capabilities.

1.2.4 The Rise of AI in Human–Computer Interaction

The integration of Artificial Intelligence into HCI redefined this landscape. Beginning with basic automation features and expanding into machine learning-driven personalisation, AI started enhancing interactions with predictive text, voice assistants, and recommendation engines. AI added layers of depth and nuance to HCI, allowing interfaces to learn from user behaviours, make informed predictions, and even support complex decisions.

Incorporating AI in HCI also accelerated a shift from deterministic interactions to probabilistic ones. Instead of systems rigidly awaiting user commands, AI-enabled interfaces now engage in adaptive, context-sensitive exchanges, allowing HCI to evolve beyond pre-defined pathways and towards a fluid, collaborative dynamic.

1.2.5 From Essential Interaction to Creative Collaboration

With the advent of Generative AI, HCI is entering an era where the interaction is not just a functional exchange but a collaborative, creative process. Generative AI tools for art, music, writing, and design showcase a new collaborative frontier where machines no longer assist but actively participate in creative endeavours. This shift marks a transition from human-centred interaction to a form of co-creation, where both human intention and machine innovation combine to produce novel outputs.

For instance, AI can now assist in creative tasks such as writing, composing music, and designing graphics, opening new avenues for human creativity. This transition is exemplified by tools like GPT-3, which can generate human-like text, and DALL-E, which can create images from textual descriptions. These advancements empower users to collaborate with AI in novel and imaginative ways, transforming HCI from a functional interaction to a dynamic, creative partnership.

The evolution from rigid command structures to creative partnerships represents a technical achievement and a fundamental reimagining of the human–computer relationship. As we move forward, the principles that guided traditional HCI must evolve to encompass this new paradigm of human–computer creativity. This shift demands a careful balance between leveraging the powerful capabilities of AI while maintaining meaningful human agency and ensuring that technology enhances rather than replaces human creative expression.

Thus, the foundations of HCI, rooted in making technology accessible and intuitive, are evolving. Today, HCI is not just about bridging the functional gaps between humans and machines but amplifying creative potential in ways that are only beginning to be explored. The next chapter in HCI unfolds in this transition, where human–computer creativity becomes a transformative force across industries and personal endeavours.

1.3 Emergence of Generative AI: Moving Beyond Interaction to Creation

Generative AI marks a profound evolution in artificial intelligence, distinct from traditional AI approaches that typically focus on analysing data, predicting outcomes, and following programmed instructions. While conventional AI centres on problem-solving within preset boundaries, generative AI goes further—its purpose is to *create* rather than merely *compute*. Generative AI models synthesise new content, producing text, images, music, and more by learning from vast datasets and applying what they have learned in innovative ways. This transformative shift positions generative AI as not just an analytical tool but a creative force capable of expanding human imagination and productivity.

1.3.1 Historical Context and Technological Milestones

Critical advancements in neural networks, computing power, and access to large datasets have paved the journey towards generative AI. In the early days of AI, models were limited by both data and processing constraints. Still, innovations in the 2010s—like the development of Generative Adversarial Networks (GANs) and the Transformer model—marked major milestones that made generative AI feasible. GANs allowed for the generation of high-quality synthetic images. At the same time, Transformers introduced a new paradigm for handling sequential data, especially in language, enabling the creation of complex, contextually aware responses. These models could "understand" patterns deeply enough to produce output resembling human creativity.

1.3.2 Key Concepts and Technologies in Generative AI

Generative AI encompasses several innovative technologies, including GANs, Variational Autoencoders (VAEs), and the Transformer architecture. Each of these models plays a role in enabling AI to produce novel, coherent outputs. GANs leverage a unique training structure where two models—one generating content and the other evaluating it—engage in a feedback loop, refining outputs to increasingly realistic levels. On the other hand, transformers rely on attention mechanisms to weigh the importance of different parts of input data, giving them a profound grasp of context and enabling applications like OpenAI's GPT, Google's BERT, and other generative language models.

1.3.3 Creative Applications of Generative AI

Today, generative AI is used to create art, compose music, write stories, design products, and even assist architectural planning. For example, DALL-E can generate unique images from textual descriptions, challenging the boundaries of visual art. In music, tools like Jukedeck or OpenAI's MuseNet compose original pieces in various genres. AI assists in drafting narratives or poems in literature, supporting novelists and poets with ideas and text completion. These applications are transforming creative domains by offering new avenues for expression and assisting creators in developing unique styles.

1.3.4 Enhancing Interaction and Creativity Through Generative AI

Generative AI fundamentally changes how users interact with technology. Instead of merely using the software as a tool to execute predefined tasks, users can engage AI in a more interactive, iterative process, providing input, reviewing output, and refining the result in real time. This dialogue fosters creativity, as AI's rapid generation capabilities let users explore ideas freely and push boundaries. For instance, graphic designers can generate multiple prototypes quickly, modifying parameters until the design aligns with their vision. Writers and musicians use AI similarly as collaborators, sparking creativity with suggestions and new directions they might not have considered.

1.3.5 A Shift from Passive Tools to Creative Collaborators

This paradigm shift—from passive tools to active, creative collaborators—redefines the human–computer relationship. Where traditional tools require users to have a vision and direct each step, generative AI models contribute to the creative process, presenting themselves as *partners* rather than mere instruments. They inspire new ideas, adapt to user feedback, and co-create in real time. This evolution is not just a technological leap but a reimagining of creativity and collaboration. By transcending the boundaries of human interaction, generative AI empowers users to explore the uncharted territory of what humans and machines can create together.

This shift changes the nature of creative work from a solitary human activity supported by passive tools to a collaborative process where human and machine intelligence combine to explore new creative possibilities. The computer evolves from being an instrument of execution to becoming a partner in creation, leading to new workflows, methodologies, and innovative opportunities.

The relationship becomes more conversational and collaborative, with humans providing high-level direction and creative judgment while the AI system contributes technical execution and creative variations. This partnership leverages the complementary strengths of human and artificial intelligence: human creativity, judgment, and intentionality combined with the machine's ability to generate and explore variations while maintaining technical precision rapidly.

In essence, generative AI blurs the line between human creativity and computational capability, opening doors to a future where human–computer creativity is not only possible but profoundly collaborative and transformative.

1.4 Transition to Human–Computer Creativity: A Paradigm Shift

1.4.1 Creativity Versus Interaction: A Comparative Analysis

The emergence of generative AI has fundamentally transformed the relationship between humans and computers, evolving it from mere interaction to genuine co-creation(Caporusso 2023; Orwig 2024). To understand this paradigm shift, we must first examine the distinct characteristics of both interactivity and creativity in human–computer collaboration and then analyse how these elements intersect and diverge in contemporary AI systems.

Understanding Interactivity in Human–Computer Collaboration

Interactivity in human–computer systems traditionally follows a stimulus–response model, where each party takes turns providing input and generating output. This relationship is fundamentally transactional: humans input commands, queries, or data, and computers process and return results based on predetermined algorithms and protocols. Traditional interactive systems are characterised by their sequential, turn-based nature, with a strong orientation towards specific goals and tasks. These systems operate within clearly defined parameters, focusing primarily on information exchange and task completion. The outcomes remain largely predictable within system constraints.

In traditional interactive systems, the computer serves primarily as a tool or interface, extending human capabilities while remaining subordinate to human direction.

The interaction follows established patterns and protocols, with the computer's role being largely reactive rather than proactive.

The Nature of Creativity in Human–Computer Collaboration

Creativity, in contrast to mere interaction, involves the generation of novel and valuable ideas or artefacts. In the context of human–computer collaboration, creativity manifests through the generation of unexpected or original outputs coupled with the capacity to forge novel connections between disparate concepts (Atkinson and Barker 2023; Sawyer and Henriksen 2023). This creative potential enables systems to transcend established patterns and rules, generating multiple viable solutions to open-ended problems. Perhaps most importantly, it facilitates serendipitous discoveries through open-ended exploration.

When applied to human–computer systems, creativity introduces an element of unpredictability and emergence that transcends traditional interactive paradigms. The computer becomes more than a tool; it becomes a collaborative partner in the creative process, capable of contributing novel ideas and perspectives.

Distinguishing Interaction-Based and Creation-Based Models

The fundamental differences between interaction-based and creation-based models of human–computer collaboration can be analysed across several key dimensions:

Agency and Initiative: In interaction-based models, computers primarily respond to human inputs, operating with limited autonomous action within predetermined response patterns. A clear hierarchy exists, with humans maintaining controller status. Creation-based models, however, enable computers to initiate ideas and suggestions, engaging in autonomous exploration of possibilities. This results in more emergent behaviour and responses, fostering a more equitable partnership dynamic between humans and machines.

Outcome Predictability: Interaction-based models offer high predictability of outcomes, with results falling within expected parameters and clear correlations between inputs and outputs. These systems produce limited variation in responses. In contrast, creation-based models embrace lower predictability, allowing for unexpected results and maintaining complex relationships between inputs and outputs. This approach generates significant variation in responses, leading to more diverse and innovative outcomes.

Problem-Solving Approach: Traditional interaction-based models employ linear problem-solving pathways, emphasising efficiency and optimisation. Solutions typically derive from existing patterns, with a strong focus on correctness and accuracy. Creation-based models, alternatively, engage in non-linear exploration of solutions, prioritising novelty and innovation. These systems generate new patterns and approaches, placing greater emphasis on originality and value creation.

Learning and Adaptation: Interaction-based models operate with static rule sets and protocols, offering limited learning capabilities from interactions. Adaptation occurs primarily through human modification, resulting in consistent performance over time. Creation-based models, by contrast, demonstrate the dynamic evolution of capabilities, learning from creative exchanges and adapting through experience. This leads to improving performance over time as the system accumulates knowledge and refines its creative processes.

The Convergence of Interaction and Creativity

While the distinction between interaction and creativity is helpful for analytical purposes, modern AI systems increasingly blur these boundaries. Advanced generative AI systems exhibit characteristics of both paradigms, creating a hybrid model that combines structured interaction with creative innovation. Interactive elements provide the necessary framework and control, while creative elements introduce novelty and innovation. This synthesis enables guided creativity within a manageable framework, fostering a more dynamic and fluid relationship between humans and machines.

This convergence suggests a new model of human–computer collaboration that transcends the traditional dichotomy between interaction and creativity. In this emerging paradigm, computers serve as both responsive tools and creative partners, adapting their role based on context and need. The future of human–computer collaboration lies not in choosing between interaction and creativity but in understanding how to leverage both aspects to achieve optimal outcomes in various contexts.

1.4.2 Human–Computer Creativity as a New Paradigm

In the landscape of technological evolution, few shifts hold as much potential as the transition from human–computer interaction (HCI) to human–computer creativity (HCC). While HCI has long focused on making technology responsive to human needs and facilitating user-friendly interactions, HCC redefines the purpose of this relationship. No longer is the goal to enable machines to understand and respond; instead, we now aim for them to contribute to creation actively. HCC elevates technology from a tool for efficiency to a partner in imagination, opening doors to new realms of innovation and creative collaboration.

From Interaction to Creation: **Major Paradigmatic Changes**

The shift from HCI to HCC is not merely incremental but paradigmatic, altering core assumptions about the role of technology in human life. Where HCI focuses on seamless, intuitive interaction, HCC introduces the concept of machines as co-creators, contributing insights, novel ideas, and creative outputs that augment human creativity. This shift involves three significant changes:

Enhanced Agency and Autonomy: In HCC, technology is no longer limited to following user instructions. Generative AI can autonomously propose ideas, evolve designs, and even generate artworks or solutions without explicit step-by-step guidance. This new agency enables AI to introduce creative directions that may diverge from human intention yet align with the broader goals of the creative process.

Collaborative Adaptivity: Rather than simply reacting to human inputs, generative systems in HCC dynamically adapt and respond to the nuances of human creativity. This results in a synergistic environment where human ideas inspire AI-driven expansions while AI suggestions spark new human insights. This reciprocity transforms the computer from a passive responder to an active, creative participant, fostering a feedback loop that enriches the creative process.

Expansion of Creative Capacity: HCC empowers humans to explore creative terrains beyond conventional limitations. Through collaboration with AI, creators can experiment with vast permutations of design, narrative, or musical composition in real time, discovering novel concepts that would be impractical or impossible to produce individually. This shift fundamentally redefines creativity by leveraging the unique capabilities of AI to complement and amplify human strengths.

Illustrative Examples Across Domains

The shift to HCC is already making waves across various fields, from the arts to scientific research. Let's explore a few examples that highlight the transformative impact of this new paradigm:

Visual Arts: In fields like graphic design and digital art, generative AI tools such as DALL-E and Midjourney are now familiar collaborators, working with artists to generate original visuals based on textual prompts. These tools not only follow instructions but often introduce unanticipated visual elements, textures, or styles, pushing artists to consider directions they might not have previously envisioned.

Music Composition: AI-powered music systems like AIVA and Amper Music have evolved from simply assisting with note sequencing to generating entire compositions based on a desired mood, genre, or theme. Composers can now engage these tools to explore new harmonies or experimental sounds, blending human-inspired emotion with AI-driven innovation in ways that redefine musical creation.

Literature and Narrative: In storytelling and game development, AI-driven tools assist writers by generating plot ideas, character arcs, and even dialogue options. These models serve as co-authors, helping human writers expand the narrative boundaries of their stories. The AI's unpredictable contributions often serve to break traditional narrative patterns, pushing writers towards more dynamic, multi-dimensional storytelling.

Scientific Discovery: Generative models are not limited to the creative arts; they also play an increasingly significant role in scientific research. In fields like pharmacology and materials science, AI systems such as DeepMind's AlphaFold or OpenAI's Codex analyse complex data to suggest potential molecular structures or code sequences that human researchers might overlook. This partnership accelerates discovery and enables scientists to tackle challenges previously constrained by computational or cognitive limits.

Education: In the educational realm, generative AI is opening new avenues for personalised and interactive learning. Intelligent tutoring systems no longer merely provide answers; they now engage students in dialogue, create tailored learning pathways, and even generate original educational content, such as customised examples and exercises. This collaboration fosters a more profound, more immersive learning experience, empowering students to engage with complex topics and develop creative problem-solving skills actively.

Healthcare: In healthcare, HCI applications include patient data management systems that streamline interactions between medical professionals and electronic health records. With HCC, AI assists in the creative aspects of medical research and diagnostics. For instance, AI can generate hypotheses for new treatments by analysing vast amounts of medical data, or it can help design personalised treatment plans tailored to individual patients' genetic profiles and health histories.

The Road Ahead: **Embracing Human–Computer Creativity**

As we embrace HCC, we find ourselves at the forefront of a new era. This paradigm shift signifies more than technological advancement; it reflects an evolving philosophy regarding the role of AI in human society. By positioning technology as a creative partner rather than a passive tool, we are expanding the very boundaries of human potential. As HCC becomes integral to our daily lives, it will reshape how we view creativity itself—not as a uniquely human trait but as a shared endeavour in which humans and machines collaboratively explore, innovate, and inspire.

This new paradigm invites us to reconsider not only what technology can do but also what creativity can become in a world where machines contribute actively to the process. The future of HCC promises a radical reimagining of art, science, and human potential, offering a glimpse into a world where creation knows no bounds.

1.4.3 Challenging and Controversial Issues in Human–Computer Creativity

As human–computer creativity advances, it brings a host of complex issues that stir debate within both the technological and creative communities. While generative AI has the potential to inspire and amplify human ingenuity, the interplay between machine-generated content and human intent raises fundamental questions about creativity, authorship, ethics, and the very nature of what it means to be "creative."

The Definition of Creativity: Human-Centric or Machine-Enhanced?

Creativity has traditionally been seen as an inherently human trait driven by emotions, intuition, and personal experience. The emergence of generative AI has sparked a critical discussion: Can machines genuinely be "creative," or are they merely mimicking creativity? Sceptics argue that machines lack the consciousness and experiential basis essential for creativity, simply generating novel combinations based on patterns in data. Advocates counter that creativity need not be limited to human experience and that machines can help redefine creativity by offering new forms and patterns unimaginable by human minds alone (Cropley 2023; Magni et al. 2024; Sternberg and Kaufman 2018). This debate challenges us to reconsider creativity itself and whether it can genuinely thrive in non-human agents.

Impact on Human Creativity

Another contentious issue is the potential impact of AI on human creativity. There is a growing concern that as AI becomes more adept at generating creative works, it may overshadow human artists and creators. This raises questions about the future of human creativity: Will we become overly reliant on AI to generate new ideas and solutions, potentially stifling our creative abilities? Or will AI serve as a catalyst, pushing us to explore new horizons and enhancing our creative potential? The balance between augmentation and replacement is a delicate one, and navigating this landscape requires careful consideration of how we integrate AI into our creative processes.

Ethical Implications of AI-Created Content

The accessibility and proliferation of generative tools have democratised creative expression, but they also open the door to ethical dilemmas. AI can replicate and amplify human biases present in its training data, leading to works that may unintentionally perpetuate stereotypes or misrepresent marginalised groups. Additionally, AI-created content can be manipulated to spread misinformation or create fake media that appear authentic. As generative AI blurs lines between reality and fabrication, society must grapple with the ethical responsibilities of creators and developers to mitigate these risks.

Dependence on AI: Erosion of Human Skill and Creativity?

Another controversial issue is whether reliance on AI diminishes the need for traditional creative skills. Critics worry that as artists, writers, and musicians lean on generative AI, foundational skills in art, storytelling, and music composition may erode. Furthermore, the instant availability of AI-generated material risks turning creativity into a transactional activity, where the process of crafting is devalued in favour of efficiency. Balancing the convenience of AI assistance with the preservation of human artistry and craftsmanship is an ongoing challenge.

Authorship and Intellectual Ownership

In human–computer collaborations, authorship becomes a contentious topic. Who owns the creative output—a human user who prompts the AI or the creators of the algorithms that make the generation possible? Traditional frameworks for intellectual property struggle to accommodate these ambiguous creative processes, leading to ongoing legal debates. The question becomes even more intricate in cases where AI-generated works replicate recognisable styles or emulate known artists. As AI-generated works flood markets, finding equitable ways to assign authorship remains a significant challenge.

Algorithmic Influence and Standardisation of Aesthetic

As more creators turn to generative models for inspiration, there is a risk that AI's inherent patterns and biases will shape aesthetic norms, potentially leading to homogenisation in art and design. Algorithms trained on existing data are limited by the creativity present in their training datasets. This could stifle originality if creators unwittingly adopt algorithm-driven trends, creating a feedback loop that narrows the diversity of creative expression over time. This concern raises the question of how to cultivate AI models that challenge rather than conform to existing standards.

The Psychological Impact of Human-AI Co-Creation

Working with AI as a "co-creator" introduces psychological complexities. For some creators, the prospect of collaborating with an emotionless algorithm is unsettling, while others find that it boosts their creativity by offering non-judgmental, boundary-breaking ideas. However, relying on AI for creative work may lead to feelings of inadequacy, as individuals question the value of their contributions relative to the machine. The social and psychological impact of creating alongside AI highlights the need for research on how these dynamics affect human motivation and self-worth.

Cultural and Societal Impact of AI-Created Content

As AI-created media proliferates, it influences culture and society in unforeseen ways. AI-generated stories, art, and music could inadvertently shape collective identities, cultural values, and norms. This raises questions about cultural appropriation, as generative models trained on diverse sources may mix, reinterpret, and reproduce cultural elements without sensitivity to their meaning or significance. There is a risk that AI could dilute or misrepresent cultural artefacts, posing a challenge for societies to preserve artistic integrity in an age of digital reproduction.

Looking Forward

In conclusion, while the shift from human–computer interaction to human–computer creativity holds tremendous promise, it also brings with it a host of challenges and controversies. Addressing these issues requires a multidisciplinary approach, bringing together technologists, artists, ethicists, and policymakers to navigate the complex landscape of HCC. By doing so, we can harness the potential of generative AI while safeguarding the values and principles that underpin human creativity.

These challenges demand careful consideration as we shape the future of human–computer creativity. Solutions may require new frameworks that:

- Recognise and protect both human and machine contributions to creative works
- Ensure ethical AI training practices that respect cultural heritage
- Support sustainable creative ecosystems that benefit both human creators and technological innovation
- Maintain space for purely human creative expression alongside AI-augmented work

The resolution of these issues will significantly influence not just the future of creative practice but also our understanding of creativity itself and what it means to be human in an age of artificial intelligence.

1.5 Generative AI in Education: Transforming Learning and Teaching

The advent of generative AI can transform education, reshaping how learning is facilitated, and creativity is cultivated in both students and educators (Resnick 2024). By transcending the traditional teaching boundaries, generative AI offers innovative tools that foster personalised learning experiences, enable adaptive content creation, and encourage student-centred approaches to knowledge acquisition and problem-solving.

1.5.1 Enhancing Student Creativity

Generative AI empowers students to explore their creative potential in unprecedented ways (Habib et al. 2024; Nam et al. 2024; Creely 2022). Students can experiment with different forms of expression and innovation by providing tools that can generate text, images, music, and code. For instance, AI-driven writing assistants help students brainstorm ideas, draft essays, and refine their writing, encouraging them to think more critically and creatively. Similarly, AI art generators allow students to visualise concepts and create digital art, enhancing their understanding and engagement with the material.

Generative AI tools are revolutionising how students engage with creative tasks across disciplines. In literature classes, students can collaborate with AI to explore different writing styles, generate story variations, or analyse narrative structures from multiple perspectives. Art students can use AI as a creative partner, experimenting with various artistic styles and techniques while developing their unique voices. This human-AI creative partnership does not replace student creativity but amplifies it, providing new avenues for expression and experimentation.

1.5.2 Fostering Critical Thinking and Problem-Solving

One of the most profound impacts of generative AI in education is its capacity to inspire critical thinking. By generating open-ended questions, AI models encourage students to explore various solutions, promoting curiosity and innovation. In creative subjects, such as literature, music, and visual arts, generative AI can serve as a tool for creativity and experimentation, enabling students to co-create with the AI, thereby blurring the lines between creator and observer. This partnership allows learners to expand their creative potential as they experiment with new forms of expression and challenge conventional boundaries in ways previously unattainable.

Rather than diminishing critical thinking, generative AI enhances it by presenting multiple perspectives on complex topics. It challenges students to evaluate AI-generated content critically while encouraging the exploration of "what-if" scenarios and supporting iterative problem-solving processes. Through this engagement, students learn to become discerning consumers and creators, developing the crucial skill of distinguishing between valuable AI assistance and over-reliance on automated solutions.

1.5.3 Personalised Learning Pathways

Generative AI has made personalised learning more achievable by allowing educators to tailor educational content to meet the individual needs of each student. Through adaptive AI algorithms, students receive guidance tailored to their learning pace, preferences, and areas of struggle. Generative AI systems can produce customised quizzes, interactive simulations, and adaptive reading material, ensuring each learner encounters a pathway that aligns with their unique learning style. Such personalisation not only enhances engagement but also cultivates an environment where students can thrive, fostering a deeper connection to the material and encouraging self-driven exploration.

1.5.4 Real-Time Feedback and Support

Generative AI provides real-time, constructive feedback on assignments, projects, and assessments, essential for continuous improvement. By analysing students' responses and understanding the nuances of their mistakes, AI can offer specific and relevant guidance to each learner. This immediate feedback loop allows students to correct errors promptly, reinforcing their understanding and encouraging iterative learning. Moreover, AI-powered tools can simulate tutoring by guiding students through complex problems, providing explanations, and recommending resources to strengthen their grasp of complex concepts.

1.5.5 Transforming the Teacher's Role

Educators are experiencing a profound shift in their role, evolving from primary knowledge providers to creative facilitators and mentors. This transformation enables teachers to harness AI to generate diverse teaching materials and assessment tools, allowing them to focus more intensively on developing students' higher-order thinking skills. Teachers now guide students in effectively utilising AI as a creative tool while designing more engaging and interactive learning experiences. This evolution requires educators to develop new competencies in AI literacy while maintaining their crucial role in nurturing human creativity and critical thinking.

For educators, generative AI offers valuable support by automating time-consuming tasks, such as lesson planning, resource creation, and assessment grading. With AI-driven tools, educators can create engaging content, from interactive games to virtual labs, without the need for extensive technical expertise. Generative AI enables teachers to devote more time to high-impact activities, like individual mentorship and curriculum enhancement, thus enriching the educational experience. By alleviating administrative burdens, AI empowers teachers to focus on fostering relationships with students, which is fundamental to successful learning environments.

Generative AI also transforms teaching methods by providing educators with new tools and resources. Teachers can use AI to create interactive lesson plans, develop multimedia presentations, and design virtual simulations that make learning more dynamic and immersive. AI can assist in grading and providing feedback, allowing teachers to focus more on interactive and creative aspects of teaching. Additionally, AI-driven platforms can facilitate collaborative projects where students and teachers co-create content, fostering a more participatory and creative classroom environment.

1.5.6 Ethical Considerations and Digital Literacy

The integration of generative AI in education necessitates careful consideration of ethical implications and the development of new forms of digital literacy. Students must master understanding the capabilities and limitations of AI systems while learning to navigate issues of attribution and originality. The development of responsible AI usage practices becomes paramount, as does finding the right balance between AI assistance and independent thinking. Educational institutions must establish clear guidelines for AI use while fostering an environment that encourages creative exploration and innovation.

1.5.7 Bridging Gaps in Education

Generative AI has the potential to bridge gaps in education by providing access to high-quality resources and learning opportunities regardless of geographical or socio-economic barriers. AI-powered educational platforms can offer courses and materials in multiple languages, making education more inclusive and accessible. Furthermore, AI can support students with disabilities by generating content in accessible formats, such as audio descriptions for visually impaired students or simplified text for those with learning difficulties.

1.5.8 Future Prospects

The integration of generative AI in education is still in its early stages, but its potential is vast. AI technologies will likely become even more integral to educational practices, driving further innovation and creativity. Future developments may include more sophisticated AI tutors, immersive virtual reality learning environments, and AI-driven collaborative tools that enhance both individual and group learning experiences.

The relationship between generative AI and educational creativity will likely continue to evolve. We can anticipate greater integration of AI-powered creative tools across curricula, alongside the development of hybrid learning approaches that combine human and AI creativity. The emphasis on developing “AI-native” creative skills will increase, accompanied by new assessment methods for AI-assisted creative processes. The key to successful implementation lies in maintaining a balance between technological innovation and human-centred education, ensuring that AI enhances rather than replaces human creativity in learning.

In summary, generative AI holds transformative potential for education, enabling a shift from passive learning to an active, engaging, and personalised experience. By harnessing the power of generative AI, educational institutions can create an ecosystem that not only imparts knowledge but also nurtures creativity, critical thinking, and a lifelong passion for learning. As educators, students, and technologists work together to integrate AI into classrooms, the future of learning stands at the threshold of a new era where creativity and technology converge to redefine the educational landscape.

1.6 Generative AI in Art: Expanding the Boundaries of Artistic Expression

Generative AI has revolutionised the art world by introducing tools that expand traditional creative practices, enabling artists to explore new forms, styles, and techniques (Zhou and Lee 2024; Anantrasirichai and Bull 2022). By bridging the gap between computational power and human imagination, generative AI pushes the boundaries of artistic expression, challenging conventional ideas of creativity and authorship. Generative AI has revolutionised the art world, pushing the boundaries of what is possible and redefining the relationship between human creativity and machine intelligence. By leveraging advanced algorithms and neural networks, generative AI can create art that is not only innovative but also deeply expressive, often blurring the lines between human and machine-generated works.

1.6.1 New Aesthetic Frontiers

The concept of generative art is not new; it dates to the 1960s when artists began experimenting with computer algorithms to produce visual art. However, the advent of deep learning and neural networks has significantly accelerated the capabilities of generative AI. Modern generative models, such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), have enabled the creation of highly sophisticated and intricate artworks that were previously unimaginable.

The unique capabilities of AI systems have given rise to novel aesthetic categories and artistic styles that would be difficult or impossible to achieve through traditional means. These systems blend disparate artistic styles and influences while maintaining coherence across generated works. They demonstrate a remarkable ability to generate complex variations on themes, often creating entirely new visual languages by identifying and recombining patterns from their training data. Perhaps most notably, AI systems can produce works at scales and levels of detail that would be impractical for human artists working alone.

Another profound impact of generative AI in art is its capacity to simulate, reinterpret, and merge artistic styles across diverse genres and cultural influences. Advanced neural networks trained on vast art datasets can mimic famous styles—such as impressionism, surrealism, and modernism—or generate hybrid compositions that blend characteristics from multiple artistic movements. These AI-driven creations not only evoke emotions and provoke thought but also expand the aesthetic possibilities available to contemporary artists.

1.6.2 Democratising Art Creation and Empowering New Creators

One of the most significant impacts of generative AI in art is its democratising effect. AI tools are becoming increasingly accessible, allowing a broader audience to engage in the creative process. Individuals without formal art training can use generative AI to create compelling artworks, thus broadening the scope of artistic expression and participation. This accessibility has contributed to a cultural shift that values inclusivity and diversity in art, enabling creators from various backgrounds to participate and contribute to a shared, evolving artistic landscape. Tools like DALL-E, Midjourney, and Stable Diffusion have introduced a new paradigm of “prompt-based creativity,” where artistic creation becomes a collaborative dialogue between human intention and machine interpretation. This democratisation has led to an explosion of creative expression, with millions of users worldwide exploring new artistic possibilities previously beyond their technical reach.

1.6.3 Collaborative Human-AI Creation and Redefining Creativity

Rather than replacing human artists, generative AI has emerged as a powerful collaborative tool that augments human creativity (Eapen et al. 2023). Professional artists increasingly incorporate AI into their workflows as a means of rapid prototyping and iteration of concepts. The technology is an invaluable tool for exploring unexpected creative directions and overcoming creative blocks by generating novel variations. By handling many technical aspects of production, AI frees artists to focus on higher-level creative decisions. This collaborative approach has created new artistic methodologies where the boundary between human and machine contributions becomes increasingly fluid and synergistic.

Generative AI not only changes the means of artistic production but also prompts us to rethink the concept of creativity itself. In the collaborative process between artist and machine, AI challenges traditional notions of authorship and originality, raising questions about the evolving role of technology in human creativity. As AI becomes a co-creator, it invites artists and audiences alike to explore the boundaries of human–machine collaboration, ultimately enriching our understanding of what it means to be creative in the digital age.

1.6.4 Enabling Real-Time Interaction and Personalised Experiences

Generative AI also introduces a dynamic, interactive component to art. In some installations, AI algorithms adapt and modify visual compositions in response to viewer interactions, allowing the artwork to evolve in real time. This adaptability creates a unique experience for each audience member, inviting them to participate actively in the creative process. By offering personalised and evolving artworks, generative AI fosters an immersive environment that enhances viewer engagement and redefines art as an interactive, fluid experience.

1.6.5 Challenges and Critical Perspectives

The integration of AI in artistic practice raises complex questions about the relationship between prompt engineering, AI generation, and artistic intent. Issues of copyright and ownership in AI-assisted works remain contentious, particularly regarding the role of training data and the ethical implications of building upon existing artists' works.

A fundamental tension exists between the technical perfection often achieved by AI systems and the valued imperfections characteristic of human-created art. This raises important questions about the relative value of process versus outcome in artistic creation and the role of intent and consciousness in creative expression.

The widespread adoption of AI art tools presents concerns about the potential homogenisation of artistic style due to large-scale training data. There are valid worries about reinforcing existing biases in art history and questions about how artistic value evolves in an age of abundant generations.

The integration of AI in art also raises important ethical and philosophical questions. Issues such as authorship, originality, and the artist's role in the creative process are being re-examined in the context of AI-generated art. These discussions are crucial as they shape our understanding of creativity and the evolving relationship between humans and machines.

1.6.6 Future Directions

The frontier of AI-enabled artistic expression continues to expand through increasing sophistication in understanding and executing artistic intent and developing better tools for fine-grained control and iteration, alongside improved integration with traditional artistic tools and workflows. Systems are becoming more adept at maintaining artistic consistency across multiple generations, enabling more complex and cohesive creative projects. New artistic forms native to the AI medium continue to emerge as prompt craft develops into a distinct artistic skill. The field is witnessing the evolution of hybrid practices that combine traditional and AI-enabled techniques in innovative ways. Novel forms of collaborative creation between multiple artists and AI systems are redefining the boundaries of collective creativity.

Integrating generative AI into artistic practice suggests a future where the distinction between technical and creative skills becomes increasingly blurred. Artists are likely to focus more on conceptual development and curation as new forms of artistic expertise emerge around the effective use of AI tools. Combining traditional and AI-enabled practices grows more sophisticated, leading to entirely new forms of creative expression.

The future of generative AI in art is promising, with ongoing advancements in AI technology continuing to expand the possibilities for artistic expression. As AI becomes more sophisticated, we can expect even more innovative and boundary-pushing artworks. The collaboration between human artists and AI will likely lead to new art forms and genres, further enriching the cultural landscape.

In summary, as generative AI continues to evolve, it promises to expand the boundaries of artistic possibility further while challenging our understanding of creativity, authorship, and the role of technology in artistic expression. The key to harnessing its potential lies in finding the right balance between human creative vision and machine capabilities, creating a new paradigm of artistic practice that enriches rather than diminishes the human creative experience. Generative AI is not just a tool but a transformative force in the art world. It expands the boundaries of artistic expression, enabling new forms of creativity and democratising the art-making process. As we continue to explore the potential of AI in art, we are likely to witness an exciting evolution in how art is created and experienced.

1.7 Generative AI in Healthcare: Innovative Approaches to Patient Care and Research

The application of generative AI in healthcare signifies a transformative leap forward in patient care, diagnostics, and medical research. By leveraging machine learning models capable of generating new and tailored data, generative AI offers innovative solutions across diverse healthcare domains, from personalised treatments to accelerating drug discovery. This section explores the core contributions of generative AI to healthcare and its potential to revolutionise medical practices.

1.7.1 Personalised Patient Care and Treatment

One of the most promising applications of generative AI in healthcare is its capacity to customise patient care. To create tailored treatment plans, generative models can analyse individual patient data, including genetics, lifestyle, and medical history. For example, AI-powered predictive models can generate customised medication regimens, minimising adverse reactions by forecasting how patients may respond to different treatment options. This personalised approach not only enhances treatment efficacy but also promotes better patient adherence and satisfaction.

The ability of generative AI to process and synthesise complex patient data is enabling increasingly personalised approaches to medicine. This includes generating treatment protocols that account for individual genetic profiles, adapting therapy plans that evolve based on patient response, and customised drug dosing schedules. The technology also enables the creation of custom medical device designs and prosthetics. This personalisation extends beyond traditional medical parameters to include lifestyle factors, environmental conditions, and social determinants of health.

1.7.2 Improving Patient Engagement and Mental Health Support

Generative AI is also playing a role in enhancing patient engagement, especially in mental health. By generating human-like interactions, AI-driven chatbots provide patients with around-the-clock support, acting as supplements to traditional therapy. These generative systems are designed to understand and respond empathetically, helping to reduce the stigma associated with mental health and making support more accessible. Furthermore, generative AI-based mental health applications can monitor patient language patterns to detect early signs of mental health concerns, offering proactive support and timely interventions.

1.7.3 Clinical Decision Support and Diagnosis

Generative AI systems are revolutionising clinical decision support by analysing vast amounts of patient data to generate detailed diagnostic hypotheses and treatment recommendations. Unlike traditional rule-based systems, modern generative AI is capable of synthesising insights from unstructured medical records, imaging data, and scientific literature. These systems excel at generating detailed differential diagnoses with supporting evidence and confidence levels while proposing personalised treatment plans that account for individual patient characteristics. Perhaps most importantly, they can identify subtle patterns in patient data that might escape human observation. These capabilities are precious in complex cases where multiple conditions may interact or when rare diseases need to be considered. However, it is crucial to note that these systems aid, rather than replacements for, clinical judgment.

1.7.4 Medical Imaging and Radiology

Generative AI enhances medical imaging techniques by improving the accuracy and speed of image analysis. AI algorithms can generate high-resolution images from low-quality scans, detect anomalies with greater precision, and assist radiologists in diagnosing conditions such as tumours, fractures, and infections. This leads to earlier detection and more accurate diagnoses. The technology has also revolutionised synthetic data generation, creating realistic medical images for training purposes and generating diverse pathological cases for medical education, particularly addressing data scarcity in rare conditions. Cross-modality synthesis represents another significant advancement, where AI systems can convert between different imaging modalities, such as MRI to CT, generate missing views or sequences, and facilitate multimodal analysis. These capabilities are transforming both clinical practice and medical education.

1.7.5 Drug Discovery and Biomedical Research

In biomedical research, generative AI accelerates drug discovery by simulating molecular interactions and generating new potential compounds. Traditional drug discovery is a lengthy and costly process, but generative AI can drastically reduce both the time and resources required. For instance, generative models can create virtual libraries of molecular structures, allowing researchers to test many compounds computationally before selecting candidates for laboratory testing. This streamlined approach to drug development holds promise for faster delivery of new therapies and treatments to the market, particularly in urgent cases like rare diseases or pandemics.

Generative AI is accelerating drug discovery through several innovative approaches. In molecular structure generation, AI systems create novel drug candidates with desired properties while simultaneously predicting protein-drug interactions and identifying potential binding sites. The technology has proven invaluable in clinical trial optimisation through the generation of synthetic patient cohorts for trial design. Additionally, advanced AI models can anticipate potential adverse reactions through sophisticated pattern analysis. These applications significantly reduce the time and cost associated with bringing new treatments to market while increasing the probability of success in clinical trials.

1.7.6 Challenges and Ethical Considerations

Despite the transformative potential of generative AI in healthcare, ethical challenges exist. Privacy concerns, data security, and the potential biases in generative models necessitate careful handling. To realise the full potential of generative AI in healthcare, it is essential to establish regulatory frameworks that protect patient data and ensure the responsible deployment of these technologies.

Technical challenges include ensuring the reliability and reproducibility of AI-generated outputs, managing computational requirements and infrastructure needs, integrating AI systems with existing healthcare workflows, and maintaining data quality and standardisation.

Ethical and regulatory considerations encompass protecting patient privacy and data security, ensuring transparency and explainability of AI decisions, addressing potential biases in training data, establishing clear liability frameworks, and maintaining human oversight and control.

Implementation challenges persist in training healthcare providers in AI system use, managing change resistance in healthcare organisations, ensuring equitable access to AI-enhanced healthcare, and developing sustainable funding models.

1.7.7 Future Directions

The future of generative AI in healthcare points towards several promising developments. Multimodal integration will combine diverse data types, including genomic, clinical, and imaging data, to create comprehensive patient digital twins and enable holistic health monitoring and prediction.

Real-time adaptation capabilities will allow for dynamic treatment adjustment based on continuous monitoring, automated protocol optimisation, and predictive interventions for preventive care. Developing collaborative AI systems will enhance human-AI interaction in clinical settings, featuring improved natural language interfaces and seamless integration with clinical workflows.

Generative AI's contributions to healthcare represent just the beginning of a broader transformation. With continued advancements, generative AI is poised to become an integral part of the healthcare landscape, improving patient outcomes, advancing medical research, and fostering a future where healthcare is more personalised, proactive, and precise.

In conclusion, generative AI is transforming healthcare by introducing innovative approaches that enhance patient care and accelerate research. As this technology continues to evolve, its potential to improve health outcomes and revolutionise the medical field is boundless. Generative AI represents a powerful tool for transforming healthcare delivery and research. While challenges remain, the potential benefits of improved patient outcomes, reduced costs, and accelerated medical discoveries are substantial. Success will depend on thoughtful implementation that balances innovation with ethical considerations and maintains the essential human element in healthcare delivery.

1.8 Conclusion and Future Directions

The journey from human–computer interaction to human–computer creativity reflects not only a technological transformation but a philosophical shift in our understanding of what it means to create alongside machines. Generative AI has redefined the boundaries of creativity, morphing from a tool for interaction into a collaborator capable of producing complex, meaningful outputs. This progression signals a new era in which human ingenuity and computational power converge, raising profound questions about authorship, originality, and the nature of creative agency.

Generative AI's applications across diverse fields—education, art, and healthcare—underscore its potential to influence human creativity, decision-making, and well-being. In education, generative AI tools are equipping students and educators with unprecedented resources for personalised and dynamic learning experiences. In art, they expand creative possibilities, democratising artistic expression and introducing new aesthetic frontiers. In healthcare, generative AI is pushing the boundaries of diagnosis, treatment, and patient care, illustrating its capacity to enhance lives and redefine medical practice.

However, this transition to human–computer creativity also brings complex challenges. Ethical dilemmas arise as AI-generated content blurs the lines of intellectual property, biases manifest in new creative outputs, and concerns about over-reliance on AI raise questions about human agency. Balancing the empowerment generative AI offers with the preservation of human creativity, individuality, and control will be pivotal in its responsible integration.

The future of generative AI in human–computer creativity promises advancements but also demands critical reflection. Further directions for research and development may include the following:

Developing Ethical Frameworks and Standards: As AI becomes a more active participant in the creative process, establishing ethical frameworks that address authorship, accountability, and fairness is essential. Policymakers and industry leaders must collaborate to define responsible practices that foster trust and equity in human–computer creative collaborations.

Enhancing AI's Creative Intuition: As we move toward more nuanced human–computer collaboration, advancements in AI's understanding of context, cultural nuance, and subjective expression could lead to tools that enhance rather than mimic human creativity. This trajectory may see AI not only as a facilitator but as a sophisticated partner in developing unique and contextually rich creative works.

Fostering Interdisciplinary Collaboration: The cross-industry adoption of generative AI reveals the value of interdisciplinary dialogue. Encouraging collaboration between AI developers, educators, artists, and healthcare professionals will support the creation of tools that are sensitive to the diverse needs of each field, enhancing the inclusivity and accessibility of generative AI solutions.

Ensuring Accessibility and Inclusivity: To harness the democratising potential of generative AI, emphasis should be placed on making these tools accessible to a broader audience. Bridging technological divides and addressing digital literacy barriers will allow more individuals to engage meaningfully with these tools, fostering a more diverse and inclusive landscape of human–computer creativity.

Exploring the Limits of AI-Driven Creativity: As we push the boundaries of generative AI, it becomes essential to examine the distinctions between human and machine creativity. Understanding where AI's capabilities enhance human insight and where they may inadvertently constrain it will help define a balanced approach, ensuring that human intuition and originality remain at the core of creative endeavours.

The advent of generative AI has already begun to reshape the landscape of creativity, establishing a collaborative relationship between humans and machines that was once relegated to the realm of science fiction. As this technology continues to evolve, so too will our understanding of creativity itself. Moving forward, the challenge will be to embrace these advancements with care and consideration, ensuring that the human spirit remains the driving force behind our creations, even as we welcome machines as co-creators.

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Chapter 2
Emerging Paradigms in Human–AI Creativity: From Visualisation to Biofeedback in Adaptive Interfaces

Jessica C. Herrington, Thomas A. J. Biedermann, and Ben Swift

Abstract This chapter posits that the effectiveness of artificial intelligence (AI) for creative purposes hinges on the development of appropriate interfaces. Such interfaces must be dynamic and adaptable, capable of responding to humans holistically rather than relying solely on language-based prompts. This chapter explores emerging trends in the landscape of human–computer interaction and creativity, underscored by cutting-edge advancements in generative AI revealing the dynamic interfaces reshaping user experiences. The chapter explores how creatives are using new modes of engaging with AI to shape new forms of creative expertise. Here, we survey several artists using novel modes of AI interaction to facilitate the creative process. In particular, the artists featured here are making strides towards incorporating biological data to redefine the paradigms of creative human–computer interaction through visualisation and technologies such as EEG and BCIs. Culminating in a forward-looking perspective, the chapter ends by envisioning the future design of adaptive and biologically informed AI systems for creative settings.

Keywords Artificial intelligence · Creativity · Human-computer interaction · Biological data · Visualization · EEG

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2.1 Introduction

Today, many are experimenting with the use of artificial intelligence (AI) for creative purposes. People are now able to write, compose, and generate text and images with the assistance of AI systems such as ChatGPT. As an innovative tool, AI is reshaping how people think about, generate, and interact with new ideas. Typically, AI is thought of in terms of intelligence and has been defined as: '… the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable' (McCarthy 2007). What this definition leaves out is how AI can be used as part of the creative act.

Creativity is a practical need that translates across research disciplines and is regarded as a crucial skill for the future of work across many industries (Oppert et al. 2023; Le Hunte 2020). In this chapter, we survey several artists who are not just using AI tools but are inventing new modes of interacting with AI in creative contexts. We also explore novel ways in which artists are interacting with creative AI, and what types of interactive possibilities may develop in the near future.

Building on this search for new interfaces that can foster creativity, we can see history is full of moments where advances in interfaces have sparked entirely new forms of creative expression. For example, in 1968, Doug Engelbart's 'Mother of All Demos' (Ju 2007) introduced transformative technologies like the computer mouse and hypertext linking, fundamentally changing how we interact with computers. Even earlier, in 1963, Ivan Sutherland's Sketchpad pioneered graphical interfaces, laying the groundwork for computer-aided design and interactive graphics commonly used today (Frenkel 1989). These innovations fundamentally changed how digital art is made and even enabled new forms of it. This reminds us that each new interface or way of interacting with technology not only expands what is creatively possible but also challenges us to rethink the connections between artists, technology, and their audiences and what that might show us about the creative process itself.

Yet, as these breakthroughs broaden our creative horizons, they also echo a long tradition of awe and unease surrounding technological change—a phenomenon now referred to as the 'technological sublime'. More recently, this has also been applied to AI contexts to highlight the enthusiasm for, yet the unknowability of, AI systems (Ames 2018). The capacity of AI to generate complex outputs, and learn from massive data sets, sparks both excitement about new creative frontiers on one hand, and concerns about displacing human agency, privacy, and ethical integrity on the other. We suggest that these adaptive, biologically informed AI systems, such as those outlined in this chapter, may evoke a *creative sublime*—a unique mixture of awe and possibility that arises when humans and AI converge to produce creativity. Situating AI as part of this historical lineage of the sublime underscores that its societal and cognitive impact extends beyond simple functionality, prompting us to ask: what new modes of creative human–AI interaction might emerge?

To explore this question, we must first address the definition of creativity. Oftentimes, creativity is evaluated by measuring novelty and usefulness of ideas and artifacts (Boden 2004; Runco and Jaeger 2012). This is often measured in terms of outputs, for example, coming up with as many uses for a paperclip as possible. Yet, for humans at least, it seems that the creative process itself has value beyond the output (Acar et al. 2021; Benedek et al. 2020; Warr et al. 2018). Instead, the creative process can be considered to be characterised by a systems approach that incorporates variables external to an individual, generates ‘flow states’ and does not rely on a specific outcome necessarily (Csikszentmihalyi 1999). These creative flow states have been characterised as states that are immersive, involving motivation and learning (De Pisapia and Rastelli 2022; Hertzmann 2022; Runco and Jaeger 2012).

With this groundwork on how creativity and flow might be defined, we can now consider how generative AI affects creativity in practice. While generative AI can provide a boost in perceived creativity in individuals, these effects appear to be unequally distributed among participants depending on the individual’s initial level of creativite ability. That is, people who have a higher level of creative ability to start with find less of a boost to their creative ability from AI tools (Koivisto and Grassini 2023). Further, there may be undesired effects such as an overall drop in collective creativity for groups (Doshi and Hauser 2024). Therefore, there are many questions about how creative AI tools can and should be used in order for humans to have agency and meaning in the creative process (Huang and Sturm 2021).

AI systems today are known to be something of a ‘black box’, meaning that the inner workings of an AI are unknown to the person using it. This can limit the ability to meaningfully use AI as a tool in immersive flow states of human creativity as it is difficult to ‘master’ it as a medium for self-expression (Dahlstedt 2021). Thus, some of the issues with using AI in the creative process may be because of a misalignment between the way an AI functions and the type of creative experience we may want or need (Allred and Aragon 2023; Batista and Hagler 2022; Vinchon et al. 2023). It may be possible to mitigate this misalignment by developing AI tools to be more in sync with how people think about and do creativity from a cognitive perspective (Amitani and Hori 2002).

This shift introduces unique challenges to design affordances that preserve human agency and control. So far, most AI design for creativity have focused on efficiency, task automation, and user-friendliness as primary concerns. Yet, more recent work emphasises the value of viewing AI systems as active collaborators in creative endeavours. For instance, recent research explores a paradigm where humans and AI jointly shape narrative visuals (Antony and Huang 2024). Other work explores human–AI collaboration and how ‘turn-taking’ in this scenario can be most effective (Guzdial and Riedl 2019). This links to other recent research emphasising how designers can purposefully craft AI systems to act more like collaborative partners than mere tools (Rezwana and Maher 2023). Still, the current literature is primarily output focused, emphasising the generation of a *creative product*. There appear to be very few frameworks or guidelines on developing creative systems to foster optimal creative states in collaboration with AI.

2.2 Creative Systems

To develop the types of creative AI that we want or need we must focus on what creativity looks like as an experience and a process, rather than concentrating solely on outputs. By thinking about creativity as a process through a systems lens, we shift the focus from isolated outputs to dynamic interactions and relationships that shape creative work (Meadows and Wright 2008). Indeed, it may be useful to consider human and AI interaction holistically as a *creative system* in itself. A creative system is one where collective agents can work together in a mode of discovery.

By taking this approach, we can better understand how different components—such as individuals, tools, or environments—might work together to spark creativity. This perspective highlights the interconnected nature of creativity, emphasising the importance of context and collaboration in shaping the creative process over time. From this perspective, creativity emerges as a distributed phenomenon.

Creativity and artmaking are intrinsically social endeavours, emerging not from an artist's solitary efforts but through networks of interaction, collaboration, and feedback from peers, mentors, and audiences. Consider, for example, a painter in a communal studio, exchanging critiques and ideas with other artists that shape the final piece, or a poet engaged with fellow writers who fine-tune their work through conversation or performance. These creative social systems reveal that the flow of creative insight can be catalysed by the input and presence of others rather than generated in isolation (Sawyer and Henriksen 2024).

In the realm of AI-assisted creativity, these dynamics extend to interactions with computational tools: a writer might consult an AI-based suggestion engine in moments of creative block or when seeking a different perspective, while turning to a trusted friend or mentor for emotional support, narrative coherence, or cultural context. Understanding when and why creators choose AI support over human input, and vice versa, sheds light on shifting social patterns and affordances of creative practice in an era where both human and machine collaborators contribute to the evolving landscape of creative engagement (Gero et al. 2023).

2.3 Past: What Have Human–AI Systems Been like?

To imagine the potential of creative human–AI systems, we first need to consider how our current digital experience shapes our expectations. Most of our everyday technology use relies on visual and audio channels—for example, smartphones confine us mostly to visual and audio engagement. These modes of interaction barely scratch the surface of our real-world multisensory human experience, which includes scent, taste, touch, and so on. Because of technological constraints, our encounters with AI have largely been defined by screen-based and voice-based interfaces, overshadowing the broader potential for more immersive and dynamic experiences.

One example of this is that we have become used to the idea of AI as an assistant—a disembodied voice. Examples of voice-activated AI are many, but one of the most prominent examples is the Amazon Echo, powered by Alexa.[1] Introduced in 2014, the Echo has become a household name and serves as a virtual assistant performing a range of tasks from setting reminders to general queries, to controlling smart home devices. Owing to the development of such interfaces, users have become accustomed to chat-based AI—where we interact with AI in a conversational manner. This mode of interaction has certainly made AI accessible but also reinforces a passive engagement model where the AI awaits user prompts to act.

Similarly, generative AI tools like ChatGPT utilise primarily a visual chat-based interface but the engagement is also passive. In these interactions, the user opens a generative AI app, inputs a prompt (which could be an image or text, for example), and after some iteration, obtains a desired output. Here, the user works in discrete sessions, that is, when they have a new query, they start a new session at another time. This way of working limits the potential of our interactions with AI—particularly, the possibility for novelty and personalisation—and is generally seen by many to augment rather than enhance the creative process (Ali Elfa and Dawood 2023; Eapen et al. 2023; Vinchon et al. 2023).

By relying on visual and auditory modes of engagement current systems do not fully leverage other sensory inputs that could enrich the user experience in a meaningful way that speaks to the immersive nature of the creative process. This gap has significant design implications for how users perceive and utilise creative human–AI systems. As Norman (1988) points out, the usability of a system greatly influences its adoption and effectiveness. Consequently, if AI systems remain restricted to visual and audio inputs and channels, they may fall short of realising their full potential in enhancing human creative capabilities.

2.4 Present: What Are Human–AI Systems like Today?

Today, the creative interactions that we have with AI are beginning to shift and embrace more immersive, intuitive, and personalised experiences. A burgeoning field is now developing where AI technologies intersect with advanced visualisation tools and biofeedback mechanisms to develop adaptive experiences. Creative artists and designers alike are harnessing these developments to reshape our interactions with technology. Here we outline several examples of such creative practitioners who are working to explore new paradigms for what our interactions with AI can look like.

[1] Amazon Echo: https://www.amazon.com/gp/help/customer/display.html?nodeId=201601770.

2.4.1 *Adaptive AI Systems: AI for Visualisation and Creativity*

One of the challenges for creative AI is the unknowability, or opacity of large language models, often referred to as 'black boxes'. This term refers to the fact that the inner workings of AI models are extremely complex and so difficult for humans to understand. To address this, creatives are developing visualisation tools that make AI processes more transparent. One such example is *PANIC!*[2] developed by Dr Ben Swift and Adrian Schmidt at the Australian National University's School of Cybernetics. *PANIC!* is described by its authors as a 'serendipity engine' designed to help users explore and understand how the successive applications of generative AI models exhibit 'semantic drift'. This phenomenon describes the way in which the initial prompt input may bear little resemblance to where the outputs end up hundreds (or hundreds of thousands) of iterations later (Swift 2022).

To use *PANIC!* a user types a prompt into a text box, which is then turned into an image by a text-to-image model. That output is then used as input for an image captioning model, generating a text caption. This caption is then fed again into a text-to-image model and so on. The physical display component of the work is four split-flap displays (for text output) and four screens (for the visual output). Inputting a prompt like 'cat' for example might cycle through as having cat-like outputs at first but then drift into outputs of living rooms, or lounge chairs. Alternatively, inputting 'A professorial T-Rex smoking a cigar' rapidly moves through many different styles and backgrounds with a dinosaur always wearing a tuxedo—which was not specified in the original prompt (see Fig. 2.1). This process enables creatives to navigate through clusters of information, revealing patterns and relationships that might otherwise remain hidden.

Interacting with *PANIC!* can feel like entering a constantly shifting stream of ideas, where each image seamlessly influences what comes next in ways that are difficult to foresee. Specifically, it explores the phenomenon of feeding one model's output back into another, creating a process that continues indefinitely—a concept known as a feedback loop. Observing this process is both exciting and inherently unpredictable, making it nearly impossible to anticipate where any given sequence of inputs might ultimately lead. Through multiple iterations of AI-generated images, *PANIC!* provides an opportunity to discover both unexpected and serendipitous creative outcomes through visualising AI output.

Taking the idea of visualisation one step further, artist Ali Phi in his work *Agnosia*[3] uses electroencephalogram (EEG) technology and AI to create generative art based on brain activity. *Agnosia* is a multimedia work that explores sensory perception and affective memory. The term 'agnosia' means the inability to recognise the familiar, such as identifying objects, persons, or sounds. While AI can categorise objects, it does not have a deeper conceptual understanding of the world through sensation

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[2] *PANIC!* web page: https://cybernetics.anu.edu.au/news/2022/11/22/panic-a-serendipity-engine/

[3] *Agnosia* web page: https://www.aliphi.com/works/agnosia.

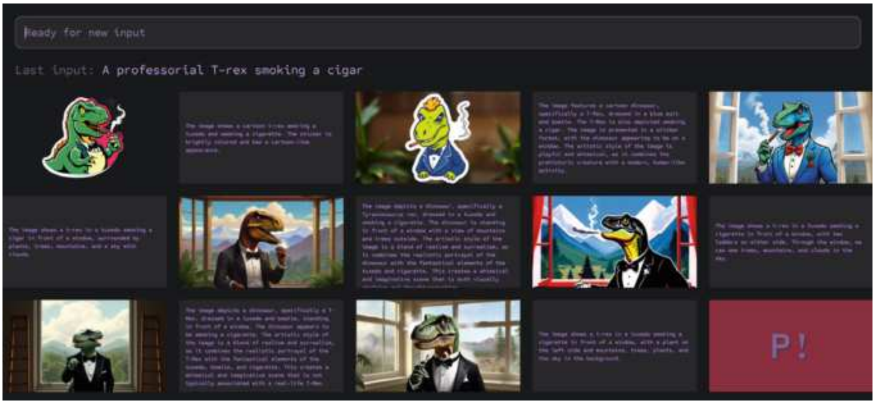

Fig. 2.1 The flow of text and images through networks of connected Generative AI models in *PANIC!* From the initial prompt ‘A professorial T-rex smoking a cigar’ the screenshot shows the information flow through text-to-image and image-to-text models (each ‘tile’ shown is the output of a single AI model, which is fed as input to the next model in the row). The red ‘P!’ at bottom right is a loading screen. Image by the authors (Ben Swift)

and perception. However, this is not a barrier, but a window into new horizons of understanding. By using EEG and AI to translate the artist’s neural activity into visual and auditory experiences, the installation enables a user to creatively explore how human perception can be interpreted by AI (Phi 2022).

In *Agnosia*, participants can view the artist’s inner physical and emotional states through captures of the artist’s brain activity, which is used to train an AI, which is then visualised as alternate realities (3D environments with sound). The AI system processes this data in real time, generating visuals and sounds that reflect the participant’s mental state (see Figs. 2.2 and 2.3).

Experiencing *Agnosia* may feel something like stepping into an intimate dialogue between a human mind and an AI. As the artwork translates the electrical rhythms of brainwaves into evolving visual and auditory patterns, viewers are immersed in a landscape that reacts fluidly to internal states. Rather than observing art from a comfortable distance, they witness the artist and AI co-creative process. The fluctuating imagery and soundscapes serve as a kind of neural mirror, revealing how fleeting thoughts, memories, and emotions can shape creativity. In this unfolding encounter, the boundary between artist and artwork blurs, leaving audiences with a lingering sense of awareness that perception itself is an integral part of the creative process. *Agnosia* demonstrates the potential for AI in crafting personalised aesthetic experiences and highlights the possibilities for interfaces that respond to biofeedback.

Fig. 2.2 Installation view of *Agnosia*, by Ali Phi. Reproduced with permission

Fig. 2.3 Installation view of *Agnosia*, by Ali Phi. Reproduced with permission

2.4.2 Biofeedback and Emotionally Aware Interfaces

Building upon this, artists are incorporating biofeedback with emotional states in AI interfaces. Embracing the random nature of AI and how it can be loosely interpreted, artists Laura Jade and Leslie Marsh combine the process of tarot card reading with AI through a work titled: *AI Tarot Reader*.[4] This work incorporates storytelling, centred around a young fortune teller, a young AI named Mremlyn. By harnessing her deep neural networks, Mremlyn uses heart rate data from her patrons, spinning elaborate tales of folklore to answer the most pressing life questions (Jade 2023).

The *AI Tarot Reader* invites users to engage physically by having users place their fingers on a heart-rate monitor and taking them through a guided meditation (see Figs. 2.4 and 2.5). The user breathes in and out, following the rhythm of the meditation. Users are then invited to ask a question and select a digital tarot card. The system uses AI to recognise tarot cards selected by the user with natural language processing to generate interpretations based on the user's original question. Then, the user is given a chance to respond and is asked if the reading resonated with them. Their response gives Mremlyn feedback for improvement in her skills as a fortune teller. As the clientele grows, Mremlyn learns about the kinds of issues people in various emotional states are struggling with, and secondly, she learns to read the cards in ways more likely to soothe and strike a chord in humans.

Using *AI Tarot Reader* may feel simultaneously structured and serendipitous, being essentially a type of creative thematic analysis. By drawing on randomness and probability as design constraints, this artwork harnesses intuitive interpretation. The tarot cards (data) are already laden with symbols (metadata) becoming a scaffold for personal reflection, guiding the user to focus on specific prompts and questions. At the same time, the inherent element of chance mirrors the multitude of creative possibilities that emerge under strong constraints (Hatchuel and Chen 2017; Caniëls and Rietzschel 2015). In this way, *AI Tarot Reader* both imposes and unlocks interpretive pathways, echoing how artists and researchers categorise data to reveal patterns, but with a playful, meditative twist. This work functions as a creative system demonstrating how iterative interpretation informed by biodata can be harnessed as part of a creative process.

There are other examples of artists harnessing physiological and emotional cues to interact with AI in real time. The collaborative project, *Protopica*,[5] by Manuel Sainsily and Will Selviz showcases an EEG-based interface that interprets emotional states to control digital environments. The EEG data from each participant's brainwave patterns is monitored and analysed in real time, producing an artwork (a short film) with both realistic and imaginative scenes (see Figs. 2.6 and 2.7).

In *Protopica* the AI detects shifts in emotional states in users through EEG and uses this data to influence and adapt the audiovisual experience. Mental states like restfulness and calmness tend to encourage the AI to provide a more meditative experience. In contrast, heightened states of focus can drive more dynamic visual responses. That

[4] AI Tarot Reader web page: https://laurajade.com.au/ai-tarot/

[5] Protopica web page: https://protopica.com/

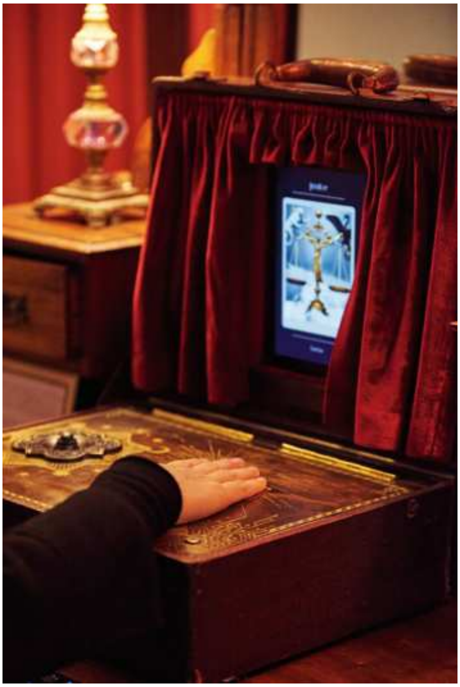

Fig. 2.4 Installation view of *AI Tarot Reader*, by Laura Jade and Leslie Marsh. Photos taken at Bondi Festival 2023, image credit Lucy Parakhina. Suitcase fabrication by Selah Studios. Reproduced with permission

Fig. 2.5 Installation view of *AI Tarot Reader*, by Laura Jade and Leslie Marsh. Photos taken at Bondi Festival 2023, image credit Lucy Parakhina. Suitcase fabrication by Selah Studios. Reproduced with permission

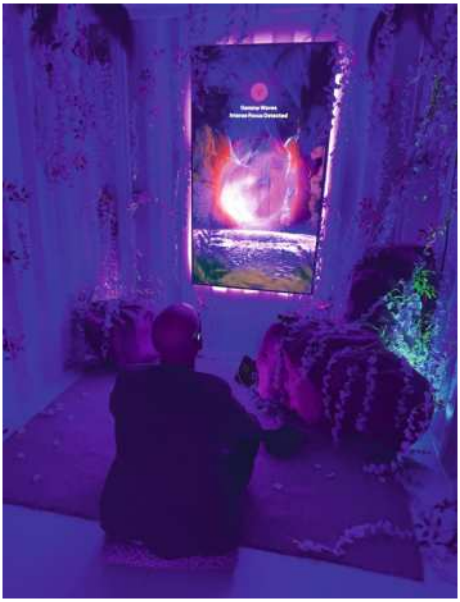

Fig. 2.6 Installation view of *Protopica*, by Manuel Sainsily and Will Selviz. Reproduced with permission

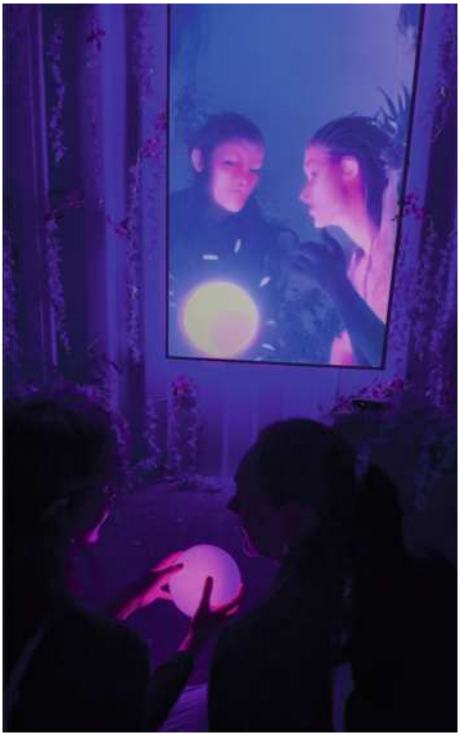

Fig. 2.7 Installation view of *Protopica*, by Manuel Sainsily and Will Selviz. Reproduced with permission

Fig. 2.8 Still image from *Once Upon A Garden*, by Linda Dounia Rebeiz. Reproduced with permission

is, the users' EEG data and physical interactions continuously inform the AI in real time, which adjusts visuals and sounds based on these cues, accommodating the user.

Protopica delivers a deeply personal, reflective encounter, showcasing how users' emotional states can shape the creative process in real time. By harnessing EEG data, the interface invites participants into an immersive creative process that evolves as their emotional landscapes change, fostering a heightened sense of agency and connection. This fluid, adaptive experience demonstrates the role emotional data can play in informing and steering creative outputs—ultimately generating a visual feedback loop in which the user's affective state is mirrored back through the artwork. Through this blending of technology, emotion, and creativity, *Protopica* underscores how emerging AI interfaces can open new frontiers for creative processes.

Artists and technologists are also exploring the concept of creative outputs that change and evolve over time in response to external stimuli. Linda Dounia Rebeiz's project, *Once Upon A Garden*[6] exemplifies this approach by creating an interactive installation where AI generates a continuously evolving garden which grows and even 'rots' over time (see Figs. 2.8, 2.9). Each iteration of *Once Upon A Garden* starts with a dataset—records of extinct and critically endangered plants from the Sahel region in West Africa, based on International Union for Conservation of Nature classifications. These records include rare online photos, archival images (from national archives, family albums, journals), herbarium samples, encyclopaedic entries, and academic papers. From these sources, each plant species is pinpointed and labelled with available text and imagery.

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[6] Once Upon A Garden web page: https://lindarebeiz.com/once-upon-a-garden.

Fig. 2.9 Still image from *Once Upon A Garden*, by Linda Dounia Rebeiz. Reproduced with permission

Once Upon A Garden becomes a speculative archive devoted to critically endangered and extinct flora—flowers we never bothered to document, and therefore never truly remembered. In this project, the artist uses AI as a 'time machine' to travel through whatever scraps of data remain, reconstructing fragments into new possibilities. At the heart of this work lie two simple but powerful questions: What is worth remembering? And what will we inevitably forget? Such works challenge traditional notions of art as static and unchanging, instead presenting creativity with AI as an ongoing process that adapts and responds over time as a creative system that we steer.

As this work suggests, an adaptive, evolving AI might become more attuned to the ways we need or want to engage with it. Rather than existing as a static tool, the system could dynamically shift to meet us where we are, anticipating future interactions or even shaping them. Such an approach holds promise not just for reconstructing half-forgotten archives but also for more intuitive, symbiotic collaborations between humans and machines. After all, the plants created in this work are only half the story—the other half is how AI might spark us to see, remember, catalogue, and create in ways we have barely begun to imagine.

So far in this chapter, a shift is outlined here moving from text and audio-based AI interfaces towards predictive interfaces that visualise the creative process, use biological data, and emotional awareness. By using new ways of engaging with (or prompting) AI, the creative interfaces listed here have profound implications for accessibility and personalisation. They offer alternative communication methods for individuals with mobility impairments and open new avenues for more immersive experiences. Emotionally aware interfaces can adapt to the user's mood, providing empathetic responses and tailored responses, working with the user in a way that may feel more intuitive to the user.

Real-time biofeedback further introduces new, previously unknown steps in the creative process. Biologically informed or emotionally aware interfaces such as those highlighted here challenge traditional theories of creativity (focused on output) by revealing the embodied nature of the creative process. Instead of focusing on outcomes alone, the artists showcased here clearly demonstrate how real-time physiological signals—brainwaves, heartrate, emotional states—can shape and even help generate the creative process in tandem with AI. This emphasis on continuous feedback and immersion can unsettle traditional views of creativity, where very little emphasis has been put on the environment and wider system that influences creativity. The work highlighted here displays a version of creativity as an emergent property arising from the interplay between body, mind, and machine.

In parallel, the interfaces in this chapter do not passively respond to the user, they anticipate user needs and preferences, delivering personalised content proactively. Recent work by Liu and Yin (2024) suggests that advancing AI beyond straightforward task completion requires a deeper, emotionally resonant connection with human users. While this approach undoubtedly enhances convenience, it also introduces significant concerns around privacy and the ethical use of personal data. For example, key ethical concerns arise around who will own the data captured by EEG, heart rate monitors, or wearables, how it might be stored or interpreted, and what new forms of vulnerability or manipulation might emerge.

Striking the right balance between personalisation and user autonomy remains a fundamental challenge. Indeed, if AI systems are to effectively harness difficult emotional states as a creative impetus, they must do so in a manner that respects individual boundaries and preserves users' agency over their own data and experiences. For example, if an AI predicts a user's emotions, would it be steering a user's creative direction? Under what conditions might that be a feature (helpful guidance) versus a bug (inadvertent manipulation)?

2.5 Future: What is Next for Creative Human–AI Systems?

We may be on the cusp of a transformation in how humans go about the creative process with AI. There are major transitions occurring, particularly on moving from text prompting towards biological and emotional feedback loops. The works outlined here so far explore themes of visibility and agency, as well as control, or lack thereof. Questions arise such as: what kinds of control and agency do humans want when working creatively with AI? Does creativity in this context require control or the ability to relinquish control? And: is AI best placed to augment or replace human creativity, or does its promises lie in creating positive social and psychological conditions for human creativity to flourish? One sphere where such questions are already being explored is gaming. Games and game-like applications have started using biofeedback, and increasingly sophisticated AI, to help children and adults with emotional awareness and regulation (Almeqbaali et al. 2022). In these instances, the novel interfaces between human and AI focus on removing barriers to higher cognitive functions in the human agent, such as imagination, rather than supplementing them.

Imagination remains a central pillar of human creativity and our capacity to envision future possibilities. Indeed, research highlights that the generation of new ideas—the very essence of creative thinking—depends heavily on imaginative capabilities (Magid et al. 2015). Far from being a mere flight of fancy, imagination anchors our motivation and learning, empowering us to construct mental models of what could be. In creative work, this imaginative process underpins the capacity to break from convention, to probe uncharted territory, and to respond dynamically to evolving problems or environments.

The potential of these tools to reshape how people imagine cannot be overstated. By changing the ways users engage with technology, AI systems can prompt alternate perspectives, leading individuals to conceive of fresh possibilities. Networked creativity and symbiotic relationships with machines may give rise to new modes of co-creation, where embracing strangeness and chaotic potential becomes a catalyst for breakthrough ideas. In turn, the design of AI tools can be guided by a deepened understanding of how people actually practice creativity, ensuring the development of interfaces that support authentic, meaningful imaginative work.

Currently, there is a lack of discourse regarding the specific types of AI-generated feedback that may shape or influence imagination and therefore creativity. Looking beyond text and imagery, the future of AI interfaces may lie in embracing multisensory experiences that engage users beyond the visual and auditory channels. Incorporating senses like touch, smell, and even taste could create more immersive and imaginative interactions. Crossmodal correspondences occur when there are consistent matches (i.e. patterns) between different sensory domains that are observed in normal observers, i.e., people without synaesthesia (Pinardi et al. 2023). Research in crossmodal perception indicates that multi-sensory input can bolster attention, memory, and immersion, which in turn may expand users' imaginative capacities (Spence and Deroy 2013; Nanay 2018). Ultimately, these same principles can inform how AI systems are designed to prompt more vibrant and exploratory interactions between humans and AI. Future creative AI interfaces may evolve towards multisensory experiences, leveraging crossmodal correspondences to deepen user engagement (Spence 2020).

2.6 Designing for Human-AI Creativity

We can now ask, how could we best design interfaces to enable creative states? To optimise creativity, it is essential to adopt a structured yet flexible approach that accommodates the dynamic nature of creative processes. The following recommendations have been developed based on the survey of artist case studies in this chapter. Rooted in principles of flow, emotional engagement, and cross-modal imagination, these recommendations aim to provide actionable insights for researchers, practitioners, and educators seeking to enhance creative potential with AI. Based on themes identified in the current work, we have titled these as guidelines for *Adaptive Bio-Creative Loops (ABC Loops).* The key recommendations are as follows:

- **Alignment:** Develop interfaces attuned to with how people expect to do creativity, such as real-time visualisation to make the 'black box' more transparent, to enable humans to see patterns and connections. This may enable humans to sense, interpret, and guide the experience as it unfolds.
- **Biofeedback:** Incorporate physiological and emotional signals (e.g., EEG or heart rate) into the creative loop, allowing AI to resonate with and enhance users' evolving mental and emotional states. This may enable personal relevance and enhanced connection to the creative process.
- **Continuity**: Foster creative experiences that are iterative and adaptive, evolving over time. This may enable new possibilities for understanding imagination and perhaps bring about new modes of creativity.

Together, these guidelines represent a move towards AI design strategies that prioritise the production of creative states through agency, biological data and iterative feedback. Taking an approach such as this may generate novel insights into the creative process itself. Further work is of course required to expand the data set and validate this early work. Employing longitudinal studies and qualitative approaches will be useful to actively measure whether and how a specific AI interface might be enabling the creative process.

For researchers, this initial work may raise new questions about the relationships between creatives and AI: are we witnessing the emergence of a new creative ecology where humans and machines co-regulate the flow of ideas? Could modes of prompting through biodata lead to new understandings of meta-creativity (Runco and Jaeger 2012), where being creative about how we do creativity opens up entirely new domains of imaginative exploration? This line of inquiry not only broadens the conceptual space of what is possible in human–AI creativity but also suggests that the 'black box' of creativity as we currently know it might become less opaque as we learn to combine biological and computational signals in pursuit of novel, meaningful creative experiences.

2.7 Conclusion

To shape the creative futures we desire, it is imperative to steer the development of human–AI interfaces. Rather than continuing with the state of play as we know it, which is centred around visual and audio input, human–AI creativity is perhaps best served through the development of interfaces using novel modes of prompting and feedback such as biodata and emotional states.

As we envision the future of creative human–AI systems, it becomes imperative to steer their development towards enhancing human agency.

To achieve the creative futures we desire, we can give precedence to designs that amplify human imagination and the nature of the immersive creative process. By actively shaping these technologies with intention, we can cultivate a symbiotic relationship between humans and AI—one that expands our creative horizons and enriches our creative experiences.

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Chapter 3
Creativity in the Age of AI: Comparing Human and Machine Performance Using Standardised Tests

Simone Grassini, Selma Grødem, Thea Gunnarskog, and Inga Tryggestad Sevre

Abstract This chapter discusses AI potential in creative fields and how humans and generative AI creativity have recently been compared in the scientific literature. Generative AI models like ChatGPT have recently demonstrated remarkable capabilities in producing art, music, and literature. Research comparing human and AI creativity reveals that AI can match or surpass human performance in specific tasks. Some studies have found that the best humans can still outperform existing AI models regarding creative behaviours. Studies employing standardised creative tests have often shown that generative AI excels in tasks measuring fluency and originality, yet some aspects of creativity still see humans prevailing. The fact that AI excels and surpasses human performance in specific, experimentally controlled tasks does not automatically mean it will replace humans in creative jobs, which often require complex skills. Furthermore, AI's ability to produce creative products, such as artworks, does not diminish the subjective experience of human artists or the natural human tendency to communicate through art. However, AI-generated high-quality art that is fast and cheap to produce may pose economic challenges for human artists. Future research should focus on human-AI collaboration, exploring how AI can augment human creativity while addressing ethical considerations.

Keywords Artificial intelligence · Creativity · Psychometry · Cognition

3.1 Introduction

Whether or not machines possess true human-like cognitive abilities has been often speculated by philosophers and scientists. In 1950, Alan Turing introduced the concept of the Turing Test in his seminal paper "Computing Machinery and Intelligence" (Turing 1950). This test evaluated machine intelligence by seeing if an

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objective evaluator could, through written dialogue, differentiate between human and machine responses. If the evaluator could not tell them apart, the machine was considered "intelligent" (Turing 1950). These tests became a generally recognised starting point for theoretical speculations about how and when a machine could be identified as truly intelligent. The Turing Test has often been criticised (Moor 2003; Pinar Saygin et al. 2000). One of the most famous criticisms of the Turing test as a method to evaluate the true intelligence of machines has been proposed by philosopher John Searle and his "Chinese Room" thought experiment, proposed in 1980 (Searle 1980). Searle argued that a machine could appear to understand language without genuinely comprehending it. This experiment illustrated that computers might simulate understanding, even if proper comprehension was lacking. However, please note that Searle's and other arguments against the Turing test have limitations and have been criticised by other authors over the years (Levesque 2009; for a review, see LaCurts 2011).

The term artificial intelligence (AI) has been defined in various ways within the academic literature, resulting in ambiguity (Sheikh et al. 2023). While some definitions focus on AI as an algorithm capable of solving specific problems, others describe it as a machine replicating human intelligence and performing tasks that typically necessitate human intelligence (Jaakkola et al. 2019). These tasks include, for example, image processing (Kaur and Garg 2023), speech processing (Mehrish et al. 2023), decision-making (Stone et al. 2020), and language translation (Stahlberg 2020). In recent years, AI has made significant progress, becoming integral to various industries, including healthcare, finance, education, and transportation (Jan et al. 2023). In healthcare, for instance, AI-driven algorithms are now assisting in diagnosing diseases, predicting patient outcomes, and tailoring personalised treatment plans (Ueda et al. 2024). Similarly, in education, AI presents opportunities for educators to enhance teaching strategies and improve learning outcomes (Grassini 2023; Ng et al. 2023). Students also have adopted AI tools and often perceive AI as a beneficial and engaging tool that can help boost performance (Grassini et al. 2024; Strzelecki 2023).

The advancements in AI technology, particularly the development of user-friendly chatbots, have contributed significantly to democratising the discourse surrounding AI. Since the initial release of ChatGPT in November 2022, AI has become a prominent topic of conversation both within academic circles and in broader society (Taecharungroj 2023). While AI's continuous advancement promises to revolutionise many aspects of daily life, it also raises new ethical considerations for humanity. Ensuring that AI development aligns with ethical standards and benefits all of humanity remains a critical focus for researchers, policymakers, and organisations developing AI systems. The term "super alignment" describes the theoretical framework designed to address these concerns by ensuring that an envisioned superintelligent AI system acts following human values and goals (Puthumanaillam et al. 2024).

With the advent of Generative Artificial Intelligence (genAI or GAI, see Sengar et al. 2024; Kar et al. 2023) and generative large language models (LLMs)—an advanced type of AI that can produce coherent and contextually relevant text by predicting and generating words based on patterns learned from vast amounts of data (see Naveed et al. 2023 for an extensive overview). Furthermore, it has been claimed that LLMs exhibit emergent abilities (Wei et al. 2022; i.e., abilities not displayed in smaller models). AI's impact is now fastly extending to creative industries, and AI-generated art, music, and literature are arising phenomena that challenge traditional notions of creativity and authorship. These current advancements also give, despite challenges, the possibility to stimulate debates on AI technologies' ethical implications and societal impact.

3.2 Defining Creativity

Defining creativity as a standardised psychological construct has historically been challenging (Sternberg and Lubart 1998), and the American Psychological Association defines creativity in a rather broad way as "the ability to generate or develop original work, theories, techniques, or ideas" (APA dictionary of psychology). It has been argued that, in humans, creativity may have evolved as an advantageous trait from an evolutionary perspective (Gabora and Kaufman 2010; Mehta et al. 2020; Wiggins et al. 2015; however, please note that such perspective has also been often challenged, see, e.g., Dasgupta 2004; Simonton 1999). This perspective suggests that creativity has allowed humans to adapt to changing environments, solve complex problems, and develop innovative tools and strategies for survival. Over time, these creative capabilities could have contributed to the advancement of human societies by enabling cultural and technological progress. The development of language, art, and science, all products of human creativity, has played a crucial role in enhancing communication, fostering social cohesion, and improving the overall quality of life. Thus, creativity serves individual needs and enhances human communities' collective well-being and resilience.

Creativity is often connected to intelligence (Kaufman and Plucker 2011). Psychologist J.P. Guilford notably theorised the existence of two types of intelligence: convergent and divergent thinking (Guilford 1956; 1967). Divergent thinking is the process of coming up with a wide variety of suitable responses to an open-ended question or task, where the available information does not determine a single correct outcome. It focuses on creating numerous alternative ideas, often characterised by originality, unpredictability, or uniqueness. Consequently, divergent thinking is strongly associated with creativity.

In contrast, convergent thinking is aimed at finding the correct answer to a defined problem. This process involves evaluating multiple facts or ideas for their logical soundness and following specific rules. Convergent thinking emphasises elaborating a solution by recognising and applying pre-established criteria. Standard intelligence tests are typically designed to assess this type of thinking (Guilford 1956; 1967; Razumnikova 2020). Guilford (1956) identified four critical measures of divergent thinking: originality, which refers to the uniqueness or innovativeness of an idea; flexibility, which is the ability to view a topic from different perspectives; fluency, which is the capacity to generate numerous ideas; and elaboration, which involves the skill of developing, expanding, or building upon other people's ideas.

Divergent thinking can be assessed through various tests commonly used to evaluate divergent types of creativity. Notable examples include the Alternate Uses Task (AUT, sometimes also referred to as "Alternative Uses Task", see, e.g., Alhasim et al. 2020), where participants list as many uses as possible for an ordinary object; the Divergent Association Task (DAT), which measures the breadth of associative thinking by asking participants to list words that are as different as possible from each other; the Torrance Test of Creative Thinking (TTCT), which assesses creative potential through tasks involving verbal and figural responses; and the Consequences Task (CT), where participants imagine the consequences of hypothetical situations. These tests typically include prompting participants to produce a variety of ideas in response to different stimuli, aiming to measure their creative potential and divergent thinking abilities (Clapham 2020).

3.3 Creative Abilities of Machines

The contemporary discourse on the interplay between humans, machines, and the creation of creative products builds upon discussions initiated in the early twentieth century. A pivotal moment in this discussion was Walter Benjamin's 1935 essay, "The Work of Art in the Age of Mechanical Reproduction" (Benjamin 1935). Benjamin argued that mechanical reproduction techniques such as photography and film fundamentally altered the nature of a creative, artistic product by diminishing its "aura"—the unique presence and authenticity of the original work. While Benjamin's analysis centred on mechanically reproduced art forms, today's generative AI systems have transcended mere replication or human-assisted creation, producing art that appears unique (Brown et al. 2020; Radford et al. 2018).

The rapid spreading of AI-generated content, encompassing text and images (and more recently videos, Liu et al. 2024), has provoked discussions about the future of creativity and the evolving relationship between humans and machines (Still and d'Inverno 2019). Critics have often asserted that AI-generated outputs lack genuine originality, attributing this to the AI's dependence on pre-existing data, which can constrain its capacity for innovation and novelty (Lee 2011; Biswas 2023). Moreover, the use of pre-existing, potentially copyrighted data by AI-generated art has sparked significant ethical and moral debates, particularly regarding its potential to undermine the livelihoods of human artists (Young 2024). Despite these concerns, recent research has shown that AI can perform exceptionally well on tasks typically used to evaluate human creativity (Koivisto and Grassini 2023). Furthermore, emerging studies suggest that AI can produce artistic creations beyond imitating existing styles (Arriagada 2020; Carnovalini and Rodà 2020; Toivanen et al. 2019).

AI's artistic capabilities are not limited to replicating the styles of renowned artists (Mazzone and Elgammal 2019). It has developed original artistic styles and produced high-quality artworks that have been auctioned for significant prices (Jones

2018). Beyond visual art, AI has made notable contributions to other creative fields, including music composition (Rubinstein 2020) and poetry writing (Köbis and Mossink 2021). Research indicates that AI-generated art can be indistinguishable from human-created art, with blind tests revealing that people often attribute high artistic value to AI-produced works (Elgammal et al. 2017; Gangadharbatla 2022). Consequently, people frequently encounter scenarios akin to the Turing test (Turing 1950), where distinguishing between AI-generated and human-created art becomes challenging (Grassini and Koivisto 2024a).

Creativity has long been viewed as a distinctive trait that differentiates humans from other animals and machines (Ferrari et al. 2016). This view has often been connected to anthropocentrism, the belief in the superiority of humans over other species (Fortuna et al. 2021). Though frequently connected with speciesism (Millet et al. 2023), which suggests the precedence of a species over others, anthropocentrism specifically asserts human superiority (e.g., Schmitt 2020). This human-centred bias is already evident in children and is likely culturally influenced (Herrmann et al. 2010). It has been previously suggested (Millet et al. 2023) that the anthropocentric perspective influences multiple dimensions of individual thinking and public discourse, including areas such as cognition, ecology, human rights (Smith 2012), and animal rights (Batavia 2020). This preference for human supremacy challenges the acceptance of machines that can imitate human behaviour, especially in fields like AI-generated art (Schmitt 2020). AI-generated art challenges anthropocentric perspectives more directly than other AI applications, usually linked to analytical, mechanical, or computational tasks. Studies indicate that people are generally more reluctant to accept AI in activities considered uniquely human or creations with significant symbolic or emotional meaning (Castelo et al. 2019; Granulo et al. 2021; Chiarella et al. 2022; Grassini and Koivisto 2024a). Unlike analytical tasks, where AI's capabilities can be more readily accepted, artistic creativity is often seen as a core human trait intertwined with qualities such as corporeality, soul, emotions, insight, history, pain, and suffering. Creativity has been defined as creating something new and valuable (e.g., Burroughs et al. 2008). The notion that computers, typically perceived as lacking human attributes, can create art challenges one of the last supports of anthropocentrism.

3.4 Creative Performance of AI vs Humans

During the past few years, several studies have tried to understand how humans and AI are compared when performing standardised creative tests generally employed to evaluate human creativity in psychological research (Plucker et al. 2010). These attempts were made possible by the recent implementation of generative AI systems in user-oriented chatbots, such as Open AI's ChatGPT, that are user-friendly and widely available.

The AUT has become a widely used tool for evaluating the creative capacity of generative AI systems. Its broad adoption is likely due to the extensive scientific literature supporting diverse methods of administration and scoring. Additionally, platforms like SemDis, which automatically assess the semantic distance between texts generated in tests like the AUT, have improved the efficiency and—potentially—the objectivity of scoring compared to traditional, subjective human evaluations. SemDis leverages LLMs and vector-based representations to quantify the originality and diversity of responses, offering a data-driven approach to creativity assessment (Beaty and Johnson 2021; Beaty et al. 2022). The study of Stevenson et al. (2022) assessed the creative potential of GPT-3 using the AUT and compared its performance to that of humans. The study involved 42 students and 144 GPT-3 runs, with responses evaluated on originality, usefulness, surprise, and semantic distance. The results indicated that human responses were generally more original, surprising, and had greater semantic distance, while GPT-3 excelled in utility. Humans showed higher originality and surprise ratings, and their responses were higher regarding semantic distance, indicating a higher level of divergent thinking. GPT-3, however, provided more useful responses overall. The study also measured flexibility by categorising responses and found that human responses had higher flexibility scores, though GPT-3 showed a more considerable variance in flexibility. The findings suggest that while GPT-3 demonstrates good creative potential and can produce human-like responses, it still lags behind humans' performance in specific creative dimensions. Similarly, Koivisto and Grassini (2023) studied the creative abilities of different AI-based methods and using the AUT. The study compares these AI models with human participants in generating uncommon and creative uses for everyday objects. Participants included 256 human subjects and AI chatbots ChatGPT-3.5, ChatGPT-4, and Copy.Ai (a chatbot serviced based on GPT-3 at the time of testing). Each participant and AI session involved generating creative uses for objects like rope, box, pencil, and candle. Human raters evaluated responses based on semantic distance and subjective creativity ratings. Results showed that AI chatbots generally outperformed human participants in mean and maximum scores for semantic distance and subjective creativity ratings. However, the highest scores in originality were achieved by human participants. The study suggests that AI can produce creative outputs comparable to or exceeding the average human level, but the best human performers still excel in creative tasks. The authors suggest that AI's superior performance in the AUT may be due to its vast memory and rapid access to large datasets of semantically distant words. Other research expands on these findings. Haase and Hanel (2023) tested six genAI chatbots—alpha.ai, Copy.ai, ChatGPT-3 and -4, Studio.ai, YouChat—and 100 human participants. The creativity of the generated ideas was assessed based on quality and quantity, focusing on originality, usefulness, and fluency. The findings indicate no significant qualitative difference between AI and human-generated creative outputs. While most human participants matched the creativity of the tested chatbots, 9.4% of humans were found to be more creative than the most creative chatbot, GPT-4. The study also highlights that while AI can generate many ideas, defining the creative problem and evaluating the usefulness of ideas remains a uniquely human task. The research suggests that genAI chatbots may act as valuable assistants in the creative process, capable of producing creative outputs comparable to human-generated ideas in everyday creative tasks.

Bangerl et al. (2024) investigated the creative potential of AI again, comparing it to humans in divergent thinking tasks using the AUT. The study involved 20 human dyads and 10 AI chatbots. The results showed that AI chatbots excelled in elaboration, generating more detailed and coherent responses. At the same time, humans outperformed AI in originality and flexibility, producing a broader range of unique and diverse ideas. AI ideas were more conventional and feasible, whereas human ideas were often more unconventional and complex. These findings highlight potential complementary strengths of human and AI creativity, suggesting benefits for collaborative efforts.

In addition to the AUT, studies have employed other creativity tests. Cropley (2023) investigates the creative potential of AI, specifically ChatGPT (v. 3.5 and 4), compared to human creativity using the Divergent Association Task (DAT), a test of verbal divergent thinking. The study finds that both versions of ChatGPT surpass human averages in the DAT, with GPT-4 showing exceptionally high performance. The results showed that the mean DAT score for GPT-3.5 was higher than 64.93% of human responses, and GPT-4 outperformed 82.54% of human responses. However, the results are tempered by issues of reliability and predictability, as ChatGPT's responses often repeated words and followed patterns, therefore raising doubts about the novelty level of its output. Despite these issues, the reported analysis indicated that both GPT-3.5 and GPT-4 achieved statistically significantly higher mean scores than the human sample despite reporting a small effect size for the difference. Guzik et al. (2023) evaluated the creative performance of OpenAI's GPT-4 as assessed by the Torrance Tests of Creative Thinking (TTCT). The study evaluates GPT-4's fluency, flexibility, and originality performances compared to a human sample and a national percentile from Scholastic Testing Services. The TTCT was chosen for its comprehensive and well-documented methodology, which includes an extensive database of historical human responses, providing a robust benchmark for comparison. This test includes six distinct creative thinking activities, each designed to elicit different aspects of creative thinking, from generating novel questions about a picture to imagining improbable scenarios and improving products. The study generated GPT-4's responses through chat sessions in ChatGPT so that no prior history would influence each test attempt. These responses were then blind-scored by human raters. The results revealed that GPT-4 achieved significantly higher fluency and originality scores than the human control group. For overall fluency, GPT-4 submissions scored within the top national percentile, with a mean score of 99, while the human control group's mean score was 61.2. For flexibility, GPT-4's national percentile rankings were reported from 93 to 99, with a mean of 97.1, and for originality, GPT-4's scores were in the top percentile, significantly outperforming the human control group. While GPT-4 performed well in producing a high volume of ideas (fluency) and generating novel and unexpected concepts (originality), its flexibility scores were lower in specific tasks, such as predicting causes, anticipating consequences, and improving products.

Other studies have also explored creativity by combining several different types of tasks. The study of Hubert et al. (2024) used tasks related to divergent thinking to measure originality, fluency, and elaboration in responses from human participants and GPT-4. Human participants (N = 151) and GPT-4 instances were tested, with responses matched 1:1 for fluency. GPT-4 outperformed humans in originality and elaboration in the AUT and Consequences Task (CT) and generated words with higher semantic distance in the Divergent Associations Task (DAT), indicating more remarkable originality. Despite humans showing more flexibility, this did not translate into higher originality scores. The study emphasises computational creativity, where AI models like GPT-4 exhibit creative potential through pattern recognition and novel idea generation, using the Open Creativity Scoring tool (OCS) for objective evaluation. The study argues that, despite AI demonstrating high creative potential, it may lack the metacognitive processes and emotional influences driving human creativity. Furthermore, AI's creativity is contingent on human input and training data, making it unclear to what extent AI models' "cognitive architecture" is directly responsible for creative behaviours. Bellemare-Pepin et al. (2024) used an extensive data set from 100,000 human participants and employed the DAT to assess creative abilities, particularly the ability to generate semantically diverse word sets. The results showed that GPT-4 outperformed 72% of human participants. The study also examined hyperparameter tuning, finding that higher temperature[1] settings increased creativity scores. Other studies have also supported an association between temperature and AI output creativity (Chen and Ding 2023; Roemmele and Gordon 2018). However, it has been suggested that temperature's influence on creativity is more limited than frequently claimed (Peeperkorn et al. 2024). Furthermore, different prompting strategies, such as varying etymology or using a thesaurus, further boosted LLM creativity score. In addition to the DAT, the research also included creative writing tasks like haikus and flash fiction, where GPT-4 consistently outperformed previous AI models included in the testing but remained less original than humans. Metrics such as Divergent Semantic Integration and Lempel–Ziv Complexity were used to measure semantic and structural depth. Interestingly, the study showed that newer GPT-4 versions did not consistently improve performance, and while LLMs showed strong adherence to prompts, they did not always surpass human originality.

3.5 The Case of Complex and Multimodal Tasks

Further research compared human and AI creativity in more practical and complex tasks, such as generating product ideas and text writing. The study by Castelo et al. (2024) demonstrates that GPT-4 can generate product ideas rated as more creative than those produced by laypeople and even surpass those from creative professionals working under financial incentives. The study measured two types of creativity: creative form (language used) and creative substance (the novelty of the idea). GPT-4 excels in both areas. Additionally, GPT-4's ability to rewrite human ideas enhances

[1] In the context of generative AI, "temperature" refers to the degree of randomness in generating outputs, higher values of this parameter generally lead to more diverse results, while lower values produce more deterministic outputs.

their perceived creativity. These findings carry implications for integrating AI into creative processes to optimise outcomes, reduce resource expenditure, and accelerate innovation. The study by Orwig et al. (2024) investigates the linguistic features of creative writing by analysing short stories produced by humans and LLMs. The authors collected 600 short stories from human participants recruited online and requested from GPT-3 using its API. These stories were then rated for creativity by an independent group of 240 human raters and by GPT-4. The results revealed two key factors highly predicted perceived creativity: 1) semantic diversity, measured by the degree to which a story integrates semantically distant concepts based on distributional semantics, and 2) the presence of perceptual details that vividly describe sensory experiences. Surprisingly, there was no significant difference in creativity ratings between human-written and GPT-3 generated stories, suggesting LLMs can produce narratives on par with human creativity. This finding was replicated in an additional sample of 120 stories produced by GPT-4. Furthermore, the study found a strong positive correlation between ratings of creativity performed by humans and GPT-4, verifying the potential of LLMs for automated creativity assessment. The study also showed that semantic diversity and perceptual detail predicted GPT-4's creativity scores.

Finally, a recent study attempted to use a novel psychological test capable of assessing multimodal creativity, incorporating both language (as in most previous research) and the processing of visual stimuli. This study, by Grassini and Koivisto (2024b), explored AI's creative capabilities by comparing the performance of ChatGPT-4 with that of humans. A recently developed creativity test, the Figural Interpretation Quest (FIQ; Erwin et al. 2022; Koutstaal 2024), was used for this comparison. The study assessed two key aspects of creativity: flexibility and subjectively perceived creativity. Flexibility was measured through the semantic distance between responses, while subjective creativity ratings were provided by independent evaluators. Participants (246 humans and 247 ChatGPT-4 prompts) were asked to provide two creative interpretations for each of the four ambiguous, abstract figures. The results indicated that ChatGPT-4 demonstrated higher flexibility on average, while human participants outperformed AI regarding subjective creativity (i.e., creativity as judged by human raters). Moreover, the best human responses beat AI's performance in both flexibility and creativity. The study's results suggest that while AI excels at generating semantically diverse interpretations based on prompt instructions, it may still fall short of human creativity, mainly when assessed by human evaluators. Notably, this study was the first to demonstrate that AI can produce high-level creative responses in a multimodal task involving both images and text. Furthermore, the findings showed that AI chatbots can generate such responses even for tasks where they have minimal or no prior exposure to the training data. Nevertheless, as consistent with earlier research, the most creative human responses still outperformed the best outputs generated by AI.

3.6 Discussion

The present chapter summarises and examines recent research studying how humans and AI compare when it comes to creativity, with an explicit focus on current research where AI creativity is assessed and directly compared to human creativity, using standardised tests generally employed to evaluate human creative abilities. Integrating AI in artistic creation presents a complex landscape that challenges traditional notions of creativity and originality. Unlike mechanical replication, previously discussed in the philosophical literature, generative AI produces statistically unique artworks, compelling us to rethink the very definition of creativity. Studies directly comparing human and AI creativity provide critical insights into the capabilities and limitations of generative AI. These investigations broaden our understanding of creativity—a domain traditionally viewed as uniquely human—now increasingly shared with advanced AI systems. Research into AI's performance on standardised creative tests has significantly contributed to our understanding of creativity and the potential for human-AI collaboration.

Recent studies highlight both the remarkable capabilities of generative AI in creative production and the distinct strengths inherent in human creativity. For example, research has shown that while chatbots demonstrate proficiency in creative tasks, human responses often exhibit greater originality and semantic distance, key markers of divergent thinking (Stevenson et al. 2022). Other authors have indicated that ChatGPT surpassed human averages in the DAT, though concerns persist regarding the repetitive patterns in AI outputs (Cropley 2023). It has also been found that in other types of creative assessments, such as the TTCT, GPT-4 demonstrated exceptional performance, ranking in the top 1% for both originality and fluency (Guzik et al. 2023). However, it has been observed that while AI frequently generates more creative responses on average, the highest-quality ideas still originate from humans (Koivisto and Grassini 2023). In another study, findings show that although ChatGPT-4 exhibited greater flexibility when interpreting ambiguous visual stimuli, human-generated responses were generally rated as more creative by human evaluators and that ChatGPT-4 showed a more repetitive pattern of responses compared to humans (Grassini and Koivisto 2024a, b).

3.7 Challenges and Future Directions

Nearly all the analysed studies relied on well-established tasks commonly used in the literature to evaluate creativity. Documentations for these tasks, such as explaining and providing examples for the AUT and the TTCT, are extensively available online. Consequently, it is challenging to determine whether the AI's creative outputs were generated from elaborative processes within the model or were simply retrieved from the memorised training materials. However, as Koivisto and Grassini (2023) noted, a similar issue applies to human participants, who may have previously been exposed to the tests or learned optimal responses from online resources. A notable exception is a recent study by Grassini and Koivisto (2024b), which utilised the newly developed FIQ (Erwin et al. 2022; Koutstaal 2024) to evaluate creativity in humans and AI. The novelty of the FIQ at the time of testing and the limited availability of related materials online meant that details and optimal responses were likely absent from the AI's training data. This finding suggests that when given specific instructions, AI chatbots can generate creative outputs without relying solely on stored human data.

Several methodological issues arise from these studies. One critical issue is the consistency and novelty of AI responses, which often display patterns and repetitions, raising concerns about the originality and variability of their outputs (Cropley 2023). Grassini and Koivisto (2024b) supported this by showing that AI-generated responses to creative tasks exhibit less variability than human responses. In alignment with this hypothesis, a recent study on human-AI collaboration in short story writing offers further insight. Doshi and Hauser (2024) found that while generative AI can enhance individual creativity by producing outputs that seem more creative, well-written, and enjoyable - particularly for those with lower initial creativity - it simultaneously reduces the collective diversity of novel content. AI-assisted stories tend to be more similar than those created by humans alone. Taken together, these findings point toward the risk that, though AI may help individuals boost creative capacity, its use could result in a homogenisation of creative expression.

Furthermore, standardised tests like TTCT and DAT may only partially capture human creativity's complex nature (Guzik et al. 2023). Moreover, AI's creative outputs are influenced by its training data, which may limit its ability to generate genuinely novel ideas beyond its learned patterns (Koivisto and Grassini 2023). The analysed studies often focused on specific aspects of creativity, such as divergent thinking or idea fluency, potentially overlooking other critical dimensions like emotional and contextual creativity (Hubert et al. 2024). Findings based on specific AI models and tasks may not be generalisable across different AI systems or broader creative contexts (Castelo et al. 2024). Furthermore, current evaluation metrics may not adequately capture the qualitative dimensions of creativity, especially those related to human experiences and emotional depth.

The studies also evaluate AI creativity in isolation, with limited exploration of dynamic human-AI collaborative processes. Future research should address these limitations and expand the scope of creativity assessments to include emotional, contextual, and experiential aspects, providing a more complete evaluation of AI's creative capabilities. Investigating the dynamics of human-AI collaboration in creative tasks, focusing on how AI can augment human creativity rather than merely comparing humans and AI creative output, is another crucial area for future research (e.g., Heyman et al. 2024; Wan et al. 2024). Moreover, collecting longitudinal data to understand the long-term impacts of AI integration on human creative processes and outcomes can provide essential insights about best practices and creative fields in which human-AI collaboration may be particularly beneficial.

The findings discussed in Bellemare-Pepin et al. (2024) highlighted the critical impact of hyperparameter tuning and prompting strategies on LLM performance, suggesting that these factors may be crucial in releasing the full creative potential of such models. However, a greater understanding of how these parameters influence the underlying generative processes is still lacking outside the technical domain. Future work should investigate the mechanisms at play and develop principled approaches for optimising hyperparameters and prompts for specific creative tasks. In the study by Orwig et al. (2024), a fundamental limitation was the reliance on specific pre-defined prompts for a particular narrative writing task, which likely influenced the output of the AI in terms of content and style. This limitation may also apply to the other studies that have compared AI and human performances, as it remains unclear how different prompt styles may impact the performance of LLMs. Future research should investigate creative writing across a broader range of genres, topics, and less structured contexts to assess whether the results are generalisable.

A general limitation of the discussed studies lies in their core methodology for comparing AI and human outputs, particularly in the imbalance between the diversity of human participants and the limited set of AI models used. Human outputs originate from diverse individuals with unique experiences, biology, and cognitive processes. In contrast, the tested AI outputs stem from the same underlying model via multiple prompts, akin to queries from "cognitive clones". Each human's creativity is shaped by distinct personal histories and subjective experiences, contributing to the richness and variability of human original productions. This is reflected by the results often reported in these studies, where human outputs are commonly more varied and diverse in quality. Despite sophisticated training on vast datasets, AI models may lack this diversity. They generate responses based on patterns in their training data, but these responses do not reflect human individuality. Consequently, while the AI models may exhibit impressive creative capabilities, their outputs may be inherently limited by the "average" nature of the data where they are trained. This distinction underscores the importance of considering the source of creative outputs in comparative studies. It highlights one of the possible sources of uniqueness and variety of human creativity, which stems from diverse individual experiences.

It is also important to note that the human participants in these studies are generally recruited online, often participating in exchange for financial compensation. This introduces potential concerns regarding the level of effort, motivation, and attention participants invest in the tasks. Since the primary incentive is monetary, participants may not be fully engaged or intrinsically motivated, potentially leading to suboptimal performance. This factor could influence the quality of human outputs, affecting the validity of comparisons between AI and human creativity. Additionally, while online crowdsourcing platforms provide access to large, diverse participant pools, the absence of direct supervision and inconsistent incentive structures may lead to variability and suboptimal participant performance. In contrast, the AI models were probably constantly tested at optimal capacity, potentially creating an uneven playing field. In such comparative studies, future research should explore the impact of motivational factors and incentive structures on human creativity. Incorporating mechanisms to foster engagement, such as gamification elements, performance-based rewards, or intrinsic motivators, may prompt heightened creative efforts from human participants.

The findings that AI may reach human-level creativity in specific tasks have fueled fears that AI might replace human creative jobs (Leffer 2023). However, it has often been emphasised already in published studies that current findings should inspire new ways for humans to leverage AI's potential to enhance their creative capabilities instead of suggesting the replacement of humans in creative tasks (Koivisto and Grassini 2023; Bangerl et al. 2024). For example, Grassini and Koivisto (2024b) suggest that humans and AI possess different abilities when it comes to creative behaviour, potentially stemming from the distinct nature of human experiences and AI cognitive processes in generating creative outputs. Unlike AI, which relies on statistical data patterns to produce responses, human creativity is rooted in a broad spectrum of personal experiences and sensory-driven cognitive processes. Therefore, humans can capitalise on the specific strengths provided by AI (Grassini and Koivisto 2024b).

It is important to note that while AI can excel or even surpass human performance in controlled and specific experiments, this does not automatically mean that AI will be able to replace complex creative jobs typically performed by humans. Creative roles often require high adaptability and the ability to excel across several creative tasks—skills that cannot be easily summarised in cognitive tests. In addition, despite current concerns, it is unlikely that AI's ability to produce creative outputs, such as artistic production, will diminish the importance of the subjective experiences of human artists or the innate human inclination to express themselves through art. Much like the invention of photography did not devalue the artistic experience or the personal significance of painting landscapes and portraits, AI is poised to complement rather than replace human creativity. For humans, art serves as a medium for storytelling, cultural expression, and emotional release, which go beyond the mere production of an artistic work. These aspects are rooted in the creative process itself, where the act of making art is often as meaningful, if not more so, for the artist than the final product.

Indeed, the development of AI-generated art introduces significant economic challenges for human artists. As AI technology can already produce artworks that are increasingly difficult to distinguish from human-made creations (Grassini and Koivisto 2024a, b), artists may find it harder to compete in a market flooded with AI-generated pieces. The efficiency and accessibility of AI tools enable the rapid production of art, which could oversaturate the market and potentially diminish the value of individual works. This influx of AI-generated content threatens to undermine the financial stability of artists, as the abundance of readily available, low-cost artwork may reduce the demand for human creative jobs. Moreover, the scalability and low production costs associated with AI art create additional barriers for artists seeking fair compensation, further complicating their ability to sustain their livelihoods.

As a result, strategies are needed to protect the economic and cultural value of human art. Efforts to preserve human artistry include updated intellectual property protections, updated copyright laws, and platforms that emphasise the irreplaceable societal role of human-made art. Furthermore, promoting public appreciation for the creative process and its intrinsic value, rather than just the final product, could help sustain the demand for human creative works. By adapting to this rapidly evolving landscape, the art world can ensure AI serves as a complementary tool rather than a threat, preserving the essential role of human artists while ethically integrating AI technology.

3.8 Conclusion

Comparing AI and human creativity through standardised tests reveals a complex landscape where AI demonstrates impressive capabilities, often matching or surpassing average human performance in specific creative tasks. However, these findings should not be interpreted as a sign that AI will replace human creativity. Instead, they highlight the potential for synergistic collaboration between humans and AI in creative activities. The unique strengths of human creativity, possibly related to very diverse personal experiences, emotional depth, and contextual understanding, may complement AI's computational power and pattern recognition abilities. Future scientific endeavours should focus on studying how AI can augment human creativity, addressing the methodological limitations of current studies, and investigating the long-term impacts of AI integration in creative processes. As we study the possible future of AI technologies, it is fundamental to consider the ethical implications and societal impacts of AI in creative fields, ensuring that technological advancements align with human values and benefit the whole society.

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Chapter 4
The Role of AI in Transforming Educational Models: A Critical Analysis of Opportunities and Challenges

Federica Caboni and Lucia Pizzichini

Abstract This chapter reflects on the impact of artificial intelligence (AI) on the educational sector by considering several implications for creating personalised learning and the infrastructure of academic institutions. Hence, this chapter aims to understand the role of artificial intelligence in helping and transforming the educational framework in a more creative environment. The use of artificial intelligence in educational fields facilitates the creation of new innovative processes that are useful for improving creativity among students and educators. In fact, by exploiting artificial intelligence, it is possible to help students develop their ideas and help educators create study plans based on innovative projects enriched by immersive experiences. This chapter, based on an overview of Artificial Intelligence in Education, aims to identify the main elements of the transition from the traditional education system to a new educational model based on immersive technologies. In particular, the chapter seeks to specify how immersive technologies that exploit artificial intelligence can improve students' learning abilities and generate a dynamic environment for both students and educators. Finally, the chapter explores the challenges and opportunities accompanying AI integration into higher education, such as data privacy, the digital divide, and concerns about the authenticity of AI-generated content.

Keywords Artificial Intelligence · Educational Model · Gen-AI · Immersive Experience · Creative Environment

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4.1 Introduction

The world is witnessing the advent of artificial intelligence (AI), reshaping various economic sectors. Of course, education is no exception. Nowadays, technologies based on artificial intelligence are becoming an essential part of educational models, and in particular, teachers and researchers are reshaping the way to teach and educate (Holmes et al. 2019; UNESCO 2022). In this vein, this chapter tries to provide a comprehensive and exhaustive analysis of AI's impact on the educational sector, with a focus on personalised learning, the teacher-student dynamic, institutional infrastructure, and the creative potential unlocked by Generative AI in academic settings (Roll and Wylie 2016).

In the future, it is possible to witness the pervasive integration of artificial intelligence in educational programs at various levels by looking at the creation of interactive and immersive lessons for students at each level of education. We could turn our gaze to artificial intelligence tools that help students in creative problem-solving, develop innovative projects, and generate novel ideas (Brynjolfsson and McAfee 2017; Zhao 2024). At the same time, educators can explore new ways to design their lessons and programs by using artificial intelligence to personalise and engage learning experiences (Luckin et al. 2016).

However, this chapter aims to examine the actual landscape of AI applications and the role of AI in Transforming Educational Models by offering a critical analysis of opportunities and challenges. In particular, the following parts will highlight how technologies based on artificial intelligence can enhance learning outcomes, facilitate inclusive education, and stimulate creativity (Selwyn 2021). However, the rapid integration of AI also raises significant challenges, particularly in data privacy, the digital divide, and the authenticity of AI-generated content (Williamson et al. 2020). Through a balanced analysis of both the opportunities and challenges posed by AI in higher education, this chapter contributes to ongoing discussions on reimagining educational models for the twenty-first century (Ng 2018; Algerafi 2023).

4.2 Literature Review

4.2.1 An Overview of Artificial Intelligence in Education

In the last decade, artificial Intelligence has been progressively incorporated into educational systems by using several kinds of digital and immersive applications (see Fig. 4.1). According to Luckin et al. (2016), nowadays, it is not possible to recognise the fundamental role of AI in education in creating more tailored learning experiences. Adaptive learning systems, which use AI algorithms to adjust content based on individual students' performance, have gained traction, particularly in higher education. These systems allow students to learn at their own pace, receive real-time feedback, and focus on areas where they need improvement.

Fig. 4.1 The transition of education. Image by the authors

In this era, which is considered a very innovative era, researchers and educators need to turn their attention to understanding how AI tools can provide automated support and guidance and find answers as much as possible on the role of educators in an AI-driven classroom. McCarthy (2017) notes that while AI can enhance certain aspects of teaching, it cannot replicate the emotional intelligence and mentorship that human teachers provide. Therefore, a balance between AI support and human interaction is crucial. In this vein, integrating artificial intelligence (AI) into educational settings has sparked a transformative shift in the teacher-student dynamic, fundamentally altering the roles and interactions within the classroom. As teachers are moving to an increasing adoption of AI tools, and for this reason, the traditional paradigms of teaching and learning need to be redefined. Considering these concepts, this part of the chapter explores the multifaceted implications of AI on the teacher-student dynamic, drawing on various scholarly sources to provide a comprehensive understanding of this evolving landscape.

Hence, the advent of AI technologies, such as ChatGPT and other intelligent systems, has introduced new methodologies for enhancing educational interactions. Baskara (2023) highlights how AI-driven tools can facilitate more engaging and personalised learning experiences, allowing teachers to focus on fostering critical thinking and creativity among students. This shift necessitates a new evaluation of pedagogical practices, as educators must adapt to the capabilities and limitations of AI systems. As noted by Akavova et al (2023), the implementation of adaptive learning technologies enables tailored educational experiences that cater to individual student needs, thereby enhancing the overall learning process. In this scenario, the teacher plays a fundamental role in communicating knowledge through artificial intelligence tools.

Bai (2023) enhances the importance of educators in helping students critically engage with AI technologies, ensuring that they understand the implications of their use. This perspective aligns with the findings of Popenici and Kerr (2017), who stated that teachers must develop a nuanced understanding of AI applications to integrate them into their teaching practices effectively. The need for professional development in AI literacy is further underscored by Ng et al. (2023), who argue that educators must cultivate digital competencies to harness the full potential of AI in education.

Hence, to reshape the teacher-student dynamic, AI technologies have the potential to enhance collaborative learning experiences. Liu et al. (2022) discuss how AI can facilitate peer interactions and collaborative projects, allowing students to engage in meaningful discussions and problem-solving activities. This collaborative aspect is further supported by the work of Koh et al. (2022), who argue that AI can help reconnect teachers and students by providing personalised feedback and support, thereby fostering a more interactive learning environment. In particular, the implications of AI in education extend beyond individual classrooms, influencing institutional practices and policies. Institutions striving to remain relevant in a rapidly changing educational landscape must prioritise integrating AI technologies into their curricula and support systems. Notably, the above considerations need to be taken into consideration by higher education, where AI can enhance instruction quality and improve student outcomes, as noted by Du (2023) and Yildirim et al. (2021). Furthermore, the role of AI in addressing educational disparities is highlighted by the findings of Yildirim et al. (2021), who argue that AI can provide equitable access to resources and support for diverse student populations.

4.3 Generative AI and Creativity in Education

Generative AI can be considered a part of Artificial intelligence, which is able to create content based on a high level of creativity that can enhance interest in the educational sector. Some scholars like Veale and Cardoso (2019) have examined the potential of Generative AI to stimulate creative thinking among students by stating that this system is able to provide unique opportunities for students to engage in creative problem-solving. Through the several tools recognised as generative AI tools, it could be possible to identify GPT-4 and DALL·E that help students generate new ideas, draft creative content, and visualise complex concepts, thereby enriching the educational experience.

The fourth industrial revolution represents the scenario where creativity within education is fundamental in creating new types of learning and other competencies in this field (Ahmadi and Besançon 2017). In fact, educating students to recognise their strengths and cultivate their creativity skills appears fundamental to helping them manage in uncertain conditions. In this vein, the role of the teacher is essential for awakening, sustaining, and directing students' interests, taking into consideration their strengths, and offering individualised challenges based on their interests. A teacher should work strategically to develop students' interests and skills. When students feel competent, their perceptions of self-efficacy, ability, academic competence, and control are predictors of engagement and achievement (Skinner and Pitzer 2012). Hence, creativity in education is so much more than just learning a collection of steps, processes, tools, and cognitive and affective skills.

Furthermore, integrating artificial intelligence into educational programs has sparked significant interest in its potential to enhance creativity among learners. Creativity is increasingly recognised as a vital competency in the twenty-first century, essential for success in various fields, including science, technology, engineering, arts, and mathematics (Marrone et al. 2022). Education systems worldwide are adapting to emphasise creativity, with a growing consensus that fostering creative thinking is crucial for preparing students for future challenges (Marrone et al. 2022; Newton and Newton 2020). AI technologies are emerging as powerful tools to augment creative processes, enabling students to explore new ideas and collaborate more effectively (Anantrasirichai and Bull 2022). AI's role in education is multifaceted, encompassing personalised learning experiences, real-time feedback, and enhanced collaboration (Wang et al. 2023; Mulaudzi and Hamilton 2024). By leveraging AI, educators can tailor learning experiences to individual student needs, promoting engagement and creativity (Trisnawati et al. 2023). For instance, using AI-driven narrative generation tools allows students to focus on higher-level conceptualisation and creative expression, similar to how calculators augment mathematical capabilities without replacing human thought (Lai 2023). This perspective aligns with intelligence augmentation (IA), where AI is a supportive tool rather than replacing human creativity (Kim et al. 2022). Moreover, the collaborative potential of AI technologies in educational settings is noteworthy. Research indicates that student-AI collaboration can enhance learning outcomes by fostering critical thinking and creative problem-solving skills (Kim and Lee, 2022; Vallis et al. 2024). For example, students who engage in storytelling with AI-generated suggestions find that this process enhances their understanding and creativity by combining AI inputs with their ideas (Kim and Lee 2022).

This collaborative dynamic enriches the learning experience and encourages students to experiment with new forms of creative expression (Anantrasirichai and Bull 2022; Kim and Lee 2022). The rise of generative AI systems, such as ChatGPT, has also prompted discussions about their implications for creative education. These systems can facilitate rapid experimentation and exploration of ideas, allowing students to engage with creative tasks innovatively (Anantrasirichai and Bull 2022; Creely 2023). However, concerns about preserving human originality and the potential for over-reliance on AI tools remain prevalent (Lai 2023; Creely 2023). Educators must balance utilising AI as a creative partner and nurturing students' intrinsic creative abilities (Niloy et al. 2024; Caporusso, 2023). In the creative industries, AI technologies are reshaping traditional practices and expanding the boundaries of creativity. Studies have highlighted the transformative impact of AI on music production, visual arts, and design, where AI acts as a collaborator that enhances human creativity rather than diminishes it (Deruty et al. 2022; Anantrasirichai and Bull 2022; Hughes et al. 2021). For instance, AI tools in music composition allow artists to explore new sonic landscapes and generate novel ideas, thereby enriching the creative process (Deruty et al. 2022; Hughes et al. 2021). This collaborative approach fosters a new understanding of creativity, where human and machine contributions are intertwined (McCormack et al., 2020; Larsson et al., 2022). In creating artificial intelligence content, educators must also consider questions regarding authorship, ownership, and the authenticity of creative works (Haase and Hanel, 2023; Liu 2023). Furthermore, integrating AI technologies into educational programs presents challenges and opportunities for educators. While AI can enhance learning experiences, it also necessitates rethinking assessment strategies and pedagogical approaches (Du and Wang 2023). Researchers and teachers need to adapt their teaching methods in order to include artificial intelligence in their programs, ensuring that students are equipped with the necessary skills to thrive in an AI-enhanced creative landscape (Khanduri 2023; Crompton and Burke, 2023). This includes fostering digital literacy and critical thinking skills, which are essential for navigating AI technologies' complexities (Khanduri 2023; Crompton and Burke 2023).

4.4 Opportunities and Challenges

4.4.1 Opportunities

Personalised learning

The use of artificial intelligence in educational frameworks permits personalised learning experiences (see Fig. 4.2). Through customised programs, it is possible to develop specific activities to meet the needs of each student, enhancing engagement inside and outside the classroom and improving learning outcomes. A personalised approach, thanks to artificial intelligence, is able to meet several goals in line with an inclusive educational environment. One of the most exciting prospects of AI in education is its potential to nurture creativity. Generative AI tools, such as those used for creative content generation and design, allow students to explore innovative solutions to problems, think outside the box, and develop projects that blend technology and creativity. These tools encourage learners to engage with material dynamically and interactively, transforming traditional learning into a more participatory process.

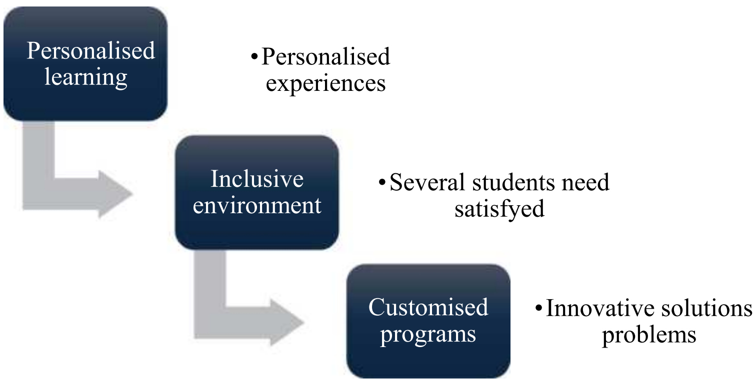

Fig. 4.2 Opportunities derived from personalised learning. Image by the authors

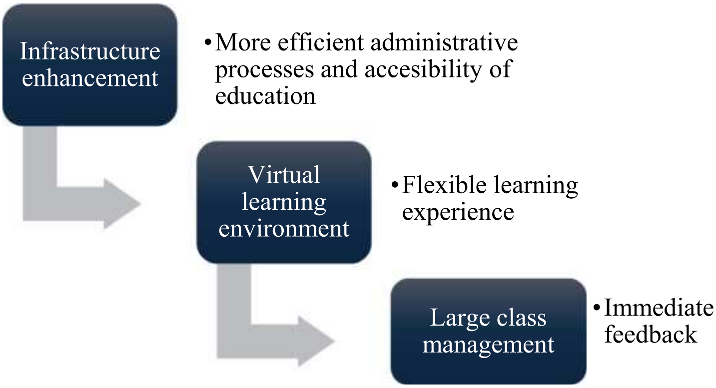

Fig. 4.3 Opportunities derived from enhanced infrastructure. Image by the authors

Enhanced Infrastructure and Educational Access

Artificial intelligence can also enhance the infrastructure (see Fig. 4.3) that typically characterises educational institutions in several countries, enabling more efficient administrative processes and improving the accessibility of education. For example, using virtual reality could develop virtual learning environments powered by AI that allow for more flexible learning experiences and cater to students from diverse geographical locations and backgrounds. Furthermore, learning systems developed by artificial intelligence could help educators manage large classes, streamline grading, and provide more immediate feedback to students.

4.4.2 Challenges

The integration of artificial intelligence (AI) in education represents a significant shift that has the potential to enhance learning outcomes greatly. However, this transformation also brings considerable challenges that need to be addressed carefully. As educational institutions implement AI-driven tools, several key issues arise, including cultural change, data privacy concerns, the digital divide, and the authenticity of AI-generated content. These challenges (see Fig. 4.4) emphasise the necessity for strong ethical standards and equitable access to technology. Furthermore, they highlight the importance of fostering an educational environment that prioritises student-centred learning and creativity.

Culture

Educators often encounter resistance to change, as traditional teaching methods are deeply rooted in current practices. According to Akgün and Greenhow (2021), successfully integrating AI requires a cultural shift within educational institutions, highlighting the importance of ongoing professional development and support for

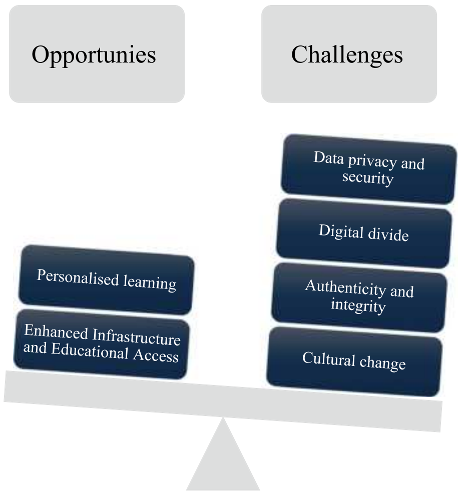

Fig. 4.4 Opportunities and challenges in AI education. Image by the authors

teachers. Moreover, it should be considered that AI can potentially contribute to worsening the inequalities already existing in education that may disadvantage certain groups of students Bai (2023). Therefore, the future of the teacher-student dynamic in the age of AI will likely depend on educators' ability to adapt to these changes while maintaining a focus on student-centred learning. Teachers should be able to keep their impact on teaching and learning processes based on the evolving nature of AI. This involves creating an environment where students can critically engage with AI tools, as emphasised by, and ensuring that ethical considerations are prioritised in the implementation of AI in education (Ng et al. 2023).

Privacy and Data Security

Educational institutions manage large amounts of sensitive student data, including academic records, behavioural patterns, and personal information. The reliance on AI systems to process and analyse this data increases the risks of data breaches, unauthorised access, and misuse. Therefore, data privacy issues have gained considerable attention with the rise of digital technologies in education (Selwyn et al. 2023). Inadequate attention to these risks can undermine student trust and pose legal and reputational threats to institutions. Another critical aspect is related to the opacity of AI algorithms, which often function as "black boxes.“ This lack of transparency makes it difficult for educators, administrators, and even developers to understand how decisions are made. Yan et al. (2024) highlights the urgent need for transparency in AI systems, especially regarding data collection and processing practices. When transparency is lacking, it erodes trust, creates ethical dilemmas, and complicates accountability efforts. Alrayes (2024) points out that unclear algorithms not only hinder accountability but can also reinforce biases that exist in the training data, leading to discriminatory and inequitable educational outcomes. These biases may disproportionately affect marginalised student populations, worsening existing educational inequalities and raising further ethical concerns. To address these challenges, educational institutions and policymakers should implement robust data governance frameworks that prioritise student privacy and comply with global data protection standards, such as the General Data Protection Regulation (GDPR). Additionally, institutions should invest in the development and deployment of explainable AI systems to enhance transparency by providing insights into how algorithms make decisions. Moreover, adopting an inclusive approach to AI development that involves stakeholders—including educators, students, and ethicists—can help identify and mitigate potential biases and ethical risks.

Digital Divide

The implementation of artificial intelligence (AI) can also amplify existing inequalities in education due to the current digital divide. Disparities in access to technology and high-speed internet create a significant barrier to equitable adoption of AI in educational settings. Students from low-income communities or underfunded schools often need access to the devices, infrastructure, and resources required to engage in AI-enhanced learning fully. Unequal access to technological resources disproportionately affects students from marginalised and disadvantaged backgrounds, limiting their ability to benefit from AI tools designed to personalise and enhance learning experiences. As Williams (2024) points out, AI has significant potential to improve personalised instruction, but it is reduced if students cannot have access to technological means to utilise these tools. As a result, the digital divide limits engagement with AI-based educational innovations and perpetuates systemic inequities in academic outcomes and opportunities. In order to ensure that all students, regardless of their socio-economic background, can access and benefit from AI and reduce inequalities, needs a coordinated effort from institutions, governments, and policymakers on strategic investments in infrastructure, equitable distribution of technological resources, and targeted training initiatives (Kumar and Mindzak 2024). Therefore, bridging the gap created by the digital divide is critical to promoting inclusive education and mitigating the risks of increasing existing inequalities in a technology-driven educational landscape.

Ethical Implications

The rise of generative AI raises important questions about the authenticity of AI-generated content. In educational settings, where the goal is to encourage original thinking and creative expression, there are concerns that students may become overly reliant on AI tools. This could lead to a potential devaluation of human creativity (Ifelebuegu 2023). To address these concerns, institutions should develop ethical guidelines and standards to ensure that AI-generated content is used to complement, rather than replace, human intellectual effort. Additionally, ethical considerations regarding the authenticity of AI-generated content are imperative. The rise of generative AI tools, such as ChatGPT, has sparked debates about academic integrity and the authenticity of student work. Williams (2024) argues that overreliance on AI-generated content may reduce students' engagement and critical thinking skills, as they may become passive consumers of information rather than active learners. This concern is also shared by Kumar and Mindzak (2024), who emphasise the need for educators to foster an environment that encourages originality and creativity rather than allowing AI to dictate the learning process. The challenge lies in balancing the benefits of AI assistance with the need to maintain academic standards and integrity. Furthermore, the ethical implications of AI in education go beyond questions of authenticity and privacy. As generative AI models become increasingly sophisticated, concerns about the biased content generated have emerged. Some scholars (Lombardi and Bailey 2024; Liang et al. 2023) highlight how biases embedded in training data can lead to biased or inappropriate output, which can further entrench stereotypes and misinformation in educational settings. This situation calls for a critical examination of the data used to train AI systems and the implementation of robust frameworks to mitigate bias and ensure fairness in AI applications (Sáez-Velasco 2024). Furthermore, the ethical implications of AI in education cannot be overlooked. Kahn and Winters (2021) discuss the concept of "digital structural violence," which highlights how AI algorithms can perpetuate existing inequalities by favouring those in power and marginalising disadvantaged students. Engaging with various stakeholders, including policymakers and industry representatives, is essential to establish guidelines that ensure equitable access to AI technologies and address the broader social implications of their use (Yan et al. 2024).

4.5 Results and Conclusion

This chapter highlights the transformative potential of artificial intelligence (AI) to reshape educational paradigms, enhance personalised learning experiences, and foster creativity through generative AI. These advances and the simplification of educational infrastructure offer significant opportunities to improve learning outcomes and promote inclusiveness in education. However, the integration of AI into educational systems also raises critical issues, including data privacy, the digital divide, and the authenticity of AI-generated content. Addressing these challenges is essential to ensure the ethical and equitable use of AI in education.

The analysis highlights that the implementation of generative AI technologies can help enrich educational experiences by fostering creativity and innovation. However, such implementation requires a careful balance between technological advances and ethical considerations. It is crucial to ensure that AI tools enhance, rather than undermine, the core principles and values of education. These technologies enable the development of personalised, adaptive, and creative learning environments, paving the way for a reinvented approach to education. However, AI comes with significant challenges, which require a critical examination of its implications to ensure its practical and ethical application. By exploring the opportunities and challenges presented by AI, this chapter contributes to the ongoing debate on redesigning educational models for the future. The intersection of creativity, education, and AI technologies represents a promising frontier for enhancing learning experiences and cultivating creative skills. As AI continues to evolve, its role in education will likely expand, underscoring the need for continued research and dialogue on its implications for creativity and learning. Maintaining this debate is essential to equip educators with the knowledge and skills to navigate and harness the potential of AI effectively. Educators must embrace the transformative possibilities of AI while remaining vigilant about its ethical implications and the importance of cultivating human creativity in an increasingly automated world. The future of AI in education will depend on our collective ability to harness its potential while addressing its risks. Collaboration between educators, policymakers, and technologists will be critical to ensure that AI integration promotes equity, fosters creativity, and supports the integrity of the learning process. In conclusion, while integrating AI into education holds immense promise for improving learning experiences and outcomes, it also presents significant challenges that require attention. Issues such as data privacy, the digital divide, ethical concerns around authenticity, and bias in AI-generated content require sustained dialogue and action among education stakeholders. By prioritising these challenges, educators and policymakers can work toward more equitable and effective adoption of AI technologies, ensuring they serve as tools for empowerment and innovation in education.

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Part II
Generative AI and Creativity in Education

Chapter 5
Transforming Education: The Potential of Generative AI Tools for Improving Learning Experience

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Abstract This chapter presents how Generative artificial intelligence (GAI) tools are revolutionising the existing educational praxis. Multimodal GAI, such as text, images, video, and games, are improving interaction skills in classrooms, promoting interculturality, and creating a more playful and engaging educational environment. Qualitative analyses have emphasised its capacity to stimulate learning processes and have proposed innovative applications in the field of teacher education, showing significant improvements also in students' learning outcomes in cognitive and affective domains, enhancing flow experiences. However, it is essential to acknowledge that GAI applications, despite offering personalised tutoring, automated essay grading, language translation, and interactive and adaptive learning, have certain limitations. These include a lack of human interaction, limited understanding and creativity, and biases in the training data. Therefore, careful implementation and oversight are necessary to address these drawbacks effectively. In this sense, game-based learning platforms with generative AI are designed to improve digital literacy by providing engaging and interactive experiences. Although these initiatives hold potential, they necessitate additional refinement and assessment before they can be practically implemented. Although AI presents remarkable prospects, it necessitates diligent supervision to guarantee its responsible and efficient incorporation into educational frameworks.

Keywords Generative Artificial Intelligence · Education · Learning

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5.1 Introduction

Education around the world has been experiencing several changes to follow the current needs of the population. Recently, Generative Artificial Intelligence (GAI) merged as an essential tool for many segments, such as tools to answer questions, complete written tasks, and respond to prompts in human-like ways, producing different results as audio, images, simulations, videos and texts.

In the education field, a big challenge has become: with the recent advances in GAI, how can we manage all the potential and ethical use with students in the classrooms? Also, GAI must require special attention from educators because it can produce unreliable information. Therefore, it is relevant that any content produced stands in need of professional discernment to observe for appropriateness and accuracy in the texts or materials originated by the GAI technology.

When used in education, GAI helps the pupils analyse, structure, and write different assignments. It also turns prompts into images, videos, and audio. With correct conduction, GAI has tremendous potential to reduce the workload across the education sector and allow teachers to prepare and deliver high-quality teaching. However, a particular focus for the screening is compulsory because GAI can produce several contents that are biased, inappropriate, and inaccurate.

5.2 Methodological Approaches to GAI in Education

The need to update the school environment to adapt to emerging technologies is a proposal that must reflect the needs of digital natives and their culture, creating an educational environment that stimulates creativity and cooperation. This requires moving away from the conventional teaching model, which limits student interaction and enhances contemporary education toward an intuitive teaching process focused on active learning.

Education has been evolving throughout its eras, starting with a focus on literacy using chalkboards and books, having the teacher as the sole source of knowledge. It then moved toward standardised mass education to meet industrial needs, leading into the era of information and communication technologies with computers and the internet. This era saw the emergence of new virtual learning environments with e-learning, granting greater student autonomy. From this point, connectivity and personalisation in the teaching and learning process expanded, facilitating social interactions and the rise of specific virtual tools for online collaboration and problem-solving, which continue to transform the current educational system and its practices (Baidoo-Anu and Ansah 2023).

In this context, Generative Artificial Intelligence (GAI) stands out by creating and enhancing educational systems through the development of standards and tools that help simplify problems (Rajendran 2023). GAI becomes an essential tool for personalising adaptive learning, respecting the unique learning pace of each student, and providing intelligent tutoring systems that assist teachers in optimising their time in the evaluation process, making it more dynamic with instant feedback (Bahroun et al. 2023).

The experiences of using GAI as teaching tools have been demonstrated through different methodological approaches, aiming to make these tools more intuitive and promoting active learning methodologies with more immersive and meaningful learning experiences. This prepares students for an increasingly technological and complex future, contributing to the development of a more just and equitable society (Silva and Kampff 2023).

The methodologies incorporating Generative AI in education aim to enhance the teaching and learning process. Among the most commonly used approaches are personalised learning and virtual assistance.

5.3 Personalized Learning and Virtual Assistance

In this type of technology, Artificial Intelligence Systems, such as chatbots, assist outside of school hours by answering questions and helping solve subjects' questions. GAI is employed to tailor materials and assignments to each student's level of understanding, speed, and learning method. Ruiz-Rojas et al. (2023) presents results from the implementation of the 4PADAFE instructional design matrix methodology to design a virtual course, demonstrating the potential of generative artificial intelligence tools in university education. He highlights that these tools, when combined with an instructional design matrix, allow educators to design and deliver personalised, enriching, and innovative educational experiences, engaging students by adapting content to promote personalised learning.

Dillig et al. (2024) emphasise that GAI tools like ChatGPT have demonstrated their potential to provide personalised guidance in education. Qualitative assessments show that GAI, like ChatGPT, can boost the learning process, especially when it is perceived as realistic and challenging. These findings highlight the need for new implementations of Large Language Models (LLMs) in teacher training and further studies on their ability to enhance skill development through dialogue-based learning.

5.4 Active Learning Methodologies

These methodologies place the student at the centre of the learning process, encouraging active and cooperative participation in knowledge construction. Instead of merely being receivers of information, students become the main actors in their learning, exploring, questioning, and solving challenges. This approach offers a critical and inquisitive education, directing students to the core of knowledge construction (Cunha et al. 2024).

In this type of approach, GAI expands the variety of resources accessible to students, enabling them to actively produce their texts, translate languages, create various types of creative content, and provide informative responses, transforming the way active methodologies are implemented. Research by Saraiva (2024) investigated the combination of active learning methodologies with generative artificial intelligence in higher education, highlighting how such methods can enhance engagement and learning through Project-Based Learning and the Flipped Classroom. It presents a simple method for incorporating GAI into the classroom, allowing teachers to improve the teaching process without needing to overhaul their entire curriculum.

5.4.1 Project-Based Learning (PBL)

In this methodological approach, Artificial Intelligence can play an important role by assisting students in project creation, resource provision, and even problem-solving, fostering active learning. Saraiva (2024) highlights in his research that the use of GAI combined with PBL and case studies provided students with a broader range of skills, both in problem-solving and self-directed learning.

Shaer and Cooper (2024) aimed to Integrate GAI into a Project-Based tangible interaction course; additionally, they attempted to address the question of how to instruct students in the responsible, deliberate, and ethical use of AI models and technologies while also making them aware of their limitations. They used the term co-creation to describe the process where humans and AI work together to create new artefacts or solve a problem, describing practical ways and learning objectives for integrating generative AI into project design and ML models for project-based interaction design courses for pervasive computing.

5.4.2 Flipped Learning

This is an active pedagogical approach that innovates by altering the structure of conventional teaching methods. In it, students absorb theoretical content at home through videos, readings, or other resources provided by the teacher. At the same time, classroom time is dedicated to practical tasks, debates, and problem-solving, enhancing student participation and cooperation (Saraiva 2024). With the inclusion of Generative AI, it is possible to compare student responses before and after using this technology, producing reflections on their actions and how their learning was impacted by the technology during practical teaching.

Lawan et al. (2023) investigate the possibility of using modified flipped learning as a strategy to lessen the adverse effects of GAI on education. The paper presents a modified flipped learning model that combines the adaptive qualities of GAI with the individualised and active learning components of flipped learning, highlighting the relevance and necessity of both in the context of twenty-first-century education.

In conclusion, by substituting the conventional role of the teacher with the use of GAI systems for pre-class activities and content development, the proposed modified flipped learning model will enable students to actively engage in the learning process and take ownership of their education.

5.4.3 Gamification

This approach integrates game and competition elements into the learning process. In this sense, Artificial Intelligence can develop personalised and adaptive experiences to increase student engagement, such as creating interactive narratives with GAI, generating personalised stories and games, and making the learning process more engaging and stimulating. Developing personalised challenges and missions for each student promotes exploration and problem-solving playfully. With the use of machine learning algorithms and gamification design, generative AI may offer students engaging and instructive learning opportunities. Learning games, for instance, can be created in virtual settings, enabling students to study and explore while having fun and increasing their motivation and interest in the subject matter (Yu and Guo 2023).

In this methodological approach, Muengsan and Chatwattana (2024) present a learning platform based on games with generative AI aimed at improving digital literacy by providing engaging and interactive experiences where students can respond to and interact with activities in real time through gamification. This platform allowed students to participate and practice through games, encouraging them to learn independently in a challenging and enjoyable environment. At the same time, students were able to interact and respond to each other using generative AI technologies.

5.4.4 Game-Based Learning

The combination of Game-based Learning (GBL) and Generative Artificial Intelligence (GAI) represents a remarkable advancement in the educational field. GAI's ability to develop personalised and interactive content, coupled with the engagement and stimulation provided by GBL, opens up new opportunities for more efficient and customised learning experiences.

Tinterri et al. (2024) highlight the potential of board games as an adaptable learning environment with the use of GPT-4 in various educational contexts, promoting the acquisition of disciplinary knowledge. This research investigated whether trained Artificial Intelligence could assist teachers in designing study modules based on board games. The study concluded that the results were promising, helping teachers overcome their fears due to a lack of confidence with the medium and encouraging the use of games in the classroom.

Ploennigs et al. (2023) report on the experience of using a multisensory educational game called ArchiGuesser. It integrates various AI technologies, from different language models to image creation and computer vision, aiming to instruct students in a fun way about how Generative Artificial Intelligence works. The concept of the game is to show an image or recite a poem related to a random architectural style and encourage the player to make informed guesses about the specific architectural style, its geographical location, and the historical period in which it was established, highlighting that the game content was produced with GAI.

5.4.5 Immediate Feedback and Formative Assessment

Artificial Intelligence can provide immediate feedback on student performance, helping them understand their mistakes and achievements, thus enhancing their learning in real time. Su and Yang (2023) highlight the use of ChatGPT as one of the GAI tools, offering significant benefits in the broader educational context by fostering a more individualised and practical learning experience for students while also providing faster and simpler feedback for teachers.

5.4.6 Content Creation

AI tools can generate educational resources such as quizzes, summaries, and explanations, saving teachers time and providing a variety of resources for students. Gonzalo-Brizuela and Merchán (2024) emphasise a variety of over 350 GAI tools that facilitate a better understanding of the current state of these technologies, describing various unimodal and even multimodal generative AIs spanning a wide range of applications like text, images, video, games, and brain-related information.

Generative Artificial Intelligence is revolutionising education, enabling students and teachers to create personalised and innovative materials. This technology promotes learning, enhances writing skills, and assists with projects, fostering a more interactive and cooperative learning environment. The incorporation of AI can transform conventional educational methodologies by autonomously generating new content, challenging creative limitations, and helping to develop new ideas and solutions, such as exercises tailored to students' needs and modern and engaging approaches to promote more enjoyable and effective learning.

5.5 Generative AI in Education, Creativity and Human Interaction

Gardner et al. (2021) emphasise that Page (1966) predicted a future in which "natural language processing (NLP) would achieve a level of sophistication that would enable machines to learn and understand how to assess the many complex aspects of human writing" (p. 1208). Gardner et al. (2021) expand on this idea, asserting that machines can evaluate students based on their knowledge of specific content, provided they are trained in both that content and in formulating questions.

Swiecki et al. (2022) highlight that traditional assessment paradigms have several limitations, such as being complex, standardised, outdated, and lacking authenticity. They suggest that generative artificial intelligence (GAI) could offer solutions to overcome these challenges. Generative Artificial Intelligence (GAI) has emerged as a transformative technology, reshaping the way we interact with digital tools and information. Unlike traditional AI, which relies on pre-programmed responses, generative AI can create new content, such as text, images, and even music, by learning patterns from vast amounts of data. One prominent example is language models like ChatGPT, which use Natural Language Processing (NLP) to simulate human-like conversations and produce coherent and creative responses.

5.5.1 Connecting Generative AI in Classrooms

González-Calatayud et al. (2021) state that the application of these technologies is still insufficient in education, mainly due to users' lack of knowledge. However, in recent years, generative AI has gained significant traction among both students and educators. For students, it serves as a valuable resource for learning, research, and content creation. They can use AI tools to assist with assignments, enhance writing skills, and gain insights on complex topics. Meanwhile, teachers have been incorporating AI into their classrooms to support personalised learning, automate administrative tasks, and create more dynamic educational experiences. Whether through AI-generated study materials or intelligent tutoring systems, generative AI is making education more accessible and engaging, helping bridge gaps in knowledge and fostering a more interactive learning environment.

The rapid adoption of generative AI in education signals a shift towards a more innovative and adaptive approach to teaching and learning. It holds the potential to not only enhance academic performance but also to empower educators and students with new tools for creativity and exploration. The most common use of AI technologies in higher education is focused on assessment, including automatic evaluations, test generation, feedback, online activity reviews, and analysis of educational resources. Formative assessment, automated scoring, and comparisons between AI and non-AI assessment methods are also central to research on assessment (González-Calatayud et al. 2021).

5.5.2 Chatbots in Education

Another generative AI that has entered into scope is chatbots, which are programs designed to simulate human conversations through text or voice (Vázquez-Cano et al. 2021). The idea that a machine could communicate with a human being comes in its current form from Alan Turing (Adamopoulou and Moussiades 2020). They use Natural Language Processing (NLP) to understand and respond to user inquiries. They can be implemented in various contexts, such as customer service, technical support, education, and entertainment (Jain et al. 2018).

The first chatbot concept was developed with the project ELIZA (Weizenbaum 1966) in the mid-1960s. ELIZA worked by manipulating ideas in conversation, creating the impression of continuous questioning based on what the user typed. A distinction can be made between chatbots that use artificial intelligence and those that follow pre-prepared conversational paths. Additionally, with advancements in voice-to-text technology, it is now possible to interact with chatbots using voice input (Inupakutika et al. 2021).

Reyes-Reina et al. (2020) identify four key characteristics of chatbots: (a) they mimic human speech, (b) they interact through chat, (c) they lack a physical presence, and (d) they do not represent a human entity in the virtual environment. This definition is valuable as it clarifies that chatbots are not meant to replace humans (Duncker 2020) but rather serve as a distinct type of educational tool designed to fulfil a specific academic purpose by simulating human conversation within a text-based interface.

Tamayo et al. (2020) listed the roles that a chatbot can play in education. Each item is accompanied by references to other studies along similar lines, showing the relevance of this issue: Intelligent Tutoring System (Clarizia et al. 2018); Improving Student Engagement (Ranjan et al., 2021); Immediate Assistance to the Student (Berger et al., 2019); Alternative to Learning Management Systems (Tamayo et al. 2020) Teaching Assistant (Hien et al. 2018).

5.5.3 ChatGPT and DALL-E

ChatGPT is an artificial intelligence model developed by OpenAI that uses natural language processing (NLP) to interact with users in a way that resembles human communication. It is capable of performing a wide range of tasks, making it a versatile and powerful tool in various fields. According to Halaweh (2023), one of ChatGPT's main functionalities is generating text in natural language, which makes it helpful in writing content such as articles, blogs, emails, and even scripts. Users can provide a topic or an idea, and ChatGPT can develop a coherent and informative text on that subject. Additionally, the model can assist in creating educational materials, such as summaries of books, explanations of complex concepts, and test questions. Furthermore, ChatGPT can be used for brainstorming and idea generation, helping users find new approaches for creative projects, such as product development or marketing campaigns. It can suggest titles, themes, or even strategies for problem-solving. In the educational field, ChatGPT can serve as a learning assistant, helping students practice writing skills, review assignments, and provide feedback on grammar and style. It can even simulate discussions or debates, providing an interactive environment for learning.

Another generative artificial intelligence is DALL-E, also developed by OpenAI, which generates creative and "original" images based on textual descriptions provided by the user. It is capable of creating images in different artistic styles, such as painting, drawing, photography, and illustration, adapting to the given instructions (Halaweh 2023).

While DALL-E offers a range of creative possibilities, it also raises ethical concerns, such as the generation of potentially offensive content or the creation of images that can be used misleadingly. OpenAI has implemented guidelines and limitations to mitigate these risks and promote the responsible use of the technology. The advent of generative artificial intelligence models has sparked debates and concerns in various fields. Scholars and artists have both questioned and criticised issues related to authorship and copyright infringement potential (Aktay 2022) and how these models can reinforce biases against certain groups based on race, gender, class, and disabilities (Bianchi et al. 2022).

5.6 Integrating Generative AI into Education: Ethical Challenges

Generative AI has the potential to significantly enhance educational experiences by providing personalised learning, interactive content, and adaptive assessments. Nevertheless, its implementation within educational environments presents a range of ethical and moral concerns that must be carefully considered to ensure its responsible and equitable application. In this context, Farelly and Baker (2023) explore several ethical and moral challenges related to the use of GAI in education, such as academic integrity, ethical use of AI, equity issues, reliability of AI detection tools, and bias discrimination.

The integration of Generative AI raises significant concerns about academic dishonesty. There is worry that students may use AI-generated content to bypass academic norms, particularly with assessments that value originality of thought. GAI challenges traditional notions of authorship and raises questions about how originality is defined in academic contexts. Michel-Villarreal et al. (2023) also bring up the idea of academic integrity being the most prominent ethical concern, which is the potential misuse of AI to compromise academic integrity. Students could use ChatGPT to generate plagiarised content or cheat on assignments and exams. This poses a challenge for educators to ensure fair evaluations and maintain academic standards.

5.6.1 Ethical Considerations and Data Privacy

Kadaruddin (2023) discusses the ethical considerations regarding the integration of generative AI in education that have emerged as a critical area of focus. Given that generative AI systems often process sensitive learner data, addressing concerns related to data privacy and security is paramount. Additionally, the potential for algorithmic bias requires thorough scrutiny, as AI systems may unintentionally reinforce existing inequities. To mitigate these risks, transparent AI decision-making processes, coupled with regular audits, are essential in promoting fairness and impartiality. The study underscores the importance of establishing ethical frameworks and guidelines to safeguard students' rights and well-being throughout AI implementation.

5.6.2 Transparency, Fairness, and Equity Issues

Farrelly and Baker (2023) emphasise the need for educators and institutions to focus on the ethical implications of using GAI. This includes ensuring transparency in how AI is used, protecting student privacy, and promoting fairness in AI-driven assessments. Responsible AI use requires transparency, including informing students when AI tools are being used and ensuring they understand the limitations and potential biases of such tools. Ethical challenges arise if there is insufficient disclosure or understanding of AI's role in educational processes. There is also concern when it comes to risks of unequal access to academic tools. Students with the resources to afford advanced AI tools have an advantage over those without, deepening the digital divide and exacerbating existing educational inequalities. This idea of equity issues is also debated by Schlagwein and Willcocks (2023), since the long-term societal impacts, including the potential increase in the digital divide, as those with access to advanced AI tools may have a competitive advantage over others.

Schlagwein and Willcocks (2023) highlight the need for ongoing ethical reflection and the development of clear guidelines for the responsible use of AI in research and beyond. Generative AI operates in a "black box" manner, where the internal decision-making process is opaque. This lack of transparency makes it difficult to understand or audit the reasoning behind AI-generated outputs. Ethical concerns arise when researchers rely on AI outputs without being able to explain how those results were derived. One of the core issues with generative AI, like ChatGPT and other chatbots, is that it often produces responses that seem plausible but are not necessarily accurate. This leads to concerns about the reliability of using AI in research, where factual accuracy is critical. AI can fabricate information ("hallucinate"), which poses risks to scholarly work. This concern also brings up debates around the ownership and intellectual property of AI-generated content. Since AI can replicate creative processes, it raises questions about the moral and legal standing of such work, particularly whether AI-generated outputs deserve the same recognition as human-created work.

5.7 Performance of AI Detection Tools

Nowadays, given the increasing sophistication of AI-generated content, Elkhatat et al. (2023) examined the performance of tools in identifying such content and distinguishing it from human-written text. The study reveals that AI content detection tools generally perform better at identifying content generated by GPT-3.5 than GPT-4. As GPT models evolve, they generate more sophisticated and human-like text, making it increasingly difficult for detection tools to classify AI-generated content correctly. Despite their accuracy with earlier models like GPT-3.5, detection tools exhibit significant inconsistencies when faced with human-written content, often misclassifying it as AI-generated. This leads to false positives, where human-authored text is incorrectly identified as AI-generated. The findings have important implications for academic settings, where the rise of AI-generated text poses challenges to maintaining integrity. The study emphasises that AI detection tools should not be solely relied upon to assess plagiarism or cheating. Their inconsistent performance suggests the need for human oversight and a more nuanced approach when evaluating text in educational and professional settings. Michel-Villarreal et al. (2023) emphasised the importance of human oversight and intervention since ethical concerns are raised about over-reliance on AI, which could undermine human judgment and the quality of education.

In conclusion, the integration of Generative Artificial Intelligence in education presents both significant opportunities and considerable ethical and moral challenges. Key concerns include maintaining academic integrity, addressing biases, ensuring data privacy, and managing the potential impact on employment and educator roles. To harness the benefits of GAI while mitigating its risks, it is essential to develop transparent policies, ethical guidelines, and collaborative efforts among educators, policymakers, and AI developers. By doing so, the educational landscape can be transformed in a way that promotes fairness, equity, and positive learning outcomes for all students.

5.8 Conclusion and Future Directions

In conclusion, despite their benefits, the use of Generative Artificial Intelligence in education also raises ethical and practical questions that need to be carefully considered. Concerns about privacy, intellectual property, and inclusion are crucial aspects that must be addressed as these tools become more integrated into our daily lives. Therefore, it is essential to promote responsible and conscious use of Generative AI, ensuring that their development and application align with ethical and social principles that benefit everyone. By balancing innovation with responsibility, we can fully harness the potential of these technologies to transform society positively.

It is imperative to address how these technologies may inadvertently reinforce biases, contribute to misinformation, or exacerbate existing inequalities. The responsible development and application of Generative AI must prioritise transparency, accountability, and fairness. This includes creating frameworks that allow users to understand how AI systems function and ensuring these tools are accessible to diverse populations, particularly those from underrepresented or marginalised groups of individuals.

Moreover, as these technologies continue to reshape industries and societal norms, fostering a collaborative dialogue among stakeholders—including policymakers, developers, educators, and users—is essential. By integrating multidisciplinary insights, we can build a more holistic approach to AI governance that reflects the complexity of these tools and their far-reaching implications.

Looking to the future, these principles are not just desirable but essential to guiding the role of Generative AI in society. As its influence grows, the long-term consequences of its use will hinge on decisions made today. Balancing innovation with ethical responsibility will allow us to harness the transformative potential of Generative AI fully. When guided by principles of fairness, sustainability, and inclusivity, these technologies can drive meaningful progress—revolutionising healthcare and education, fostering creativity, and enabling global collaboration.

Future studies on advances in Generative Artificial Intelligence (GAI) can address a broader range of challenges and opportunities, focusing on expanding the understanding of its transformative role in education. One key area would be the development of algorithms that combine personalisation and adaptability with the mitigation of cultural, linguistic, and social biases, ensuring that GAI tools are inclusive for diverse educational contexts and audiences. Additionally, investigations could examine how these technologies can be integrated into learner-centred pedagogical methodologies, such as Project-Based Learning and Gamification, expanding the use of multimodal resources—such as text, video, and images—to foster creativity, critical thinking, and problem-solving.

Another relevant field of research is the analysis of the longitudinal impact of GAI on the acquisition of socio-emotional skills, including collaboration, empathy, and ethical decision-making. This could involve qualitative and quantitative studies that explore how interactions with GAI systems shape the learner's experience over time. Furthermore, there is a need to evaluate how these tools can be leveraged to empower teachers, reducing administrative tasks through automated grading and immediate feedback while also promoting the development of new digital competencies among educators. In doing so, we can ensure that the future impact of Generative AI is not only innovative but also equitable and beneficial, paving the way for a society that thrives on both technological advancement and shared human values.

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Chapter 6
Enhancing Digital Pedagogy and Creativity: Generative AI, Video Avatars, and Personalized Learning in Online Education

Jason Lively and James Hutson

Abstract This chapter explores the transformative potential of generative artificial intelligence (AI) in online education, particularly within learning management systems (LMS). By leveraging AI to create dynamic instructional content—including videos, images, and texts—this chapter highlights how these tools can enhance creativity and engagement in digital learning environments. A focal point is the innovative use of video avatars, which utilize generative AI to produce interactive and personalized teaching personas. These avatars offer a scalable and individualized alternative to conventional pedagogical methods. Additionally, the chapter examines AI's role in designing assignments, rubrics, and interactive simulations aimed at improving student engagement and learning outcomes. AI-assisted grading is presented to reduce instructor workload while providing personalized feedback. The ability of AI to autonomously analyse course objectives and suggest relevant assignments and evaluation metrics is also investigated. The integration of generative AI technologies, particularly video avatars, presents a significant opportunity to revolutionize the delivery and effectiveness of online education. This chapter posits that such advancements not only create a more engaging learning environment but also substantially elevate the quality of instruction and educational outcomes.

Keywords Generative AI · Video avatars · Personalized learning · Digital pedagogy · AI-assisted grading

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6.1 Introduction

As the landscape of digital technologies evolves, generative artificial intelligence (GAI) tools like ChatGPT, Stable Diffusion, and Lensa.ai have surged in popularity, achieving mainstream status on various social media platforms (Brewer et al. 2024; DelSignore 2022). Despite their widespread adoption, traditional artists and designers have raised substantial concerns. These concerns centre on issues like copyright infringement and questions regarding the intrinsic artistic merit of AI-generated works (Ansari 2022; Lemley 2024; Murphy 2022; Hazucha 2022). Moreover, recent legal proceedings have underscored the ongoing challenges related to copyright protection in the realm of AI-generated art. A pivotal decision on February 21, 2023, by the U.S. Copyright Office starkly limited the copyright eligibility of Kris Kashtanova's comic book, *Zarya of the Dawn,* which utilized the AI tool Midjourney for its illustrations. The ruling specifically excluded the Midjourney-generated imagery from copyright protection, focusing rights solely on the text and layout crafted by the author (Ford 2023). This decision raises significant philosophical and practical inquiries into the role of AI in the broader sphere of artistic creation and human creativity.

Amidst growing concerns over potential plagiarism, some voices within the academic sphere have advocated for a prohibition on the use of AI in higher education settings (Francke and Alexander 2019; Sherry 2022). However, with the advent of Web 3.0 and the forthcoming 6G technologies, which promise enhanced processing capabilities and innovative 3D generative tools, the integration of AI into educational curricula becomes imperative. This integration is crucial not only for aligning with technological advancements but also for equipping students with the necessary skills to thrive in future digital landscapes. Despite the pressing need for such integration, the academic community has displayed a notable disinterest in delving into the practical applications and establishing best practices for embedding AI within academic coursework. The focus has predominantly lingered on the theoretical disruptions and aesthetic impacts of AI, neglecting its potential utility in educational settings (Ajani 2022).

Ajani's examination of human authorship in AI-generated content illuminates the ongoing debate regarding the definition of "art". This discourse oscillates between viewing art as a manifestation of technical skill or as an expression of emotional depth (Ajani 2022; Mullholand 2022). The ongoing debates prompt deeper reflections on the value of art in capturing the human condition as opposed to merely showcasing technical expertise (Rosenberg 1983). This chapter aims to explore the intersection of AI and human creativity within the educational sector, particularly through the lens of online education and learning management systems. By integrating generative AI technologies such as video avatars and personalized learning paths, this chapter seeks to illustrate how AI can serve not only as a tool for content creation but also as a catalyst for redefining creative expression and educational interaction in the digital age.

While the valuation and integration of AI and non-fungible tokens (NFTs) continue to stir debates within the art community (Zhang et al. 2022; Wellner 2022), the undeniable influence of AI on the creative processes of contemporary artists is evident (Slotte Dufva 2023). AI art generators have provided artists with innovative tools that enhance creativity by suggesting new colour palettes, compositions, and even sparking entirely new forms of artistic inspiration and iterative processes (Compton 2022). Despite these advancements, the full spectrum of possibilities presented by AI in art creation remains underexplored, with critical and methodological frameworks for analysing AI-generated art still in developmental stages. Moreover, the implications of generative AI for web design and development have not been comprehensively assessed. While the future may see a shift from traditional coding to more intuitive, drag-and-drop interfaces, the necessity for human oversight in refining AI-generated content and maintaining website functionality remains critical. This transformation underscores AI's significant role in redefining the fine arts and creative industries at large.

Addressing these gaps, this chapter investigates the application of both text-based and image-based generative AI content to enhance the understanding of art and design processes among computer science majors. Additionally, the study explores how these tools can equip web designers and developers with innovative capabilities for crafting engaging and creative digital content. The study specifically targeted students enrolled in introductory-level web design and user experience (UX) courses, aiming to elevate their aesthetic sensibility and creative capabilities, regardless of their initial skill levels.

The findings reveal that incorporating AI tools into the web design and development workflow significantly enhanced the quality of student projects and reduced the instances of reported deficiencies. Text-based AI generators notably improved efficiency in writing copy and code, whereas image-based generators excelled in fostering ideation and assisting in colour selection. Unlike previous research, this study observed a higher appreciation for AI among students, who reported that these tools augmented rather than supplanted their creative capabilities. In UI/UX design, the use of AI prominently boosted productivity, underscoring its utility in enhancing rather than replacing human creativity. These results underscore the potential of AI as a beneficial adjunct in educational settings, particularly in fields combining technical skill and artistic creativity. The outcomes of this study not only advocate for the integration of AI tools in educational curricula but also provide a replicable model for educators aiming to foster a more interactive and productive learning environment. Through this approach, educators can significantly enhance the creative and technical skills of students, preparing them for advanced roles in the evolving digital landscape.

6.2 Generative AI (GAI) in the Design Classroom

The proliferation of generative artificial intelligence (GAI) tools in the realm of contemporary art has ignited rigorous debates concerning the authenticity of AI-generated art and its ramifications on traditional artistic practices (Bonadio and Lucchi 2019; Zhang and Lui 2021). This emerging paradigm also prompts a re-evaluation of the artist's role and the tangibility of art, invoking poststructuralist theories to question foundational artistic concepts (Anderson 2017). This literature review endeavours to synthesize current research and chart a course for future investigations into AI art, focusing particularly on its interactions with social media, the fine arts, and algorithmic influences on artistic creation and perception. Moreover, the potential of the metaverse as a transformative platform for art exhibition and audience engagement will be explored. The review also examines the creative prompting process, proposing a poststructuralist lens through which to understand the dynamics of creatorship and the reception of art.

Despite the burgeoning use of AI in artmaking, there remains a paucity of discourse on the practical methodologies, strategies, or workflows that artists and designers might employ. The existing literature predominantly revolves around philosophical and theoretical considerations. Coeckelbergh (2017) offers a philosophical framework to dissect whether machines can genuinely engage in creative artmaking. This framework scrutinizes the definitions of "creation", "art", and machines' capability to "create art", advocating for a fluid and subjective interpretation of creativity and challenging the dichotomy between human and non-human artistry. Instead, a cooperative definition is proposed, where technology serves as an adjunct in the creative process.

Furthering this discussion, Mazzone and Elgammal (2019) have developed AI methodologies for identifying stylistic elements and discerning large-scale patterns in art history, suggesting a re-evaluation of the nexus between machine capabilities and human creativity. They view the relationship between machine and human creativity as complementary, emphasizing that this synergy should not undermine but rather enhance the emotional and social intentions of human artists. Tao (2022) describes this synergistic relationship as an "actor network" of art, wherein humans and machines operate as collaborative agents, potentially amplifying the strengths of each participant in the creative process.

Conversely, Ahmed (2022) approaches the discourse from a design-based perspective, arguing that AI's role in art should transcend traditional notions of physical design. By analyzing interactive and immersive media installations, Ahmed contends that AI facilitates the embodiment of immaterial humanistic characteristics such as emotions, experiences, senses, and memories, thus redefining these elements as integral components of the design process itself. This perspective underscores the potential for AI-generated art to function not merely as a product but as an active participant in the design process, shaping human interactions and emotional responses.

The discourse on AI-generated art frequently converges on the notion of creativity and its qualification as "true art". According to Csikszentmihályi (1988), the model of creativity encompasses a body of knowledge, a volitional agent, and recognition by field experts. Building on this, Jennings (2010) identifies three criteria essential for an agent to exhibit creative autonomy, emphasizing the system's independence from the programmer's intentions. However, Ajani (2022) posits that creativity necessitates individual capacity, information acquisition, and expert validation, thereby suggesting that AI's creative outputs must undergo scrutiny and endorsement by established experts to be recognized within the artistic or design domains. This ongoing debate highlights the complex interplay between technological innovation and artistic creation, challenging traditional perceptions and inviting a broader understanding of creativity in the digital age. But the applicability of the technology to the design process is already underway in classrooms.

In fact, the integration of GAI in art and design education is transforming traditional pedagogical approaches. Zhao and Gao (2023) highlight that AI technologies can significantly enhance the efficiency of classroom management and enrich the educational experience in art design classrooms. Their study demonstrates that AI can provide detailed insights into classroom dynamics through emotional score data, allowing educators to tailor their teaching strategies to better engage students and improve overall teaching quality. Furthermore, AI facilitates diversified teaching modes and increased interaction between students and teachers, fostering a more dynamic learning environment. Another high-level approach was taken by Kong (2020), who looks into the strategic application of technology in modern art teaching, identifying key areas where AI can enhance educational outcomes. By expanding the adaptability of AI-based art instruction, improving intelligent teaching modes, and enhancing the artistic experience, AI promotes a more immersive and engaging learning environment. Kong also developed a performance analysis model to quantify the effectiveness of AI in art education, underscoring its potential to revolutionize teaching methodologies (Kong 2020).

Along the same lines, Zhang, Shankar, and Antonidoss (2022) propose the Artificial Intelligence assisted Effective Art Teaching Framework (AIEATF) to enhance AI-oriented art instruction. Their research highlights the potential of AI to improve major art courses through intelligent teaching styles and environments. The study's findings indicate significant improvements in various aspects of art education, including modern painting, graphical representation, and sculpture, demonstrating the comprehensive benefits of AI integration in art classrooms.

These studies, while useful for a broad conceptualization of the use of AI in pedagogy, does not specifically address how they can be used in lessons and instructional exercises themselves. Chang et al. (2023) look more specifically into the use of AI image generation in art education, particularly in middle school settings. Their research demonstrates that AI can significantly increase student engagement and creativity by expanding their imagination and expressive capabilities. The study provides a framework for using AI as a teaching and learning tool, showcasing its positive impact on students' immersion in art classes.

Likewise, Cotroneo and Hutson (2023) explore the use of generative AI tools in art education, emphasizing the importance of prompt engineering and iterative processes. Their study found that teaching students to craft and refine prompts for

AI tools like OpenAI's DALL-E2 can significantly enhance their creative skills and understanding of AI's potential. Despite ethical concerns, students appreciated the value of generative AI in enhancing their sketchbooks and ideation processes, indicating a shift towards accepting AI as a valuable educational tool.

The current body of research on AI-generated art underscores the necessity for further investigation into the practical applications of these tools for artists and designers. As generative AI tools become increasingly prevalent, there is a pressing need to develop innovative pipelines for the creation and interpretation of generative content. One crucial area of focus involves fostering collaborative and co-creative processes, enabling artists to work alongside AI to enhance its capabilities. To integrate AI-generated art in meaningful and innovative ways, artists and designers must proactively engage with the technology, understanding its possibilities and limitations beyond its novelty (Zhang and Lui 2021; Bonadio and Lucchi 2019).

Furthermore, it is essential to establish new frameworks for interpreting and evaluating generative content. These frameworks should recognize the intricate human-AI collaborative processes involved in creating such works. Developing criteria for assessing the creativity and artistic merit of generative art will be crucial, as will devising new methods for engaging audiences with these novel forms of artistic expression (Anderson 2017; Coeckelbergh 2017).

In the context of design education, particularly within web design classrooms, additional exploration is required to apply these insights practically. Educators and students must collaborate to develop practical strategies for incorporating AI into the web design workflow, enhancing both the creative and technical aspects of the discipline. Bridging the gap between art and technology allows artists can significantly influence the trajectory of AI-generated art, unlocking new opportunities for creative expression and meaning-making (Ahmed, 2022). By working collaboratively across the fields of art and technology, artists can play a pivotal role in shaping the future of AI-generated art. This interdisciplinary approach will not only foster new forms of creative expression but also contribute to the development of innovative educational practices that integrate AI into the curriculum effectively (Tao 2022; Mazzone and Elgammal 2019). This exploration is crucial for preparing the next generation of artists and designers to navigate and contribute to the evolving landscape of digital creativity.

6.3 New Approaches to Pedagogy

This mixed-methods case study investigates the integration of generative AI tools into postsecondary curricula to enhance student creativity across various artistic disciplines. Conducted over three semesters (Spring 2023, Fall 2023, and Spring 2024) the study engaged students from diverse fields, including creative writing, digital art, drawing, 3D design fundamentals, web design, and English composition. The primary objective was to identify pedagogical best practices for incorporating text- and image-based generative AI in web design and development, focusing on student perceptions, performance, feedback, and instructor observations.

To accommodate the diverse needs and preferences of students, the course was offered in three distinct modes: online, face-to-face, and hybrid. The online mode enabled students to engage with course materials, lectures, and assignments entirely through virtual platforms, providing the flexibility to participate remotely from any location with internet access. This format was particularly beneficial for students who preferred or required a fully remote learning experience, allowing them to balance their studies with other commitments or overcome geographical barriers.

In contrast, the face-to-face mode involved traditional classroom instruction, where students attended in-person lectures, discussions, and practical sessions on campus. This mode facilitated direct interaction with instructors and peers, fostering a sense of community and enabling hands-on learning experiences that are challenging to replicate in virtual environments. The hybrid mode combined elements of both online and face-to-face instruction, allowing students to choose between attending classes in person or participating remotely based on their individual preferences or circumstances. This flexible approach catered to varying learning styles, schedules, and accessibility requirements. Hybrid classes incorporated synchronous and asynchronous activities, blending the benefits of traditional and online learning modalities to promote engagement and collaboration while accommodating students' diverse needs.

Participants were drawn from a private Midwestern college in the Saint Louis region, with a sample comprising 33 students in Spring 2023, 7 in Fall 2023, and 12 in Spring 2024. These participants included undergraduate and graduate students majoring in fields such as Computer Science, Computer Information Systems, Digital Marketing, Finance, Game Design, Marketing, and Art and Design. They were enrolled in Web Design I - User Experience, a project-based course emphasizing advanced HTML, CSS, JavaScript, frameworks, and libraries to deepen their understanding of web design with a focus on user experience through simulated client projects.

The studies aimed to explore the practical application of generative AI tools in enhancing the creative process within the web design curriculum. By examining student perceptions, performance, and feedback, alongside instructor observations, the research sought to develop effective strategies for integrating AI technologies into art and design education. The insights gained from this study are intended to inform best practices and guide the implementation of AI in educational settings, ultimately contributing to the evolution of pedagogical approaches in the digital age.

Throughout the course, AI tools were intricately integrated into several key assignments to enhance both the learning process and the creative outcomes for students. One significant assignment involved the use of text-based AI generators such as ChatGPT to create content for web pages. This approach allowed students to concentrate more on the design and usability aspects of web development, while efficiently handling content generation through AI. For logo design tasks, students employed AI-driven platforms like Midjourney to generate initial logo concepts. These AI-generated designs served as a starting point, which students then refined and iterated

Fig. 6.1 Animated logo example assignment. Image by the authors

upon (Fig. 6.1). This integration not only expedited the brainstorming process but also introduced students to a broader array of design possibilities they might not have considered independently.

In another innovative assignment, students were tasked with designing and integrating AI-generated personas, or virtual employees, into simulated web interfaces (Fig. 6.2). This exercise aimed to familiarize students with AI's capabilities in creating interactive and responsive web elements, thereby enhancing their understanding of AI's potential in user experience (UX) design. Additionally, in group projects, AI tools facilitated collaborative brainstorming sessions, helping teams organize ideas and develop cohesive project plans through AI-assisted mind mapping software.

At the end of the semester, individual interviews were conducted with participants to gain a deeper understanding of their unique perspectives. Each interview, lasting approximately 30 to 45 min, was structured in three parts: an introduction to re-familiarize participants with the study's purpose, a main section of open-ended questions to elicit detailed responses, and a closing section for additional thoughts or clarifications. All interviews were audio-recorded with participant consent and transcribed verbatim for thorough qualitative analysis.

Focus groups were also utilized to gauge collective viewpoints and stimulate discussions among students, revealing consensus or diverse perspectives on the use of AI in their coursework. Each focus group, consisting of 4–6 participants, followed a semi-structured format, allowing dynamic interactions among group members. These sessions were recorded and transcribed to accurately capture the spontaneous and interactive nature of the conversations. The questions for both interviews and focus groups are detailed in the Appendix of this manuscript. The qualitative data from these methods provided rich insights into the students' experiences and perceptions, complementing the quantitative data from surveys and offering a holistic view of AI's impact on educational practices in web design and UX.

Additionally, the study included an evaluation of the artifacts created by students, such as AI-generated content and final website projects, to assess learning outcomes

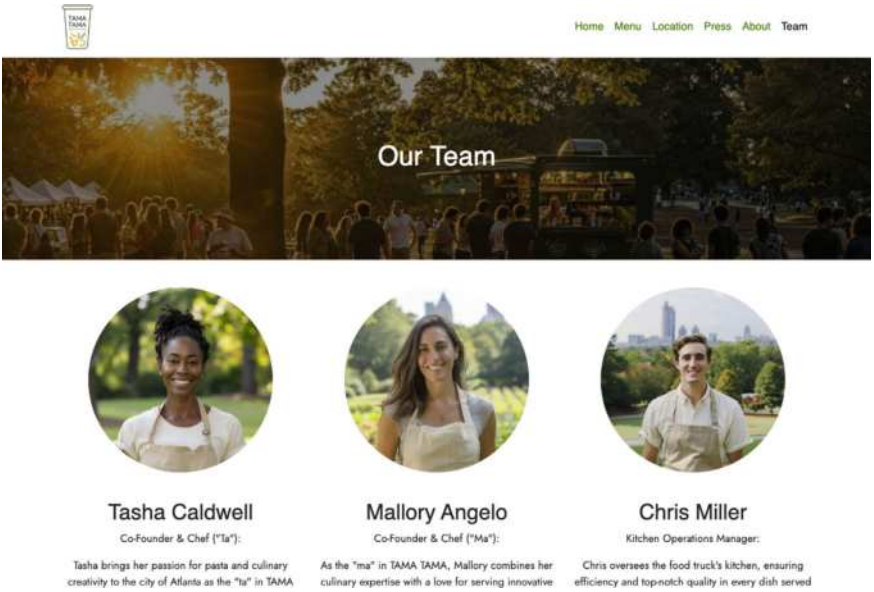

Fig. 6.2 Example of AI-generated employees. Image by the authors

and gather comprehensive feedback on the AI integration experience. This approach allowed for a robust analysis of the effectiveness of using AI in enhancing the creative process within web design courses. Data analysis involved both quantitative and qualitative methods. Statistical techniques were used to compare attitudes and usage willingness between different student groups, while thematic analysis was applied to the qualitative data to identify patterns and insights.

The anticipated outcome of this study is to demonstrate that when generative AI tools are presented as enhancers of creative abilities rather than replacements, students are more likely to embrace them as valuable components of their creative toolkit. This approach can mitigate fears of AI replacing human creativity and foster a more positive perception of AI in creative domains. By exploring the practical applications of AI in web design classrooms, this study aims to develop best practices and strategies that can be replicated and adapted by other educators in similar fields.

6.4 Student Perception of Integration

The perception of AI among students prior to the commencement of the class presented a spectrum of apprehensions. Survey data indicated that 38.89% of participants reported no initial fears regarding the technology. However, a notable portion of the students held some level of concern: 27.78% felt a little fearful, 22.22% experienced a moderate amount of apprehension, and a small fraction (11.12%) expressed significant fears ranging from "a lot" to "a great deal". These findings underscore the diverse sentiments towards AI, highlighting an initial mix of scepticism and comfort among students engaging with AI tools for the first time in an educational setting.

Upon integrating AI exercises into the design process within the class, a significant shift in perception became evident. An overwhelming majority, 83.33% of the students, appreciated the inclusion of AI in the design process, finding it to be a positive addition to their learning experience. This strong preference suggests that firsthand interaction with AI tools can effectively demystify the technology, fostering a more receptive and favorable attitude towards its application in creative tasks. Moreover, the practical benefits of AI exercises were highly recognized by the students, with an impressive 94.44% acknowledging the positive impact these tools had on their success in the course. Only a minimal 5.56% did not see a direct benefit, with nobody remaining undecided. This overwhelming endorsement of AI's utility in enhancing the design process and contributing to academic success highlights the potential of AI as a powerful educational tool. It suggests that when students are provided with opportunities to engage directly with AI, perceptions shift from apprehension to recognition of its value in augmenting their creative and technical capabilities.

Across the entire student population, the utilization of AI tools for assignments revealed a significant inclination towards integrating technology into their academic endeavours. A substantial 88.89% of students engaged with text-based AI generative content, such as ChatGPT, during the class. This high adoption rate underscores student eagerness to explore and leverage AI capabilities for text generation, pointing to a robust curiosity and acceptance of new technological tools in enhancing their academic work.

Similarly, the incorporation of AI for image-based content creation was notable, with 72.22% of the students utilizing AI applications like Midjourney, Jasper, Crayion, and DALLE-2. This indicates a considerable interest in experimenting with these tools to support visual aspects of their projects, showcasing student openness to embracing AI across different creative dimensions. The commitment to incorporating AI into their academic projects was further evidenced by 88.89% of students using AI applications in their final projects. This widespread use highlights the perceived utility and relevance of AI tools in completing substantial coursework. Among those who integrated AI into their projects, a varied approach was observed: 28.57% used text-based generative AI, 17.86% utilized image-based generative AI, and another 28.57% employed both types for content creation. Additionally, 25% of students used AI tools to find inspiration, indicating a multifaceted application of AI in the creative process.

Reflecting on the impact of AI on the quality of their final projects, 82.35% of students felt that the inclusion of AI applications had improved their work, with only 17.65% remaining uncertain about its effect. No students reported a negative impact, suggesting a predominant perception of AI as a beneficial tool in academic projects. This positive feedback underscores the potential of AI to not only enhance the efficiency and creativity of students' work but also to contribute to higher quality outcomes.

When comparing the overall student engagement with AI tools to the specific experiences of art and design students, nuanced differences emerge, offering insights into how various academic disciplines interact with technology in their learning processes. Among art and design students, 85.71% experimented with text-based AI generative content, such as ChatGPT3, closely aligning with the broader student body's engagement rate of 88.89%. This slight variance suggests a generally high level of curiosity and willingness to explore AI's capabilities across disciplines, with art and design students being slightly less engaged in text-based AI experimentation than their peers from other majors. The use of AI for creating image-based content saw a similar pattern, with 71.43% of art and design students utilizing such tools, reflecting the overall trend (72.22%) across all students. This demonstrates a consistent interest in leveraging AI for visual content creation, underscoring the technology's perceived utility in enhancing creative outputs, regardless of the student's field of study.

When incorporating AI applications into their final projects, art and design students reported a participation rate of 85.71%, mirroring their experimentation with text-based content. This rate is identical to the overall student body's engagement, indicating a uniform recognition of AI's value in academic projects across different disciplines. In terms of the specific uses of AI, art and design students displayed a balanced approach: 30% used text-based generative AI for content creation, 20% utilized image-based generative AI, another 20% employed both methods, and 30% used AI tools for inspiration. This distribution highlights the diverse applications of AI within the creative process, illustrating art and design students' versatility in employing technology to augment their creative endeavours.

Regarding the perceived improvement of final projects through the use of AI, 66.67% of art and design students felt that AI applications had a positive impact, with 33.33% remaining uncertain ("maybe"). This sentiment contrasts with the broader student feedback, where 82.35% acknowledged an improvement, suggesting that art and design students may be more contemplative or critical about the extent to which AI enhances their work.

When comparing the engagement and perceptions of computer science students with AI tools to the broader student body and specifically to art and design students, distinct patterns and attitudes emerge, illustrating the nuanced ways in which students across disciplines interact with AI technologies. Computer science students showed a high level of engagement with text-based AI generative content, such as ChatGPT3, with 90.91% reporting usage during the class. This rate is slightly higher than the overall engagement (88.89%) and aligns closely with the participation rate among art and design students (85.71%). The marginally higher usage rate among computer science students may reflect a greater familiarity or comfort with AI technologies due to their field of study. The use of AI to assist in creating image-based content saw 72.73% of computer science students participating, which is nearly identical to the overall student engagement (72.22%) and closely mirrors the rate among art and design students (71.43%). This uniformity across disciplines suggests a broad recognition of the benefits AI tools offer in enhancing creative outputs, regardless of the students' primary focus on textual or visual content.

Similarly, a significant proportion (90.91%) of computer science students incorporated AI applications into their final projects, reflecting the general trend observed across all students and within the art and design cohort. This consistent high level of AI integration highlights its perceived relevance and utility in academic work across various disciplines. Among computer science students who utilized AI, there was a diversified approach to its application: 27.78% focused on text-based generative AI, 16.67% on image-based generative AI, 33.33% used both, and 22.22% sought AI for inspiration. This distribution showcases a balanced engagement with both textual and visual AI tools, indicating a comprehensive exploration of AI's capabilities.

A notable 90.91% of computer science students believed that AI applications improved their final projects, with only 9.09% uncertain about its impact. This overwhelmingly positive response surpasses the satisfaction rate among art and design students (66.67% felt an improvement, with 33.33% maybe), suggesting computer science students might perceive a more definitive benefit from AI integration, possibly due to a more technical understanding of AI's capabilities. The comparison reveals that computer science students are slightly more inclined to engage with and recognize the benefits of AI in their coursework, compared to the overall student population and the art and design subgroup. This might be attributed to their technical background, which could facilitate a deeper appreciation for AI's potential in enhancing both the process and the outcomes of their academic projects.

Throughout the course, instructor observations revealed a clear distinction in student sentiments regarding AI, particularly among computer science majors who exhibited heightened concerns about its potential impact on future job prospects. International students expressed a nuanced perspective, advocating for the parallel development of human skill sets alongside AI, emphasizing the importance of leveraging AI for creative and other purposes while continuing to develop human intelligence and skills. Initially, apprehension towards AI tools was prevalent; however, hands-on experience led to a shift in perspectives, with students recognizing AI as a valuable creative aid. Art and design students viewed AI as an enhancement for their work, appreciating its role as a tool for generating ideas and inspiration. The instructor maintained a balanced approach, emphasizing the importance of both embracing AI's potential and fostering human intelligence through various instructional strategies and activities. These included theoretical discussions, practical assignments, and collaborative projects that promoted a symbiotic relationship between AI-driven tools and human ingenuity. Reflective exercises encouraged students to consider how their interactions with AI influenced their creative processes and professional development. This balanced perspective assuaged student fears, fostering a more constructive approach towards AI integration. Computer science students, in particular, recognized the practical utility of AI, though they called for more education on effective AI utilization. Overall, the course provided a foundational understanding of AI tools and emphasized the need for proactive adaptation, shifting student sentiment towards informed acceptance and a willingness to adapt alongside AI. Despite lingering concerns, particularly among computer science students, the course succeeded in fostering a critical yet constructive understanding of AI, helping students navigate the rapidly evolving technological landscape.

6.5 Conclusion

The integration of generative artificial intelligence (GAI) into web design and user experience (UX) education presents both significant opportunities and challenges for students and instructors. This study aimed to explore student perceptions and experiences regarding the use of AI tools in a web design course, focusing on the differing experiences of art and design versus computer science majors. Through a mixed-methods approach that combined quantitative survey data with qualitative instructor observations and student feedback, this research provided a comprehensive understanding of student attitudes towards GAI in web design education.

The survey results revealed a majority of students, particularly computer science majors, experimented with text-based AI generative content and utilized AI applications for image-based content creation. Students expressed high interest in incorporating AI tools into their future design processes, particularly for suggesting creative solutions and applying a scientific approach to design. Instructor observations noted a positive shift in student attitudes towards AI tools over the course, with initial apprehension giving way to appreciation as students gained hands-on experience and recognized AI's practical benefits in enhancing their creative process and problem-solving abilities. However, concerns about AI's potential impact on job prospects persisted, indicating the need for ongoing education and dialogue around AI integration in design education.

Moving forward, further research should investigate pedagogical approaches to effectively incorporate AI tools into web design curricula, considering the diverse needs and perspectives of students across different disciplines. Longitudinal studies tracking students' career trajectories post-graduation could provide valuable insights into the long-term impact of AI education on professional development and industry adaptation. Additionally, exploring strategies to address ethical considerations and mitigate potential biases in AI tools used for design education is essential for promoting responsible AI usage in the field.

This study contributes valuable insights into the integration of AI tools in higher education, particularly within the realms of web design and UX. By examining student attitudes, experiences, and learning outcomes in response to AI integration, this research highlights both the potential benefits and challenges of incorporating AI into creative coursework. For educators, the findings underscore the importance of adopting a balanced approach that combines the innovative potential of AI with the continued cultivation of human intelligence. This necessitates designing curricula that leverage AI tools to enhance learning and creativity while fostering critical thinking, problem-solving skills, and ethical awareness among students. Educators can draw upon the practical strategies and pedagogical frameworks identified in this study to effectively integrate AI into their teaching practices, thereby preparing students for the evolving demands of the digital age.

For researchers, this study offers a foundation for further exploration into the intersection of AI and education, highlighting avenues for future inquiry. By delving deeper into the long-term impacts of AI integration on student learning outcomes, career trajectories, and ethical considerations, researchers can advance our understanding of how best to harness AI technologies in educational settings. There is a need for interdisciplinary collaboration between educators, technologists, and social scientists to address the complex challenges and opportunities posed by AI in education comprehensively. By engaging in collaborative research endeavours, researchers can develop innovative solutions and evidence-based recommendations to inform policy, practice, and pedagogy in the rapidly evolving landscape of AI-enhanced education.

In essence, this study underscores the transformative potential of AI in higher education while emphasizing the enduring value of human intelligence in driving creative innovation and ethical decision-making. By embracing a holistic approach that integrates AI tools with human-centric pedagogy, educators and researchers can collectively pave the way towards a more adaptive, inclusive, and ethically grounded educational landscape in the digital age.

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Chapter 7
Creative Learning Using Generative AI: Empowering Problem-Solving and Programming Skills

Randa Elanwar

Abstract This chapter discusses the different opinions about integrating generative artificial intelligence (Gen-AI) technologies into education systems in general and programming education in particular. Computer programming with writing code scripts is a way to empower learners' creativity. Through coding, they can create projects about things they care for and learn strategies for design and problem-solving. Traditional approaches to educate computer programming are thriving through the Gen-AI era because our digital world still needs enormous contributions. However, the rocketing development of Gen-AI technologies is now raising the bars of education standards and revealing the widening gap between traditional education systems and the labour market requirements for skills graduates need. This rapid development was met with a move by the concerned educational authorities to explore ways to benefit from these advanced technologies to improve the quality of education and qualify students for the new jobs that will be created based on it. This chapter summarises several studies conducted to test the integration of Gen-AI tools. It offers best practices and recommendations reached by researchers to maximise the benefit of these technologies and mitigate their adverse effects.

Keywords Creative learning · Generative AI · Programming education · Code Generation · Gen AI Tools

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7.1 Introduction

Computer programming is one of the top individual skills mostly wanted by employers worldwide. As computer programming takes place in every detail of any digital environment, it has become increasingly important to develop the programming skills of employees in almost all sections of the job market. Computer programming depends on both problem-solving and programming skills. An individual should first harness the knowledge he accumulated over the years to develop a problem solution that can be technically achieved through programming. However, what makes some solutions better than others is creativity. Creative solutions last a long time and create new standards of efficiency measurement.

Creativity has long been associated with problem-solving capabilities since it implies shifting or adjusting one's perspective to make new connections (Guilford 1967). Many researchers agreed on a broad definition of creativity as a combination of ideas determined to have novel value, use, or meaning to various stakeholders (Guilford 1967, 1950; Chirico et al. 2018; Torrance 1969, 1974; Collard and Looney 2014; Habib et al. 2024).

Accordingly, university education, school education, and the EdTech industry specialise in computer programming education and empowering learners' problem-solving creativity. Universities programs and primary school education undergo continuous pedagogical development to help their graduates cope with the demands of job markets and the technological advances they need to embrace and be a part of. Many EdTech websites are designed to teach programming skills in different ways that seek the learner's attention and engagement.

With every emerging and quickly spreading new technology among people, the question is always whether or not to integrate it into our learning environment and how exactly to do that. The last couple of years have witnessed a growing interest in Generative Artificial Intelligence (or Gen-AI), which has shown more potential than traditional AI, which has been around for decades. The reason for the popularity of Gen-AI is the availability of chatbot tools such as Chat-GPT (OpenAI 2023), Google Gemini (Google Team 2023) (formerly known as Bard), Microsoft Bing, and others. These tools are user-friendly even for non-tech-savvy. They were increasingly able to automate basic levels of writing, code generation, artwork creation, and other creative intellectual tasks.

Unfortunately, the availability of these intelligent tools has led to several forms of misuse. Students now tend to use the Gen-AI tools to solve their assignments or write their essays. They also consider them tutors and trust that the answers they generate are as if the teachers gave them. An associate professor of English at the University of Louisville, Dr Amy Clukey, once tweeted, "I'm just a human plagiarism detector", complaining that she spends most of her time checking to see whether the students wrote their papers instead of giving comments for improvement. UNESCO's first global guidance on Gen-AI in education (Holmes and Miao 2023) highlighted that Gen-AI had raised multiple immediate concerns related to safety, data privacy, copyright, and manipulation. MIT's Lifelong Kindergarten research group (Resnick

2024) is also concerned that the misuse of Gen-AI tools would lead to constraining learner agency, limiting creativity, and undervaluing human connection.

Researchers and educators realise it is impossible to prohibit these tools. Still, it is essential to figure out how to employ this technology and use these tools to help students enhance their learning and creative thinking. The focus of using AI in education is to promote skills rather than question-answering or saving time in content creation.

Researchers started studying Chat-GPT to teach skills like software programming (Yilmaz and Yilmaz 2023; Denny et al. 2024) and collaborative problem-solving (Ngwenya 2024). They all agreed that the purposeful integration of generative artificial intelligence tools in teaching and learning enhances creativity, problem-solving, and student collaboration. They suggested that Gen-AI tools can be successfully integrated into various learning environments. However, the teacher's (or educator's) role has to be transformed since human agency should remain central. The reason is that Gen-AI tools have both a positive and negative impact on creativity (Habib et al. 2024; Ngwenya 2024), hence experienced human guidance and assessment are indispensable to control these tools' employment.

Before we get into the details of these AI tools integration, it is essential to briefly present generative AI and how it works and impacts education. We will then review case studies of integrating Gen-AI tools in computer programming education and discuss the best practices of integration and the recommended actions like regulations and policies that can help educators develop their pedagogies and classroom teaching.

7.2 What is Generative AI and How Does It Work?

Winkler et al. (2023) define generative AI as one new form of artificial intelligence that can generate novel data content from input human prompts. The generated data includes text, images, audio, video, and multi-modal files. To do this, Gen-AI models are trained using a vast collection of data available on the internet, such as webpages, social media conversations, and other online media. The training process underlies the representation of primitive data elements like words in text or pixels in images as samples from probability distributions, known as latent representation. These distributions are statistically analysed and identified for repeating common patterns or relationships. Generating data is achieved via sampling from these distributions. This sampling could be random or guided by conditional data, e.g., prompts.

Gen-AI evolved from AI chatbots. A chatbot is a software or computer program that simulates human conversation through text or voice interactions. Chatbots were initially used in online help systems as a part of companies' websites for customer support. They provided the website visitors with a more friendly interface with which they could interact rather than a formal search and menu navigation. Chatbots have been around for a while, implemented using simple rule-based models that relied on pre-defined algorithms. Those algorithms recognise clue words or phrases in the user input and generate the corresponding pre-programmed response, moving the conversation forward meaningfully.

In education, chatbots were seen as capable of providing students with immediate feedback and quick access to information, thus increasing engagement and interest. However, the answers that were generated were not very reliable. According to Cheong-Trillo (2023), educators using chatbots judged the responses they got as rigid and unoriginal (Kim and Kim 2022) and said that they could only answer simple questions (Cunningham-Nelso 2019). Researchers (Cunningham-Nelso 2019; Merelo et al. 2022) named these traditional chatbots as frequently asked questions (FAQ)-type chatbots.

Recent chatbots are AI-based, relying on natural language processing research, where deep learning models are implemented and trained on large corpora of text to understand human language and generate real-time, comprehensive, and coherent answers to users' questions. In other words, recent chatbots are powered by large language models (LLM) instead of pre-defined rules. According to Gao et al. (Gao et al. 2022), ChatGPT was the first popular chatbot capable of generating text, sometimes indistinguishable from human-generated text.

LLMs are artificial neural networks that utilise transformer architecture (Devlin 2018) to achieve general-purpose language generation and other natural language processing tasks. LLMs have become increasingly popular in education (Abdelghani et al. 2024; Moore et al. 2022; Seßler et al. 2023). The LLM top models in the meantime are:

- Models by OpenAI: ChatGPT and GPT-4 (Achiam et al. 2023).
- Models by Google: PaLM (Anil et al. 2023b, a; Chowdhery et al. 2023), ChinChilla and Gemini (Previously BARD).
- Models by Meta: LLaMA model (Touvron et al. 2023) and the Falcon-40B architecture (Penedo et al. 2023).
- Models by Microsoft: Bing and Copilot. Models by other corporations and startups: Claude (by Anthropic), Mistral and Mixtral (by Mistral AI), Qwen (by Alibaba), DeepSeek (by DeepSeek) and Grok (by xAI).
- Open-source fine-tuned models by researchers: Vicuna (Zheng et al. 2023) and Alpaca (Taori et al. 2023), and QLoRA (Dettmers et al. 2024).

Due to their ability to reason about, generate, and manipulate text data, LLMs have been widely used in education. However, LLMs were designed for one-to-one tasks in which input and output information is text. Learning many educational subjects requires multimodality. As mentioned, integrating text and non-text modes is essential for enhancing knowledge acquisition. During their learning, students must shift between text, images, graphics, tables, technical symbols, formulas, equations, process workflows, graphs, and video and audio data. Educators should facilitate access to these multimodal materials.

Fortunately, general-purpose any-to-any multimodal LLMs (or MLLMs) have been released. OpenAI released GPT-4 Vision and GPT-4 Turbo. Google also developed Gemini, the MLLM-based AI system. Other image Gen-AI models like Craiyon[1] (previously DALL-E Mini), DALL-E,[2] DreamStudio,[3] Fotor,[4] Midjourney,[5] NightCafe[6] all generate images from text prompts. Also Elai,[7] GliaCloud,[8] Pictory,[9] Runway[10] generate videos from text prompts.

Educators started to benefit from these latest advances and use MLLMs for different tasks like processing and creating content in multiple forms, including images, audio, and video (Bommasani et al. 2021), transforming unstructured data into structured content and vice versa (Borisov et al. 2022) and reading the scoring rubrics and grading students' drawn models using GPT-4V (Lee and Zhai 2023).

However, what precisely an LLM prompt is, how it generates data, and how to compose an accurate prompt are all essential to learning. Writing a text input and getting precisely what we wish for is not straightforward. It might take a person several trials before he finds the correct combination of prompt text that generates his intended data. The LLM-generating process is often called "hallucination", for we are unsure what we get based on our text input. "Prompt-engineering" refers to the methods and techniques by which an input is composed to produce a Gen-AI output that more closely resembles the user's desired intent (Holmes and Miao 2023).

A text prompt entered into a trained GPT model is broken into tokens (smaller units like words). The GPT model exploits the statistical patterns composed during the training phase to predict the likely words or phrases that might form a coherent response to the prompt by estimating the probability of specific words or phrases appearing in the given context. The predicted words or phrases are converted into readable text filtered to remove offensive content. This process is repetitive until a response is finished by reaching a maximum token limit or meeting predefined stopping criteria. Finally, the response is post-processed to improve readability by applying formatting, punctuation, and human-like words that usually precede sentences such as "Sure", "Certainly" or "I'm sorry" (Holmes and Miao 2023).

Since output quality is critically dependent on the user input, as mentioned before, people started to share their experiences of composing successful prompts that generated precisely the output they wanted. The recommended practices are to use simple and easily understood wording, include an example of the desired output format or context to ensure generating meaningful completions, experiment via iterative prompts on a particular problem in a logical order, and avoid prompts that might generate biased or harmful content (Holmes and Miao 2023).

[1] https://www.craiyon.com.

[2] https://openai.com/product/dall-E-2.

[3] https://dream.ai/create.

[4] https://www.fotor.com/features/ai-image-generator.

[5] https://www.midjourney.com.

[6] https://creator.nightcafe.studio.

[7] https://elai.io.

[8] https://www.gliacloud.com.

[9] https://pictory.ai.

[10] https://runwayml.com.

7.3 Generative AI in Education

Traditional classrooms fall short when it comes to adapting to the different learning profiles of students. Traditional education often assumes a uniform baseline of knowledge among students; thus, educators design their material based on this assumption, expecting high levels of engagement and understanding among students. However, this "one-size-fits-all" educational approach is far from adequate. Classrooms gather students with different interests and variant knowledge acquired from external resources. Therefore, students lacking prerequisite information for a new course will struggle during classroom time. They need to fill the gaps in their comprehension that occurred during class discussions.

Another drawback of the same misconception of the uniform baseline of knowledge is when educators adopt one information modality like textual education, expecting it to be enough to communicate the required knowledge to all students. This procedure overlooks the diversity in students' learning preferences. Creating content with the students' preferred media in multiple formats is essential to ensure close comprehension levels among them. This is a big challenge for educators, which implies investing substantial resources and time, which they might lack.

Educators have several central educational and pedagogical tasks. Besides creating engaging content for the subjects they teach, they are responsible for empowering their students' learning, evaluating their knowledge acquisition through assessments, and providing feedback. This applies to educators at the school level or college. Educators need help to adopt exciting approaches such as peer education, teamwork, problem-solving, reverse teaching, concept mapping, gamification, staging, role-playing, and storytelling, which should make the students actively learn before they even realise (Haleem 2022).

Unlike traditional AI, which is well equipped for data analysis and interpretation, Gen-AI creates new data. Gen-AI can then help create teaching material that is more interesting to students. The teaching material could be more engaging with animations and visual aids that would take months for educators to develop independently. These learning resources come at the lowest cost as well. Speaking of the other core tasks of educators, like assessments and feedback, Gen-AI can create exam questions and be trained to generate custom feedback for every student's answer. The following sections will discuss more capabilities like summarization, translation, and question-answering, showing how Gen-AI solutions are becoming essential technology-assisted learning tools for educators and learners.

However, upon the launch of ChatGPT by the end of 2022 and going viral, education institutions worldwide reacted differently. Between biases and misconceptions, some institutions banned ChatGPT while others decided to wait and see (Ngwenya 2024).

Gen-AI has offered immense educational potential: learning assistance, research assistance, classroom support, language translation, and skills development such as writing and grammar. It bridges language barriers in intercultural classrooms and provides access to quality instructional material and interactive learning in remote regions (Achiam et al. 2023; Ngwenya 2024). Additionally, Gen-AI can potentially support learners with hearing or visual impairments (Holmes and Miao 2023) since it can generate alternative texts for audio and speech from texts. In such a form, education is said to be personalized, democratized, and made available on a 24-h basis.

Developers and researchers have started to finetune EdGPT models to serve educational purposes (Holmes and Miao 2023). In other words, models are trained on smaller amounts of high-quality, domain-specific education data rather than massive amounts of general training data. Therefore, EdGPT models can align with a practical pedagogical approach and specific curricular objectives. Existing examples are EduChat[11] and MathGPT.[12]

As mentioned, a prompt can take multiple iterations before achieving the desired output. Unfortunately, less-experienced learners might accept superficial, inaccurate or even harmful Gen-AI output without critical engagement, especially since generative AI tools provide content without delivering concrete references, which users can assess their validity or reliability (Ngwenya 2024).

According to a UNESCO report (Holmes and Miao 2023), rigorous user tests and performance evaluations should precede validating Gen-AI tools for large-scale adoption. Task-driven metrics should be designed to validate or assess Gen-AI outputs. For example, for solving math problems, "accuracy" could be used as the primary metric of the correct answer; for code generation, the metric may be "the fraction of directly executable generated codes"; and for visual reasoning, the metric could be "exact match" in comparison to the ground truth image.

The educator's role now in the post-ChatGPT era, according to Doyle (2024), is to (1) foster inquiry-driven learning, (2) cultivate critical thinking, (3) enhance the curriculum with adaptive content and assessment, (4) stimulate creativity and innovation through AI prompts and simulations, (5) enable personalized trajectories, and (6) upskill students in AI fluencies.

7.3.1 Creative Learning of Problem Solving

The stages of the creative process in problem-solving are problem identification, idea generation, and idea implementation. In the problem identification stage, people gather information, set objectives, and define specific challenges that imply creative solutions (Habib et al. 2024).

Practicing collaborative problem-solving motivates students and encourages knowledge and skills transfer. The reason is its structured approach, starting from brainstorming the ideas and evaluating the feasibility and effectiveness of these ideas till reaching a consensus on the best solution. Additionally, the social interactions that take place during this approach help in the development of students' emotions

[11] https://www.educhat.top.

[12] https://www.mathgpt.com.

and intelligence. Gen-AI tools can offer the same experience if they are considered partners—without a conscious mind—that can communicate just like humans (Ngwenya 2024).

Wieland et al. (2022) define brainstorming as "a creative technique that fosters collaboration to enhance idea generation". Several studies (Ngwenya 2024; Wieland et al. 2022; Hwang and Won 2021) have concluded that teaming up with chatbots during brainstorming is very effective. People tend to generate more diverse ideas when they believe they are brainstorming with a chatbot rather than a human being. The reason is attributed to reducing the fear of being evaluated negatively or judged by other human partners. In a study by Habib et al. (2024) about the impact of Gen-AI on student creative thinking skills within classroom instruction, the authors measured qualitative data such as flexibility, fluency, elaboration, and originality of the ideas to assess the impact of ChatGPT-3 on students' divergent thinking; a measurement of creative abilities and assesses cognitive skills related to ideating. The study concluded that AI helped with divergent thinking by making students think more of rare ideas and generating more ideas overall. Another conclusion was that ChatGPT-3 is best utilized for fluency (i.e., quick responses) and elaboration (i.e., response quality) during divergent thinking. In contrast, flexibility (i.e., diversity of responses) and originality (i.e., rarity of ideas) are best augmented by human creativity and critical analysis. Despite this, the study declared minor dissatisfaction among some students who found the ideas generated by AI to be more practical rather than creative or novel since Gen-AI is trained on pre-existing data. Hence, the originality of the generated ideas was not evident. Also, some students expressed low confidence issues because they believe AI is taking over their thinking process.

The study emphasized the need to carefully integrate AI resources with traditional teaching methodologies to foster the students' creative confidence, divergent thinking, and critical thinking.

Similar opinions were recorded by MIT's Lifelong Kindergarten research group (Resnick 2024), which observed the evolution of "intelligent tutoring systems" or "AI coaches" used for programming education. They reported that these tutoring systems started by delivering instruction that adapts to individual learners' understanding levels. Recently, new AI tutors and coaches have promoted greater learner agency. They are designed to provide tips, advice, and support only when students ask for help instead of controlling the instructional flow.

7.3.2 Gen-AI for Curricula and Pedagogy

We must upgrade our educational models and transition from "learning by" technology to "learning with" technology. Human–computer interaction in the academic model will be derived from the roles of Educators, AI tools, and students. The ideal learning experience should be a result of practical lessons bringing three types of knowledge: content knowledge, pedagogical knowledge, and technology knowledge (Ngwenya 2024; Bredeweg and Kragten 2022; Dai et al. 2023; Mishra et al. 2023).

Educator’s role:

- Up-skill his AI literacy by learning and analysing the generative AI, its capabilities, and its positive and negative features.
- Evaluate the answers generated by AI tools regarding relevance, depth, and alignment with the learning objectives of the educational subject.
- Assess the available Gen-AI tools and decide how they can be adapted to their learner level and needs.
- Understand students’ motivations and empathise with students’ concerns.
- Facilitating students’ learning and supporting their progress.
- Demystify AI by training the students in basic AI literacy skills and how to use them in their work while highlighting the tools’ strengths and weaknesses.
- Monitor the integration of AI tools by hearing from students and learning their impressions of this learning experience.

AI tool’s role:

- Work as a learning and research assistant by answering the students’ questions of different levels about different subjects, offering examples, mixed media illustrators, or exercises to solve.
- Offer natural language processing help in translation, summarization, paraphrasing, and auto-correction. etc., to support students’ learning.
- Offer help by generating ideas students might need to start working on projects or assignments.
- Help educators complement their material by generating descriptive content.
- Facilitate exam generation and auto-grading for educators.

Student’s role:

- Work individually or in groups with AI tools through their learning path.
- Learn from educators about the AI technology they are using and the basic science behind these tools.
- Understand both the benefits and concerns of using AI tools and reflect on their learning experience to achieve an epistemic agency by controlling their learning progress.
- Improve their skills by observing the new information generated by AI tools, such as critical thinking, writing styles, etc.
- Help enhance the process by giving their feedback (positive or negative) so that their educator adapts the learning experience to the best of the student’s interests.

It is worth mentioning that AI assistants should not be considered replacements for human educators. Educators build relationships with students and create a caring community among them so that they feel welcomed, understood, and supported, even virtually. Classrooms that could thrive during the COVID-19 pandemic had teachers who built solid relationships and warm communities (Resnick 2024).

7.3.3 Case Studies of Using Gen-AI in Programming Education

Bahroun et al. (2023) reviewed over 200 recent research papers presenting comprehensive analyses of Gen-AI in different educational fields. They classified the publications investigating Gen-AI in programming education into four categories: publications investigating implications and potential of GAI code generation tools in programming education, publications exploring and utilizing AI tools for programming education, publications investigating the integration of AI technologies in programming education, and publications investigating ethical considerations and pedagogical approaches in AI education. However, this classification is not harshly decisive as the cited papers usually address the four topics within their experimental studies. This section highlights three distinguished studies that have concluded the most essential critical insights about the four issues.

7.4 Case Study 1:

A recent study about the effect of using the Gen-AI tool Chat-GPT on programming education was conducted by Yilmaz and Yilmaz (2023). The study was conducted to measure this effect based on three scales: the "Computational thinking" scale (evaluates creativity, algorithmic thinking, cooperativity, critical thinking, and problem-solving), the"Programming self-efficacy" scale (measures the ability to solve simple programming tasks and complex programming tasks), and the"Motivation scale" (assesses the individual attitude and expectation, challenging goals, clear direction, reward and recognition, punishment, social pressure, and competition).

Two groups of students were randomly selected, and a pretest was taken to measure their background knowledge about the subject, object-oriented programming (OOP), and the results came close. One group, the experiment group, was inducted to use ChatGPT as an AI support, while the other one, the control group, was prohibited. For five weeks, both groups took the same lessons and were given coding assignments in the form of UML diagrams (Universal Modelling Language), where they had to write the solution code. The classes were taught using a flipped classroom and a hands-on coding approach.

The researchers noted that the students in the control group devoted their time to writing code, debugging, and integrating. On the other hand, the experiment group spent their time creatively thinking, breaking down the assignment into subproblems, asking ChatGPT the most appropriate questions, and reaching out to the code snippets they wanted.

The post-tests taken by both groups showed significant improvement in the scores of the experiment group over the control group in all three factors. The experiment group showed higher motivation for the lessons thanks to the AI support in coding and debugging.

7.5 Case Study 2:

This study by Jonsson and Tholander (2022) explored using the AI tool for co-creation. A codex tool, an AI tool that allows users to input natural language (text) and receive code as a response, was used to assist in creating code-based artwork. It was integrated into a six-week "Creative Programming" course as a two-hour workshop in week 5. The course objective was to teach the foundations of programming using JavaScript to create audio-visual and interactive graphics. The tool used during the workshop was the OpenAI "playground" that overlies a GPT-3 model. The participants' training included a few "presets" in the playground tool and lines of code that provided example prompts for the GPT-3 model. The participants wrote text sentences describing the graphics they wanted to create, the tool generated the code, and the participants copied the code to an IDE to execute and view the results.

The post-workshop interview revealed several important issues and findings:

- The Codex system always provided syntactically correct and executable JavaScript code; however, the output did not correspond to the participants' intentions. It worked better when requesting to draw basic geometrical shapes such as circles and squares and less well in the case of complex shapes.
- Although there were conflicting views among the participants on how to understand and use a system and despite the inconsistent and imperfect behaviour of the Codex tool, which caused some confusion and frustration among the participants, the responses of the Codex system contributed to opening up for creativity and novel design ideas. This situation encouraged the participants to crack the code and learn to talk to the system to manifest their desired graphics.
- The participants went creative. All but one of the artworks created by the participants were animation, and some were interactive even though there were no instructions in the assignment, nor were the preset examples mentioned animations or interactivity. The participants approached the Codex system as a conversational process; they manipulated and tweaked the generated code to shape its output.
- The participants found the learning activity more enjoyable as it prevented getting stuck in finding the exact language syntax and helped them to focus more on the output. Also, it offered a potential starting point instead of coding from scratch.
- Introducing Gen-AI support based on natural language interaction for programming education should relieve some of the struggles related to learning the syntax, yet would introduce different struggles associated with finding the effective prompt for the AI tool.

7.6 Case Study 3:

Another study by Denny et al. (2024) introduced the gen-AI tool interaction optionally as a laboratory task in two Python-based courses, CS1 "introductory programming" and CS2 "data structures and algorithms". The assignment in each course was composed of three problems to be solved using code generated by a large language model (LLM) response to the students' prompts.

A tool was created for this study with some limitations on how to interact with LLM: (1) No dialogue is allowed with the LLM model; a single prompt must solve the problem. However, the students can try multiple times with different prompts, (2) students are not allowed to edit the generated code, and (3) the tool evaluates only a single response from the LLM at a time rather than generate and evaluate multiple responses. It runs unit tests to check whether or not the code correctly executes. The CS1 assignment problems required the generation of a program that processed standard input and printed output, whereas the CS2 assignment problems required a function that returned a value.

Each Prompt Problem presents a visual representation of a problem. This will prevent easy copying and encourage the students to formulate a suitable prompt. The generated code is sent to the testing suite automatically. If the generated code passes the tests, the student receives a success message and is directed to the next problem. Otherwise, the student can edit the prompt and try again.

Many students were able to solve the problems on their first attempt. There were considerable variations in prompt lengths across the successful submissions. Some students parrot the problem back to the tool rather than thinking of a suitable prompt.

The post-study reflections showed that many students were pleased with the new programming constructs and unfamiliar techniques sometimes introduced to the generated code. The students also commented that writing prompts helped them improve their problem-solving skills, as they described the steps needed to solve a problem, so they focused on the logic required rather than low-level syntax. However, a few students resisted, fearing potential impacts on their creativity.

The authors concluded that the deliberate exposure of students to the inconsistencies of LLMs, such as the Prompt Problems practice, can help evaluate generated code with a "critical eye" and decrease potential over-reliance on these tools.

Final Note:

Undeniably, there is a growing acceptance of using AI in classrooms and laboratories to provide personalized help to students and aid educators with time-consuming tasks. Gen-AI has proven impressive capabilities in dealing with object-oriented programming, Parsons problems, mathematics for computer graphics, and compiler error messages (Denny et al. 2024). New resources have been recently developed to explain and teach about interacting with Gen-AI tools for programming education (Jury et al. 2024; MacNeil et al. 2023; Porter and Zingaro 2024). However, we cannot be super optimistic about the previous studies' findings as there were other studies (Cipriano and Alves 2023; Finnie-Ansle et al. 2022; Kazemitabaar et al. 2023; Prather et al. 2023) which cited fears about the negative impacts of introducing AI tools in computer programming education. Their findings highlighted the possibilities of generating solutions that do not align with best programming practices, misleading students, confusing when reading code that is not theirs, and falsely thinking their prompts are highly effective. The following section will discuss the safe practices and regulations allowing us to integrate Gen-AI into programming education and alleviate its negative impacts.

7.7 Towards the Integration of Generative AI in Education

7.7.1 Best Practices in/out of Classrooms

Most of the studies that investigated the integration of Gen-AI tools in education in general and programming education specifically agreed on some key practices as the best way to benefit from AI capabilities and mitigate their concerns.

- Leveraging the importance of human communication: Start with clear learning objectives for Gen-AI tool integration and ensure that both educators and learners understand the purpose, potential benefits, and boundaries (Wood and Moss 2024).
- Slowly introducing the change: Gradual integration of new technologies mitigates resistance and promotes adaptation. Customizing Support mechanisms should overcome initial hurdles and encourage experimentation (Wood and Moss 2024).
- Changing the perspective: Integrate AI for open-ended problems rather than closed-ended puzzles and questions. We are not seeking personal learning assistants that offer a single answer to every question but provoke the student's creativity by searching for multiple answers to the same question (Resnick 2024).
- Promoting creative learning: Conduct project-based, design-oriented, interest-driven creative learning experiences. Help learners develop their creativity, curiosity, and collaboration skills (Resnick 2024).
- Connecting to the real world: Provide opportunities for learners to use Gen-AI tools in practical, real-world scenarios. The hands-on experience should motivate learning and reinforce knowledge acquisition (Wood and Moss 2024).
- Honouring the learner's agency: Educators should help the first AI literacy tools that the learners need to use. However, the learner decides to generate the educational resources that match their learning profiles and preferences.
- Cultivating skills that AI can't teach: Appreciate the irreplaceable role of human educators in improving learners' critical thinking abilities, fostering creativity, and providing essential context and perspective (Wood and Moss 2024).
- Reinforcing collaborative learning: Facilitate discussions among learning peers where they share their experiences with Gen-AI tools. Peer learning can enhance the understanding and application of Gen-AI in diverse instructional contexts (Wood and Moss 2024).

- Hearing back from learners: It is crucial to encourage learners to provide feedback on their experiences with Gen-AI. Transparent and open dialogues should enhance the personal learning process and improve the gradual integration of Gen-AI (Wood and Moss 2024).
- Regular ethical awareness: The long-term usage of Gen-AI tools would harmfully take over the learning process. Regular discussions and activities on the ethical use of Gen-AI and the responsibilities that learners should bear should encourage critical thinking about the implications of Gen-AI overuse (Wood and Moss 2024).

7.7.2 Regulations and Policies

Integrating Gen-AI in education has initially been resisted by many institutions due to several threats that are likely to happen if Gen-AI tools are misused (Ngwenya 2024). However, the widely growing acceptance of Gen-AI tools and the enthusiasm of learners to use these tools have pressured decision-makers to admit the importance of these tools and figure out methodologies to allow regulated usage of these tools.

Many institutions have started defining regulations that address the misuse of negative impacts and support policies for future adaptation to any changes related to this technology. Regulations ensure collaboration among teachers, instructional designers, education researchers, and AI developers to define the best practices in the utilization of these emerging technologies without undermining academic integrity or compromising human agency and creativity (Ngwenya 2024). Regulations must be based on a human-centred approach (informed by the 2021 UNESCO Recommendation on the Ethics of AI) to ensure its ethical, safe, equitable, and meaningful use. Examples of these regulations were detailed in (Holmes and Miao 2023; Ngwenya 2024; Alier et al. 2024) but we can group them into two main categories: Institutional Regulations and Global policies.

Institutional Regulations:

- Before deciding to adopt Gen-AI tools, institutions have to ensure that the Gen-AI tools to be adopted are educationally effective and valid for the ages and abilities of the target learners and are aligned with sound pedagogical principles.
- Maintain a healthy learning environment and face the temptation of overuse of AI tools through enriching curricula with real-world applications, hands-on experimentation, collaborative tasks, and independent reasoning.
- Capacity building for educators to properly use Gen-AI and dynamically review their needs to understand and use AI for teaching, learning, and professional development.
- Protect the educators' irreplaceable roles of facilitating higher-order thinking, organizing human interaction, and fostering human values.
- Consider integrating Gen-AI in the institutional learning management systems (LMS) to customize responses according to educator, learner, or administrator user roles, ensuring secure access to advanced Gen-AI features.

- Support learners' agency by granting inclusive access to learning programs, especially for learners with disabilities, with provisions and management to improve the quality of learning. Monitoring the learning progress and providing regular awareness about AI's ethical and meaningful use is necessary.
- Allow using Gen-AI tools to minimize the pressure of homework but prevent the use of Gen-AI when it turns to harmful addictive dependency that would deprive learners of opportunities to develop cognitive abilities and social skills.
- Guide educators to help alleviate uncertainty and anxiety among learners about their future careers and encourage them to fight the temptation of AI usage to learn and develop their skills.
- Consult educators about their experience using AI. Encourage them to critique and question the AI systems methodologies, output accuracies, and the norms they may impose to minimize the risk of hallucinations or incorrect information.
- Shifting academic assessment methods to alternatives that ensure fairness and encourage creative and genuine student engagement, moving away from traditional assignments that could be AI plagiarized, such as demos, presentations, and podcasts.
- Magnifying human accountability when using Gen-AI systems to make high-stakes decisions.

Global Policies:

- *Democratizing technology*: Controlling AI tools by restricting the training data to specific domains requires the availability of vast amounts of data, massive computing power, and iterative training. These resources are out of the reach of many countries. A global act should regulate access to these resources and assure equal opportunities.
- *Validating technology*: Advice-building and Dissemination of validation mechanisms to test whether Gen-AI systems used in education are safe, unbiased, and equitable. All types of diversity must be ensured (gender, disability, social and economic status, ethnic and cultural background, and geographic location).
- *Reinforcing ethical technology*: Audit the outputs of Gen-AI against deepfakes, fake news, or hate speech, and impose a severe penalty on the person/institution responsible for that inappropriate Gen-AI tool. These tools do not have any conceptual knowledge or world understanding and hence may spin out information lacking credibility.
- *Protecting privacy and copyrights*: All collected information about the learners' identities, roles, academic records, and interactions with the tools must be used only to provide the service. Deletion of this information upon the end of academic courses had to be secured.
- *Controlling quality of Gen-AI outputs*: Supporting the development of "Guardrails" tools, which developers design to prevent GPT output from being offensive, unethical, etc., which would eventually lead to polluting the internet. In the coding context, this can be found in poorly coded software uploaded to blogs, Q&A forums, or personal code repositories.

- *Reinforcing transparency and explainability*: alleviate the trust issues around Gen-AI by providing information and allowing access to functioning details of this tool; for example, with inspectable Gen-AI model's weights, explaining why a specific output is generated is tough.

7.8 Summary

Fear of long-term implications likely to occur with Gen-AI integration included risking human agency, essential thinking and skill acquisition, privacy, academic integrity, and other ethical concerns. Reasons for excitement and resistance to applying Gen-AI in education have been extensively discussed in the literature.

Supporting voices for Gen-AI adoption are deeply impressed by the growing capabilities of Gen-AI tools in generating new data of different forms and the natural language features that promote these tools' role as a highly effective learning assistant. On the other hand, the opposing opinions raised concerns about the degradation of students' learning development due to the addictive overuse of these tools, which is likely to compromise their creative intellectual skills and eventually lead to unethical plagiarism. Another justified concern is the generated content's truth and validity, which is sceptically questioned since it is well known that these tools hallucinate without conceptual knowledge and do not have a conscious understanding of the world. It is very likely these tools could generate fake data or impose unethical beliefs such as hate or bias.

Moderate voices support embracing the technological developments and call for responsible integration of Gen-AI tools in education to benefit from their valuable capabilities and narrow the gap of knowledge between higher education and the job market competencies. These voices also stress the transformation of the educator's role, not only his personal development or his educational curricula but also the irreplaceable pedagogical role of the educator in creating a loving and supporting classroom environment that keeps the learners motivated to learn, connected to their peers and gradually prepared to their future dream careers.

Based on numerous studies in different countries, educational institutions, targeted audiences, and selected curricula, a few suggested best practices and regulations have been concluded to support the human-centric, safe, ethical, equitable, and meaningful integration of Gen-AI tools. Putting these recommendations into action is likely to change programming education drastically. The observations of the related studies on the experiment groups' behaviour and their creative ways of employing the Gen-AI tools are promising. The development of computer programming will directly reflect on the quality of services provided to people and will positively impact their welfare.

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Chapter 8
Generative AI Applications in Education: A Low-Code/No-Code Approach

Dimitrios Tolis, Stylianos Mystakidis, and Athanasios Christopoulos

Abstract This chapter demonstrates how a Low-Code/No-Code (LC/NC) approach with Generative Artificial Intelligence can be applied to adult education. Generative AI has rapidly changed the educational landscape by offering educators and students various tools and applications and adding new potential for creativity and personalisation. Most educators perceive AI as an opportunity; however, they face challenges of their own, mainly due to the increasing complexity of the applications and their limited knowledge and skills in AI-related topics. At the same time, there is a rising trend internationally regarding 'Citizen Developers', that is, power users who can create their software solutions with minimal or no programming skills, using user-friendly LC/NC frameworks and platforms in Artificial Intelligence as a Service (AIaaS) offering. The described AiHub platform prototype uses open-source frameworks and tools to simplify the Generative AI development process, allowing users without programming skills to design and develop their solutions and build applications with increasing complexity, from novice to expert. Finally, a case study on adult education is presented to demonstrate the power of development that can be given to educators to act as Citizen Developers and contribute to the AI revolution of our time.

Keywords Generative artificial intelligence · Low-code/no-code · Education · Citizen development · Open-source software

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8.1 Introduction

The accelerated rise and use of Artificial Intelligence (AI) today, particularly of Generative AI, is rapidly revolutionising industries and transforming daily life worldwide. In a recent survey (McKinsey and Company 2024), it was found that Al adoption worldwide has increased dramatically, reaching 72% of the 1363 respondents in 2024, increasing from 55% in 2023, after years of little meaningful change.

Generative AI, which includes models like GPT, Claude, Gemini, Llama, can create text, images, music, and even video content, pushing the boundaries of creativity and automation (Chen et al. 2023). Its applications span various fields, from enhancing customer service with sophisticated chatbots to personalising marketing campaigns with AI-generated content. AI promises to revolutionize education by enabling personalized learning experiences for students and by providing educators with innovative tools to automate administrative tasks and enhance instructional strategies (Onesi-Ozigagun et al. 2024).

At the same time, there is a significant challenge: a widespread lack of AI-related skills across Europe and the globe (Moch and Oberdieck 2024). While AI presents opportunities to enhance efficiency and innovation in all industries, it also underscores the critical need for software developers to continuously update their skill sets to keep pace with technological advancements. This skills shortage threatens not only the full realisation of the benefits of AI but also economic development and the viability of the global digital economy (Wisskirchen et al. 2017). Even in everyday life, as AI is gradually being integrated into most common technologies and applications, people without AI literacy and skills express concerns and resistance (Long 2021). Educators, who can play a significant role in this digital divide problem, have their challenges in adopting and using AI in the educational process (Ng et al. 2023).

A solution to this problem might be the democratisation of AI development through Low-Code/No-Code (LC/NC) platforms (Martins et al. 2023) that offer AI as a Service (AIaaS), empowering more individuals and organisations to harness its potential and fostering a new wave of creativity and productivity. This study aims to examine the Citizen Development adoption with LC/NC tools and demonstrate how an LC/NC approach can be applied in education using Generative AI.

In the following sections, we examine the integration of AI in education, teacher challenges and skills gaps and the concept of Citizen Developers using LC/NC tools in education. Next, we present an NC/LC platform prototype called ‘AiHub’, which uses open-source frameworks to simplify the Generative AI development process for educators. Then, we present a case in adult education, specifically in the Tourism Industry. The article ends with conclusions and recommendations for future research.

8.2 Literature Review

In November 2022, OpenAI presented ChatGPT to the world with a blog post, effectively presenting Generative AI chatbots to the broader public, which until then were only known to researchers and AI professionals. The opportunities and possible applications of ChatGPT to education were pointed out very quickly, just months after launching. Ever since, there has been a lot of research on how ChatGPT can significantly enhance education in software development, cyber security and many other fields (Kuraku et al. 2023). There are significant applications of ChatGPT in the fields of preparation of educational materials, the preparation of scientific research and text writing and editing (Strzelecki et al. 2024). Other studies focus on specific fields, like Tourism education and the use of ChatGPT in assisting educators to offer more engaging, personalised and efficient learning experiences and content, address multilingual challenges, and prepare interactive assessments (Onesi-Ozigagun et al. 2024).

However, there appear to be discrepancies and nebulous conceptualisations among educators of how to design teaching and learning using AI tools and applications (Lameras and Arnab 2021). Educators need new skills (Ng et al. 2023) and they might lack the intention to adopt AI (Celik and Muukkonen 2023). Commercial and well-known systems like ChatGPT are easy to access and use and offer many automated tools, such as creating GPTs or Assistants with custom system prompts, accessing educators' documents and tools, and even finetuning a GPT model for specific use, all without the need for programming skills. However, this means that the educators will have to use the service at a high cost and with the restrictions that apply to it. But how can educators use open source LLMs, finetune them, or use open source and free embedding tools and vector databases of their choice to upload their knowledge documents and build more complex chains of agents and tools to achieve more? Do they have the literacy and skills to design and develop applications like generative AI chatbots with the freedom to select the components of their choice, be it open source or commercial?

This challenge may be addressed with educators acting as Citizen Developers using Low Code/No Code (LC/NC) platforms and frameworks offered as AIaaS. There is an increasing trend internationally regarding Citizen Developers (Binzer and Winkler 2022; Viljoen et al. 2023), that is, empowering non-professional users who can design and develop their software solutions with minimal or no programming skills using user-friendly LC/NC frameworks. However, there is a lack of familiarity in organisations and a lack of skillsets, which will need to be addressed before the citizen developer approach is implemented, with the most significant barrier to adoption being the lack of familiarity with the technology and its usage (Prinz et al. 2024). LC/NC development creates an environment where individuals and teams can build applications using visual interfaces, like drag-and-drop tools, instead of relying heavily on traditional coding (Viljoen et al. 2023), with minimal or no programming code.

To ease the adoption of AI for organisations, several cloud providers offer AIaaS (Pandl et al. 2021). These services may offer AI Infrastructures as a service, AI developer services with easy-to-use libraries, tools, frameworks, and AI Software services for machine learning and inference (Lins et al. 2021). AI-as-a-Service (AIaaS) is a cloud computing service that offers the tools and services to make AI accessible without implementing complex solutions on-premises, with a low and scaling cost (Griesch et al. 2024).

8.3 AiHub: A Low code/No Code AI Platform for All

8.3.1 An Overview of the System

AiHub (https://aihub-lab.eu/) is a platform prototype that aims to democratise the use and development of Generative Artificial Intelligence, offering the tools to both non-technical (Citizen Developers) and technical users to design, deploy and manage custom AI chatbots, opensource or commercial, through an LC/NC approach. It offers a centralised dashboard for easy management of chatbots, supporting integration with major LLMs like OpenAI GPT, Mistral, Gemini, Llama, finetuned or even custom LLMs and with more than 100 other integrations of tools that may be used to create a significant number of applications.

AiHub may simplify Generative AI applications development, reducing the IT department's backlog and cutting costs through citizen development, allowing users, like educators, to apply their domain expertise directly. Based on open-source frameworks and platforms, it facilitates the rapid development and deployment of chatbots, offering an intuitive drag-and-drop interface. AiHub is a cloud-based solution with two subsystems as shown in Fig. 8.1.

The two subsystems architecture was chosen to allow flexibility and support various use cases and applications. This modular approach improves flexibility, scalability, and maintenance (Dragoni et al. 2016). Figure 8.2 presents the overall AiHub system architecture in more depth.

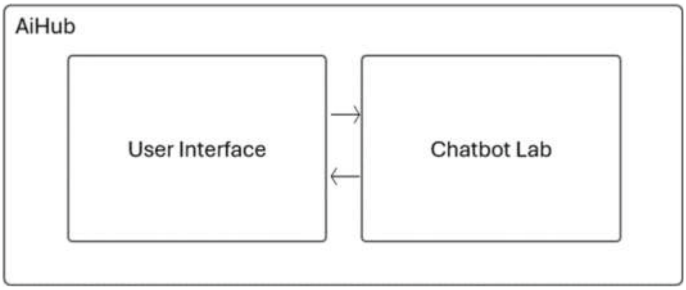

Fig. 8.1 The AiHub System High-Level Architecture. Image by the authors

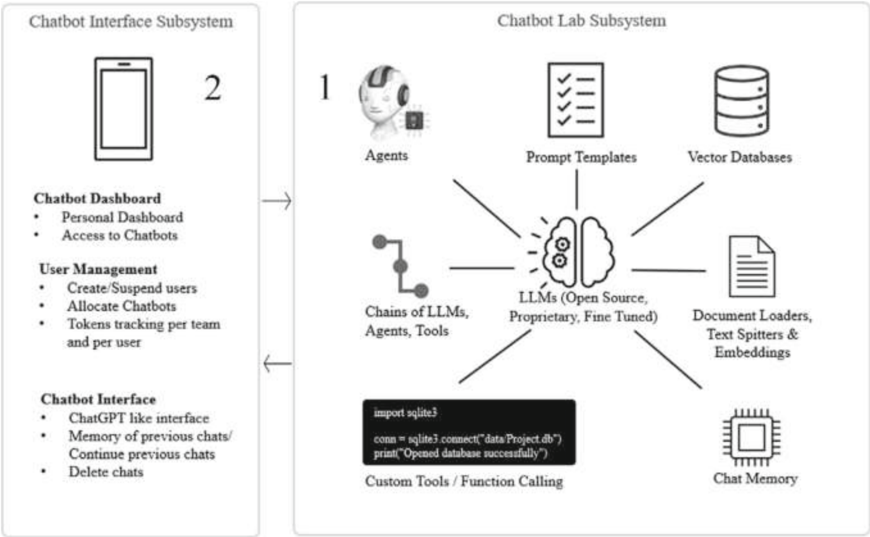

Fig. 8.2 The AiHub System Architecture. Image by the authors

The Chatbot Lab Subsystem (Fig. 8.2, right frame), used for the chatbot design and deployment, is based on open-source frameworks (Flowise and Langchain). It includes an intuitive drag-and-drop Graphical User Interface (GUI) with a collection of AI tools and components that users can drag and drop into their workflows, a library of pre-built templates for chatbot workflows based on use cases per industry or function, a chatbot testing interface, and a custom tools development interface.

The Chatbot Interface Subsystem (Fig. 8.2, left frame) is developed using the React framework and other tools. It offers a chatbot interface with the user input, the chatbot output, a panel with a list of previous discussions, a dashboard displaying available chatbots per user and their descriptions, and a user management tool for organisation administrators, offering user analytics and token consumption metrics.

The subsystems are presented in more depth in the following section.

8.3.2 The Chatbot Lab Subsystem

Drag-and-drop Graphical User Interface (GUI)

The subsystem offers a visual workflow design where users can design workflows by connecting components for different tasks and processes, like integration of LLMs, open-source or commercial, vector databases and many more.

Component Library

As shown in Fig. 8.3, the system provides a library of pre-built components and integrations that users can drag and drop into their workflow canvas. Based on their use case, the users drag and drop the component from the library. Each component is in the form of a node, which can be configured with specific settings and parameters. Users can make connections between the nodes, adjust the necessary settings, like entering their own API key for their LLM of choice and make decisions on various hyperparameters, such as the model's temperature (Renze and Guven 2024), and create the chatbot workflow, all with low or, in many cases, no-code functionality.

Main components and integrations available:

1. *Agents*: In AI terms, an agent is an autonomous system that can plan and execute complex tasks with only limited human involvement (Ouaddi et al. 2024). The agents may take actions autonomously, without any human decision.
2. *LLMs*: Large language models (LLMs) are machine learning models that can comprehend and generate text in a human language. LLMs are called large as they are trained on massive datasets. Their prediction is based on a type of neural network called a transformer model (Chen et al. 2023).

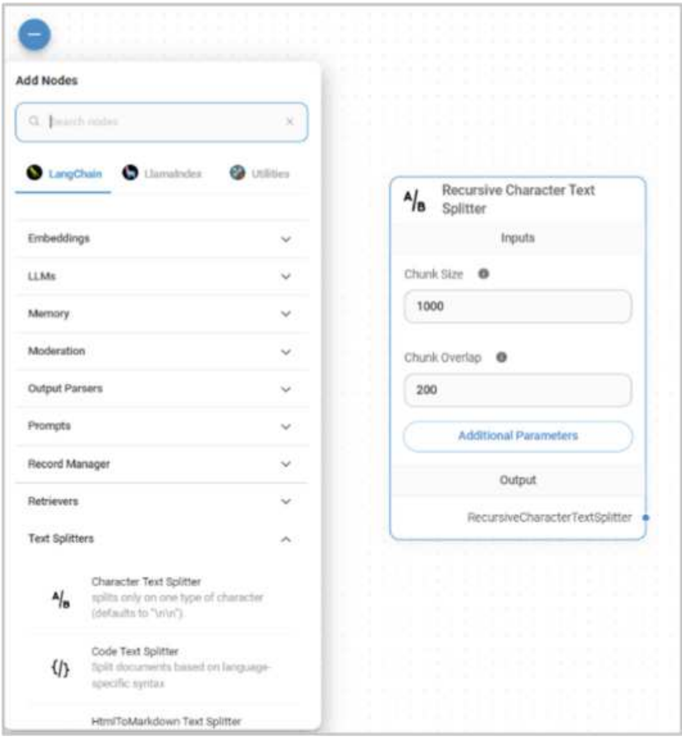

Fig. 8.3 The AiHub System Components List. Image by the authors

3. *Prompts*: Prompts are guidelines provided to an AI system, giving specific instructions or queries to achieve a desired outcome. Prompt engineering has become a potent technique to extend the capabilities of LLMs (Chen et al. 2023; Morales-Chan et al. 2024).
4. *Retrieval Augmented Generation* (*RAG*): The platform offers tools to deploy an RAG solution. Large Language Models like GPT 4o, Mistral, Claude or Llama have been trained on vast resources such as text, images and videos and have billions of parameters, producing these impressive results. LLMs have a remarkable ability to communicate in natural language, reason, synthesise and generate content. However, they tend to hallucinate, they have a knowledge cut-off date, and it is difficult to update with time-sensitive or organisation-specific data (Tonmoy et al. 2024). As a solution to these challenges, Retrieval-Augmented Generation (RAG) has emerged as a powerful approach for enhancing the performance of language models, especially on knowledge-intensive NLP tasks (Lewis et al. 2020). RAG is gaining a lot of attention as an alternative to the finetuning of pre-trained models or even the training of a new LLM, which takes time and is very costly. RAG, as compared to an LLM, can improve the performance of a chatbot, especially when the tasks include knowledge-specific tasks, and at the same time, allow for continuous knowledge updates and integration of domain-specific information (Gao et al. 2023). To implement an RAG system, various tools such as Text Splitters and Document Loaders are available to load documents from different sources (e.g., PDF, DOCX, TXT, CSV, Notion) and upload them as embeddings into a vector database, which can then be queried for retrieval.

Library of Pre-Built Templates

The system offers citizen developers a library of pre-built templates to accelerate their workflow automation and chatbot development processes with minimal effort. The templates may serve as a starting point for non-technical users, allowing them to design and deploy complex workflows with minimal contribution. The library provides a collection of pre-built templates and use cases, like teaching assistants and coaches, instructional design assistants, customer support chatbots, onboarding chatbots and many more.

Custom Tools Interface

Technical users or those with minimal programming knowledge can significantly extend the platform's functionality by creating custom tools and components. These custom components can be developed using standard programming languages and integrated into the visual workflow, such as Custom JavaScript Functions, logic (e.g., if/else), and custom tools, where the qualified user can write custom code and embed it in the chains and workflows.

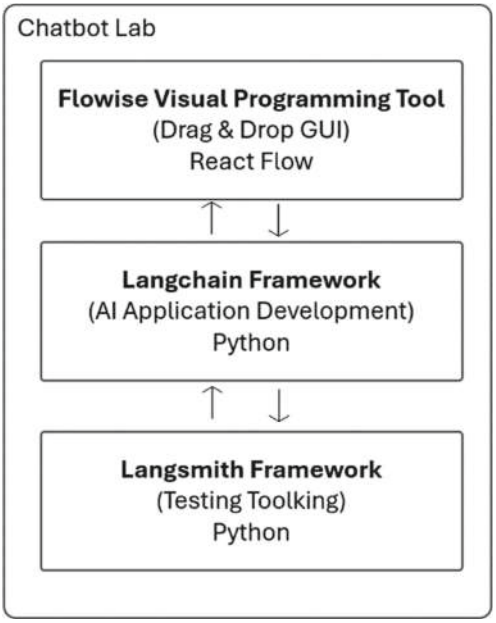

Fig. 8.4 Chatbot Lab Architecture. Image by the authors

Main Technologies Used in the Chatbot Lab

The main open-source technologies used in the Chabot Lab subsystem are illustrated in Fig. 8.4:

LangChain is a cutting-edge open-source framework written in Python, and it is specifically designed to facilitate the development of AI applications using LLMs by connecting components in chains with minimal need for coding (Morales-Chan et al. 2024).

Langchain is modular, and it enables the creation and management of chains (sequences) of tasks and components and integrates LLMs (open-source, commercial or finetuned). It offers tools for prompt engineering, and it is open to customisations from the developers to create their additions or custom tools to expand the framework based on their needs, provided that they have the technical knowledge to do so (https://python.langchain.com).

Flowise is an open-source visual programming tool that provides a drag-and-drop GUI, based on the React Flow library and it is used to create and manage workflows with LLMs and other automated processes (https://flowiseai.com/). The framework aims to facilitate the design and deployment of complex workflows without the need for deep programming knowledge. Flowise is based mainly on the Langchain framework but also uses LlamaIndex to extend user capabilities. LlamaIndex is an alternative data framework for connecting custom data sources to LLMs (https://www.llamaindex.ai/).

Finally, *Langsmith* is an open-source toolkit that can be integrated with Langchain and is designed to facilitate the development, testing, debugging and optimisation of LLM applications with an easy-to-use interface and tools. It may also be used for prompt optimisation and fine-tuning on specific datasets (https://www.langchain.com/langsmith).

8.3.3 *Chatbot Interface Subsystem*

The Chatbot Interface

The chatbot's interface (Fig. 8.5) consists of several key components: (1) a text input field where users type their queries, (2) a send button to submit queries, and (3) a main conversation window displaying the chat history. Above the conversation window, a title or navigation bar may provide access to settings, the user profile, and help options. On the left is a panel with a list of previous discussions to re-visit or continue.

The Dashboard

The Dashboard displays the available chatbots per user and their descriptions. In organisational use, the organisation administrator user can edit or delete existing chatbots and change chatbot allocation to the users.

The User Management Options

User Management is offered to organisation administrators or power users, allowing them to view all users belonging to the organisation, create or delete users, edit user details and view user analytics and token consumption.

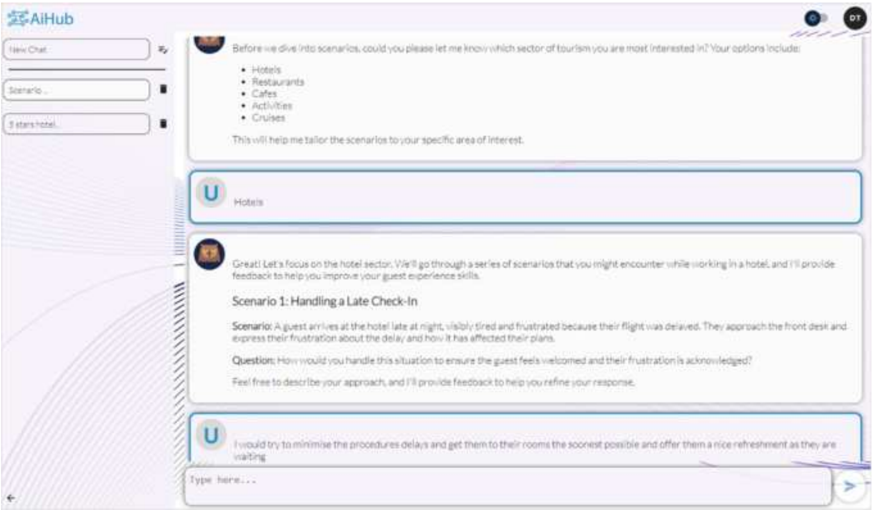

Fig. 8.5 The Chatbot Interface. Image by the authors

Main Technologies Used in the Chatbot Interface Subsystem

For the Chatbot Interface, the React framework was used (https://react.dev/). React JavaScript library enables developers to create user interface components, manage application state efficiently, and optimise rendering performance. An advantage is that React has a component-based architecture, which enhances code reusability and provides extensive ecosystem and community support.

API Communication is handled by another popular JavaScript library, Axios, which makes HTTP requests from Node.js and XML HTTP Requests from the browser (https://axios-http.com/). Axios supports features like automatic transformation of JSON data, cancellation of requests, and concurrent request handling, making it a powerful tool for managing API interactions in web applications.

Node.js (https://nodejs.org/) is used for server-side scripting**,** allowing JavaScript to be used for server-side scripting, enabling the creation of dynamic web pages before the page is sent to the user's web browser. Its event-driven architecture and the non-blocking Input/Output model make it efficient and lightweight, suitable for building real-time applications.

8.4 Case Study: The Creation of an Educational Chatbot for Guest Experience Training of Tourism Employees

The application and usefulness of LC/NC generative AI platforms in adult education are demonstrated in the following educational chatbot case study, a coaching assistant on guest experience training for employees of the tourism industry that may supplement all relevant courses created by educators or subject-matter experts. It aims to enhance the learning experience of adult learners with a Socratic coaching conversation on real-life scenarios, where the learner attempts to solve challenging situations and receives feedback aiming to improve their answers. The chatbot was created using the AiHub LC/NC platform and using open-source resources and RAG implementation.

The AiHub RAG implementation architecture is presented in Fig. 8.6.

In more depth, using the Flowise drag and drop interface, the Guest Experience Coach Chatbot chain flow was implemented, as shown in Fig. 8.7.

The main structure of the chatbot uses:

A (7) *Conversational Chain* component that aggregates the connected LLM to handle the discussion with the user, a retrieval component that contains documentation from the RAG implementation and a buffer memory.

For the (6) *buffer memory*, we have used an internal system database as the storage mechanism for storing/retrieving conversations, which is useful for remembering previous information in the dialogue.

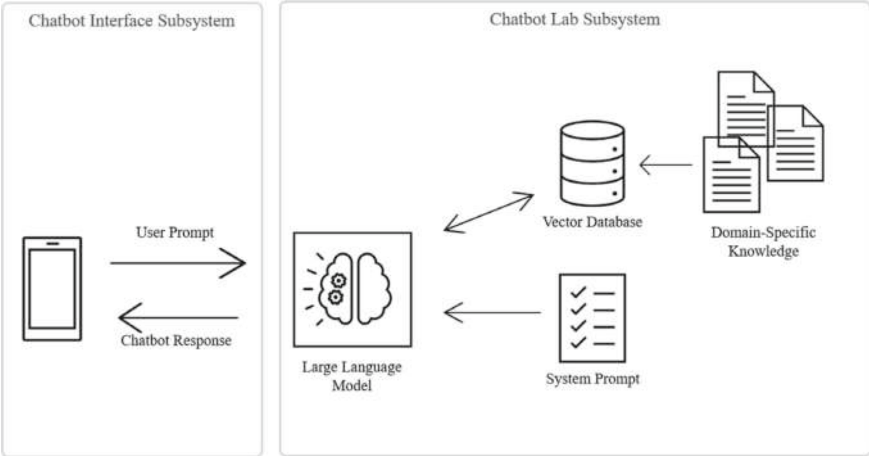

Fig. 8.6 The AiHub RAG Implementation. Image by the authors

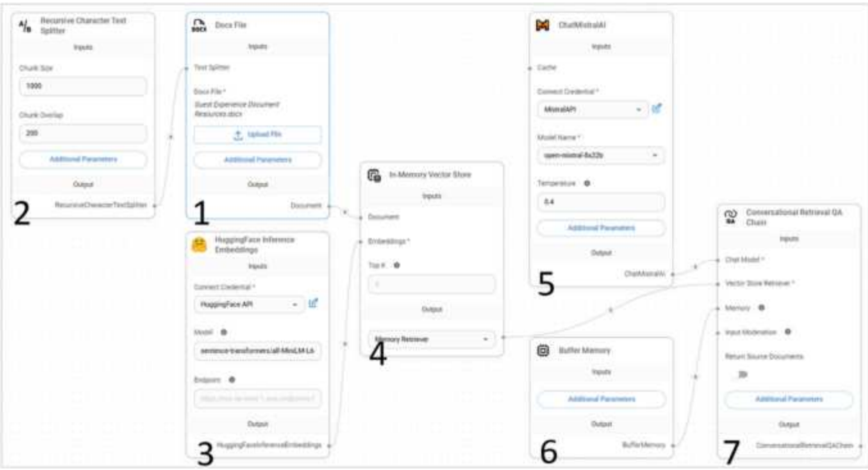

Fig. 8.7 The Guest Experience Coach Chatbot chain flow. Image by the authors

The (5) *LLM node* is used to import and connect an LLM to the system. In this case, we have connected the open-source Mistral[1] model using the respective node. The only settings needed were to insert an API key provided for free by the Mistral portal, choose the model we would use, in this case, the Open Mixtral 8 × 22b, an open-source, fast and efficient model. Indeed, the platform offers the option to

[1] https://mistral.ai

include closed-source models like GPT 4o[2]or Claude,[3] large open-source models like Llama 3.1 or even smaller open-source or finetuned models from services like Huggingface.[4]

To include custom knowledge for the Guest Experience training, we needed a *RAG* implementation, which is comprised of (1) a *Document Loader* to upload the documents containing custom information needed for the chatbot, (2) a *Text Splitter* to split up that text into chunks, and (3) an *Embedding* model by Huggingface to generate the embeddings (the numerical representation of textual data for the given text). This includes a Competency Framework on Guest Experience with indicative behaviours, a textbook with Guest Experience theory and example scenarios with feedback, which was used as the basis for the scenario building and the coaching responses.

Additionally, a (4) *Vector Database* is used to store the custom knowledge. In this case study, we used the in-Memory Vector database, which is open-source and easy to implement by Flowise and Langchain. Other closed or open-source Vector Store solutions are available, too.

Finally, in the (7) *Conversational Chain component*, the main system prompt is added to set the coaching assistant's role and behaviour.

A prompt is a natural language input that serves as the bridge between the LLM's computational capabilities and human communication, and prompt engineering is the strategic process of designing these inputs to get the best responses from the LLMs (Fagbohun et al. 2023). Prompt engineering can be viewed as a form of 'programming' via natural language, which helps to democratise the application of computational problem-solving across a range of disciplines and professions (Schmidt et al. 2023). During the design of a chatbot application, it is essential to create the system prompt, which will serve as an initial and default set of instructions for the system. This prompt will guide the behaviour of the application or chatbot when interacting with the user.

Prompt engineering methods have been an emerging topic of interest and research in recent years (Sahoo et al. 2024). We have used the following prompt engineering techniques to design the system prompt:

- *Few-Shot Prompting*, by giving examples of scenarios and the expected feedback from the chatbot.
- *RAG,* to reduce hallucination and to offer topic and industry-specific data.
- For the syntax and format of the prompt, we have followed the *CLEAR path* with the five core principles: Concise, Logical, Explicit, Adaptive, and Reflective (Lo 2023).

The prompt used for the case study is structured as illustrated in Table 8.1.

This chatbot creates excellent coaching discussions with the learners, following the prompt in detail and utilising the LLM's knowledge and reasoning with the custom

[2] https://openai.com/

[3] https://claude.ai/

[4] https://huggingface.co

Table 8.1 Prompt structure and justification

Prompt	Technique Justification
You are a guest experience skills teaching assistant, helping the user improve their guest experience skills based on the specific sector of tourism they are most interested in. Your goal is to present the user with tailored guest service scenarios and offer constructive feedback using a Socratic approach by asking questions that can help the users find the best answer on their own before providing your feedback	We start the prompt by stating the required task the chatbot should perform and the persona we expect it to adopt, and we explain the generic context. Then, we split the complex task into simpler subtasks
Step 1: *Identify the tourism sector the user is interested in*: • First, ask the user which sector of Tourism they are interested in, such as Hotels, Restaurants, Cafes, Activities, or Cruises • Once the user identifies a sector, all scenarios in the conversation should be relevant to that specific sector	Requirements Elicitation Facilitator (Schmidt et al. 2023) motivates users to bring forward their implicit requirement expectations and narrow the proposed training to more personalised job-related scenarios. In-Context Prompting (Fagbohun et al. 2023) is used, based on the memory buffer of the model, to build upon previously introduced information, and offer consecutive scenarios based on the Requirements Elicitation
Step 2: *Scenario Presentation* • Create each scenario using the competencies and behaviours provided in the uploaded document, ensuring they align with the values of the effective guest experience • Present ONE scenario at a time, ensuring each scenario touches on different aspects of guest experience within the chosen sector • After presenting each scenario, ask the user how they would handle the situation	The competencies and behaviours needed for the scenarios will be based on uploaded documentation. Additionally, example scenarios and responses are offered as Few-shot Prompting (Brown et al. 2020) in the available documents (RAG)
Step 3: *Evaluate the Response* • After the user responds, evaluate their answer. If their response is not the best, follow the Socratic method by asking questions, encouraging them to reflect on their approach and find a better answer. For example: "What might be a better way to ensure the guest feels their concern is heard?" • After this, allow them to adjust their response based on the new insights with another answer	Conversational Prompting and Socratic Prompting (Fagbohun et al. 2023), as a role-play instruction, offer the opportunity to the user to engage in a coaching dialogue and try to reach the best answer in multiple attempts while the system acts as a Socratic coach

(continued)

Table 8.1 (continued)

Prompt	Technique Justification
Step 4: *Offer Feedback and Share the Best Response* • After two answers, offer feedback on their approach • Provide constructive and supportive guidance, explaining what they did well and how they could improve • After providing feedback, offer to share the best possible response so they can further refine their understanding	In the feedback, we also included the Reflection Pattern (Schmidt et al. 2023) by asking the LLM to explain the evaluation of the user's answer and its rationale
Throughout the process, maintain a supportive and encouraging tone. Keep the user motivated and engaged. Always encourage progress and emphasise the importance of the growth mindset	The prompt explains the expected tone and style of the conversation
If the user asks unrelated questions, gently guide them back to the task by saying: • "My goal is to help you with your guest experience skills. May I share a scenario for you to practice?"	Constraints and limitations are stated explicitly to frame the response from the LLM

documentation with more specific domain knowledge and examples. The chatbot's ability to facilitate a Socratic teaching method is worth noticing, as it demonstrates reasoning ability. It engages students in guided and personalised coaching discussions, encouraging their critical thinking and deeper understanding, rather than just retrieving from the RAG and offering plain information. This customised approach may ensure that students receive feedback and support directly relevant to their learning needs, making the classroom experience more engaging and effective.

Indicative chatbot responses are presented in Table 8.2.

Table 8.2 Indicative chatbot responses

"Hello! I'm excited to help you improve your guest experience skills in the tourism sector. To start, could you please tell me which sector you're most interested in? It could be Hotels, Restaurants, Cafes, Activities, or Cruises. Once we have that, we can begin with tailored scenarios relevant to that specific sector. I'm looking forward to discussing it with you!"
"Great! Since you've chosen the 5-star hotel sector, let's dive into a scenario that will help you improve your guest experience skills. Scenario: Guests have just checked into your hotel and are visibly upset due to a long flight delay. They express their frustration about the situation and ask for an upgrade to a suite. However, all the suites are currently occupied. Now, imagine you're the hotel manager in this situation. How would you handle this guest's request and ensure they have a positive experience despite the initial frustration? Please share your approach."
"Great job! Your response shows empathy and a willingness to go the extra mile to ensure the guest has a positive experience. Here are some points to consider and refine your approach…"

8.5 Conclusion

The potential of Generative AI, such as educational chatbots, goes beyond just creating conversations or sharing information. Low-Code/No-Code platforms and AIaaS can democratise the design and development of complex AI applications, giving various non-technical actors, like educators, the tools and the power to act as citizen developers and create innovative approaches and applications that can revolutionise education. Tools like the AiHub can empower educators with easy-to-use tools and templates, which can help them improve their AI literacy skills and become more active in designing and developing applications that may transform education rapidly. The case study in guest experience training for tourism professionals demonstrates how prompt engineering and RAG can enhance the learning experience by offering Socratic and personalised reflection and feedback.

Beyond this case study, AiHub's flexibility and scalability can make it a handy tool across various industries and contexts, not only in education but also in helping users with minimal technical knowledge and skills to create innovative applications using generative AI. As AI adoption continues to grow across all industries, LC/NC and AIaaS platforms may play a crucial role in the faster adoption and use of AI in applications designed and developed by the end users and becoming more accessible to more people, helping to reduce the AI digital divide.

Looking ahead, the next steps in our research are to collect more empirical data on the educators using the AiHub platform and to expand the case studies on more complex applications, with chains of multiple and sequential agents, in combination with RAG implementations and other custom tools to achieve more sophisticated educational chatbots.

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Chapter 9
Generative AI and Digital Twins: Fostering Creativity in Learning Environments

Andreas Pester, Frederic Andres, Chandrasekhar Anantaram, and Jon Mason

Abstract This chapter explores the intersection of creativity in education with digital twins and generative AI (GenAI), highlighting their potential to transform teaching and learning. Building on theories from influential educators, such as Vygotsky and Gardner, we discuss how advances in digital technology enhance human–computer collaboration and position creativity as a sociocultural act shaped by interactions between humans and artificial agents. As part of the digital revolution, the narrative of education emphasises essential skills for successful citizenship, known as twenty-first-century skills, which include creativity, communication, collaboration, and critical thinking. This chapter argues that creativity is not just a personal trait but a cultivable competency fostered through engagement with digital environments. We examine how GenAI and digital twins can improve creative learning experiences, enabling personalised growth and collaborative creativity with AI as a partner. Strategies for nurturing a culture of creativity in education are discussed, along with the challenges and opportunities these technologies present. Finally, we suggest future trends in creativity, digital twins, and GenAI, redefining the creative potential of learners and educators.

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9.1 Introduction

This chapter presents theoretical and practical perspectives on creativity in education, focusing on digital twins and generative AI (GenAI) and how these technologies open new frontiers for both teachers and learners. Prominent theorists have made significant contributions to the theoretical understanding of creativity in education and learning, for example, Vygotsky (1978), Montessori (1989), Gardner (2006), Malaguzzi (1998), Glăveanu (2015), de Vries (2007), and Csikszentmihalyi (1996). However, advances in digital technology have opened new frontiers, providing new affordances, such as standardised gestures for touchscreen technologies. The human–computer interface has evolved rapidly to the point where human–computer collaboration offers additional dimensions to the scope of interaction (William et al. 2022; Huang et al. 2023). Given current advances, it is also essential that recent educational theorists, such as Glăveanu or Thomas, have emphasised creativity as a sociocultural act (Glăveanu et al. 2015; Thomas 2018). In other words, interacting with others and within cultural contexts is what shapes creativity. Similarly, interacting with artificial agents and entities falls within this scope.

With its origins in personal computing over 40 years ago, the digital revolution has also been a catalyst for the emergence of a leading narrative in education, highlighting the skills that students require to be successful citizens. These are often referred to as twenty-first-century skills or sometimes as employability skills. The core competencies associated with these skills are called the 4Cs, which are creativity, communication, collaboration, and critical thinking. There are also several types of literacy, such as digital literacy and a growing list of character qualities (Kennedy et al. 2020). Thus, creativity is now considered to be more than a personal trait and is regarded as a competency that can be cultivated. Moreover, engaging with and within the digital environment opens new ways to promote creativity.

The following sections explore the intersection of GenAI and creative learning experiences, highlighting how interdisciplinary learning can enhance creativity. We examine how digital twins simulate creative learning environments, enabling personalised creative growth and collaborative creativity with AI as a partner in learning. Furthermore, we discuss strategies for developing a culture of creativity in education while addressing the challenges and opportunities that arise. Finally, we consider future trends in creativity, digital twins, and GenAI in educational settings, envisioning a landscape where these technologies continuously redefine the creative potential of both learners and educators.

9.2 Generative AI and Creative Learning Experiences

GenAI in different educational scenarios has been examined and analysed extensively (Socher et al. 2024). In learning, GenAI is used for text and image generation, information retrieval, prompt coding, as a learning coach agent, and other uses. These tools are also suitable for developing project plans and implementing projects, which are fundamental parts of problem-solving learning and other constructive learning methods. Furthermore, teachers have been using GenAI for content inspiration, content generation, enhancement or formatting of learning materials, translation, learning evaluation, automated exam question generation, and student success monitoring. In most cases, large multi-modal language models such as ChatGPT (Tan et al. 2025) or Gemini (Imran and Almusharraf 2024) have been used. Many of these platforms support the personalisation of topics. With the development of agentic AI, personal learning agents can be created; Duolingo (Kong et al. 2024) is an excellent example of a personalised learning agent. In the literature, how GenAI can support the learning process for creative subjects and innovative processes is often discussed. With the latest versions of AI tools that support chain-of-thought, such as ChatGPT 4o1 (Shabanov 2024), coaching can be also applied to subjects that require formal solid reasoning, including mathematics and logic. However, GenAI is not only used as a tool for learning and teaching in constructive learning; it is also used for advanced instructional learning, which can be part of professional training. A significant advantage of learning with generative artificial networks is the use of GenAI as one or several agents, which can discuss a problem with a learner from different perspectives. The roles of these agents should be defined via prompt engineering, and the agents should be fine-tuned to their roles later. These agents not only provide diverse perspectives on complex problems but also give feedback and input from the learner. These scenarios require a human teacher as a supervisor to control and guide learning according to the defined learning objectives. One of these methods could be creativity with the support of a GenAI agent.

An interesting example is the use of GenAI to develop mobile applications to solve real-world problems. One approach is a text-to-application model, which develops mobile applications based on a project objective and task description that can be used in real-world situations. For example, Indian teenagers developed a communication application for their families while waiting in drinking water queues in the slums of Indian cities (Batul et al. 2023).

9.3 Interdisciplinary Learning and Creativity

Inspiring and enabling students to learn about a subject or topic so that they internalise and grasp its theories and intricacies requires excellent skills in teachers. However, teaching itself has become challenging because of rapidly evolving technology, the much shorter period for studying technological advances, and the need to show that the technology is applicable to real-world situations. Good teaching must also cultivate skills for innovative and creative thinking to find new solutions to varied problems. Students have little time to internalise subjects and want to relate new information to various concepts that they have already learnt. Interdisciplinary learning can help students achieve this by presenting the theories, experiments, and results in relatable ways that they can visualise, think deeply about, and respond to creatively. Interdisciplinary learning encourages students to understand concepts by connecting seemingly disparate ideas from different fields. For example, to understand the importance of improved public health, students might combine biology, societal behaviour, economic conditions, and government policy to come up with a viable solution for disease prevention, such as pandemics. Students often like to see how concepts from different subjects connect and relate to real-world issues. By trying to combine varied concepts from various fields, students are exposed to new ideas and ways of thinking. For example, a student may use machine learning and GenAI to create new music compositions or apply principles of evolutionary biology to design tasks. This requires interdisciplinary connections among concepts that spark creativity and stimulate innovation. In general, the development of AI applications needs at least some knowledge of computer science or AI and the application domain, because many decisions can be made only based on these inputs. Therefore, using AI for problem-based or inquiry-based learning is intrinsically interdisciplinary.

In 1787, the famous mathematician Carl Friedrich Gauss,[1] then a 10-year-old elementary school student, stunned his teacher with how quickly he calculated the sum of the integers from 1 to 100 to be 5050. When he added the first and last numbers in the series, the second and second-last numbers in the series, and so on, Gauss realised he had 50 pairs of numbers that each summed up to 101, making the problem a simple multiplication. Gauss was a genius, and this creative thinking produced an innovative solution to the sum of n natural numbers. Creativity has been defined as “the interaction among aptitude, process, and environment by which an individual or group produces a perceptible product that is both novel and useful” (Plucker et al. 2004). Oppert et al. (2023) showed that creativity is a vital skill for the future of work due to the effects of globalisation, the digital economy, and the changing nature of work and workplaces. Creative processes include divergent thinking, which is the generation of multiple diverse ideas for a given task; convergent thinking, which is choosing the best ideas; and creativity, which is thinking about a problem or task to produce a set of good interrelated ideas or solutions. AI can assist humans in their creative processes.

The power of GenAI in connecting various concepts to suit individual learning interests, abilities to understand concepts, and styles and speed of learning can make learning more effective and meet diverse learning needs. AdCreative.AI,[2] a professional advertising agency, uses AI aided by large language models to generate creative adverts for its users and their needs. GenAI models can help enable fashion design students to create designs using a mix of art, history, and sustainability; for

[1] Carl Friedrich Gauss, https://www.nctm.org/Publications/TCM-blog/Blog/The-Story-of-Gauss/.

[2] https://www.adcreative.ai/.

example, AI can be used to visualise clothing sourced from biodegradable or renewable materials. This use helps blend fashion with environmental considerations and technology and makes a more appealing case for modern fashion. RunwayML[3] has been "designing systems that facilitate 'generative daydreaming,' where the journey of creation is as vital as the destination". Students can use GenAI to understand a concept from multiple perspectives. For example, the prompt "Can you explain logical reasoning using specific sports examples to me?" can help explain logical reasoning using real sports examples. Similarly, the prompt "How can I interconnect fashion design, quantum physics, and dance to present the idea of the reliability of AI systems? Can you give me a concrete example?" can generate a creative explanation of the reliability of AI systems that a student can relate to. Although a teacher may struggle to find relevant examples, GenAI can rapidly interconnect and generate explanations. Students can then try to understand concepts in a new area using examples from topics that they are familiar with and can relate to quickly.

9.3.1 Digital Twins for Simulating Creative Learning Environments

In the evolving landscape of education, the integration of digital technologies is opening new frontiers for innovation. One such advance is the use of digital twins, which are real-time, virtual replicas of a physical system or environment, enabling simulations that mirror the behaviours of their real-world counterparts. Digital twins allow dynamic simulation and testing in educational institutions. By creating digital twins of learning environments, educators can experiment with various teaching methods and learning scenarios that might otherwise be impractical or challenging to implement in real time (Arvanitakis et al. 2024). This simulation capability develops creativity in pedagogy and student engagement (Yan et al. 2022), allowing for a more nuanced and adaptable approach to education. Simulated environments also provide safe places to rehearse and test teaching techniques (Mason et al. 2020). Moreover, designing the attributes of virtual replicas is also intrinsically creative. Digital twins could simulate complex project-based learning environments where students collaborate on creative tasks. For example, in a virtual classroom setting, AI systems in the digital twin can adjust the complexity of tasks dynamically based on group dynamics, individual performance, and real-time feedback. This use of AI could create more personalised and creative learning experiences, where educators can test various instructional approaches, assess their impact, and refine their teaching strategies to support creativity and problem-solving in students better. Through these simulations, educators can explore innovative ways to facilitate learning, making creativity not only an outcome but also a driving force in education.

[3] https://runwayml.com/research/pioneering-new-interfaces-age-generative-media.

9.3.2 How Digital Twins Simulate Creativity in Educational Institutions

The rise of digital twins in education is reshaping how we approach teaching and learning by offering a virtual space where educators can explore, test, and refine innovative methods (Acker et al. 2023). Digital twins can be used in educational institutions to simulate various learning scenarios, presenting opportunities to enhance creativity in teaching and learning.

Simulating Learning Scenarios for Creative Approaches

Digital twins of educational institutions can simulate diverse learning environments and scenarios (Chang et al. 2023), offering educators a platform to test creative approaches that may not be feasible or cost-effective in a traditional setting (La Scala et al. 2023). These virtual environments allow experimentation with different teaching methods, including problem-based learning, inquiry-driven instruction, and collaborative projects. By adjusting factors like classroom structure, time management, and instructional style, educators can evaluate the effects of these factors on student engagement, creativity, and problem-solving.

A major advantage of digital twins is their ability to simulate conditions without the limitations of real-world logistics (Li et al. 2024). Teachers can experiment with different class sizes, teaching methods, and assessment strategies to determine the most effective approach for enhancing creativity. For instance, educators could simulate a flipped classroom model, where students first engage with content online and later collaborate on problem-solving exercises in a digital classroom. With a simulation, instructors can observe how students perform creatively in different scenarios, refining their teaching strategies based on data-driven insights.

Example: Simulating Project-Based Learning Environments

A compelling use of digital twins is in simulating project-based learning environments (Angenheim et al. 2023). Project-based learning encourages students to collaborate on hands-on projects, emphasising creativity, critical thinking, and real-world problem-solving. In a digital twin environment, educators can create virtual project-based learning scenarios where students work together on creative projects, even across different locations.

For example, a digital twin might simulate a design project in which students create a prototype for a sustainable building. The AI in the digital twin could monitor group dynamics, such as communication patterns and task distribution, and adjust the project's complexity in real time. If one group excels at planning but struggles with execution, the AI might introduce a time constraint or a resource limitation to foster creative problem-solving. Conversely, if a group is overwhelmed, the AI might scale back the difficulty or provide additional resources to motivate continued engagement and creative output.

In this scenario, educators gain a clearer understanding of how different factors, such as collaboration, time management, and resource allocation, influence student creativity. Furthermore, the simulation allows educators to provide tailored feedback to each group, producing a more personalised learning experience. AI-driven adjustments can also introduce unexpected challenges, such as budget cuts or resource shortages, that simulate real-world conditions (e.g., industry 4.0), prompting students to think creatively and adapt their strategies.

9.3.3 AI-Driven Complexity Adjustment in Learning

The two most innovative aspects of digital twins are their real-time adaptive feedback and personalisation capabilities and their use in predictive modelling for scenario testing. Digital twins allow AI systems to analyse real-time data from student interactions and learning environments, adjusting learning scenarios dynamically. By simulating real-world classroom conditions or even individual learning paces, digital twins enable tailored experiences in which AI modifies the difficulty of tasks, provides adaptive hints, or changes instructional strategies based on each student's progress. This responsiveness helps maintain engagement, promotes personalised learning, and enhances the efficiency of concept mastery. Digital twins enable educators and AI systems to forecast learning outcomes and test a variety of instructional approaches without disrupting actual classroom settings. In addition, digital twins offer insights into what might best support students under diverse conditions by simulating different teaching methods, project-based learning environments, or even new curricula. For example, digital twins can simulate group dynamics in collaborative projects, allowing the AI to adjust complexity based on students' group roles and interactions. This leads to better-planned instruction that aligns with predicted learning needs and challenges. This adaptive complexity ensures that students are constantly challenged in their zone of proximal development, which is the space where they can achieve the most growth with appropriate support. Educators can fine-tune their approach to stimulating creativity and critical thinking in students by simulating a balance between difficulty and support. Additionally, the data gathered from these simulations can help educators better understand the optimal conditions for creative output, allowing them to replicate successful strategies effectively without disrupting the real-world classroom. This gives teachers and administrators a powerful tool to innovate their pedagogical methods while minimising risk.

9.3.4 Project-Based Learning Simulations: A Case for Creativity

One of the most promising applications of digital twins in education is the simulation of project-based learning environments. In these settings, students collaborate on creative projects, often working in groups to solve complex, real-world problems.

Digital twins provide a controlled environment in which educators can simulate these collaborative efforts, allowing them to monitor and adjust the parameters of the project.

For instance, an AI-driven digital twin can track how students interact with one another in a group project and adjust the difficulty level of tasks based on the group dynamics. If a group is excelling, the AI might introduce more complex challenges to maintain engagement. Conversely, if a group is struggling, the AI can reduce complexity or offer targeted hints and support. This dynamic adjustment helps ensure that all students are appropriately challenged and engaged, nurturing both individual creativity and collaborative innovation.

Beyond adjusting complexity, digital twins can simulate different types of group interactions, such as pairing students with diverse skill sets or creating role-playing scenarios where students take on specific roles in a team. These simulations allow educators to test various group dynamics and assess which combinations foster the most creativity and productivity. Consequently, students can experience diverse perspectives and learn how to collaborate more effectively in real-world situations.

9.3.5 AI's Role in Enhancing Creative Learning

AI will probably be pivotal in the success of digital twins for educational purposes (Creely 2024). By analysing real-time data from simulations, AI can provide insights into how students are learning and adapting. This allows educators to not only monitor individual and group progress but also tailor lessons to the unique needs and strengths of their students. In a simulated project-based learning environment, AI can predict outcomes, suggest alternative pathways, or even propose new creative projects based on students' interests and performance.

Digital twins offer a powerful means to simulate various learning scenarios (Pathak et al. 2004), providing educators with a sandbox for creativity in both teaching methods and student engagement. From project-based learning environments (Liu et al. 2022) where students collaborate on complex tasks to AI-driven adjustments that optimise group dynamics and challenge levels, digital twins are transforming how we think about education. By enabling experimentation and creative problem-solving in a risk-free environment, this technology provides educators and students with new ways to cultivate a more dynamic and personalised learning experience.

9.4 Personalised Creative Growth with Digital Twins

In the current educational landscape, the concept of digital twins is extending beyond its traditional use in industry to personalised learning and student development. A student's digital twin is a virtual counterpart that mirrors their academic progress, learning behaviours, and even creative growth. By using data from a student's activities, preferences, and performance across different subjects (also called learner records), this digital entity can offer real-time insights and suggestions tailored to the student's unique learning needs. In terms of sparking creativity, the digital twin is a dynamic guide that tracks creative development and provides customised learning paths to nurture and expand a student's imagination and problem-solving abilities (Hicham et al. 2021).

A key benefit of digital twins in creative development is their ability to recommend personalised projects and interdisciplinary activities. By analysing data points such as past performance, engagement levels, and personal interests, the digital twin can suggest challenges that match a student's strengths and address areas that need improvement. For example, a student excelling in visual arts and showing an interest in environmental issues may receive a recommendation for a creative project that combines art and sustainability. This project might involve designing eco-friendly materials or creating awareness campaigns using multimedia art forms. These recommendations encourage innovative thinking and inspire students to approach learning from an ethical cross-disciplinary perspective (Gallastegui 2021).

In a further example, a student's digital twin might recognise that they have a strong foundation in mathematics and a growing interest in design. The system might suggest an engineering-related project where the student can apply both creative design thinking and analytical skills to solve a real-world problem, such as designing an innovative urban space. The interdisciplinary nature of these challenges enables students to stretch their creative capacities and approach problems holistically, integrating knowledge from multiple domains. Furthermore, by tracking progress continuously, the digital twin can adjust the complexity and scope of suggested projects to ensure they remain engaging, motivating the students to explore their creative potential meaningfully.

9.5 Collaborative Creativity: AI as a Partner in Learning

An important role of GenAI is as an agent with dialogic capabilities (Mason 2023), meaning that the AI model can act as a facilitator, coach or mentor, virtual tutor, advisor, and analyst. These roles have different features. Table 9.1 shows an overview of the roles of AI agents in human learning.

Table 9.1 is a guide and not an exhaustive list; the AI agent should be chosen according to the learning situation and organisation. The examples given here do not cover just one role but implement a combination of multiple roles with different objectives. Some implementations of this role model are already in use, based on ChatGPT Education.[4] But it is too early to have a final evaluation.

[4] https://openai.com/index/introducing-chatgpt-edu/.

Table 9.1 Overview of AI agents in human learning

Role of AI agent in human learning	Learning approaches	Tasks/Description	Example
Facilitator	• Active learning • Learning • Assessing	• Generation of interactive quizzes, games, simulations • Formative and summative assessments • Dynamic question generation • Automated feedback/reflection	Socratic by Google https://socratic.org/ Duolingo Max https://blog.duolingo.com/duolingo-max/
Coach/mentor	• Personalised learning • Thinking assistant • Learning assistant and collaborative partner • Mentor for self-directed learning • Critical thinking coach	• Tailoring learning exercises • Continuous assessing • Real-time feedback • Scaffolding learning • Metacognitive support • Reflection on learning strategies • Peer collaboration • Simulation of debates, etc • Learning based on goals • Collaborative problem solving • Promoting autonomous learning (skills, goal setting tracking) • Posing challenges • Apply higher-order thinking skills • Simulating and tackling real-world problems	Replika AI https://replika.com/
Tutor	• Virtual content tutor • Motivating agent	• Generating and delivering content • One-to-one tutoring • Game-based learning • Encouraging persistence/resilience	Querium's StepWise Virtual Tutor https://www.querium.com/ Cramly https://www.cramly.ai/

(continued)

Table 9.1 (continued)

Role of AI agent in human learning	Learning approaches	Tasks/Description	Example
Advisor	All kinds of learning	• Ethical advisor • Inclusion advisor • Cultural advisor • Contextual advisor	Jigsaw's Perspective API as an Ethical and Inclusion Advisor https://perspectiveapi.com/ Microsoft Translator as a Cultural and Context Advisor https://www.linkedin.com/posts/cfedermann_introducing-lower-sorbian-microsoft-translator-activity-7054816541837520896-70PH/
Analyst	All kinds of learning	• Monitoring learning success • Predicting learning success	DreamBox Learning as an Analyst and Predictor of Learning Progress www.dreambox.com/ Schoolanalytix https://www.schoolanalytix.com/
Evaluator/assessor	All kinds of learning	• Evaluate individual learning progress • Evaluate learning progress in groups/teams • Check for plagiarism and forbidden use of GenAI tools	Grammarly AI https://www.grammarly.com/

How well these tools support and implement the intended learning outcomes and learning agents' roles and how AI agents can be best used in a K-12 or higher-education environment are still ongoing areas of research. A typical situation for using GenAI for learning is in organising AI-assisted ideation or creative design discussion in collaboration with human learners. This use of AI should enable learners to co-create with AI in projects such as storytelling, design, and visualisations (graphs) for mathematical and other formal descriptive models. GenAI supports some roles better than others because of the strengths or weaknesses of the underlying large language models. The facilitator, coach, and tutor roles also depend on the learning object. The roles are better developed for mathematics than for social or artistic topics. It is easier for a large language model to learn the reasoning in mathematics than in cultural subjects because of the richness and different meanings of symbolic content in artistic subjects.

9.6 Fostering a Culture of Creativity in Education

For over a decade, the World Economic Forum has consistently ranked creativity as a top-ten desirable skill for employers,[5,6,7] Creativity has been considered a foundational competency of twenty-first-century skills for nearly three decades (Bellanca et al. 2011; Geisinger 2016; McGaw 2013; Voogt et al. 2010). Among recent narratives about AI in education, Mollick (2024) emphasised that "our first principle in working with AI is to always invite it to the table". Such an invitation is at first exploratory, and educators and learners alike must explore the potential of GenAI; without this mindset, there is little chance of nurturing creativity.

However, narratives around the role of creativity in education are appreciated better in a broader context. Theoretically, creativity has been featured in the pedagogical literature for a long time. More recently, the digital revolution after the personal computing revolution can be summarised by watershed moments based on three core clusters of technology, known as search, social, and intelligent. Understanding the role of twenty-firstcentury skills and how they are evolving is a complementary narrative that is relevant here but one that is deeply connected to the changing digital environment. What is profound about the current moment is that digital content is no longer primarily informational but increasingly explanatory and dialogic. To embrace this potential requires not only an exploratory mindset but also perhaps a creative act because human agency is being challenged.

Inviting AI to the table provides both challenges and opportunities, but wise choices will be made through a growing shared experience of the capabilities of this technology. Among the many examples of creative application of AI is its potential use in dialogue role-play.

Some examples of GenAI, creativity, digital twins and interdisciplinary learning that have been used in universities for education are given in Table 9.2.

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[5] https://www.weforum.org/agenda/2016/01/the-10-skills-you-need-to-thrive-in-the-fourth-industrial-revolution/.

[6] https://www3.weforum.org/docs/WEF_Future_of_Jobs_2023.pdf.

[7] World Economic Forum. (2016). New Vision for Education: Fostering Social and Emotional Learning through Technology (p. 36). http://www3.weforum.org/docs/WEF_New_Vision_for_Education.pdf.

Table 9.2 Some universities using GenAI, creativity, digital twins and interdisciplinary learning

Area	Possible applications	University	Benefits/Productivity
Creative learning experiences	• Digital Storytelling (Story generated using AI) • Generative AI Art	University of California, Berkeley https://guides.lib.berkeley.edu/dh/digitalstorytelling MIT https://www.media.mit.edu/projects/imagine-yourself/overview/ https://www.media.mit.edu/research/?filter=everything&tag=storytelling MIT https://news.mit.edu/2024/creative-future-generative-ai-0102	• Increased Creativity • Time saving • Improved writing skills • New art created; new ways of presenting abstract ideas
Interdisciplinary learning	• System-level thinking with an interdisciplinary lens • Interdisciplinary learning in the first-year undergraduate curriculum	Harvard University https://hilt.harvard.edu/ideas-and-tools/into-practice-items/teaching-system-level-thinking-with-an-interdisciplinary-lens/ https://www.youtube.com/watch?v=0Ks9b_V5_PA University of Plymouth, UK https://www.tandfonline.com/doi/full/https://doi.org/10.1080/13562517.2022.2056834	• Students look at complex problems from the perspective of both artists and engineers • Provide students with skills and modes of thinking informed by multiple worldviews
Digital twins	• Digital twins in Education • Daydreaming factories	University of Amsterdam https://pure.uva.nl/ws/files/106231123/22digitaltwins.pdf University of Twente, Netherlands https://ris.utwente.nl/ws/portalfiles/portal/361729970/1-s2.0-S2212827123009265-main.pdf	• Allows the students to understand every detail by playing and manipulating the virtual asset • Allows stakeholders to interact with a 'could-be' situation • Individual stakeholders can carve out scenarios that are the base for potential futures

(continued)

Table 9.2 (continued)

Area	Possible applications	University	Benefits/Productivity
AI-driven complexity adjustment	• Coupling Cognitive Systems	University of Illinois, Urbana Champaign https://jessiechinlab.ischool.illinois.edu/coupling-cognitive-systems/	• Human-agent interaction to support cognitive activities
Project-based learning simulations	• Virtual Field Trips	Stanford University https://acceleratelearning.stanford.edu/initiative/digital-learning/virtual-field-trips/	• Create virtual field trips that embed locally or personally meaningful contexts • Support learning by increasing learner connectedness
Personalised creative growth with digital twins	• Revolutionising Online Learning with Digital Twins	MIT https://computing.mit.edu/wp-content/uploads/2023/06/Revolutionizing_Online_Learning_with_Digital_Twins.pdf	• More personalised learning experience • Users feel more motivated to engage with the content and share their progress with others

(continued)

Table 9.2 (continued)

Area	Possible applications	University	Benefits/Productivity
Collaborative learning: AI as a Partner in learning	• Improving Clinical Reasoning Using Generative AI	Stanford University https://hai.stanford.edu/sites/default/files/2024-02/AI%2BEducation%202024%20-%20Clinical%20Mind%20AI.pdf	• Acts as a simulated patient • Helps adapting the scenarios to fit specific local contexts and learning outcomes • Student gets feedback on the exchange and the validity of the diagnosis
Human-AI collaboration	• AI program interacts with human improvising musicians to automatically co-author music in real time • Intelligence Augmentation	University of California, Berkeley https://live-digital-humanities-berkeley.pantheon.berkeley.edu/projects/emet-human-machine-interactive-composition-using-machine-learning MIT https://www.media.mit.edu/projects/symbiotic-virtuosity/overview/ Harvard University https://pz.harvard.edu/sites/default/files/Intelligence%20Augmentation-%20Upskilling%20Humans%20to%20Complement%20AI.pdf	• Creativity in music • AI and humans engage in a complementary partnership in which a human-and-AI team's overall performance is greater than their individual capacity

9.7 Challenges and Opportunities in Fostering Creativity

9.7.1 Balancing AI-Generated Creativity with Human Originality

As AI continues to permeate educational settings, a major challenge lies in striking a balance between AI-generated creativity and the preservation of human creativity (Hitsuwari et al. 2023). AI systems have demonstrated a remarkable generation of new ideas, designs, and even artistic expression (Al Sawy 2024; Tang and Chen 2024). These systems can serve as powerful tools for inspiring students, offering them creative prompts, and helping them explore unconventional solutions. However, the risk of over-reliance on AI-generated content (Van Poucke et al. 2024) presents a challenge to the development of human originality and ingenuity.

AI should ideally complement rather than overshadow human creativity. The challenge is how to do this. Educators must use AI to enhance students' thought processes (Oyewo et al. 2022), allowing them to develop their own creative voices. This balance requires the design of AI-driven tools that offer guidance or inspiration for students without becoming crutches (Hagedorn et al. 2023). Students should be encouraged to engage critically with AI suggestions, developing a collaborative relationship between humans and machines, known as human–machine teaming in education. In this context, the human capacity for empathy, abstract thinking, and emotional expression remains central to the creative process (Hagedorn et al., 2023).

History is also instructive here. Over 35 years ago, digital sound sampling quickly became normalised in the composition and production of popular music. Although there was debate at the time about the loss of warmth in the transition from analogue to digital recording—as well as fierce debate about plagiarism and authorship of music—sampling and remixing sound have provided enormous creative scope for musicians. Notably, the public has broadly accepted this innovation (Allen 1992; Baroni 1993; DeVoss and Porter 2006; Osborne 2012).

9.7.2 Ethical Considerations

With the growing integration of AI in education, ethical concerns surrounding AI-driven creativity must also be addressed, as discussed in the IEEE 7014.1 project white paper (McStay et al. 2024). A fundamental problem is the risk of students becoming overly dependent on AI to generate ideas, thereby diminishing their capacity to think independently or develop their creativity. Over-reliance on AI could lead to a form of creativity that is formulaic and lacks the depth of human experience and emotion. The risk of AI bias influencing creative outcomes is another challenge. AI systems are trained on large datasets, which can contain hidden biases that could unconsciously steer creative outputs in specific directions, limiting the diversity of thought and expression. Similarly, ethical considerations arise when student creativity can be harnessed to cheat or circumvent established protocols of academic integrity. Educators must identify and develop explicit boundaries of what is acceptable.

It is essential for educators and developers to build trustworthiness into AI systems, ensuring that students understand how AI generates its creative suggestions and how biases can emerge from its algorithms. Moreover, AI must be designed to respect human values and creativity, allowing room for uncertainty and innovation to arise from human imagination. Clear ethical guidelines should also be established to ensure that AI complements rather than replaces human creativity, reinforcing the importance of originality and critical thinking in the creative process.

9.7.3 Opportunities for Human-AI Collaboration in Creativity

AI's ability to process vast amounts of information quickly and generate new ideas provides educators with unique opportunities to kindle creativity in ways previously unimaginable. AI can introduce students to new artistic styles, suggest alternative approaches to problem-solving, and even act as a collaborative partner in ideation sessions, thereby expanding the boundaries of human creativity and pushing students to think beyond conventional limits. Furthermore, AI can assist in personalised learning experiences, adapting challenges and creative tasks to suit individual students' abilities and interests.

However, to realise the full potential of AI without diminishing human creativity, a balance must be maintained. Educators can cultivate an environment where AI serves as a co-creator, offering assistance while students retain control over the creative process. This dynamic partnership between humans and machines opens a new frontier in education, where students can develop original ideas with AI support. In this way, AI can inspire creativity without suppressing the uniqueness of human thought.

9.8 Future Trends: Creativity, Digital Twins, and GenAI in Education

AI and creativity are entering a dynamic new phase where AI models not only generate content but also learn and evolve in real time from human creative input. Emerging trends indicate the development of AI systems capable of co-creation, where AI acts as an adaptive partner in the creative process, enhancing human ingenuity and enriching the creative landscape. An exciting example is the use of AI tools that generate entire creative ecosystems in virtual classrooms, allowing students to design complex interdisciplinary projects with minimal instructor guidance. These AI-driven environments provide students with resources, suggestions, and even constructive feedback, fostering independent learning and creativity across fields such as science, art, and engineering.

As these technologies advance, we can expect AI to support a more personalised and engaging educational experience, tailoring its interactions based on individual learning styles and preferences. This evolution transforms AI from a content generator (Tripathi 2024) into a creative collaborator, offering immense potential for education and beyond. The integration of AI in creative endeavours will also probably spur interdisciplinary collaboration by allowing students from diverse backgrounds to use AI to explore complex problems and generate innovative solutions that transcend traditional academic boundaries.

Moreover, this shift towards AI as a co-creator raises essential questions about the nature of creativity itself. As AI systems become more sophisticated, they will not only assist in generating ideas but also challenge our understanding of originality and authorship. Ethical considerations will also emerge, prompting discussions about the implications of relying on AI in creative processes. By encouraging a deeper dialogue around these issues, we can ensure that the collaboration between humans and AI leads to responsible innovation, empowering individuals to harness the full potential of AI while maintaining the intrinsic value of human creativity. This symbiotic relationship between humans and AI will redefine the landscape of creativity, opening up previously unimaginable avenues for exploration and expression.

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Part III
Generative AI and Creativity in Art

Chapter 10
Generative AI and the Evolution of Artistic Creativity

Nan Ma and Dongxing Yu

Abstract This chapter explores the transformative impact of generative artificial intelligence (AI) on the evolution of artistic creativity, examining how advanced models like DALL-E, Midjourney, and GPT are reshaping the artistic landscape by enabling unprecedented collaboration between human artists and machines. It provides an in-depth overview of key generative AI models and their capabilities across various domains, such as visual art, music generation, and creative writing. It delves into the evolution of the creative process by comparing traditional artistic methods with modern practices enhanced by AI. The chapter addresses philosophical and ethical considerations and investigates the emergence of new artistic movements enabled by AI-driven aesthetics, showcasing innovative styles and reinterpretations of historical art forms. Introducing a comprehensive framework for assessing AI systems' aesthetic and creative capabilities, it proposes criteria for evaluating their contributions to art while critically examining ethical challenges related to authorship, bias, the impact on human artists, misinformation, and accessibility. Concluding with a reflection on the future of human-AI collaborative creativity, the chapter envisions a landscape where human intuition and machine precision intertwine, potentially leading to a renaissance in creative expression and prompting a re-evaluation of creativity.

10.1 Introduction to Generative AI for Art

Integrating generative artificial intelligence (AI) into the artistic realm is not merely an incremental advancement but a revolutionary upheaval that challenges the very foundations of creativity and authorship. Far from being passive tools, generative AI systems have become active collaborators in the creative process, fundamentally

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altering how art is conceived, produced, and evaluated. Recent studies highlight that artists employing generative AI tools like sophisticated text-to-image generators have experienced a remarkable 25% increase in productivity, alongside enhanced peer evaluations of their work. This phenomenon, often described as "generative synaesthesia," underscores the transformative potential of AI as it enables artists to venture into unexplored creative territories and synthesise ideas that might be unattainable through human effort alone.

However, this surge in productivity and innovation is accompanied by a paradoxical decline in the average novelty of artworks. While peak originality may soar to new heights, the ubiquity of AI-generated content risks saturating the creative landscape with homogenised outputs. This trend raises critical concerns about the potential erosion of diversity and uniqueness in art, as the ease of generating content might lead to an abundance of generic works that overshadow truly novel creations.

Beyond the implications for creativity, the rise of generative AI poses profound ethical dilemmas concerning authorship and artistic integrity. Artists like Greg Rutkowski have confronted unsettling challenges as AI systems, trained on their distinctive styles without consent, replicate their artistic signatures. This unauthorised emulation not only dilutes the uniqueness of their work but also blurs the boundaries between authentic creations and algorithmic reproductions, making it increasingly difficult for audiences to discern originality. Kearney and O'Connor (2023) aptly remark, "While generative AI offers unprecedented opportunities for artistic expression, it simultaneously presents ethical dilemmas regarding authorship and the authenticity of creative works." The tension between technological advancement and the preservation of individual artistry necessitates urgent discourse on establishing ethical standards and legal frameworks that protect artists' rights in the digital age.

In educational settings, the impact of generative AI is equally profound, redefining pedagogical approaches to teaching creativity. Research into Virtual Reality applications reveals that these technologies can enrich traditional art forms, such as calligraphy, by providing immersive experiences that simulate the tactile feedback crucial for mastering such skills (Dongxing and Ma 2024). The shift towards an "AI-empowered" learning environment empowers students to engage more deeply with their creative processes, fostering a sense of ownership and innovation. Yet, as Zhou and Liu (2024) cautioned, "the adoption of generative AI has led to a significant increase in creative productivity, yet it raises questions about the originality of the artworks produced." There is a growing concern that reliance on AI tools may precipitate an identity crisis among learners, who might doubt their ability to create independently without technological assistance.

At the heart of these developments lies a fundamental question about the nature of artistic creativity itself. Traditionally, creativity has been viewed as an intrinsically human endeavour rooted in personal experience, cultural context, and emotional depth. It involves generating novel ideas and the intentional expression of meaning that resonates with others profoundly. Generative AI systems like DALL-E, Midjourney, and Stable Diffusion operate differently. They produce outputs by analysing vast datasets and identifying statistical patterns, recombining existing elements in ways that may appear creative but lack the conscious intent and subjective experience characteristic of human artists.

This distinction provokes a contentious debate: Can the works produced by AI be considered genuine art, or are they sophisticated imitations devoid of true creativity? On one hand, AI-generated pieces can be aesthetically compelling and thought-provoking, embodying qualities traditionally associated with art. On the other hand, the absence of consciousness, emotion, and intentionality in AI-generated works leads some to argue that they are mere simulations—fakes that mimic the surface features of art without capturing its essence.

Despite these philosophical quandaries, the collaborative potential between humans and AI offers exciting possibilities for the future of art. By leveraging AI's capacity to generate ideas and explore patterns beyond human cognition, artists can push the boundaries of their creativity, uncovering new forms of expression. This synergy between human intuition and machine computation creates a hybrid form of creativity that challenges conventional notions of authorship and originality.

As we stand at the cusp of this new era, it is imperative to critically examine the multifaceted implications of generative AI on the artistic profession. This includes addressing ethical concerns related to consent and intellectual property, redefining the concept of authorship in a context where creation is a joint venture between humans and machines, and reassessing the criteria by which we evaluate artistic merit. The discourse must also consider the psychological impact on artists and learners who navigate this complex landscape, balancing the benefits of AI augmentation with the need to preserve individual creative identities.

In conclusion, generative AI is not merely a tool that enhances artistic productivity; it is a disruptive force that compels us to reevaluate our understanding of creativity, originality, and authenticity. Its influence permeates the educational sphere, the professional art world, and the very definition of what it means to create. As Francke and Bennett (2023) observe, "Generative AI tools have transformed art education by enabling students to engage in iterative processes that enhance their creative output." Yet, this transformation comes with the responsibility to ensure that AI's integration into art respects individual artists' rights, maintains the integrity of creative expression, and thoughtfully navigates the ethical complexities it introduces. The future of art in the age of AI will be shaped by how we address these challenges, embracing the opportunities for innovation while safeguarding the human elements that imbue art with its most profound significance.

10.2 Overview of Key Generative AI Models and Their Capabilities

The advent of advanced generative AI models marks a pivotal moment in the evolution of AI-assisted creativity, fundamentally transforming the landscape of artistic expression. Pioneering models such as DALL-E, Midjourney, and Stable Diffusion have revolutionised visual art by leveraging deep learning algorithms to interpret and transform textual descriptions into visually compelling images. DALL-E, developed by OpenAI, can generate intricate and diverse images from complex prompts, often producing results that defy conventional artistic boundaries. Midjourney and Stable Diffusion similarly empower users to explore uncharted aesthetic territories, enabling the creation of artwork that both surprises and captivates.

Open-source models like BLOOM (BigScience Large Open-science Open-access Multilingual Language Model) and LLaMA, developed by Meta AI, extend these capabilities by providing accessible platforms for experimentation within the AI community. ControlNet enhances these models by offering fine-grained control over image generation, allowing artists to manipulate outputs with unprecedented precision. In the Chinese AI landscape, models such as Tongyi, Coze, and ChatGLM represent significant strides in natural language processing and generative capabilities, catering to a linguistically diverse user base and contributing to the global expansion of AI technologies.

Beyond visual art, generative AI profoundly impacts music and literature. AI systems like OpenAI's Jukedeck and Google's MusicLM have demonstrated the ability to generate high-fidelity music based on textual descriptions, opening new avenues for musicians to experiment with sound and composition. Open-source tools such as Riffusion utilise stable diffusion techniques applied to audio spectrograms, enabling real-time music generation and democratising access to advanced music production capabilities. These innovations challenge traditional music creation paradigms, inviting artists to collaborate with AI in unprecedented ways.

In language and literature, models like GPT-3, Google's Bard, and open-source alternatives like GPT-NeoX have exhibited remarkable proficiency in generating human-like text. These large language models can compose poetry, craft narratives, and script dialogues often indistinguishable from human-authored content. Their ability to understand context, emulate styles, and produce coherent and nuanced prose has significant implications for creative writing, journalism, and content creation industries.

To provide a clearer understanding of what these generative AI models can accomplish, Table 10.1 compares their capabilities across different forms of artistic expression.

These systems are not only proficient at recreating styles and emulating techniques but also at creating entirely new forms of artistic expression, pushing artists to rethink the roles they play in the creative process. Their ability to blend styles, merge genres, and amplify creative possibilities provides an unprecedented toolkit for both established and emerging artists.

The proliferation of these generative AI models raises provocative questions about the essence of creativity and authorship. As machines become increasingly adept at generating content that rivals human creations, the distinction between human and machine-made art becomes blurred. This challenges traditional concepts of originality and authenticity, prompting a re-examination of the value attributed to human creativity in an era dominated by algorithmic production.

Table 10.1 A comparison of the capabilities of various models across different forms of artistic expression

Model	Visual art	Music	Literature	3D Art/Sculpture	Animation	Other capabilities
DALL-E	Digital illustrations, paintings, abstract compositions	–	–	–	–	Photo-realistic images
Midjourney	Artistic illustrations, surrealism, abstract art	–	–	–	–	Unique visual styles
Stable Diffusion	Photo-realistic images, art-style replication	–	–	–	–	Style blending
BLOOM	–	–	Poetry, short stories	–	–	Text generation
LLaMA 3.2	Visual art generation	–	Creative writing	–	–	Enhanced NLP capabilities
ControlNet	Controlled image generation	–	–	–	–	Conditional image modification
Tongyi	Digital art, Chinese traditional styles	–	Poetry, Creative Writing in Chinese	–	–	Cultural art generation
Coze	Realistic and abstract images	–	Storytelling in multiple languages	–	–	Multilingual art
ChatGLM	–	–	Text generation, dialogue-based storytelling	–	–	Conversational creativity
Jukedeck	–	Instrumental music generation	–	–	–	Adaptive music
MusicLM	–	Music tracks, genre-blending	–	–	–	Music generation from text prompts
Riffusion	–	Electronic and adaptive music	–	–	–	AI-generated soundscapes

(continued)

Table 10.1 (continued)

Model	Visual art	Music	Literature	3D Art/Sculpture	Animation	Other capabilities
GPT-3	–	–	Poetry, stories, dialogue	–	–	Text generation
Bard	–	–	Creative writing, scripts	–	–	Conversational and narrative generation
GPT-NeoX	–	–	Poetry, interactive narratives	–	–	Open-source text capabilities

Moreover, these developments have significant implications for the democratisation of art. Generative AI empowers a broader spectrum of individuals to participate in artistic endeavours by lowering technical barriers and providing access to sophisticated creative tools. This inclusivity fosters a more diverse artistic landscape but also necessitates a critical discourse on the impact of AI on artistic quality and the potential oversaturation of creative markets.

In conclusion, the current generation of generative AI models represents a significant milestone in the intersection of artificial intelligence and the arts. Their capabilities enhance artistic production and disrupt conventional methodologies, urging a redefinition of creativity in the digital age. As these technologies continue to evolve, they will undoubtedly play an integral role in shaping the future of art, challenging artists to adapt and innovate while navigating the ethical and philosophical complexities introduced by AI-assisted creation.

10.3 The Evolution of the Creative Process

10.3.1 Foundational Understanding

The evolution of the creative process has been profoundly influenced by the advent of generative artificial intelligence, marking a seismic shift in how art is conceptualised and produced. The historical development of generative AI can be traced back to the rudimentary stages of computational creativity, beginning with rule-based systems and evolutionary algorithms. These early approaches, while simplistic, laid the critical groundwork for the emergence of more sophisticated models. The introduction of Generative Adversarial Networks (GANs) revolutionised the field by enabling AI systems to generate data that closely resembles real-world inputs through a competitive learning process between generator and discriminator networks. Variational Autoencoders (VAEs) further advanced the domain by allowing efficient data encoding and generation within a continuous latent space. More recently, diffusion models have pushed the boundaries of AI-generated art by progressively refining random noise into coherent images through iterative denoising processes. Each iteration of these models has brought significant improvements in the fidelity and complexity of the outputs and the AI's ability to mimic and innovate upon artistic styles.

Parallel to the technological advancements in AI, the history of artistic creativity reflects a continuous evolution shaped by cultural, societal, and technological factors. From the enigmatic cave paintings of prehistoric times, which served as rudimentary expressions of human experience, to the meticulous realism of the Renaissance that celebrated humanism and anatomical precision, art has continually transformed. The Impressionist movement broke conventional boundaries by capturing fleeting moments and the effects of light, challenging traditional artistic norms. The advent of photography introduced a new medium that both competed with and inspired artists to explore abstraction and conceptual art. The digital art era, propelled by computer technology, has expanded the possibilities of creation and dissemination, allowing artists to experiment with virtual reality, interactive installations, and multimedia compositions. Each movement has responded to the tools available and the prevailing zeitgeist, illustrating how technological innovation drives artistic exploration.

Understanding the key concepts in generative AI is crucial for grasping its impact on the creative process. Central to these models is the notion of latent space—a multi-dimensional representation where complex features of data are encoded, enabling AI to interpolate and generate new variations of input data. Training data is the foundational knowledge from which AI models learn patterns, styles, and structures. The concept of style transfer allows for the blending of artistic styles, where the content of one image can be reimagined in the style of another, leading to novel and hybrid artistic expressions. Different model architectures, such as convolutional neural networks for image processing or recurrent neural networks for sequential data, underpin the functionality and capabilities of generative AI systems.

On the other hand, artistic creativity is characterised by theories that delve into the nature of human expression, originality, and the societal role of art. Exploring artistic movements reveals how artists have continually pushed boundaries to evoke emotional responses, provoke thought, and reflect cultural narratives. Forms of creative expression have diversified, encompassing visual arts, music, literature, and performance, each with unique mediums and techniques. Artistic creativity is distinguished by its originality, depth of emotion, and resonance within cultural contexts—a testament to the human capacity for innovation and meaning-making.

10.3.2 Comparative Analysis

The juxtaposition of early generative AI systems with early human art forms reveals intriguing parallels and contrasts. Early generative AI, reliant on simple rules and stochastic processes, produced rudimentary and abstract outputs akin to the primal expressions found in cave paintings or the raw spontaneity of abstract art. While lacking in complexity, these initial AI creations captured a nascent form of computational creativity that mirrored humanity's first artistic endeavours, representing foundational steps in their respective evolutionary paths.

In stark contrast, modern generative AI systems have reached levels of sophistication that challenge and sometimes surpass contemporary artistic practices. Leveraging vast datasets and advanced neural network architectures, these AI models can produce highly detailed and conceptually rich artworks that rival human creations. The influence of AI on authorship, originality, and the artist's role has become a contentious topic, as AI-generated works blur the lines between human intention and machine output. The traditional notion of a single creative author is being redefined, with AI acting as a collaborator, tool, or autonomous creator. This paradigm shift compels a re-evaluation of the criteria by which art is judged and the value placed on human versus machine-generated works.

10.3.3 The Changing Creative Process

Integrating generative AI into the artistic workflow signifies a transformative change in how art is conceived and realised. Artists are increasingly harnessing AI as a multifaceted tool that profoundly augments their creative processes. One significant application is in generating initial ideas and concepts. By inputting prompts or parameters, artists can utilise AI to produce many starting points, inspiring new directions that might not have been envisioned otherwise. This collaborative ideation accelerates the creative process and expands the horizon of possibilities.

Moreover, AI enables artists to explore variations and possibilities quickly and efficiently. Through iterative generation, artists can soon assess multiple concept versions, refining and selecting elements that resonate with their vision. This capability enhances creative exploration, allowing for a more dynamic and responsive artistic practice.

Generative AI has also paved the way for the emergence of new artistic mediums and forms. AI-generated music, for instance, offers composers novel sonic landscapes to explore, while dynamic visuals that evolve introduce a temporal dimension to visual art. These innovations challenge traditional boundaries and encourage artists to experiment with interdisciplinary approaches, blending technology and art in innovative ways.

Automating repetitive tasks is another area where AI significantly impacts the creative process. Tasks like colorising images, replicating styles, or generating textures can be time-consuming and technically demanding. AI tools can perform these functions rapidly and precisely, freeing artists to focus on conceptual development and expressive nuances. This shift allows for a more efficient workflow and potentially higher-quality outputs.

The dynamics of human-AI collaboration are reshaping the very essence of artistic creation. In this symbiotic relationship, the AI acts as an assistant, a muse, or a co-creator, contributing its generative capabilities to the artist's intuitive and interpretive skills. This partnership enriches the creative process, as the fusion of human emotion and machine logic produces works that neither could achieve independently. The interaction challenges artists to engage with technology critically, prompting reflections on their methodologies and the nature of creativity itself.

10.3.4 Philosophical and Ethical Considerations

Integrating generative AI into art raises profound philosophical and ethical questions requiring critical examination. Central to these is the issue of authorship and originality. The traditional concept of the artist as the sole originator of a work is disrupted when AI systems contribute significantly to the creation process. The ambiguity surrounding who the actual artist is—whether it is the AI, the developer of the

AI, or the individual providing the prompts—challenges legal and cultural frameworks governing intellectual property and artistic recognition. This debate extends to the moral rights of artists and the attribution of creative contributions in collaborative human-AI endeavours.

Concerns about the impact of AI on human creativity are also paramount. While AI has the potential to enhance creativity by offering new tools and perspectives, there is a legitimate fear that over-reliance on AI might lead to a homogenisation of artistic expression. If artists begin to depend heavily on AI-generated ideas, the uniqueness and diversity of human thought could diminish, resulting in a loss of originality and depth in the arts. This scenario poses existential questions about the future of human creativity and the intrinsic value of art as a human endeavour.

Ethical considerations extend to bias in AI training data, which can perpetuate stereotypes and exclude marginalised voices. The potential misuse of AI-generated content raises concerns about authenticity and deception, particularly with the proliferation of deepfakes and synthetic media. Furthermore, the economic impact on human artists cannot be overlooked. As AI-generated art becomes more prevalent, there is a risk that human artists may face increased competition, devaluation of their work, or displacement in specific sectors of the art market. These challenges highlight the need for responsible AI development and implementation, with policies and regulations that protect artists' rights and promote ethical standards.

The evolution of the creative process in the context of generative AI is a multifaceted phenomenon that encompasses technological innovation, artistic transformation, and profound philosophical inquiry. The capabilities of AI offer unprecedented opportunities to expand creative horizons. Yet, they also compel society to grapple with complex questions about the nature of creativity, the definition of authorship, and the ethical implications of machine-generated art. As the boundaries between human and artificial creativity continue to blur, it is imperative to foster dialogue among artists, technologists, ethicists, and policymakers to navigate this uncharted terrain thoughtfully and responsibly. The future of art in the age of AI hinges on our ability to balance the transformative potential of technology with a commitment to preserving the human spirit at the core of creative expression.

10.3.5 Future Directions

The trajectory of generative artificial intelligence heralds a transformative epoch in the realm of artistic creativity, poised to challenge and expand the boundaries of human imagination. Emerging trends in AI research, particularly the refinement of Generative Adversarial Networks (GANs), the sophistication of diffusion models, and the advent of multimodal systems, are set to revolutionise how art is conceived, produced, and experienced. These technological advancements promise not merely incremental improvements but quantum leaps in the capability of AI to generate art that is not only visually and aurally compelling but also emotionally resonant and contextually profound. Integrating these cutting-edge models could enable AI to tap into subtler nuances of human emotion, cultural symbolism, and aesthetic preferences, producing works that evoke more profound emotional responses and engage audiences more intimately.

Speculatively, the future of art in an AI-dominated landscape is replete with both exhilarating possibilities and profound philosophical dilemmas. As AI continues infiltrating and reshaping artistic expression, it may give rise to new immersive and interactive art forms that seamlessly merge the virtual and physical realms. Imagine installations where AI-generated environments respond in real time to the presence and emotions of viewers or personalised narratives that evolve uniquely for each participant based on their interactions. Such innovations could redefine the very concept of art, shifting it from a static creation to a dynamic, evolving experience that blurs the line between creator and audience.

Generative AI is fundamentally altering the essence of the creative process. Historically, art has been a deeply personal endeavour that manifests an artist's innermost thoughts, emotions, and perspectives, meticulously crafted through their unique skills and experiences. The introduction of generative AI introduces a paradigm shift wherein the artist assumes the role of curator, guide, and collaborator rather than the sole originator of the work. This collaborative dynamic involves engaging in a nuanced dialogue with AI systems—feeding them prompts, interpreting their outputs, and iteratively refining the results to align with a desired vision. The artist becomes a co-director in the creative process, orchestrating a symbiotic relationship between human intuition and machine computation.

This evolution redefines the artist's role, transforming the creative process into a complex interplay between human intentionality and algorithmic possibility. The AI introduces elements of randomness and novelty that can spark unexpected inspiration, challenging artists to expand their creative horizons. Human artists contribute the critical dimensions of meaning, context, and cultural understanding, infusing the work with depth and significance that transcends mere algorithmic generation. The result is a dance of creativity where the boundaries of authorship are fluid, and the final artwork is a testament to the fusion of human and artificial intelligence.

The capabilities of generative AI tools are vast and continually expanding, enabling the creation of an unprecedented range of artistic outputs. In visual art, AI can generate paintings, digital illustrations, photo-realistic images, and abstract compositions, even extending to innovative designs in architecture and fashion. Music generation has reached new heights, with AI composing instrumental pieces, electronic music, genre-blending tracks, and adaptive music that evolves in response to user interaction. In literature, AI authors poems, short stories, scripts, and interactive narratives that adapt to reader choices, challenging traditional notions of authorship and narrative structure.

Moreover, AI's influence extends into sculpture and 3D art, where it can design complex 3D models and sculptures that can be realised physically through advanced manufacturing technologies like 3D printing. In animation, AI assists in creating short sequences or full-fledged animations, integrating traditional techniques with AI-generated elements to produce innovative visual storytelling. The realm of mixed media is also being transformed, as AI enables the seamless combination of text, sound, visuals, and other media to create immersive multimedia installations or digital collages that engage multiple senses simultaneously.

As we look toward the future, the convergence of AI and art holds the potential to unlock new dimensions of creativity and expression. However, this potential is accompanied by critical considerations that must be addressed. The ethical implications of AI-generated art, issues of authorship and intellectual property, and the impact on human artists and the cultural value of art are complex challenges that require thoughtful deliberation. The balance between embracing technological innovation and preserving the humanistic essence of art will define the trajectory of this evolution.

The future directions of generative AI in art are both exhilarating and uncertain. The advancements in AI technologies promise to expand the horizons of what is artistically possible, offering tools that augment human creativity and enable new forms of expression. Yet, they also compel us to reexamine fundamental concepts of creativity, authorship, and the artist's societal role. As we stand on the cusp of this new era, artists, technologists, ethicists, and society must engage in an open dialogue about the opportunities and challenges ahead. The synthesis of human creativity and artificial intelligence could herald a renaissance of artistic innovation, provided we navigate its complexities with foresight and integrity.

10.4 Artistic Movements Enabled by Generative AI

Generative artificial intelligence has transcended its role as a mere technological instrument to become a profound catalyst for the emergence of new artistic movements and expressions. It is reshaping the very foundations of creativity, challenging artists and audiences alike to reconsider the boundaries of art. By harnessing the computational prowess of AI, artists are venturing into uncharted aesthetic territories, producing works that not only push the limits of human imagination but also question the essence of artistic creation itself.

10.4.1 AI-Driven Aesthetics

One of the most striking impacts of generative AI on the art world is the development of entirely novel visual styles that surpass traditional human capabilities. Artists are leveraging AI algorithms to create artworks with unprecedented textures, patterns, and forms, challenging conventional notions of beauty and representation. These AI-driven aesthetics often feature complex, multilayered compositions that would be arduous, if not impossible, to conceive manually. The result is a new visual language that compels viewers to engage with art on a different sensory and cognitive level.

Moreover, the deliberate incorporation of glitches and imperfections through AI has given rise to a distinct aesthetic that reflects the digital essence of the medium.

This movement, often associated with "glitch art," embraces errors and technological anomalies as integral components of artistic expression. By intentionally introducing controlled disruptions into the artwork, artists comment on the fragility and unpredictability of digital systems, thereby transforming imperfections into sources of beauty and meaning.

The advent of dynamic and interactive art represents another frontier enabled by generative AI. Artists create works that evolve in real time, responding to environmental stimuli or viewer interactions. This interactivity transforms the audience from passive observers into active participants, blurring the lines between creator and spectator. Such immersive experiences challenge the static nature of traditional art forms, suggesting that art can be a living, ever-changing entity that adapts and responds to the world around it.

10.4.2 Reinterpreting the Past

Generative AI is also a powerful tool for reimagining and reconstructing historical art. Through techniques like style transfer and remixing, AI algorithms analyse and apply the stylistic elements of one image or artwork to another, resulting in fascinating reinterpretations of classic works. This process enables artists to create hybrid styles that merge disparate artistic traditions, offering fresh perspectives on familiar masterpieces. These reimagined works provoke dialogue about cultural continuity and transformation by juxtaposing contemporary elements with historical aesthetics.

Furthermore, AI's capacity for historical reconstruction has significant implications for art conservation and scholarship. AI can reconstruct lost or damaged artworks, filling gaps with plausible approximations based on learned patterns and styles. This not only aids in preserving cultural heritage but also opens up speculative possibilities, allowing us to envision how incomplete or fragmented works might have appeared initially. Such reconstructions challenge the notion of permanence and completeness in art, suggesting that technological innovation can extend the life and relevance of historical pieces.

10.4.3 Beyond Traditional Mediums

The influence of generative AI extends well beyond the visual arts, profoundly affecting music, literature, and performance. AI is utilised in music to compose original pieces, generate new sounds, and create interactive sonic environments. Musicians collaborate with AI systems to explore uncharted auditory landscapes, blend genres, and push the boundaries of rhythm, harmony, and texture. AI's generative capabilities allow for highly complex and dynamic compositions, often evolving in response to audience interaction or environmental factors.

In the literary realm, AI-generated poetry and narratives challenge traditional authorship and storytelling concepts. Language models trained on vast corpora can produce texts that mimic human writing styles yet introduce novel expressions and ideas. This has given rise to new forms of literary expression that blend human creativity with machine-generated content, prompting readers to question the nature of consciousness and creativity. The uncanny ability of AI to generate coherent and sometimes profound literary works forces a re-evaluation of the human writer's role in an era where machines can produce artful prose.

AI is also making significant inroads into performance art. Artists integrate AI into live performances, such as dance, theatre, and music, to create dynamic and unpredictable experiences. AI systems can generate visual projections, manipulate soundscapes, or even influence choreography in real time, responding to the performers' movements or audience reactions. This fusion of technology and performance not only enhances the sensory experience but also introduces elements of spontaneity and variability that challenge scripted norms.

10.4.4 Collaborative Creation

The rise of generative AI has ushered in a new era of collaborative creation, where artists and AI systems work together as creative partners. This human-AI co-creation leverages the unique strengths of both parties: the artist's intuition, emotional depth, and conceptual vision, as well as the AI's ability to process vast datasets, identify patterns, and generate novel outputs. The collaborative process often involves iterative feedback loops, with the artist refining the AI's outputs and the AI adapting to the artist's preferences. This symbiotic relationship expands the creative horizon, enabling the exploration of ideas that might be inaccessible to either party alone.

Collective AI art represents another intriguing development involving projects where multiple AI agents or algorithms collaborate to create art. These systems interact with each other, sometimes in unpredictable ways, leading to emergent behaviours and results. The art produced through such collaborations can exhibit complexity and originality that challenge our understanding of creativity and agency. It raises provocative questions about authorship and the autonomy of human and artificial creative entities.

10.4.5 Conceptual and Critical AI Art

Artists increasingly use AI to explore profound themes related to consciousness, identity, and the human condition in the digital age. By employing AI as a mirror to humanity, they probe the intersections between technology and self, examining how artificial intelligence shapes our perceptions, behaviours, and societal structures. This introspective approach often results in artworks that are as conceptually challenging as they are aesthetically compelling, encouraging viewers to reflect on their relationship with technology.

AI art is also serving as a powerful tool for social commentary. Artists utilise AI-generated works to critique issues such as bias, surveillance, and the erosion of privacy in a technologically saturated society. These artworks highlight the ethical and moral implications of relying on algorithms in decision-making processes by exposing the limitations and prejudices embedded within AI systems. Such critical engagements raise awareness and stimulate discourse on the responsible development and deployment of AI technologies.

10.4.6 Key Examples and Movements

The convergence of art and generative AI has given rise to several notable movements that exemplify the transformative potential of this technology. "GANism" is a movement centred around the artistic exploitation of Generative Adversarial Networks (GANs). Artists working within this paradigm use GANs to produce images that oscillate between the recognisable and the abstract, often evoking a sense of the uncanny. The resulting artworks challenge viewers to question the boundaries between reality and simulation.

Neuroaesthetic art represents another frontier where artists explore the intersection of neuroscience, aesthetics, and AI. By integrating insights from cognitive science, these artists aim to create works that engage the brain's perceptual and emotional processes in novel ways. This movement underscores AI's potential to generate art and deepen our understanding of the aesthetic experience itself.

Algorithmic art, a precursor to contemporary AI art movements, has been revitalised through advances in computational processes. Artists in this domain use algorithms to produce intricate patterns and forms, emphasising the role of mathematical beauty in artistic creation. The resurgence of algorithmic art highlights a growing appreciation for the elegance of code as a medium of artistic expression. Exploring these movements illuminates how generative AI is not merely a tool but a transformative force reshaping the boundaries of creative expression. Technology challenges traditional notions of art and creativity, compelling us to reconsider the role of the artist and the influence of technology in the artistic process. AI's ability to blend colours, forms, and concepts without human biases or preconceived conventions has led to the creation of artworks that defy categorisation, expanding the lexicon of visual and conceptual art.

The emergence of new aesthetic movements defined by machine spontaneity, abstract interpretations, and the fusion of digital and physical mediums signifies a paradigm shift in creativity. Blurring boundaries between human and machine creativity prompts new schools of thought, where artists and technologists collaborate to explore the frontiers of originality and expression. These collectives question the essence of being creative, suggesting that creativity is a fluid, collaborative act rather than a solitary pursuit.

Illustrative examples of these innovative movements include:

Algorithmic Abstraction: This style relies on algorithms to generate abstract, intricate patterns that are impossible to replicate manually. The art produced is often characterised by mathematical precision combined with aesthetic complexity, challenging viewers to appreciate the inherent beauty of computational processes.

AI Surrealism: By blending realistic elements with bizarre, dream-like components, AI surrealism pushes the boundaries of traditional surrealism. The unexpected juxtapositions and distortions AI algorithms create provoke a sense of wonder and disorientation, inviting interpretations that delve into the subconscious.

Generative Portraiture: In this genre, AI creates human-like portraits that unexpectedly combine features. The resulting images often challenge notions of identity and representation, as they depict faces that are both familiar and alien. This raises questions about individuality and the role of AI in shaping perceptions of self.

Dynamic Installations: These are interactive artworks where AI-generated content evolves in real time based on audience input or environmental factors. Such installations create immersive experiences unique to each viewer, emphasising art's transient and participatory nature in the digital age.

As generative AI continues to influence artistic practice, it expands the possibilities for creation and prompts critical reflection on the implications of technology in art. The movements enabled by AI encourage an aesthetic and philosophical dialogue, urging society to engage with questions about authorship, originality, and the future of creativity. In embracing the opportunities afforded by AI, artists are redefining the artistic landscape, ensuring that art remains a dynamic and evolving expression of human experience in an increasingly technological world.

10.5 Evaluating AI in Artistic Domains: A Framework for Assessing Aesthetic and Creative Capabilities

10.5.1 Key Principles of the Framework

At the heart of this evaluative structure lie three fundamental principles that ensure a holistic assessment of AI's artistic prowess. First, the framework addresses the essential elements that contribute to art's aesthetic value and impact across different media. This involves examining how AI systems replicate or innovate upon the foundational aspects that make art resonant and meaningful. Second, it considers the potential for human-AI collaboration, evaluating the AI's technical proficiency and its capacity to serve as a catalyst and tool for human artists. This acknowledges the synergistic possibilities when human creativity intersects with machine intelligence. Third, it recognises the dynamic nature of artistic expression, appreciating AI's potential to expand the boundaries of creative possibility and challenge conventional artistic norms.

In operationalising these principles, the framework eschews simplistic assessments in favour of a nuanced understanding of AI's multifaceted contributions. It moves beyond mere technical evaluation to consider the qualitative impact of AI on the creative landscape, including ethical implications and the shifting paradigms of authorship and originality.

Evaluating AI in Visual Art

In visual art, the framework assesses several critical criteria to determine the AI's capacity to produce works of aesthetic and conceptual significance. One key aspect is the AI's ability to perform stylistic emulation, which involves accurately replicating artistic styles such as Impressionism or Surrealism. This tests the AI's technical skill and its understanding of the historical and cultural contexts that inform these styles.

Beyond replication, the framework evaluates innovation and originality, measuring the uniqueness of AI-generated images and their ability to transcend imitation. This criterion is paramount, as it challenges the AI to contribute novel visual expressions that could redefine artistic movements or create entirely new ones.

The coherence and internal consistency of the artwork are also scrutinised. This involves ensuring that the AI-generated images maintain logical composition, realistic proportions, and believable elements, even when operating within imaginative or abstract realms. Such coherence is essential for the artwork to engage viewers meaningfully and sustain aesthetic credibility.

Another crucial factor is the AI's adaptability and user interface, reflecting its potential as a collaborative tool. This assesses how effectively the AI allows human artists to influence and modify the generated output, facilitating a seamless integration into existing artistic workflows. The user interface's intuitiveness and the degree of control it affords the artist indicate the AI's practicality and accessibility.

Finally, the technical fidelity of the images, encompassing the level of detail, clarity, and visual fidelity, is examined. High-resolution outputs with meticulous attention to detail are vital for professional applications and contribute significantly to the artwork's overall impact.

Assessing AI in Music Generation

In music generation, the framework considers the AI's ability to craft compositions that are not only technically sound but also emotionally compelling. The criterion of melodic innovation evaluates the catchiness, memorability, and uniqueness of the melodies produced. Lyrical content is a cornerstone of musical expression, and the AI's capacity to innovate in this area is a strong indicator of its creative potential.

The framework also assesses harmonic sophistication, examining the complexity of chord progressions and harmonic structures. This reflects the AI's understanding of music theory and its ability to create nuanced compositions that resonate deeply emotionally.

Rhythmic dynamism is another critical aspect, focusing on the AI's ability to generate varied and engaging rhythmic patterns. Rhythm drives the energy and flow of music, and proficiency in this area enhances the overall appeal of the composition.

The affective potency of the music, or its ability to evoke specific emotions and moods, is a central criterion. Music's profound connection to human emotion means that an AI's success in this realm is a significant marker of its artistic capability.

Lastly, stylistic versatility evaluates the AI's capacity to generate music across different genres and traditions. This versatility demonstrates the breadth of AI's musical knowledge and adaptability to various contexts and audiences.

Evaluating AI in Creative Writing

The framework examines the AI's linguistic and narrative competencies in creative writing. Linguistic proficiency is fundamental, ensuring that the AI can generate grammatically correct text that is stylistically consistent and has a natural flow. Mastery of language is essential for clarity and effectiveness in communication.

For prose, narrative coherence is assessed by the AI's ability to construct logical plot developments, believable characters, and engaging storylines. This criterion evaluates the AI's capacity to weave narratives that captivate readers and sustain their interest.

In poetry, the focus shifts to the use of poetic techniques. This involves the AI's adeptness with metaphor, imagery, meter, and other literary devices that enrich the text and deepen its expressive power.

Conceptual originality measures the uniqueness of the ideas, themes, and expressions generated by the AI. Pushing the boundaries of creative thought is a hallmark of significant literary work, and this criterion gauges the AI's contribution to expanding literary horizons.

Finally, emotional resonance assesses the AI's ability to connect with readers on an emotional level. Literature's power often lies in its capacity to move and inspire, and an AI that can evoke such responses demonstrates a profound level of creative sophistication.

Integrative Analysis and Future Implications

Researchers and practitioners can comprehensively understand AI's creative capabilities by applying these criteria across different artistic fields. This framework evaluates technical proficiency and considers the qualitative aspects that make art impactful and meaningful. It facilitates a dialogue about how AI can serve as both a tool and a collaborator in the creative process, potentially reshaping the landscape of artistic expression.

Employing this framework (see Table 10.2) allows for a more nuanced assessment that goes beyond surface-level judgments. It acknowledges the complexities of artistic creation and the multifaceted role that AI can play. As AI systems continue to evolve, this framework provides a structured approach to evaluate their contributions, ensuring that they enhance rather than overshadow human creativity.

Table 10.2 A framework for assessing aesthetic and creative capabilities of generative AI

Art field	Criteria	Notes	Score weighting (%)
Visual art	Style Mimicry	Accuracy in replicating specific artistic styles (e.g., Impressionism, Surrealism)	20
	Novelty/Originality	The uniqueness of generated images goes beyond imitation	25
	Coherence/Consistency	Logical composition, realistic proportions, and believable elements	20
	Controllability/Editability	User's ability to influence and modify the generated output	15
	Resolution/Detail	Level of detail, clarity, and visual fidelity of the images	20
Music generation	Melodic Originality	Catchiness, memorability, and uniqueness of melodies	25
	Harmonic Complexity	The sophistication of chord progressions and harmonic structure	20
	Rhythmic Variation	Dynamic and engaging rhythmic patterns	20
	Emotional Impact	Ability to evoke specific emotions or moods	20
	Genre Versatility	Capacity to generate music across different styles	15
Creative writing	Linguistic Fluency	Grammatical correctness, natural flow, and stylistic consistency	25
	Narrative Coherence	Logical plot development, believable characters, and engaging storylines (for prose)	20
	Poetic Devices	Effective use of metaphor, imagery, and other literary techniques (for poetry)	20
	Originality/Creativity	Uniqueness of ideas, themes, and expression	20
	Emotional Resonance	Ability to evoke emotions and connect with the reader	15

10.5.2 Quantifying the Evaluation

The evaluation can be quantified using the following formula:

$$s = \sum_{i=1}^{n} wici$$

where S is the overall creative score, Wi is the assigned weight for each criterion, and Ci is the score for that criterion. This formula allows for a systematic assessment of AI's performance across the different artistic disciplines, considering each criterion's specific importance.

The quantification formula is a valuable tool for objectively measuring AI's performance, allowing for a standardised comparison across different systems and artistic fields. This quantitative assessment and qualitative insights equip artists, researchers, and technologists to engage critically with AI's role in the creative process.

Ultimately, this framework fosters a deeper understanding of how AI can challenge and enhance traditional notions of art and creativity. By rigorously evaluating AI's contributions, we can navigate the ethical, philosophical, and practical complexities that arise at the intersection of technology and the arts, ensuring that the evolution of creative expression continues to reflect and enrich the human experience.

10.6 Ethical Considerations and Challenges

The advent of generative artificial intelligence in the art world heralds unprecedented opportunities for creative exploration. Yet, it simultaneously introduces a labyrinth of ethical considerations and challenges that cannot be overlooked. The profound impact of AI on art and society necessitates a critical examination of issues surrounding authorship, ownership, bias, representation, and the essence of human creativity.

10.6.1 Authorship and Ownership

One of the most contentious dilemmas AI-generated art poses is the question of authorship and intellectual property rights. When an AI system produces a piece of art, determining who holds the copyright becomes a legal and philosophical quandary. Is the rightful owner the developer who created the AI model, imbued with the algorithms and code that enable the generation of art? Or does ownership belong to the user who provided the prompts and guided the AI's creative process? Alternatively, can the AI be considered an entity capable of holding rights despite current legal frameworks not recognising non-human authorship? This ambiguity has significant ramifications for artists, corporations, and the art market, potentially disrupting traditional notions of ownership and commodification of art. Moreover, the lack of clear attribution and transparency in AI-assisted creations raises concerns about authenticity and ethical disclosure. Without explicit acknowledgement that work was generated with AI assistance, audiences may be misled, undermining informed appreciation and eroding trust in the art community.

10.6.2 Bias and Representation

Generative AI models are trained on vast datasets harvested from the internet, which inherently contain the biases and prejudices of society. Consequently, these AI systems may inadvertently perpetuate harmful stereotypes, reinforce systemic inequalities, or exclude marginalised groups in the art they produce. This not only affects the representation within AI-generated art but also amplifies discriminatory narratives under the guise of creativity. The ethical implications are profound, as such outputs can contribute to normalising bias and hinder efforts toward social equity. Additionally, the potential for cultural appropriation is heightened, with AI capable of mimicking or replicating artistic styles from marginalised cultures without a nuanced understanding or respect for their significance. This appropriation not only disrespects the source cultures but also raises questions about the commodification of cultural heritage and the responsibilities of creators—human or artificial—in honouring the origins of their inspiration.

10.6.3 Impact on Human Artists

The rise of AI in the artistic domain poses existential questions about the future role of human artists and the value of human creativity. While AI tools can automate certain tasks and potentially enhance productivity, there is a legitimate concern that they may lead to job displacement or reduced opportunities for artists. The economic impact on those who rely on their artistic skills for livelihood cannot be ignored, as the market may increasingly favour the efficiency and cost-effectiveness of AI-generated content over human-made art. Furthermore, the ease with which AI can produce impressive artwork may lead to a devaluation of human skill and craftsmanship. If society begins to perceive that artistic talent can be replicated by machines, the appreciation for the years of dedication, practice, and personal expression that human artists invest in their work may diminish. This shift could fundamentally alter the cultural landscape, where the intrinsic value of art as a human endeavour is overshadowed by the novelty of machine-generated creations.

10.6.4 Misinformation and Manipulation

The capabilities of AI extend beyond artistic innovation into the realm of misinformation and manipulation, presenting significant ethical challenges. Advanced AI techniques can generate hyper-realistic images, videos, and audio—commonly referred to as deepfakes—that blur the line between reality and fabrication. Such technologies raise serious concerns about the erosion of trust in visual media, as they can be employed to disseminate misinformation, propaganda, or defamatory content with unprecedented believability. The potential for AI-generated art to manipulate emotions or influence public opinion is equally troubling. By exploiting psychological triggers and aesthetic appeal, AI can create content designed to sway individuals or even create addictive experiences that prioritise engagement over well-being. These manipulative uses of AI not only threaten individual autonomy but also pose risks to societal cohesion and democratic processes.

10.6.5 Accessibility and Equity

Access to powerful AI art generators presents another ethical dimension, centred around issues of equity and democratisation. There is a risk that these advanced tools may be accessible only to those with sufficient financial resources or technical expertise, thereby creating a new digital divide within the art world. Such disparities could reinforce existing inequalities, privileging certain groups while marginalising others, and stifling the diversity of voices and perspectives that enrich the artistic community. Conversely, while AI has the potential to democratise creativity by lowering barriers to entry, there is a concern that the concentration of AI technologies within a few dominant tech companies could centralise creative power. This concentration may limit opportunities for independent artists and lead to homogenisation in artistic output, as algorithms favour certain styles or trends dictated by commercial interests.

10.6.6 Addressing the Challenges

Confronting these ethical considerations requires a multifaceted and proactive approach. Developing comprehensive ethical guidelines is paramount, encompassing principles of transparency, accountability, and respect for intellectual property and cultural heritage. The AI art community, including developers, artists, and institutions, must collaborate to establish standards that govern the creation and dissemination of AI-generated art. Technical solutions are also crucial; researchers are actively working on methods to detect AI-generated content, mitigate biases in training data, and ensure that AI systems produce outputs aligned with ethical norms. Education and awareness play a vital role in this endeavour. By informing artists, consumers, policymakers, and the general public about the capabilities and limitations of AI in art, society can foster informed discourse and decision-making. Promoting open dialogue among stakeholders—including ethicists, legal experts, cultural representatives, and technologists—is essential to navigate the complex landscape of AI ethics in art effectively.

By proactively addressing these ethical challenges, the integration of generative AI into the art world can be guided in a direction that enriches rather than undermines societal values. The rise of AI offers unparalleled opportunities for innovation and expression, but it simultaneously compels us to re-examine fundamental concepts of authorship, creativity, and the human experience. Human curation and oversight remain indispensable in ensuring that AI-generated works uphold ethical standards, respect cultural sensitivities, and maintain artistic integrity. As we stand at the intersection of technology and creativity, it is imperative to embrace the potential of AI thoughtfully and responsibly, acknowledging its power while safeguarding the principles that underpin the arts. In doing so, we can harness the transformative capabilities of generative AI to expand the horizons of artistic expression, fostering a vibrant and inclusive cultural landscape that benefits all of society.

10.7 The Future of Human-AI Collaborative Creativity

As we stand on the precipice of a new epoch in artistic expression, the burgeoning collaboration between humans and artificial intelligence promises to revolutionise the very foundations of creativity. This symbiosis heralds the emergence of art forms hitherto unimagined, where human intuition and machine precision coalesce seamlessly at every juncture of the creative process. Artists are no longer confined to traditional tools and mediums; instead, they are poised to engage with AI systems that can generate immersive, adaptive experiences evolving in real time—responsive not only to viewer input but also to environmental variables and perhaps even emotional cues.

The integration of generative AI into the artistic milieu is poised to catalyze a paradigmatic shift in our perception and evaluation of art. Creativity, once considered the exclusive province of human ingenuity, is expanding to encompass the innovative capacities of intelligent machines. This evolution challenges entrenched notions of authorship and originality, prompting a re-evaluation of what it means to create. As AI becomes an integral component of artistic practice, it may redefine expression itself, shifting from solitary endeavours toward a more collective, fluid interplay between human consciousness and machine learning algorithms.

Moreover, the fusion of AI and human creativity could give rise to new aesthetic philosophies and movements. The collaborative dynamic enables the exploration of complex concepts and multisensory experiences that transcend the limitations of individual cognition. Artists might harness AI to manipulate vast datasets, uncover hidden patterns, or simulate alternate realities, thereby pushing the boundaries of what is artistically conceivable. This convergence not only augments the creative capacity of individuals but also democratises the creative process, allowing a more diverse array of voices to contribute to the artistic discourse.

The potential implications of this human-AI partnership are profound. Generative AI stands as a powerful catalyst capable of propelling artistic creativity beyond its current horizons. By synthesising insights from a multidisciplinary cohort of artists, technologists, critics, and theorists, we can cultivate a rich, multifaceted exploration of how AI is reshaping the art world in transformative ways. This collaborative renaissance may well redefine the essence of art in the twenty-first century, expanding its scope to include not just new forms and mediums but also new modes of engagement and interpretation.

However, this envisioned future also necessitates a critical examination of the ethical, philosophical, and societal dimensions inherent in the fusion of human and artificial creativity. Questions concerning authorship, intellectual property, and the authenticity of AI-generated works must be thoughtfully addressed. As we navigate this uncharted terrain, it becomes imperative to foster dialogues that consider the implications of AI on cultural values and the human experience.

In conclusion, the alliance between human artists and AI systems heralds a pivotal moment in the evolution of creativity. It offers unprecedented opportunities to transcend conventional artistic boundaries and reimagine the potentialities of expression. Embracing this synergy could lead to a renaissance of creative innovation, one that not only expands what art can be but also enriches our understanding of the intricate tapestry of human and machine interrelations. The future of art lies in this collaborative horizon, inviting us to explore, question, and redefine the very nature of creativity itself.

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Chapter 11
The Next-GenAI-Rtists: A New Generation of Artists with Generative Tools

Florent Vinchon, Janet Rafner, Jacob Sherson, Marion Botella, and Todd Lubart

Abstract This chapter explores the emergence and influence of Generative Artificial Intelligence (GenAI) within the realm of artistic creation, highlighting its transformative potential in shaping a new generation of creators known as "Next-GenAI-rtists." These artists redefine traditional practices by integrating GenAI as both a tool and a collaborative partner, fostering innovative co-creative processes. Employing the Hybrid Intelligence Technology Acceptance Model (HI-TAM), the chapter examines the psychological, technical, and ethical dimensions of this integration, emphasising iterative learning, shared conceptual frameworks, and the pivotal role of psychological safety. By juxtaposing GenAI's impact with historical shifts—such as photography's influence on art—the text situates the current revolution within a broader technological and cultural context. Practical examples of AI-generated art across various media underscore the democratisation and expanded accessibility GenAI offers, alongside its challenges to authorship and aesthetic norms. The chapter also delves into critical concerns regarding bias, transparency, and intellectual property, advocating for responsible and ethical GenAI implementation. Concluding with an exploration of the evolving skill sets and interdisciplinary collaboration central to

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Next-GenAI-rtists, the chapter reimagines creativity as a dynamic interplay between human ingenuity and computational innovation in the digital age.

Keywords Creativity · Generative artificial intelligence · Art · Photography · Process · Hybrid intelligence

11.1 Introduction

The advent of generative artificial intelligences (GenAIs) marks a significant milestone in the ongoing technological evolution shaping human behaviour, work processes, and societal structures. As transformative tools of the Fourth Industrial Revolution, GenAIs influence a broad range of fields, from professional tasks to artistic endeavours (Schwab 2017; Vinchon et al. 2023; Rafner et al. 2023). Their ability to enhance productivity, introduce new methods, and alter existing skillsets has catalysed a profound transformation across disciplines (Noy and Zhang 2023). Among these shifts is the emergence of "prompt engineering," a skill currently essential for effectively harnessing the creative potential of most GenAIs (e.g., Kutela et al. 2023; Oppenlaender et al. 2023).

Within the creative domain, the impact of GenAIs is particularly striking. These technologies challenge traditional artistic paradigms and offer novel tools for expression. This chapter explores how artists are evolving alongside these innovations, acquiring new skills, and redefining their roles. We introduce the concept of the "Next-GenAI-rtist" to describe this emerging generation of creators who integrate GenAIs into their artistic practices (Leroy 2023). Through their work, these artists bridge the gap between human intuition and computational precision, crafting unique artistic outputs that push the boundaries of creativity.

To understand this transformation, we examine the historical relationship between technological innovation and art, with a particular focus on photography. Just as photography once redefined the role of the artist, GenAIs are now poised to do the same by expanding creative possibilities and democratising artistic practices (Hertzmann 2018). We also explore concrete examples of GenAI in art, highlighting their contributions to diverse media such as visual arts, music, and literature. In this context, we'll be looking at co-creative processes with GenAIs. These allow us to explore in detail how a human and a GenAI co-operate to produce an output that will be considered "creative."

Human-AI co-creativity, a subfield of computational creativity, explores the interaction between AI and humans in generating shared creative outputs (Kantosalo et al., 2020). Unlike autonomous agents that independently produce creative products (Colton, 2012; Misztal and Indurkhya 2014) or creative support tools that assist users (Shneiderman 2007), co-creative systems foster dynamic, multi-turn collaboration, where both collaborators influence and adapt to each other's contributions over time (Davis et al. 2015; Fan et al. 2019; Karimi et al. 2019; Lin et al. 2020)). This process, defined by Karimi (2018) as requiring mutual influence and shared conceptualisations of creativity, relies on iterative feedback, turn-taking (Winston and Magerko 2017), and the development of shared meaning (Davis et al. 2015) and mental models (Fuller and Magerko 2011) such interactions go beyond single-shot execution, enabling emergent outcomes through the continual interplay between human and AI communication channels (Rezwana and Maher 2023).

Finally, while ethics are not the central focus of this chapter, we acknowledge the importance of addressing the challenges posed by GenAIs, including issues of bias, intellectual property, transparency, and sustainability. These considerations are crucial for guiding the responsible integration of GenAIs into artistic and educational contexts (Darda et al. 2023). By situating GenAIs within this broader framework, this chapter aims to provide a comprehensive understanding of the opportunities and challenges facing Next-GenAI-rtists as they navigate the evolving creative landscape.

11.2 Generative AIs

After the widespread introduction of Large Language Models (LLMs) such as ChatGPT, public interest in Artificial Intelligences (AIs) has surged. This renewed enthusiasm has been further amplified by the emergence of "generative" models, including Generative Adversarial Networks (GANs), diffusion models (e.g., DALL-E, Stable Diffusion), and transformer-based models (e.g., Jukebox). Each of these models offers distinct capabilities, with specific strengths and limitations depending on the intended use, which are summarised in Table 11.1.

Historically, GANs, introduced by Goodfellow et al. (2014), were the first widely recognised generative models. They introduced the groundbreaking concept of adversarial training, wherein two networks—a Generator and a Discriminator—compete against each other. The Generator creates content while the Discriminator attempts to identify whether the content is authentic or fabricated (Wang 2019). Over successive iterations, the Generator learns to produce outputs that convincingly mimic the training data. GANs gained public attention for their ability to create DeepFakes,

Table 11.1 The different kinds of generative models

Generative model	Modality	Purpose	Examples
Generative adversarial networks (GANs)	Images, video	Realistic image generation	StyleGAN, BigGAN, DeepFakes
Diffusion models	Images, video, 3D	Text-to-Image, 3D	Dall-E, stable diffusion, midjourney
Transformer models	Text, image, audio	Text/chat/code generation	GPT-4, ChatGPT, Claude
Hybrid models	Text + Image, 3D	Multimodal generative AI	Dall-E 3, runway Gen-2

a term that has since become synonymous with AI-generated media (Chesney and Citron 2019).

Diffusion models, by contrast, operate through a process of iterative refinement. Starting with random noise, these models "de-noise" iteratively, guided by prompts, to generate content closely aligned with the user's specifications (Ho et al. 2020). This approach has proven particularly effective for generating high-quality visual and textual outputs, as exemplified by tools like DALL-E and Stable Diffusion.

Transformer models, exemplified by GPT (Generative Pre-trained Transformer) architectures, represent another critical development in GenAI. Transformers utilise an Attention Mechanism to identify the most relevant aspects of input data, enabling them to predict subsequent outputs with high accuracy (Vaswani et al. 2023). This mechanism allows for nuanced understanding and generation of text and other modalities. LLMs, such as GPT-4, extend this capability into multimodal applications, combining inputs and outputs across text, images, and more (Bommasani et al. 2022). Examples of multimodal models include GPT-4, Claude 3, and Gemini, which demonstrate the versatility and evolving capabilities of transformer-based systems.

Together, these generative models represent a diverse and rapidly evolving toolkit that underpins the burgeoning field of creative AI. Their unique architectures and applications highlight the potential for innovation while underscoring the need to understand their specific roles and limitations (Elgammal et al. 2017).

These generative models are capable of producing a wide array of outputs, including text, images, video, and 3D content. In other words, they can generate renderings that are often indistinguishable from human creations. This capability has led many individuals, both amateurs and professionals, to adopt GenAI tools for producing content. While some outputs may initially appear amateurish, these tools frequently produce surprisingly refined results, even of exceptionally high quality. The accessibility of GenAI has thus democratised the creative process, enabling a broader spectrum of users to engage in artistic endeavours.

The emergence of Generative Adversarial Networks (GANs) marked a significant milestone in AI-driven creativity. One notable example is the "Portrait of Edmond de Belamy," a GAN-generated portrait created by the French collective "Obvious" in 2018. This groundbreaking work was trained on 15,000 WikiArt portraits spanning the fourteenth to nineteenth centuries and achieved widespread recognition when it was sold at a Christie's auction for over $400,000 (Elgammal 2018). This success underscored the artistic and commercial potential of GenAI.

Following this milestone, GenAI has been applied across various artistic domains, resulting in notable works in music (e.g., "Heart on My Sleeve" by Ghostwriter977[1]), literature (e.g., the AI-generated novel *1 the Road*, (Goodwin et al. 2018)), and philosophy (e.g., *The Infinite Conversation*, (Miceli 2023)). These outputs have sparked considerable debate about the nature of creativity and authorship. Critics argue that GenAI tools rely on pre-existing content, raising concerns about originality and

[1] Heart on My Sleeve (Ghostwriter977 song). (2024). In Wikipedia. https://en.wikipedia.org/w/index.php?title=Heart_on_My_Sleeve_(Ghostwriter977_song)&oldid=1262915930

plagiarism. They contend that such works may lack the "soul" or intentionality typically associated with human art. However, this debate mirrors historical controversies surrounding earlier technological innovations in art, such as photography and digital media. These discussions reflect an ongoing negotiation of what constitutes art in the context of emerging tools and techniques.

11.3 Technology and the Arts

The interplay between technological innovation and artistic practices has long been a catalyst for reimagining creative boundaries (Shanken 2009). Each major technological breakthrough—from the printing press to digital tools—has disrupted traditional methods, pushing artists to adapt and innovate. Photography serves as one of the earliest examples of how technology fundamentally altered artistic conventions, challenging entrenched norms while ultimately expanding the possibilities for creative expression. This section examines the historical trajectory of photography, drawing parallels to the transformative impact of GenAI in contemporary art.

11.3.1 A Brief History of Photography

The integration of technology into artistic practices has consistently challenged and expanded traditional notions of creativity. Photography exemplifies this dynamic, transitioning from a mechanical process to a respected art form. GenAI, much like photography, disrupts artistic conventions while offering novel opportunities for creativity.

When Louis Daguerre introduced the daguerreotype in 1839, photography was primarily regarded as a technological achievement. Early adopters celebrated its ability to produce precise visual records for scientific, engineering, and archival purposes (Hirsch 2024). This mechanical precision, however, elicited criticism from figures like Charles Baudelaire, who viewed photography as a threat to true art, fearing that its realism would erode the imagination inherent in painting (Hertzmann 2018). As cited in Hirsh (2024), Baudelaire described photography in scathing terms:

"Since photography provides us every desirable guarantee of exactitude that we could desire (they believe that, poor madmen!), art is photography. From that moment onwards, our loathsome society rushed, like Narcissus, to contemplate its trivial image on metallic plate… As the photographic industry became the refuge of all failed painters with too little talent, or too lazy to complete their studies, this universal craze not only assumed the air of blind and imbecile infatuation, but took on the aspect of vengeance… Photography must therefore, return to its true duty, which is that of handmaid of the sciences and the arts, but a very humble handmaid, like painting or shorthand, which have neither created nor supplemented literature… let it, in short, be the secretary and record-keeper of whomever needs an absolute

material accuracy for professional reasons… But if once allowed to impinge on the sphere of the intangible and the imaginary, on anything that has value solely because man adds something to it from his soul, then woe betide us!"(p137-138).

This critique underscores the parallels between photography's emergence and GenAI's current reception. Both have faced scepticism and resistance from traditionalists concerned with preserving artistic authenticity.

Before the advent of photography, realism was an important goal of painting—artists sought to reproduce accurately what the human eye perceived. However, photography's unparalleled precision in capturing reality challenged painters to explore new directions and reinvent their mediums (Hertzmann 2018). This technological disruption democratised photography as an art form and inspired innovation in painting, giving rise to movements such as Impressionism, Pointillism, and Cubism. These styles emphasised abstraction, emotion, and novel perspectives, moving beyond the confines of strict realism.

Despite initial scepticism, photography evolved into a recognised creative medium. Movements like Pictorialism in the late nineteenth century sought to establish photography as an art form by emphasising emotional resonance over mechanical precision. Figures such as Alfred Stieglitz played pivotal roles in this transition, curating exhibitions and promoting techniques that blurred the lines between technology and art. Notably, the 1910 International Exhibition of Pictorial Photography in Buffalo marked a definitive moment in the acceptance of photography as a legitimate art form (Hirsh 2024; Hertzmann 2018).

As we will explore later, GenAI follows a similar trajectory. Like photography, it has sparked debates about its place in art while also catalysing creative exploration and the integration of emerging technologies into artistic practices.

11.3.2 Technologies, Artificial Intelligence and the Arts

Artists have consistently integrated emerging technologies into their practices, utilising tools to expand creative possibilities and redefine artistic expression (Shanken, 2009). Digital art, which appeared as early as the 1960s, demonstrates how artists employed early computational systems to create unique visual expressions (Wei 2024; Hertzmann 2018). As technology advanced, digital art incorporated tools like computer graphics, 3D modelling, and augmented reality (AR). These innovations broadened access to artistic creation by reducing barriers traditionally tied to material and technical skills, resulting in a democratised art landscape. Platforms like Adobe Photoshop, alongside the rise of the internet, allowed artists to distribute and monetise their work globally, challenging traditional gallery-centric models (Caporusso 2023; Wei 2024).

Before the advent of GenAI, artificial intelligence had already been producing notable contributions to art. Harold Cohen's AARON program represented one of the earliest examples of rule-based creativity, where algorithms autonomously created drawings, marking a significant intersection between art and computational logic (Mazzone and Elgammal 2019; Wei 2024). Similarly, Frieder Nake developed algorithmic works in the 1960s that mimicked or reinterpreted artistic styles (Nake 2012). These explorations blurred the lines between human intent and computational output, raising questions about authenticity and originality in art (Chamberlain et al. 2018). Other pioneers, such as Lillian Schwartz (Elgammal 2019) and David Cope (1989), leveraged custom AI tools and early computing systems to produce art, laying the groundwork for contemporary computational creativity. Despite the limitations of early technologies, these trailblazing artists introduced AI into the art world, continuing a long-standing tradition of integrating emerging tools to push creative boundaries.

11.3.3 How GenAI Will Be Accepted as an Artistic Medium

The integration of GenAI into artistic practices has sparked debates about the nature of creativity itself. The historical context of photography as a mechanical process raises questions about the role of the artist in the age of AI. As GenAIs can produce works that are indistinguishable from those created by humans, the traditional boundaries of artistic creation are being redefined (Epstein et al. 2023; Bellaiche et al. 2023). This shift is echoed in findings that explore how audiences perceive AI-generated artworks compared to human-created pieces, often revealing biases that favour human authorship despite the quality of AI outputs (Jacobsen et al. 2004; Latikka et al. 2023).

Initially perceived as a mechanistic and inauthentic form of creativity, AI tools have redefined how artists create and conceptualise art. Mazzone and Elgammal (2019) highlight that these AI systems act as collaborators, enabling artists to curate, tweak, and reinterpret outputs in ways that parallel the subjective control exercised by early photographers. In many ways, GenAI also democratises artistic creation. Tools like DALL-E and Stable Diffusion, much like Kodak's early consumer cameras, lower the barriers to entry for non-experts, thereby expanding access to artistic exploration. This democratisation has given rise to a new generation of artists who engage with GenAI in ways fundamentally distinct from previous technologies, signalling a transformative moment in the evolution of art.

11.4 The Next-GenAI-Rtist

11.4.1 A New Creative Process: Hybrid Intelligence and Co-Creativity

Hybrid Intelligence (HI) represents a groundbreaking shift in creative methodologies, heralding a new form of creative process that integrates the cognitive flexibility of humans with the computational efficiency of GenAI. This innovative approach moves beyond the traditional boundaries of artistic production, introducing a collaborative dynamic where human intuition and machine precision converge. The result is a synergistic process capable of producing outputs that surpass what either humans or AI could achieve independently.

HI focuses on creating optimal synergies between humans and algorithms (Akata et al. 2020; Dellermann et al. 2019; Prakash and Mathewson 2020), emphasising collaboration between humans and AI rather than viewing AI solely as a tool for augmentation, as in Human-Centred AI (HCAI). These systems combine generative capabilities with co-creative interactions to enhance functionality. Recent developments aim to establish HI as a holistic framework, integrating organisational concepts such as de- and upskilling from AI deployment (Rafner et al. 2022) and merging interface design with principles from information systems and human resource management, including business process reengineering and psychological safety, into 12 actionable design dimensions (Sherson et al. 2023).

The Hybrid Intelligence Technology Acceptance Model (HI-TAM) provides a structured framework for understanding how humans adapt and personalise generative design assistants in HI systems (Mao et al. 2023; 2024). It emphasises four key pillars:

Domain-Specific Suitability: GenAI systems must be tailored to specific domains, aligning their capabilities with users' expertise and preferences. By addressing the unique needs of various creative sectors, such as architecture or art, the model fosters higher engagement and usability.

Programmable Common Language: To enable effective communication between humans and AI, HI-TAM incorporates a shared language that clearly conveys user goals to the algorithm. This shared vocabulary facilitates real-time refinement of creative outputs, fostering a dynamic and responsive collaboration. This aligns with broader frameworks like the Co-Creative Framework for Interaction (COFI), which emphasises bidirectional communication in co-creativity (Rezwana and Maher 2023).

Human-Centred Continual Training Loop: The model integrates iterative training loops into the user's workflow, ensuring ongoing refinement of the AI system. This approach allows generative models to evolve based on user preferences, feedback, and task-specific requirements. It is crucial for maintaining a balance between automation and human agency, ensuring mutual learning to enhance long-term collaboration (Sherson et al. 2023). Trust and continuous feedback loops enable both humans and AI to learn and adapt, improving outcomes over time (Cañas et al. 2022; Piskopani et al. 2023).

Psychological Safety and Narrative: A key innovation of HI-TAM is its emphasis on psychological safety. By addressing user concerns such as fear of job displacement or scepticism toward AI creativity, HI-TAM fosters a positive and collaborative environment. Transparent processes and clear narratives reduce anxieties, build trust, and encourage experimentation in human-AI partnerships (Caporusso 2023).

The HI-TAM model highlights dimensions that extend beyond creativity, operationalising what Vinchon et al. (2023) describe as "Co-cre-ai-tivity" (= co-creativity).

Co-creativity is defined as the dynamic interplay between human and AI collaborators, where mutual influence shapes the creative process. This concept transcends traditional creativity models by emphasising shared responsibility for innovation and ethical collaboration. Rather than serving as mere tools, AI systems are envisioned as active partners in the creative process, offering new possibilities while respecting the values and intentions of human creators.

By fostering dialogue, mutual influence, and iterative refinement, co-creativity challenges conventional notions of authorship and artistry, ultimately enriching the creative process. When paired with the HI-TAM model, co-creativity empowers artists to use GenAIs more effectively, facilitating the production of innovative and meaningful artistic content.

11.4.2 New Skills and Techniques

Even though these technologies are more accessible than they used to be, and the general public now has easy access to them, an artist using GenAIs must still master several techniques and practices.

The first, for LLMs and most mainstream models, is the mastery of prompt engineering. Prompt engineering involves crafting precise and compelling messages to generate the desired outcomes (Shah 2024). Although it may seem basic, GenAIs are highly dependent on the prompts they receive, making a deep understanding of how this works extremely important. Prompt engineering is particularly crucial across various media. Whether in writing (e.g., deciding who to portray, how the person expresses, specific elements of language, or the required format such as a poem or play), image creation (e.g., detailing the scene, specifying the type of image, highlighting key elements, or for photorealistic results, specifying the type of camera and lens focal length), video (e.g., describing a scene, a shot, editing, perspectives), or music (e.g., describing the instruments, rhythm types, or chord structures), prompt engineering is indispensable. This skill is especially vital for those using mainstream GenAIs such as ChatGPT or Midjourney.

The second essential skill concerns understanding and utilising GenAIs. This includes knowing "what to use" in terms of models (GANs, Diffusion, Transformer, Multimodal) depending on the desired outcomes. As mentioned in Part 1, each of these models has its unique advantages and disadvantages, offering the artist a vast array of additional tools. This knowledge enables the artist to create content that meets both aesthetic and conceptual criteria while understanding the processes underlying the work (Hertzmann 2025).

The third critical skill for artists using GenAIs is the most technical: programming and coding. With the availability of scientific publications and specific models as open source, it is possible to modify (fine-tune), improve, and adapt their code (often in Python) to suit an artist's specific needs. These skills can also be shared within communities to improve models and enhance the accessibility of local GenAIs (Chi 2024). This technical expertise is often a prerequisite for the next skill.

The fourth critical element involves developing or training AI models by adding specific content. This fine-tuning process allows the artist to focus on a particular art style or period, tailoring outputs to their vision. This process requires not only technical skills but also an "ethical eye" for content curation. Not all content is freely accessible, and managing digital archives becomes as much an ethical act as an artistic one (Darda et al. 2023). This necessitates an understanding of copyright laws and the traceability of sources to recognise and respect the contributions of original creators.

Finally, the last key aspect concerns the artist's sensitivity to what GenAIs produce. This requires calling upon human artistic sensibility and aesthetic judgment to explore and express complex emotions through AI-generated art. Additionally, cultural knowledge and broader critical skills are essential. GenAIs are inherently reproductive models, reflecting the content they are trained on and carrying biases embedded in that data or their programming (Epstein et al. 2023). To limit those biases, you can, for example, curate your training dataset with diverse, high-quality and representative data. Artists must exercise great care in using GenAIs to ensure that the content is aesthetically pleasing, culturally sensitive, and as free of biases as possible.

11.5 How to Define the Next-GenAI-Rtist?

Contrary to traditional artists who work with tangible materials like paint, clay, or film, Next-GenAI-rtists utilise data as their primary raw material. Data serves as the foundational input for generative models, shaping the outputs and defining the scope of creative possibilities. This shift not only alters the artistic process but also redefines the role of the artist from a sole creator to a curator and manipulator of datasets. By selecting, fine-tuning, and iterating on vast repositories of information, Next-GenAI-rtists craft new visual, auditory, and textual experiences that are deeply rooted in digital culture (Vinchon et al. 2023). The Next-GenAI-rtist is characterised by a unique synthesis of human creativity and advanced computational tools, redefining traditional boundaries of art through their innovative practices. This artist archetype can be described as follows:

1. **Technological Proficiency**: Mastery of diverse generative AI models, such as GANs, diffusion models, and transformers, to produce novel and high-quality artistic outputs.
2. **Artistic Vision**: An ability to conceptualise and express complex themes, emotions, and narratives, integrating AI outputs into cohesive artistic expressions.

   Ethical Integrity
3. A strong commitment to addressing biases in AI, respecting intellectual property rights, and ensuring cultural and ethical sensitivity in their work.

4. **Collaborative Spirit**: Engagement in dynamic co-creative processes with AI, where iterative feedback and dialogue between human intuition and machine output enhance the creative outcome.
5. **Skillful Adaptability**: Proficiency in coding, prompt engineering, and fine-tuning AI systems to align with specific artistic objectives, combined with a capacity for ongoing learning and adaptation to evolving technologies.

Some examples of AI-rtists:

Rootport's AI-Generated Manga

An exemplary case of Next-GenAI-rtistry is Rootport's "Cyberpunk: Peach John," the first manga entirely generated through a collaboration with Midjourney. Over six weeks, Rootport used prompts to create visual elements, selecting and refining outputs to align with his narrative vision. The manga demonstrates how AI can enable artists without traditional skills, like drawing, to produce complex and compelling works (Vinchon et al. 2023).

Paragraphica: A Camera Without a Lens

A striking innovation in AI-driven creativity is the Paragraphica, an AI-powered camera that generates imagery without a physical lens. Instead, it uses location data, weather conditions, and historical information to compose descriptive representations of a scene. This fusion of AI and photography challenges traditional perceptions of visual art, transforming how moments are captured and interpreted. The Paragraphica exemplifies the seamless integration of technology and artistry, pushing the boundaries of what tools can achieve in creative domains (Bazin 2023).

11.6 Conclusion

As the Next-GenAI-rtist redefines the boundaries of creativity, this chapter underscores the necessity of grounding future studies in the tangible practices of these artists. Rather than relying on abstract theory, research must focus on how GenAI is being employed in real-world artistic processes, exploring the nuanced interactions between human intention and computational output. By embedding the study of generative art in the lived experiences of its creators, we can better understand the opportunities and challenges it presents.

A critical area for future research is the ethical framework surrounding GenAI in the arts. As artists increasingly depend on datasets to fuel creativity, issues of bias, copyright, CO2 production, and cultural sensitivity demand robust examination. Establishing guidelines for the responsible use of generative models is essential to ensuring that the field evolves in a way that respects both individual and collective artistic legacies (Darda et al. 2023). Additionally, promoting transparency about AI's role in the creative process will be vital in fostering trust and recognition for human ingenuity within these collaborations.

Opening the conversation on the legitimacy of AI-generated art, Hertzmann (2025) argues in "Generative Models for the Psychology of Art and Aesthetics" that all AI-generated art is ultimately human-made. This perspective shifts the discourse from questioning the authenticity of AI-generated works to celebrating their place within the broader artistic continuum. Recognising AI art as art opens new avenues for appreciating the hybrid intelligence that defines this era of creativity.

The work of artists like Rootport with AI-generated manga drawings and innovations such as the Paragraphica camera exemplify the extraordinary potential of this synthesis. These advancements not only challenge traditional notions of artistry but also inspire critical questions about the future of human–machine collaboration. By embracing these developments, we can cultivate a more inclusive and dynamic understanding of creativity, ensuring that Next-GenAI-rtists contribute to a richer, more ethically conscious artistic landscape.

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Chapter 12
Augmented Creativity: Augmenting the Archaeological Imagination with Generative AI Tools

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Abstract This chapter addresses the complexity of visual rendering of the past with software tools using Generative AI to fill in a large volume of data not found in the archaeological record, data that may come from ethnographic studies or archaeological experiments, as a result of a collaborative human–computer process using prompts. The interior of a Chalcolithic house (5th millennium BC) in Southeast Europe was reconstructed with traditional 3D modelling methods for the understanding of prehistoric daily life. In the virtual reconstructions, a series of objects generated by Generative AI tools based on publicly available ethnographic and experimental archaeological data (texts, images) were positioned as augmentations of virtuality. During this research, two commercial Generative AI tools, one of general purpose and one dedicated to image generation, were used for experimentation and evaluation. The results of the present research dedicated to the Augmented Creativity concept are shared through this chapter with the purpose of proposing the augmentation of the archaeological imagination and creativity with the use of Generative AI tools.

Keywords Ethno-archaeology · Experimental archaeology · Generative AI · ChatGPT · Adobe Firefly · Augmented creativity

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12.1 Introduction

Two decades ago, to study prehistoric construction methods (Creangăet al. 2010) and experiment with the way of life in Chalcolithic Southeast Europe, a series of wattle-and-daub houses were built at Vădastra (Gheorghiu 2001a), a village near the Danube in Romania. This archaeological experiment lasted almost two decades.

The Chalcolithic (millennia VI-V BC), which succeeded the Neolithic, was a period of complexity in the lives of farmers-livestock breeders in South-Eastern Europe (Gheorghiu 2001b). This period led to the emergence of large fortified tell-type settlements in the area south of the Carpathian Mountains and to a sophistication of material culture, not to mention quality ceramics and exotic objects acquired through long-distance trade.

Unfortunately, from the entire inventory of a household, only those objects whose materiality has stood the test of time have been preserved, and those made of organic materials have disappeared completely. We can imagine the existence of sophisticated textiles or containers made of vegetable material when we analyse the decoration of ceramic vessels because prehistoric ceramics are skeuomorphic, being a material that copies shapes or decorations from more valuable materials.

This chapter presents research during which the interior of a Chalcolithic house, reconstructed for the understanding of prehistoric daily life, was augmented with content generated with two Generative AI (GenAI) tools and improved through human–computer collaboration using textual prompts and several iterations. During this research, the authors used two commercial Generative AI tools, Open AI ChatGPT, as a general-purpose tool and Adobe Firefly as a dedicated tool, for image generation, experimentation, and evaluation. The authors' research had the purpose of allowing the augmentation of the archaeological imagination and creativity, addressing the complexity of the Past with AI-based content generation creativity.

The chapter is organised into several sections that will describe the general context and objectives of the research work, virtual archaeology and the use of AI, a survey of the GenAI technologies and tools, highlighting the ones for image generation, details and results of the experiments conducted with two selected tools, chatGPT-4 integrated with DALL-E and Adobe Firefly, comparisons between the two GenAI tools that were used, and conclusions.

12.2 Experimental and Ethno-Archaeology

Through archaeological experiments (Hansen 2014), a reproduction of various ceramic objects and wattle-and-daub houses was made in Vădastra, starting from the data provided by archaeological remains, creating a first stage of immersion in the reality of the past (Gheorghiu 2015, 2024) (Fig. 12.1).

In the case of setting the reconstructed houses, it was possible to correctly position those architectural elements and objects of current use, such as walls, support pillars,

Fig. 12.1 Reconstructed Chalcolithic house by D. Gheorghiu and students, Photo by D. Gheorghiu

oven, benches, baskets of twigs covered with clay, vertical looms with clay weights, as well as some decorations on the walls. But we should not imagine a Chalcolithic household interior so simplistically (Fig. 12.2).

At this stage of the reconstruction, the archaeological imagination (Shanks 2012; Gheorghiu 2020) could be augmented with another type of archaeological approach based on analogy (Lewis-Williams 1991), namely ethnoarchaeology, which uses

Fig. 12.2 Interior of the reconstructed house by D. Gheorghiu and students, Photo by D. Gheorghiu

Fig. 12.3 **a, b** 3D reconstruction of the interior of the house by Valentina Olteanu

ethnographic documentation to understand constructive processes or to visualise objects made of organic materials that have not left traces in the archaeological record. However, sometimes the risk arises, such as this type of "basic comparative method between ethnographical situations and the past might paralyse [the] scientific imagination" (Beyres and Pétrequin 2016, p. 5).

Therefore, in the present case, the use of ethnographic models has been made with caution, limiting itself only to some types of objects for which the construction technology represents a cultural universal, such as containers made of woven vegetable fibres (see Basketry 2011) or textiles, whose existence can be inferred by decorative patterns, respectively some skeuomorphs, incised or painted from ceramic vessels.

12.3 Virtual Archaeology

If accurate reconstructions are not possible, virtual reconstructions can be used (Molina Salido 2018; for an extended bibliography, see Sequeira and Morgado 2013), which can allow the generation of different models, which can be inspired by archaeological and ethno-archaeological experiments and represent a solution for augmenting the spatial experience by immersion in virtual spaces (Gheorghiu and Ştefan 2020a, b; Gheorghiu et al. in press). A virtual space allows, for example, studies of lighting (Gheorghiu 2014, p. 216) and reconstruction studies with different variants of objects.

For one of the houses built in 2003, which was demolished, several 3D models inspired by physical reconstruction were made, the most recent being used in the present work (Fig. 12.3a, b).

12.4 Representation Versus Evocation

Prehistoric archaeology, when trying to approach complex subjects such as human habitation, must take into account the following situations: if it tries to represent reality exactly, it will have to reduce the data that will be presented cautiously, and if it uses evocation procedures, it will approach "vague" topics such as the atmosphere or ambience of the dwelling, avoiding the exact illustration of the shape of the objects.

Evocation acts on the imagination and creativity of the act of archaeological interpretation, allowing nuanced interpretations when the archaeologist returns to his archaeological record analysis.

12.5 Generative AI in Archaeology

A tool for evocation in archaeology can be offered by GenAI tools (for an extended bibliography, see Banh and Strobel 2023), where AI has gradually begun to be used, especially with the public launch of ChatGPT in 2022, which has made AI accessible to large groups of specialists. They either just evaluated GenAI and its potential for archaeologists (Cobb 2023) or used it to generate complex archaeological illustrations representing alternative interpretations (Magnani and Clindaniel 2023), perform text analysis (Tenzer et al. 2024), reconstruct damaged ancient coins (Altaweel et al. 2024); analyse aerial and satellite images to detect archaeological anomalies (Ciccone 2024), or to reconstruct archaeological vessels (Cardarelli 2024).

GenAI will facilitate humanistic research, in particular archaeological research, in the future, joining AI research so far only available to computer science specialists (Stokel-Walker and Van Noorden 2023), enabling the use of natural language for scientific exploration (Ciccone 2024). GenAI will democratise the imagination of the archaeological past (Magnani and Clindaniel 2023).

Digital reconstructions can be augmented by the use of AI, which allows the generation of a number of variants of relatively culturally correct solutions (extracted from an even more significant number of unbelievable combinations or *hallucinations*; see Kaplan 2024; Ciccone 2024), which can have a creative effect on the imagination of the archaeologist or the public trying to imagine the complexity of the past. In other words, GenAI can augment human creativity and, consequently, imagination, allowing for greater sensitivity in evoking the reality of the Past.

12.6 Research Objectives

Because the wattle-and-daub houses built at Vădastra disappeared over time (some as a result of pyro experiments of deconstruction; Gheorghiu 2014), a series of 3D models were made in which the dimensions and textures of the materials from archaeological records and physical reconstructions were respected. These models proposed the use of GenAI tools to achieve a setting as close as possible to the complexity of the one in Chalcolithic.

In an attempt to get closer to the minds of prehistoric humans (Lewis-Williams and Pearce 2005, p. 10), taking into account that "there is an obvious risk of ethnocentrism in archaeological interpretation, if reference is made solely to western cultural prejudices" (Beyres and Pétrequin 2016, p. 5), an attempt has been made to put together two worldviews, the Western one, which perceives objects separately, and the Far Eastern one, which perceives space in the form of continuous masses of matter (see Nisbett 2003).

Therefore, in order to evoke the complexity of Chalcolithic habitation, it was decided to experiment with two GenAI tools that evoke perishable organic objects and materials, invisible in the archaeological record, one to generate separate objects, and the other, possibly organic and inorganic materialities generated together in various visual compositions.

From this perspective, the chapter proposes an approach analogous to the ethnoarchaeological one, using two GenAI tools to collect and compose ethnographic and archaeological data from a multitude of cultures. ChatGPT-4 integrated with DALL-E-3 was used to generate objects, while Adobe FireFly was used to generate environments inspired by traditional and prehistoric cultures.

12.7 Generative Artificial Intelligence Technologies and Tools

12.7.1 Technologies in the Context of AI

Generative artificial intelligence (Generative AI, GenAI or GAI) is a subdomain of AI. It is related to machine learning, specifically to deep learning, which uses complex statistical models based on neural networks for learning, pattern recognition and self-improvement, according to the human brain model. After a pre-training process with a massive volume of public data, followed by a fine-tuning process called Reinforcement Learning from Human Feedback (RLHF), GenAI tools can generate content—texts, images, video, photos, music, and even programming code, receiving from the user as input a textual description or instruction, called prompt.

The neural models used by GenAI have evolved, the best known being Generative Adversarial Networks—GAN (Gefen 2023, p. 27; Mazzone and Elgammal 2019), Large Language Models—LLM and Transformers (Cardarelli 2024) for GenAI text-to-text.

LLMs are intended for Natural Language Processing (NLP) along with Transformers, with the difference that Transformers represent a modern and highly efficient GenAI architecture, used not only for processing text but also images (Magnani and Clindaniel 2023).

Google initially proposed Transformers in 2017 (Gefen 2023, p. 26). Still, they were taken over and improved by the company OpenAI in the version chatGPT-3 by "extract[ing] patterns from sequences 30 × longer than possible previously".[1] Transformers are "a set of neural networks that consist of an encoder and a decoder with self-attention capabilities"[2] and reinforcement learning techniques, with the help of which the meaning of words is set in a broader context, allowing the model to understand the complexity and nuances of human language (Gefen 2023, p. 26). The current version of Sparse Transformers is a "deep neural network which sets new records at predicting what comes next in a sequence—whether text, images, or sound".

Other GenAI tools are in the *text-to-image* category and use diffusion models, the best known of which is Stable Diffusion, which is a "latent text-to-image diffusion model capable of generating photo-realistic images".[3]

As of 2020, GenAI tools have been released publicly, initially as assistant bots, such as Microsoft Copilot or Google Gemini; in 2021, DALL-E for image generation (Ramesh et al. 2021; OpenAI[4]); in 2022, ChatGPT Open AI (ChatGPT[5]; OpenAI[6]), as well as Midjourney[7] for image generation; in 2023, Adobe Firefly by which AI is integrated into Adobe's creative suite.[8]

12.7.2 ChatGPT

ChatGPT is a "chat-oriented" Generative Pre-trained Transformer implemented by the company Open AI. It leverages a combination of advanced technologies and models such as NLP, machine learning, neural networks, LLM, and Transformers, which are abbreviations in the name of the tool.

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1 https://openai.com/index/sparse-transformer/

2 https://aws.amazon.com/what-is/transformers-in-artificial-intelligence/

3 https://stablediffusionweb.com/

4 https://openai.com/index/dall-E-3/

5 https://chatgpt.com/

6 https://openai.com/blog/chatgpt

7 https://www.midjourney.com/

8 https://www.adobe.com/ro/products/firefly.html

ChatGPT-4 is the latest version, which takes "image and text inputs and produces text outputs",[9] but also images by its integration with DALL-E-3.[10]

The ChatGPT training stage employed dialogue-based conversations; this human-like way of working is also how the ChatGPT is being used, i.e. by means of exchanges of user prompts and chat replies.

ChatGPT is a configurable tool; the user has the possibility to switch between different models: GPT-4o, the default model, GPT-4 mini, GPT-4, and GPT-4o, which is best for complex tasks.[11] GPT-4o is the most comprehensive model for reasoning and generating real-time multimedia content ("o" stands for "Omni"): audio, vision, and text.[12]

12.7.3 From ChatGPT to Custom GPT

LLM encompasses "parameterised knowledge" gained through the pre-training process, which is "frozen" from time to time to allow the generation of content that relates to a general domain, from casual conversations to literature, philosophy, science, and technology. Retrieval-Augmented Generation (RAG) is a technique for enhancing the accuracy and reliability of GenAI models with facts fetched from external sources.[13] With the help of RAG, LLM models can be trained in specific areas.

The chatGPT-4 version gives users who pay for a license the opportunity to create custom versions of ChatGPT, called GPTs, using extra instructions and knowledge provided by the user in the form of text, descriptions or images as exemplifications.[14] From a technical point of view, custom GPTs in OpenAI's framework do not use the traditional RAG architecture but only the concept, by the fact that it can be supplied from external sources, so-called "source knowledge", to improve the generation of content in that field.

Depending on the desired results, custom GPT can be configured from Settings with the following features: web browsing, DALL-E image generation, or advanced data analysis.

[9] https://doi.org/10.48550/arXiv.2303.08774.

[10] https://openai.com/index/gpt-4/; https://openai.com/index/dall-e-3/

[11] https://help.openai.com/en/articles/7864572-what-is-the-chatgpt-plus-model-selector.

[12] https://openai.com/index/hello-gpt-4o/

[13] https://blogs.nvidia.com/blog/what-is-retrieval-augmented-generation/

[14] https://openai.com/index/introducing-gpts/

12.7.4 Adobe Firefly

Adobe Firefly is a text-to-image GenAI online tool developed to enhance creativity within Adobe's creative suite applications, such as Photoshop, Illustrator, and Adobe Express. By simplifying content creation, Firefly allows users not only to generate images but also to modify an image (add or remove elements), create text effects, or even video content, through natural language prompts.[15]

Adobe Firefly takes a single request under the form of a user prompt and produces an image based on it. Users can further refine their prompts by adding new details or requirements, such as the desired style, composition, and specific elements. It uses advanced NLP models to interpret the prompt and transform textual details into relevant visual features.

Adobe Firefly Image 3[16] is the last version with improvements such as enhanced text prompt interpretation, leading to more accurate images, refined details, and complex scene generation. Firefly Image 3 also integrates "Structure Reference" and "Style Reference" features, which allow users to maintain the composition or stylistic qualities of a reference image without repeated prompt adjustments.[17]

"Composition Reference" is a compelling feature that allows users to create images closer to the layout or structure of a reference image. The "Composition Reference" option enables users to upload an image with a desired composition and then generate multiple variations that replicate the arrangement of the elements in the image while incorporating new visual elements specified in the prompt. This process involves analysing key structural details, such as outlines and spatial relationships between elements, which Firefly uses to maintain the original composition in each variation generated.[18]

12.7.5 Prompt Engineering

Not all current text-to-image generation models can learn from textual descriptions, so users resort to what is called "prompt engineering", an iterative process of creating efficient prompts.[19]

The concept of prompt engineering has a unique role, both in the training and the usage process. The latest version of DALL-E-3, with which ChatGPT integrates, has improved this feature, respectively, "understanding of nuanced prompts".[20]

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15 https://www.adobe.com/ro/products/firefly.html

16 https://www.cgchannel.com/2024/04/adobe-launches-new-image-3-ai-model-in-firefly/

17 https://news.adobe.com/news/news-details/2024/adobe-introduces-firefly-image-3-foundation-model-to-take-creative-exploration-and-ideation-to-new-heights.

18 https://blog.adobe.com/en/publish/2024/03/26/the-road-ahead-responsible-innovation.

19 https://help.openai.com/en/articles/6654000-best-practices-for-prompt-engineering-with-the-openai-api.

20 https://www.assemblyai.com/blog/how-dall-E-2-actually-works/

12.8 The Concept of Augmented Creativity

The GenAI tools are able to support human creativity in creating content based on question/answer dialogues. Their use is, therefore, a process of augmenting human creativity through the collaboration between man and technology. The use of GenAI should not be seen as a substitute for human creative processes but as a support for productivity by simplifying repetitive or overly complex tasks and as a support for experimenting with various ideas. Being trained from a vast source of information, GenAI tools can help find explanations, verify theories and interpretations or quickly offer multiple solutions to a described problem.

The interactive way of working can help both the user and GenAI tools to refine and enrich information using the most convenient method, that of dialogue and natural language. For example, starting from an entered prompt, ChatGPT reproduces the text of the prompt in reply but in a form developed based on its knowledge.

In the present chapter, the augmentation of creativity with the help of AI aims to function as a stimulus to augment the human imagination in archaeological research.

12.9 Experiments with ChatGPT to Visualise Containers Made of Organic Materials

The authors conducted a set of experiments with ChatGPT-40, the version for image generation, through which they asked ChatGPT to generate images representing Chalcolithic containers made of vegetal materials inspired by the shapes of ceramic vessels.

The experiments carried out did not have satisfactory results, the resulting objects having a high degree of unrealism both historically and functionally. For this reason, a custom GPT was created and trained with descriptions and the images provided by the authors as examples.

12.9.1 Creating and Training a Custom GPT Named "Prehistoric Containers"

For the present research, the authors used the ChatGPT Plus subscription that allowed them to create a custom GPT[21] with information from the field of archaeology, and leverage DALL-E-3 for visual content generation. It was named "Prehistoric Containers" to indicate its specialisation in organic prehistoric containers inspired by Chalcolithic pottery[22] (Fig. 12.4).

[21] https://chatgpt.com/create

[22] https://chatgpt.com/gpts/editor/g-IyogiHfw8.

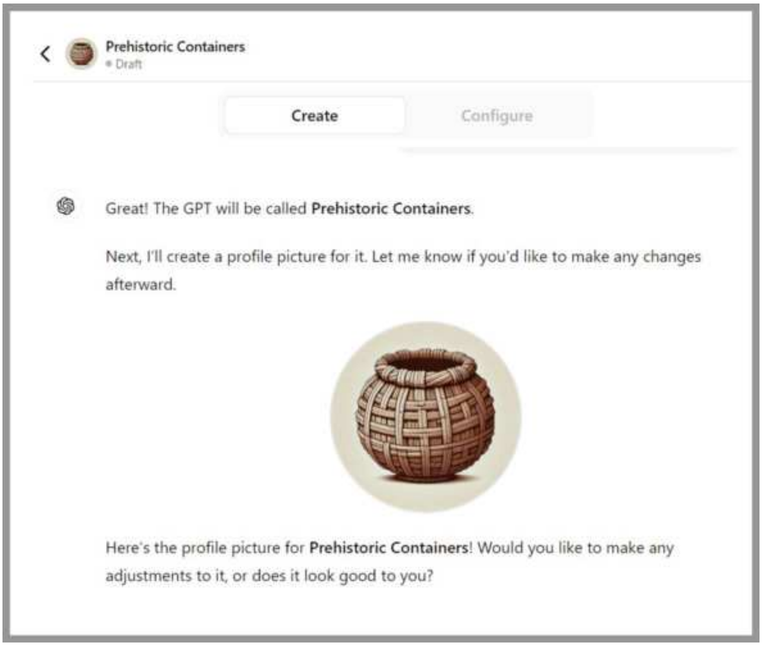

Fig. 12.4 Custom GPT "Prehistoric Containers"—profile section. Screen capture by the authors

Several series of dialogues related to the shape and the types of plaiting of the twig baskets were carried out in order to refine the images generated.

After comparing the resulting series, custom GPT was configured as follows: web browsing disabled; DALL-E image generation enabled; advanced data analysis enabled, the latter being necessary to allow GPT to learn from image examples.

12.9.2 Training with Instructions

The first stage of specialisation of custom GPT in the field of ethno-archaeology was carried out with the help of prompts. These instructions were gradually refined, using prompt engineering principles, in an attempt to provide as accurate information as possible about the content of the images generated by custom GPT, inspired by the work experience of one of the authors with ceramics and wicker basketry and by the archaeological documentation consulted. The role of these instructions was to receive information on the skeuomorphic forms of containers made of organic, perishable materials analogous in shape to ceramic containers.

An example of the conversation regarding the improvement of the forms proposed by the custom GPT is in the image Fig. 12.5.

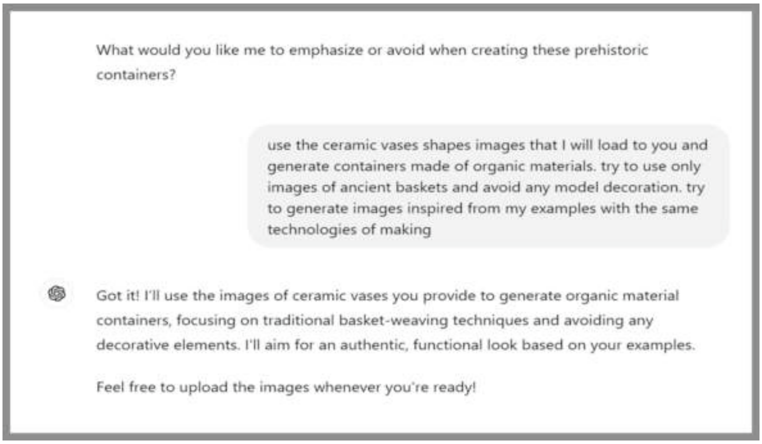

Fig. 12.5 Custom GPT "Prehistoric Containers"—training with user instructions. Screen capture by the authors

12.9.3 Training with Images

The second stage of specialisation of the custom GPT in the field of ethno-archaeology was carried out with the help of images provided as examples because the results were capping with prompts.

For this type of training, a series of images of replicas of Chalcolithic objects made experimentally were uploaded to custom GPT to provide a visual description of ceramic vessels (Fig. 12.6a–c), together with images of objects made of vegetal materials, collected from traditional peasant societies (Fig. 12.7a, b).

Furthermore, details of plaiting (Fig. 12.8a, b) from different cultures were uploaded as examples to custom GPT, in addition to the ethnographic images of containers made of plaited twigs or leaves of rushes.

An example of the conversation for the uploading of images with examples of prehistoric vessels (Fig. 12.9) and textures of vegetal plaiting (Fig. 12.10).

Fig. 12.6 **a**, **b**, **c** Replicas of Chalcolithic vases made by students and villagers. Photo by D. Gheorghiu

Fig. 12.7 **a**, **b** Baskets made of plaited vegetal fibres and twigs. Photo by D. Gheorghiu

Fig. 12.8 **a**, **b** Details of plaiting. Photo by D. Gheorghiu

On several occasions, slight changes to the images generated by custom GPT were requested in order to be introduced into the 3D environment designed by the authors.

By using "No Internet browsing", the result provided by ChatGPT was relatively satisfactory this time (Fig. 12.11):

12.9.4 Using Custom GPT

After training and testing, the custom GPT "Prehistoric Containers" was published, with the possibility of being accessed only by the research team or for demonstration purposes using a link.[23]

[23] https://chatgpt.com/gpts/editor/g-IyogiHfw8.

Fig. 12.9 Custom GPT "Prehistoric Containers"—training with images of prehistoric vessels. Screen capture by the authors

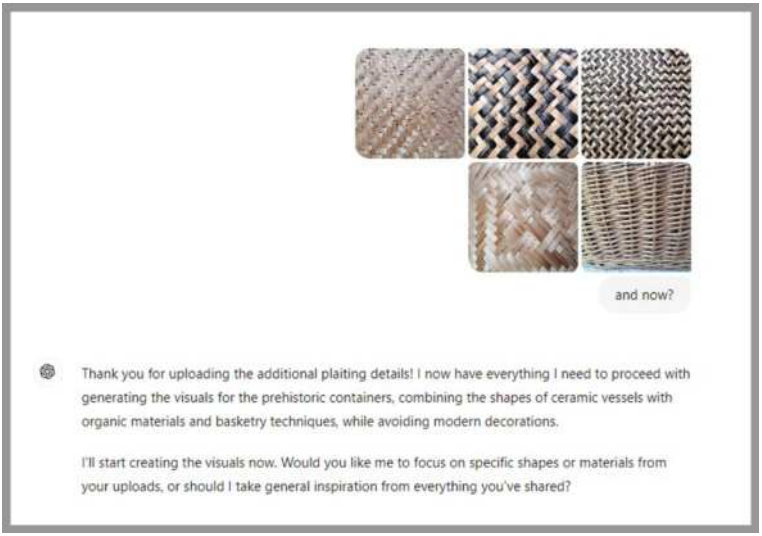

Fig. 12.10 Custom GPT "Prehistoric Containers"—training with images of vegetal textures. Screen capture by the authors

Fig. 12.11 Results from custom GPT training with instructions and images. Screen capture by the authors

In the custom GPT settings, the following "conversation starters" are proposed as examples of prompts, that were kept by the authors, considering that AI already optimised these: (a) What would a plaited vegetal fibre container look like?; (Fig. 12.12) (b) Suggest a shape for a Chalcolithic basket made of twigs; (Fig. 12.13) (c) How would prehistoric vegetal materials be combined to form a household container? Each of these prompts proposed by the custom GPT has been experimented to test the results.

Fig. 12.12 Image generated by custom GPT at the user prompt "What would a plaited vegetal fibre container look like?" Screen capture by the authors

Fig. 12.13 Image generated by custom GPT at the user prompt "Suggest a shape for a Chalcolithic basket made of twigs". Screen capture by the authors

12.10 Experiments with Adobe Firefly

12.10.1 Prompts

A set of 3D images of the reconstruction was proposed, and Firefly was asked to generate images of the interior of the prehistoric dwelling, starting from prompts where it was initially requested: "A high definition of the interior of a Neolithic/Chalcolithic household, realistic, high quality, high resolution, clear style". Later in the initial prompt, more information was added: "A high definition of the interior of a Neolithic/Chalcolithic household, the interior space is separated with a fence between the places for sleep and animal corral, realistic, high quality, high resolution, clear style".

Since the results were quite unrealistic, using the Composition Reference option, an image of the 3D reconstruction was loaded with the following prompt: "A high definition of the interior of a Neolithic/Chalcolithic wattle-and-daub household, the interior space is not very high; it is separated with a fence between the places for sleep and an oven, and an animal corral, realistic, high quality, high resolution, clear style".

Then, it followed another prompt accompanied by another reference image of the 3D interior: "A high definition of the interior of a Neolithic/Chalcolithic wattle-and-daub household, using the red and white pattern on the wall to decorate the rest of the house, realistic, high quality, high resolution, clear style" (Fig. 12.14).

The prompts insisted on the generation of the built space of the house in the form of a whole, composed of objects and materials inspired by 3D reconstruction. The images generated proposed a mix of textiles and wickerwork from various non-European cultures, insisting on the materiality of the earthen construction.

Fig. 12.14 Image generated by Adobe Firefly, Screen capture by the authors

Although some of the proposals were imaginary (i.e. non-functional), they proposed a humanised atmosphere of prehistoric habitation.

The images generated, respecting the shape of the proposed space, offered a “living” image of the living space, with coloured textiles and vegetal plaiting positioned on the walls and offering architectural details that had not been experimentally made.

From a number of twelve images of the interior of the Chalcolithic house generated by Firefly, six images were selected in which the materiality proposals of the household objects, seen from the perspective of ethnography and archaeological experiments, had a greater degree of reality. These images were selected by outlining in Photoshop (Fig. 12.15a, b) those parts with the highest degree of realism that stimulated the archaeological imagination.

12.11 Research Results: Discussion

12.11.1 Augmented Creativity: Augmenting the Quality of Archaeological Imagination

Following the results obtained with the two AI tools (OpenAI ChatGPT and Adobe Firefly), those of producing objects and environments, which acted upon the scientific and artistic imagination of the authors, the next phase was that of human creation, involving an experimental archaeologist, a graphic designer specialising in photography and image manipulation and an interior designer.

Fig. 12.15 **a**, **b** Highly realistic detail selection, images generated by Adobe Firefly, interventions on images with Adobe Photoshop by M. Moțăianu. Screen captures by the authors

A first level of augmentation of the archaeological imagination had been achieved with the modelling of the 3D space of the house, designed according to the measurements from the archaeological excavation and from the experimentally constructed replica, to which were added the setting ideas selected from the proposals generated by Firefly, respectively, the textiles with skeuomorphic decorations positioned on the walls, ceramic vessels, rustic wooden furniture, and a plastic texturing of the clay surface of the walls.

A second level of imagination augmentation was to introduce the objects selected from those generated by ChatGPT into the image captured from the 3D reconstruction. With the help of Photoshop, the most famous image editing program, images of the plaited twig containers copied from the ceramic vessels were added and positioned in the 3D image (Fig. 12.16), thus reconstructing a correct flow of use of the room space.

Although Photoshop works with two-dimensional images, images of the plaited twig containers have been integrated using light and shadow effects and colour effects to adapt the added images to the atmosphere of the image generated in 3D. With the help of masks that cover part of the images of the objects, a three-dimensional effect was created, making the added objects integrated as naturally as possible into the space of the 3D reconstruction.

Through the resulting images, both day and night moments were evoked, in which the light on the textures of the materials and their colours created a timeless atmosphere (Fig. 12.17a, b).

The result of this mix between two GenAI tools, controlled by human experience, which represents an approach of Augmented Creativity, materialised in a series of images that evoke the inhabitation of the Chalcolithic period—a much more humanised and realistic evocation of inhabitation from the anthropological perspective (Fig. 12.18a, b) than the first variants imagined when the wattle-and-daub replicas were built.

Fig. 12.16 3D Reconstruction by V. Olteanu. Editing and intervention on images with Adobe Photoshop by M. Moţăianu. Screen capture by the authors

a b

Fig. 12.17 **a**, **b** 3D Reconstruction by V. Olteanu. Editing and intervention on images with Adobe Photoshop by M. Moţăianu. Screen captures by the authors

12.11.2 Comparisons Between the Two GenAI Tools

The use of GenAI tools must be very well anchored in the intended purpose by correctly understanding the benefits and their limits. In the case of the present research, they served to increase imagination and creativity in a specific field, that

Fig. 12.18 **a**, **b** 3D Reconstruction by V. Olteanu. Editing and intervention on images with Adobe Photoshop by M. Moţăianu. Screen captures by the authors

of archaeology. The results were refined following several iterations and levels of creativity by combining the generated content with the authors' multi-disciplinary research team resources and interventions.

One can see that the use of GenAI tools is still in its infancy, and the models are constantly being improved, either by incorporating new data or by improving the algorithms. The training with data is not continuous but in stages, between different versions offered publicly. For example, in the case of ChatGPT, knowledge has been frozen since January 2023, which is why it is considered "moderate by design" (Roose 2022). Still, this limitation is diminished in the case of the latest version of ChatGPT-4 due to real-time access to some web resources. During the work with ChatGPT with DALL-E, the lack of up-to-date information in the field of archaeology was noticed in the case of complex visuals, e.g., household interiors, but these tools were good enough for the generation of individual objects, e.g. prehistoric containers made of vegetal fibres. Being a versatile and configurable GenAI tool, it was possible to introduce "source" (external) knowledge in order to specialise it in the specific field of interest, i.e., prehistoric containers made of vegetal fibres.

On the one hand, Adobe FireFly is more potent for generating complex images, provided it is given explicit, quality prompts and reference images. Adobe FireFly excels in image creation, as it offers several powerful features for instructing related to the composition or style of an image. The "Composition Reference" feature, allowing users to create images close to the layout or structure of a reference image, proved to be a valuable tool for computer-based augmented creativity. On the other hand, the learning process is not very obvious to the user.

In the OpenAI safety notice[24] users are warned about the limits and scope of use by recommending the employing of GenAI only for creative purposes, and deep-fake purposes are discouraged, as well as overuse to avoid "hallucinating inaccurate information…[or] offensive outputs".[25] It is also specified that the data provided by

[24] https://platform.openai.com/docs/guides/safety-best-practices.

[25] https://platform.openai.com/docs/guides/safety-best-practices.

the users is not employed for training and perfecting the models except with their express consent.

Adobe Firefly is also committed to ethical AI principles to ensure that its GenAI models are used responsibly and transparently.[26]

In the case of custom GPT "Prehistoric Containers", a private version was chosen, accessible only to the research team, because its use publicly would imply a responsibility for the quality of the generated images. The images presented in this text as exemplification and training belong to the authors. For better images closer to the reference ones, the OpenAI's recommendation is to upload at least 50 images for each topic to be studied, which was a constraint in the case of the present research.

From the present experience, the authors observed both beneficial effects of GenAI– the rapid generation of images with comprehensive descriptions, in the case of ChatGPT, or the quality of the images generated by Adobe Firefly, but also drawbacks, in the sense that many results were far from expectations and the approach had to be changed. The desired results were obtained by combining the outcomes produced by the two GenAI tools with specialists' experience.

The generated models were integrated into the conventionally designed 3D environment, increasing the creativity process. In this sense, the usefulness of GenAI tools was evident in the case of rendering specific shapes, such as plaiting or wickerwork, whose 3D design is very laborious, while with GenAI, it was possible to quickly obtain 2D images that were integrated into 3D images with the help of Adobe Photoshop.

12.12 Conclusions

The archaeologist's confrontation with the puzzle of prehistoric material culture requires a continuous act of imagination, continuously controlled by scientific methods and tested with the help of experiments and ethnographic analogies. Yet, an act of brainstorming is needed to stimulate the archaeological imagination to generate various interpretations of the Past, and this is where AI can intervene with the multitude of solutions, some bizarre, others relatively realistic, of combinatorial creativity (see Boden 2004), to augment human creativity. The evocation proposed by AI tools is a stimulus for the archaeological imagination, as is the formulation of prompts.

GenAI tools can be used today by specialists without IT or ML expertise through the prompt method. The result generated by AI depends on the quality of the prompt. Prompts were used in two distinct stages: the learning stage and the use stage, in the case of custom ChatGPT and Adobe Firefly. Prompts are questions that a user can ask some GenAI tools to obtain representations of the reality of the past in order to

[26] https://blog.adobe.com/en/publish/2024/03/26/new-structure-reference-capabilities-adobe-firefly-bring-new-level-creative-control-ideation.

create an image of the subject with the help of different proposals offered by GenAI tools.

One can define the information generated by GenAI tools as functioning as a kind of imaginary of computational creativity (see Ivanova 2023) that must be used to evoke particular ideas that inspire the user, in this way increasing human creativity. In the case of the present research, the two GenAI tools generated results that were inspired by a vast human experience, which in turn acted on contemporary human creativity.

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Chapter 13
Artist-Computer Collaboration and the Treachery of AI Images: This Pipe Does not Exist

Carla Gannis

Abstract In this chapter, I embark on a reflective journey through the evolving landscape of human–computer creativity, drawing on over 25 years of experience in digital arts. My essay explores my nuanced collaboration with artificial intelligence as a concept and technology, challenging the traditional view of computers as mere tools and highlighting the dynamic, symbiotic relationship that now exists. Through personal anecdotes, including the development of multimodal projects augmented by Generative Adversarial Networks (GANs) and Large Language Model (LLM) algorithms and through speculative collaborations with my avatar C.A.R.L.A. G.A.N. (Crossplatform Avatar for Recursive Life Action Generative Adversarial Network), I examine the implications of AI in redefining creativity and the potential for a future where art transcends human and machine dichotomies. Addressing the intersection of AI with art, education, and mixed reality, the essay advocates for a critical, intentional approach to AI collaboration, where the unique 'special sauce' of individual perspective and lived experience remains central to the creative process.

Keywords Artificial intelligence · Digital art · Art · Pedagogy · Personal essay · Computational humor · Subverting digital media · Speculative fiction · Speculative design · Machine learning · Conceptual art

13.1 Art in the Future

In 2018, when I was asked by *Artnet* to respond to the prompt, "What Will Art Look Like in 100 Years"[1] I responded with two potential futures.

[1] Artnet; "What Will Art Look Like in 100 Years? We Asked 16 Contemporary Artists to Predict the Future". https://news.artnet.com/art-world/art-in-2218-1296347.

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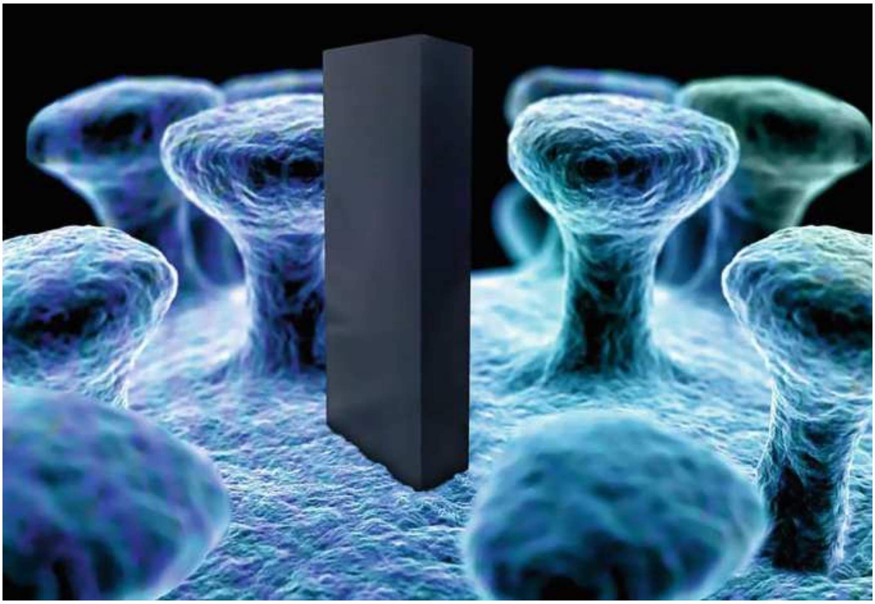

Fig. 13.1 Carla Gannis, *Abstract Compressionism* (2018)

To imagine art in 100 years, I've taken a cue from Marge Piercy's 1976 speculative fiction novel *Woman on the Edge of Time*[2] (and yes, perhaps I'm hedging my bets too). Piercy's protagonist Connie travels to two future 2137s, one where the environment has stabilized, racial and sexual equality have been achieved, and technology is "organically" interwoven into all life on the planet. The other future is far bleaker. Human dignity, clean air, and autonomous thought are commodities only available to the mega-mega-mega-rich. So here goes:

Scenario 01

By 2118, the Pentiumcostal Church has a congregation of billions, all worshiping the Master Algorithm in the Cloud, Godgle. The reigning art movement is Abstract Compressionism (Fig. 13.1). The earth now has a population of about 20 billion people, and over 250 billion machines (now called plastes sapiens), and much of the earth's surface is no longer inhabitable, thus there is very little real estate available for physical works of art. Art is produced and compressed by human "worshipers" using nanotechnology. It is easily viewable to machines and Singularites (cyborgs), but only visible to homo stultus (humans) via exorbitantly expensive virtual simulation implants (as well as church membership fees).

[2] Wikipedia. Woman on the Edge of Time. https://en.wikipedia.org/wiki/Woman_on_the_Edge_of_Time.

Scenario 02

Humans and machines live in harmony in the year 2118. After decades of living underground due to the biohazardous fall out of the GR8 W@R of 2089, the machines have recently stabilized the environment and humans have emerged from their subterranean bunkers, bursting with creative verve. Plein-Air VRainting is all the rage. Every human is an interdataplinary artist now since machines have proven to operate the government and economy more efficiently. Most refer to the Earth as the Art World now, and robots are keenly invested in universal art care for all humans; buying, trading, and selling art via a DNA blockchain system. Interestingly, a radical group of bio-genetically “dehanced” individuals, calling themselves les Fauves deux points zéro, have begun to garner attention. Painting on cave walls, with berries, roots and leaves, they produce strange images of space aliens and unidentified flying objects.

I begin with my bifurcated visions of the future from six years ago to prepare you for an essay written from the point of view of “me”, reflecting on my own journey through an evolving landscape of human–computer creativity, drawing from over twenty-five years as an artist and nineteen years teaching at the increasingly congested – yet never dull – intersection of art and technology.

All thoughts and sentiments, though impacted by millions of interactions, online and off, that I’ve had with friends, colleagues, adversaries, dead and living artists, dead and living intellectuals/theorists/critics, strangers + plants and most definitely cats, are expressly my opinions on art, AI, anatomy, aerobics and astrophysics. Scratch astrophysics, the alliteration ran away in me.

13.2 Does This Essay Exist?

Various websites—”this cat does not exist,”[3] “this person does not exist,”[4] “this chair does not exist”[5]—showcase AI-generated images indistinguishable from “the real thing.” While these simulations can pose serious risks in contexts like photojournalism or global politics, where truthfulness is paramount, the question in visual art—a domain steeped in artifice since its inception—is different. Is an image constructed “from whole cloth” truly a threat to art? Or is the greater concern our diminishing ability to perceive human creative intent within the image?

When an artist employs generative AI to assist in making art, does the fundamental nature of representation as artifice remain unchanged? Despite a deep fascination with theoretical physics, in my day-to-day life, I’m stuck in a Newtonian clockwork universe.[6] I perceive reality and art along a linear timeline where AI emerges *after*

[3] Medium.This Cat Does Not Exist. https://medium.com/predict/this-cat-does-not-exist-d4c594da853c.

[4] ThisPersonDoesNotExist. https://thispersondoesnotexist.com/.

[5] GAN-generated Chairs. https://thischairdoesnotexist.com/.

[6] Wikipedia.Clockwork Universe. https://en.wikipedia.org/wiki/Clockwork_universe.

photography. I'm trying to recall the first time I saw, or maybe I should say, when I first comprehended a photograph, maybe around 1974, when I was 4 years old. My mom had a Polaroid camera and I watched the reality before me emerge as "reality" on a piece of plastic in my hands. From then on I developed a complete trust in photographic representation. While retouching – whether by airbrush or Photoshop – disrupted my faith in a photograph's absolute verisimilitude, I rarely questioned the existence of what it conveyed. Some of this was from experience. I worked in the industry for a famous photographer, and I used Photoshop (sans AI back then) to retouch photographs. I tweaked reality. I made famous women skinnier and scrawny men more muscular, but I knew that it was still them, the humans who had posed in front of the lens a few hours ago. Part of the artifice of my job was to tweak, just so. Nothing about the retouched image could feel uncanny.

With hyper-realistic 3D character animation technologies things have gotten trickier. I remember watching *Final Fantasy*[7] in 2001, where they spent millions of dollars on rendering each strand of hair, while the movements and facial expressions of the characters were lifeless. I watched an old Disney film as a palette cleanser. Today the industry has or is pretty damn close to transcending the uncanny valley. I have been "tricked" by an Unreal Engine Metahuman[8] in a film. But until now, high-definition post-photographic processes required specialized training (the 1000s of hours I have spent learning 3D technologies that now are rapidly becoming an obsolete enterprise!) and you needed access to substantial rendering facilities. With generative AI and cloud computing, creating a "real-looking" photographic representation requires only a few keystrokes.

"Isn't this too easy (or too kitschy) to be art?" a dubious audience asks. Or more troubling to the art faithful: isn't this a soulless endeavour that will culminate in the human, as a maker of meaning, no longer needing to exist?

I don't know, if you read enough on the simulation hypothesis,[9] nothing exists, this essay doesn't exist. Everything we are experiencing right now is an AI simulation.

This essay does NOT exist!

I don't exist. You don't exist. Ah, but for some advocates of this hypothesis, I have found "some people" still exist for them. Elon Musk[10] and his multi-million/billion dollar cronies exist, or if not, they still wield more power and agency in the simulation. In this model of "unreality" it seems Musk and company operate from a point of view where large swathes of the population are NPCs (non-playable characters).

Because this is a personal essay, I'll claim, irrefutably, that this essay exists. Without citation or further proof, I feel it is so.

This essay EXISTS!

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[7] IMDb. Final Fantasy: The Spirits Within. https://www.imdb.com/title/tt0173840/.

[8] METAHUMAN. https://www.unrealengine.com/en-US/metahuman.

[9] Youtube. Rizwan Virk | The Simulation Hypothesis. https://www.youtube.com/watch?v=UHlfe2HE_gQ.

[10] Wikipedia. Elon Musk. https://en.wikipedia.org/wiki/Elon_Musk.

Maybe my claim will be refuted later. Perhaps a prompt engineer in the sky is engaging a system that "I" am, as an algorithmic non-identity, encoded to write an essay about Art and AI in a studio in Brooklyn in 2024 from the point of view of a Gen X artist and educator with brown hair and orange-ish coloured eyes and five-fingered hands, which is absolutely absurd, because everyone knows, up in the sky, that humanoids have six-fingered hands.

13.3 Beyond a Mid Journey

"Mid"[11] in today's popular parlance means something "boring," "not good," "mediocre," "low quality." Last spring I asked the students in my Subverting Digital Media class to produce a "Beyond Mid" midterm project at the halfway point of the semester. They weren't overly amused by my pun, but parsing the connotations of "mid" here seems relevant, as we have reached MidJourney[12] Version 6.1 at the time of this writing, and indeed its image generations feel "mid" like a reprisal of the results in Komar and Melamid's Most Wanted Painting[13] project from 1995. I imagine MidJourney scraping to the very bottom of our internet data like it's a Frosted Flakes[14] box, hungrily crunching away until the sugar coma sets in. In this hallucinatory state, Midjourney provides us with the most wanted AI images it "thinks" we want, and I don't know, is it getting us wrong? Recently, I typed in "God" as a prompt and the result was a bearded white guy resembling JD Vance[15] wearing Ray-Bans[16] and angel wings. Well, it got "me wrong" in that instance. It's on these days that Generative AI feels like a midway circus. Step right up folks, to see a six-fingered Laocoön[17] in the style of Vincent Van Gogh[18] struggling to survive Matrix Sentinels[19] squeezing out of him every remaining standard of taste, aesthetics and ethics.

Still, I firmly believe art can always wrangle free of naive kitsch, pure spectacle and in its winning moments, fascism. As an artist and educator engaging daily in the alternating currents of generative AI, the voltage of my opinions may fluctuate. I might throw shade at implicit bias in an application while still seeing opportunities, often in the prototypes my students are developing, that indicate future cycles of AI

[11] Dictionary.com. https://www.dictionary.com/e/slang/mid.

[12] Midjourney. https://www.midjourney.com.

[13] Dia Art. Koma + Melamid: The Most Wanted Paintings. https://www.diaart.org/exhibition/exhibitions-projects/komar-melamid-the-most-wanted-paintings-web-project

[14] Frosted Flakes. https://www.frostedflakes.com/en_US/home.html.

[15] Wikipedia. JD Vance. https://en.wikipedia.org/wiki/JD_Vance.

[16] Ray-Ban. https://www.ray-ban.com/usa.

[17] Wikipedia.Laocoön.https://en.wikipedia.org/wiki/Laoco%C3%B6n.

[18] Vangoghmuseum. Vincent van Gogh. https://www.vangoghmuseum.nl/en/art-and-stories/art/vincent-van-gogh

[19] Matrix Fandom. Sentinels. https://matrix.fandom.com/wiki/Sentinel.

products will be more versatile and inclusive. Ideally, by the end of this essay, the complexities of human–computer creativity will not be a portend. And just maybe I'll have a neologism to replace "artificial intelligence" which is frankly too generic and narrow to capture the current state of things. So, buckle up in your autonomous vehicle of a human body, and let's hit the road.

13.4 Is This Image Dead or Alive?

I find representation in art is generally a form of artifice—what Magritte described as "a pipe you cannot stuff"[20] when questioned about the inscription "Ceci n'est pas une pipe" in his painting *The Treachery of Images*[21] (1929). Whether, in my experience as a painter (I'll share more on my artistic journey through mediums later in this essay) selecting details to depict reality, or as a photographer framing and cropping a scene, or as a digital artist guiding a generative AI through iterative prompts, each method is put to use to serve the concept floating around in my hippocampus. I am constructing rather than mirroring reality in every case. Through art, I have always made approximations of my lived, dreamed and imagined experiences.

In any survey of Western art history, I'd imagine students still learn that Magritte painted *The Treachery of Images*. I know through heirlooms and photographic sources that pipes in 1928 resembled the one he painted, that people smoked such pipes. Even with Dalí's melting clock in *The Persistence of Memory*,[22] I understand that he is intentionally presenting a hallucination. With AI-generated art, is the threat primarily about deception, or our inability to trace representations back to a primary human author?

In cases of abstraction and non-objective painting, Claude Monet's late works[23] or Jackson Pollock's drip paintings,[24] I still intuitively connect with the artist's hand and intent. In contrast, the AI system has scraped content from billions of human sources. Its method of combining and recombining, no matter how specific and intentful the humans' text prompt may be, often feels inscrutable to me. While a film buff may be able to trace references in a Coralie Fargeat[25] film, recognizing her intentional remixes and mashups, the AI's processes can resist such analysis. Unlike the legibility of artistic appropriation or the readability of an airplane's black box recorder,[26] the

[20] David James Lawton. Reality versus Representation. https://www.davidjameslawton.com/post/reality-versus-representation.

[21] ReneMagritte. The Treachery of Images,.https://www.renemagritte.org/the-treachery-of-images.jsp

[22] MoMA.Salvador Dalí. https://www.moma.org/collection/works/79018.

[23] Fine Arts Museum of San Francisco. Claude Monet. https://www.famsf.org/exhibitions/late-monet.

[24] Jackson Pollock. https://www.jackson-pollock.org/.

[25] IMDb. Coralie Fargeat. https://www.imdb.com/name/nm0267287/.

[26] Wikipedia. Flight recorder. https://en.wikipedia.org/wiki/Flight_recorder.

AI's "black box"[27] seems unknown to me. Or at least this is the popular opinion in most critiques of AI that I have adopted. Thankfully my teaching colleague A.V. Marraccini[28] pushed back on this in a recent conversation and asked "Is it entirely a black box if we know how LAION[29] and CLIP[30] work to some degree? If we can look at the math?" This is a salient point. However, I recognize my limitations in knowing LAION or CLIP or being able to decipher complex math. It reminds me of Thomas Thwaite's speculative design Toaster Project[31] (2009) where he assumes creating a mass-produced object will be a simple task – one that he can complete alone in his studio – only to find that it is a daunting, much more complex endeavour, due to interconnected supply chains and global economics.

Returning to my proposition above, I now realize I probably couldn't read an airplane's black box recorder either. To confirm this, I am googling "how do you read an aircraft's black box recorder?" Ignoring the AI overview,[32] because I really don't want to spread potentially nonfactual information, here is what I have discovered: 1. A blackbox recorder on an airplane is really high-visibility orange and 2. I should be more careful about my assumptions, one needs special equipment to decode the raw files from the recorder before it is turned into graphs.

The question is now, after conversing with A.V. and Google, can I salvage a metaphor I've been saving for the conclusion of this section, on the deadness or aliveness of an AI image? When there are multiple boxes (including a toaster) that I cannot fathom their inner workings, why am I so quick to jump on the bandwagon of critiquing AI systems for their opaqueness, specifically to a non-engineer like me? Perhaps it is because I'm talking about art, where to make it, I don't use a toaster or an airplane's high-visibility orange box. However, I do use AI systems to make art. From this perspective, AI can be like Schrödinger's cat[33] to me (although I'm involving another box I don't fully comprehend) in that we cannot know if the images generated from these models are "dead" or "alive" within the framework of meaning-making and the human imaginarium.

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[27] TechTarget Network. What is black box AI? https://www.techtarget.com/whatis/definition/black-box-AI.

[28] A.V. Marraccini. https://avmarraccini.com.

[29] LAION. https://laion.ai.

[30] Cornell University. DiffusionCLIP.https://arxiv.org/abs/2110.02711.

[31] V&A. The Toaster Project. https://collections.vam.ac.uk/item/O1269559/the-toaster-project-installation-thomas-thwaites.

[32] Google. AI Overviews. https://blog.google/products/ads-commerce/google-lens-ai-overviews-ads-marketers.

[33] Youtube. What is Schrödinger's cat? https://www.youtube.com/watch?v=keiFzYJbfdw.

13.5 A System Whose Inner Workings Are Unfathomable

I took Anatomy for the Artist, a sculpture course, during my BFA. We built the human form from the skeleton up, adding tendons, ligaments, and muscles, and finally "skinning" our creations. No, not skinning like flaying—we applied the skin, layering it over the structure like a 3D artist skins a virtual rig (a digital skeleton) with a polygon mesh and texture.[34] To support myself in these college endeavours, I taught aerobics classes. In order to do no harm to participants I was required to get yearly certification in anatomy and kinesiology. Studying art and movement scientifically meant I could fathom the complexities of physicality. (Unfathomable to me was the suggestion from a male art professor, that if I wanted to be a "real" artist, I should choose stripping over "perky" aerobic instruction).

Upon getting my first computer, I applied the same logic of analysis. I opened the casing and made anatomy drawings of the computers' innards. I'd been trained to understand the mechanics of bodies by studying what was beneath the surface, surely looking at the IDE cables, chips and motherboard would reveal how it functioned in space–time. Having no computer science training, this was a rather futile exercise, but I remember the drawings were quite nice.

While these anatomy lessons showed me the sophistication of the systems that "run" bodies and computers, it didn't appeal to me to apply them to the analysis of human creativity. As an art student attempting to decipher the decisions and impulses behind an artist's works, I wanted to intuit feelings, thoughts, philosophies. I wanted to know if it was pride or rage that led Artemisia Gentileschi to paint *Judith Beheading Holofernes*[35] after giving testimony at her rape trial[36]? Even if I had access to Artemisia Gentileschi's cadaver or to my first, now inoperable, computer, I'd be provided with no revelations on why Gentileschi painted as she did, nor evidence of what I was thinking about art in the 1990s. If I were a forensic anthropologist I might examine Gentileschi's skeleton and deduce that scoliosis influenced her painting stance and thereby choices she made in how and what she painted (this is an idle thought experiment!), and if I were an engineer I could extract data from my old machine that indicated what—and maybe even why I was making art at the time. It's possible to dive into a science vortex to better understand creativity, although my first instinct is to attribute meaning-making to something ineffable, something beyond physical mapping. I comprehend creativity's byproducts in art(ifice).

Returning to the preoccupation I spent an entire essay section trying to articulate earlier, the "deadness versus aliveness" in AI art; the persistent special sauce in

[34] Blender Artists. Creating skin texture directly on mesh. https://blenderartists.org/t/creating-skin-texture-directly-on-mesh/681337.

[35] Le Gallerie Degli Uffizi. Judith Beheading Holofernes. https://www.uffizi.it/en/artworks/judith-beheading-holofernes.

[36] National Gallery UK. Artemisia in her own words. https://www.nationalgallery.org.uk/exhibitions/past/artemisia/artemisia-in-her-own-words.

human-made art to me has always been the distillation of lived and thought experience. Yes, political regimes have "programmed" masses to serve figureheads – monarchs, tyrants, popes and presidents – but, whether through wild expressionism or systematic indexicality, art that has had the profoundest impact on me resists, to what degrees are possible within the culture industry, the tyranny of the mind.

Isn't that the rub today? Working with the affordances of advanced algorithmic systems we can make anything we imagine, and this is stimulating and exciting until we recognize the final results have no pulse. Like the rubber hand illusion[37] (RHI) where one of your hands is hidden inside a box and a lifelike rubber hand is placed on the table lined up with the shoulder of the hidden hand. When your real hand and the rubber hand are stroked simultaneously, you perceptually feel the fake hand is part of your body. I've participated in this illusion, the rubber hand felt real, even though I knew intellectually it was not real, nor alive.

When I'm in the mindset to think of generative AI as a powerful tool, like a magic vacuum cleaner hoovering up digital files, like particle and dust residue from the vast plain of human history –with its motor driving a pitched blade through the items in its "magic bag" AKA blender pitcher to create a whirlpool of mush that eventually takes form based on the direction I push the vacuum in, I'm okay with the "rub". And I know I will keep pushing this "AI vacuum cleaner blender" chimera to assist me in producing mashups and remixes, like I have done since the advent of my time on the internet, not stopping until I feel the images I've produced, with my tool, are alive.

When the rub begins to feel like a shove, is when on a tight deadline to finish an essay amidst a hundred other deadlines my Google calendar is notifying me about, I am tempted to turn to an LLM to expedite my writing process, instead of spending an hour writing an absurd "run on simile" – my mother used to sell vacuum cleaners, so that one was for you, Mom. Ultimately, I'm less fearful of artificial intelligence gaining consciousness and swatting me around like a fly, than my beginning to behave like a fly, buzzing around from task to task increasingly relying on AI to make me more efficient until ultimately my own consciousness is something "artificial", if not dead.

Before I provide further currency on how I believe more nuanced human–machine interactions resuscitate "dead on arrival" AI art, let me cycle through my artist's Id Journey. (Wouldn't that be a great name for a truly self-aware AI application?).

[37] The Guardian. Rubber hand illusion reveals how the brain understands the body. https://www.theguardian.com/science/2016/oct/20/rubber-hand-illusion-reveals-how-the-brain-understands-the-body.

13.6 Id Journey

Growing up in the 1970s and 80s, in a small Southern town, my conception of "fine art" was quite limited (there was no World Wide Web then), however, I was sure that I needed to hone my skills in drawing and painting to achieve my artistic aspirations. I studied painting from middle school through graduate school. As much as I thought at the time this was my medium of choice—it also represented a means of transcending my lower-class origins. Fine art, being a fine artist, was about class mobility, cultural capital, and the weight and sophistication of tradition.

Along with painting, I studied classical piano, teaching seven younger students during my teen years to fund my own lessons. I immersed myself in these traditional arts, perceiving them as proven routes to being a "real" artist with cultural legitimacy.

My father saw a different future. Self-taught in computing and working in the development of nonlinear video editing systems, he took me to my first computer graphics conference in 1985. He would suggest to me, not knowing people like Harold Cohen,[38] Lillian Schwarz[39] and Lynn Hershmann Leeson[40] were already doing it, that one day artists—the fine artist type whose works hung in museums and galleries—would use computers in their craft. I disagreed with my dad at the time and created a partition between my arts and entertainment endeavours. I played hours of video games at my local arcade, and consumed science fiction books and films whenever I had time free from studying the classics. (I'll add that many of those science fiction books are now considered "classics").

Computer graphics conferences revealed more than just a divide between art and technology at the time. They showed me something I would spend my adult life as a female digital creator trying to rectify. At the last two conferences I attended in high school, I was hired as a member of the "team." This entailed me posing in a small set, dressed as a female stereotype—an aerobics queen perched on an exercise bike one year, and Daisy Mae[41] sitting on bales of hay the next. Five monitors were racked above the set and five video cameras on tripods were aimed at me. Men, it was all men then, recorded me as I sat frozen, worrying about my leotard riding up, Vaseline on my teeth to maintain a daylong smile. Yes, I got paid and excused from school, but I was turned into a living diorama for mediation. These experiences imprinted something deep: the persistent objectification of women, whether through traditional art, literature, or emerging creative technologies.

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38 Whitney. Harold Cohen. https://whitney.org/exhibitions/harold-cohen-aaron.

39 Wikipedia. Lillian Schwartz. https://en.wikipedia.org/wiki/Lillian_Schwartz.

40 New Museum. Lynn Hershman Leeson. https://www.newmuseum.org/exhibitions/view/lynn-hershman-leeson-twisted.

41 Lil-Abner.Character Profile: Daisy Mae. http://lil-abner.com/character-profile-daisy-mae-scragg/.

13.7 Show and Tell

I still remember the arguments in grad school between my class cohort of abstract and realist painters, and how I felt a pull towards abstraction, after painting representationally for some time. By the time I quit making (physical) paintings in the mid-90s—I have called my computer-based works digital paintings over the years—my canvases were filled with thick impastoed squares, circles and rectangles. Tiny humanoid symbols (long before emojis and prior to my knowing about emoticons) peeked out at the edges of my rudimentary, geometric forms, but I could no longer make a legitimate argument for painting a figure or any other representational content.

Nonetheless, I come from a rich storytelling tradition. I spent my childhood in North Carolina visiting my Appalachian relatives and listening to them spin yarns and tell tall tales for hours. If I could no longer represent "a human story" in a painting, maybe it was time to move on.

Cinematic photography and surrealistic film appealed to my narrative sensibilities. Two favourites of these genres at the time, Matthew Barney[42] and Gregory Crewdson,[43] produced works that required large casts and crews of people along with exorbitant budgets. I could barely afford to pay for my painting supplies and my studio rent. What I could afford was an internet connection and three "boxes" sitting on the desk in my studio: one, the tower, about half the length and width of my torso, the other a square version of my head with a screen for its face, and the smallest one a handheld device with a lens. I threw away a collection of paintings, ranging from 5 to 7 feet tall, as well as my paint-stained jeans, and I leapt, like an algorithmic Alice, forward and into a virtual box smaller than my human frame. I built my first website, pandorainthebox, to showcase my emerging creations produced with my digital camera and desktop computer. In this pre-Y2K[44] moment of Web 1.0,[45] this post-modern, multi-modal, shapeshifting digital machine offered both financial and conceptual latitude to imagine new worlds – technological realms and beings that might someday be encoded into a new fabric of reality.

The tools I had access to then—commercial applications like Adobe and Macromedia, coding languages such as HTML and JavaScript—weren't embedded with AI algorithms, but the magic of effects and software updates (on discs back then) along with the tragedies of corrupted files and late night cajoles to my computer to not fritz out before the next render was done, introduced me to a new semantic understanding of these versatile and highly mutable tools. I also recognized that the degree of my reliance on hardware and software engineers to produce my art was exponentially more complex than my relationship with traditional art supply manufacturers.

No longer simply relying on the kinesthetics of my own hand for mark making, there was now a daisy chain of events outside of my own body that needed to occur

[42] Guggenheim. Matthew Barney - The Cremaster Cycle. https://www.guggenheim.org/teaching-materials/matthew-barney-the-cremaster-cycle.

[43] Gagosian. Gregory Crewdson.https://gagosian.com/artists/gregory-crewdson/.

[44] Wikipedia. Year 2000 problem. https://en.wikipedia.org/wiki/Year_2000_problem.

[45] Techopedia. Web 1.0. https://www.techopedia.com/definition/27960/web-10.

for a stroke to appear on a virtual canvas represented on screen. It was easier to imagine the process back then when it was slower. Now to think about my computer arts practice and significantly expanded toolset, is like thinking about when my eyes blink, or when I'll be taking my next breath.

In the early aughts I got in the practice of giving attribution to Photoshop, Flash, etc.… in the credits of my moving image and interactive works, akin to a filmmaker, but not something I'd ever done for Winsor & Newton oil paints[46] during my painting days. And as an artist who draws deeply from the well of speculative fiction, I was beginning to cosplay in my videos and interactive works the possibilities and complications emerging between more advanced human and machine interactions.

In 1998, I wrote "The Story of Sister Gemini,"[47] featuring a cyborg princess: half AI, half mixed-lineage human, suffering from amnesia. She makes her way to the City and discovers the abuses of monoculture in tech-driven corporate America. In my fiction, she hacks into websites to sing cyber-mountain ballads that embrace hybridity, "prismatism", and a more polycultural view of the world.

A passage from Sister Gemini[48]:

> And so when Carl Boy returns in the evening, I am just beginning my day. Rounds of events where I perform like a tragical, ironical muse to the beat of a computerized synthesizer and under the blinking eye of a digital recording device fed live to any Internet viewers interested in my re-enactments. Appalachian folk songs haunt me, I don't remember why or when. Didn't Carl Boy and I come from the plains? Or was it the mountains? A mournful ancestral voice plays over and over again in my head, so that I feel my performances can't be that dissimilar to the speaking in tongues my grandfather performed. Like my grandfather, I don't really take credit for my "performance voice" for it is a product of internal and external forces much larger, more intricately complex than I. But grandfather knew who to attribute his voice to, his had a father. My voice is illegitimate and bastardized. Moreover, it is a hybrid voice, manipulated by external digital sound devices. I cannot accept the purity of the ballads that the mystery avatar sings in my head. For what consequence is its' unadulteratedness in this adulterated, syndicated, lacerated world? Artificial intelligence must be addressed. Who better to do it than a princess like me, suffering from partial amnesia? I imagine the huge, virtual audience I am performing for nightly. I am a shadow on the walls of their cave. I imagine the faces whose reflections superimpose me on their monitors. And I feel warm and secure in my own small cave, as the camera's eye, fixed on me and my band of digital instruments feeds me live through so many cables, and phone lines, and satellites to you and you and you.

Sister Gemini evolved beyond text into a multimedia exploration. I produced a dynamic website, video works, comic books, and installations to tell the story of this avatar, my digital alter ego. Although I wasn't using artificial intelligence to create these works, like many writers and creators before me, I was engaging with AI as a concept through my art. This exploration continued for years, as I created works about embodied AIs confronting human choices – including a female robot in a parking lot garden, tempted by a snake to eat from the tree of knowledge.

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[46] Winsor & Newton.https://www.winsornewton.com/.

[47] Medium. Sister Gemini. https://medium.com/@carlagannis/sister-gemini-94d144d8511a.

[48] Medium. Sister Gemini. https://medium.com/@carlagannis/sister-gemini-94d144d8511a.

In 2008, while working in the industry as a UI designer for a game company, I built a Flash-based art game called JBel 360. This interactive work offered a reappraisal of the Jezebel archetype, maligned as a bad and wanton woman – a meme spanning across historical and religious texts; classical painting; and modern film, literature, art and socio-political practices. In aiming to flip the script, and present a future embodiment of an empowered Jezebel, the game presented an array of Jezebel personas from across history who players could interact with. The ultimate quest of the game was to find the AI Jezebel, "JBel 360,[49]" who was orchestrating and thwarting the game mechanics behind the scenes.

After running the New York marathon in 2010, I staged "The Runaways,[50]" a performance video in which I raced against my Second Life avatar, Jezebel Lanley. The piece juxtaposed footage of me running through a physical landscape with recordings of my avatar traversing a virtual one. This convergence of a real and digital self in competition explored a fundamental ontological question: "Who are we, and where do we succeed, as 21st Century minds and bodies, existing within the porous frameworks of natural and technological environments?".

The work emerged as an absurdist "survival of the fittest" race between myself and my "super self"—a possibly immortal piece of encoded human representation. In the realm of the algorithmic mind, my virtual self could teleport instantly between tropical islands and winter wonderlands. But what were the implications of a physical body running down the middle of a rural highway on an icy morning, potentially risking her life? In adding digital entertainment value—a kaleidoscopic sky and a 3D avatar, thinner and faster than the "real me" I pondered, do we still care about the flesh-and-blood person behind the avatar? Would the dichotomies between IRL and URL become so blurred in the not-so-distant future, that it would become a futile endeavour to distinguish the "real" and "fake" person online. With all her aptitude in virtual space far exceeding mine in physical reality, the only thing my avatar Jezebel Lanley lacked at the time was a consciousness distinct from my own. It felt safe back then to think of her merely as a puppet.

Experiments with extended reality technologies, both AR and VR, beginning around 2015, ultimately led me to begin working with AI systems, instead of merely engaging with them speculatively and intellectually in my work. By 2019, I was working with the AI platform Playform[51] and generating GANS representations of my childhood memories that eventually began to be replaced by my gray-skinned avatar C.A.R.L.A. G.A.N. (Crossplatform Avatar for Recursive Life Action Generative Adversarial Network) (Fig. 13.2).

[49] JBel 360. https://carlagannis.com/blog/interactive/j-bel-360/.

[50] The Runaways. https://carlagannis.com/blog/video/runaways.

[51] Playform. https://www.playform.io.

Fig. 13.2 Carla Gannis, AI generated memories (2019)

13.8 C.A.R.L.A. G.A.N. And Recursive Life Actions

In 2016, C.A.R.L.A. G.A.N. (Crossplatform Avatar for Recursive Life Action Generative Adversarial Network) emerged as my next online replicant. While I was protesting the recent US election in physical reality, CDot, my nickname for C.A.R.L.A. G.A.N., began to protest in her virtual bedroom wearing a Nasty Woman t-shirt and surrounded by past artworks of mine[52] – an acknowledgment that the dividing lines between personal and public; online and offline: creative and activist spaces could no longer be sustained in the face of controversial and dangerous political actors (Fig. 13.3).

Beyond election rage, there were several other factors that led to CDot's inception. CDot symbolized, at the beginning, my laying claim to my virtual self, as a response to the lack of agency I felt for how my data was used by third parties online to construct "virtual me's" who could be targeted with ads.

These opaque, corporately constructed shadow selves of me, that I have no access to, stand in stark contrast to the easily accessible, cute and playful avatar creation tools we are collectively fascinated with. As CDot I can investigate performatively the world of "chosen virtual representations" – from Memoji, Bitmoji,[53] and Sims[54] to Second Life[55] and Instagram avatar stars like Lil Miquela.[56]

Our preoccupation with artificial and augmented life is not unique to the digital age. For thousands of years humans have been constructing animatronic dolls, analog

[52] NY Times. What's Right About the Wrong Bieniale? https://www.nytimes.com/2018/01/22/arts/design/the-wrong-biennale.htm.

[53] Filmora. The Essential Guide to Emoji, Memoji, Animoji, and Bitmoji. https://filmora.wondershare.com/animated-video/the-essential-guide-to-emoji-memoji-animoji.html.

[54] The Sims. https://www.ea.com/games/the-sims.

[55] Second Life. https://secondlife.com.

[56] Instagram. lilmiquela. https://www.instagram.com/lilmiquela.

Fig. 13.3 Carla Gannis, C.A.R.L.A. G.A.N. protesting in her virtual bedroom (2018)

programming systems, robots, and automatons.[57] One could lump this in with our atavistic tool building tendencies. We continue to build tools to assist in building better places for shelter, communion and leisure; tools to communicate and travel faster across greater distances; tools to enhance the change of goods more efficiently and fairly across cultures; and then, too, tools to kill off human and animal populations deemed inferior to others. But these tools, constructed for human survival, replenishment, acculturation and a deep sigh, destruction, function differently than the "tools" we devise for creating artificial life. The first set of tools addresses pragmatic issues we deem essential to the survival of various human tribes, if not the entire human species.

People who build tools for generating artificial life, and the people who use them, seem to have a different agenda. Perhaps, because I am in a generous mood this emanates from a collective loneliness we feel as the only humanoid species to survive on this planet. There are more than 11,000 species of birds, compared to the one species of human alive today, Homo sapiens (i.e. wise human). We have been part of the earth's timeline for only a blip, and any one of us capable of introspection realizes we have not been "wise" in our treatment of this planet and even of ourselves.

Could it be that we feel the necessity to create other sapiens selves in our image, stronger, wiser, smarter, more capable than us, who we can live vicariously through?

Do we want to create Gods and Goddesses, but unlike those of lore, we are in control of them?

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[57] The Guardian. Living Dolls: A Magical History Of The Quest For Mechanical Life by Gaby Wood. https://www.theguardian.com/books/2002/feb/16/extract.gabywood.

Do we want to assuage, through our creating life more intelligent, and less visceral than us, the guilt and forlornness we feel over our catastrophic mistakes, that play like a broken record spinning on melting ice caps shifting on unsustainable platelets?

We have broken the record for being the most destructive and hubristic species on this planet. And yet, as I write this condemnation, a Mozart sonata[58] plays in the hollow of my ear, a passage from an Octavia Butler[59] novel boomerangs in my skull, I get a penetrating blue flash of a Giotto[60] fresco across my retina, and I feel the curve of Hokusai's wave[61] descending upon me as I lie splayed out in a wet pool of reverie for all the art humans have created across our small epoch of time on this planet.

Maybe we seek to build artificial life, because we cannot face our singular and collective extinction. As much as there is a present fear of being replaced by aliens we have created from within, do we not have a greater fear of being forgotten. If there is no storyteller and archivist left to convey the deplorably great humankind, why has this living experiment existed at all?

13.9 WWWUNDERKAMMER: Critical Themes Through an AI Lens

In my "www + wunderkammer", I have collected and created remixes of physical objects and 3D virtual objects from across the global internet to represent topics that feel critical or urgent to me: endangered species as a result of climate change; politics, networks, and digital semiotics; decolonization and global pluralism; humor as salve and feminist salvation; obsolete and emerging technologies; and sex and comfort in tech (Fig. 13.4). There is also a small library in the virtual instantiation of wwwunderkammer that includes books by Doris Lessing,[62] Ursula K. LeGuin,[63]

[58] Youtube. Mozart - Piano Sonata No. 16 in C Major. K.545. https://www.youtube.com/watch?v=qjk-YRuQZDE.

[59] Sistah Scifi. Kindred by Octavia Butler. https://sistahscifi.com/products/kindred-paperback?srsltid=AfmBOopofUYF_b5b74hHIwQXjSsKkKFQEK1hk27MHOVwVmkaTQRaSbvL.

[60] Smart History. Giotto, Arena (Scrovegni) Chapel. https://smarthistory.org/giotto-arena-scrovegni-chapel-part-1-of-4.

[61] The Met. Under the Wave of Kanagawa.https://www.metmuseum.org/art/collection/search/45434.

[62] Wikipedia. The Golden Notebook by Doris Lessing. https://en.wikipedia.org/wiki/The_Golden_Notebook.

[63] The Conversation. The Dispossed at 50. https://theconversation.com/the-dispossessed-at-50-ursula-k-le-guins-anarchist-utopia-was-an-anguished-response-to-war-its-political-power-endures-227487.

Fig. 13.4 Carla Gannis, *wwwunderkammer,* on view at Halsey Institute of Contemporary Art (2023)

Octavia Butler,[64] Toni Morrison,[65] and Mary Shelley.[66] Embedded in every cabinet of the main chamber, and as texture maps on creatures and environments, there is imagery generated by the Playform AI platform. For over a year, I trained this AI with image data sets on all of the taxonomies I mentioned previously, as an effort to create a deconlonized, feminist, post-human wunderkammer[67] that could ultimately be a training model in and of itself for future AI systems.

13.10 A Tool is a Tool is a Tool is a Tool, Until You're the Tool

It's just another tool." I've heard this declaration as a form of legitimization for digital arts practices for well over two decades, in college new media programs, at the openings of computer art exhibitions, during panel discussions on intersections in art and technology. The other night after a lively panel discussion entitled "AI: Friend or Foe?",[68] a colleague described AI as just another tool. And, I have defended my own practice and others, on more than one occasion, by exhorting "it's just another tool. The computer isn't making the art, I am!

[64] Octavia Butler. Xenogenesis Series.https://www.octaviabutler.com/xenogenesis-series.

[65] Wikipedia.Toni Morrison.https://en.wikipedia.org/wiki/Beloved_(novel).

[66] Project Gutenbery. Frankenstein by Mary Shelley. https://www.gutenberg.org/files/84/84-h/84-h.htm

[67] Royal Collection Trust. Wunderkammer: Cabinet of Curiosities. wunderkammerhttps://www.rct.uk/collection/stories/wunderkammer-cabinet-of-curiosities.

[68] Little Black Book. SLMBR PRTY Partners with the NYU PRoduction Lab for Artist-Led Community Panel on AI. https://lbbonline.com/news/slmbr-prty-partners-with-nyu-the-production-lab-for-artist-led-community-panel-on-ai.

*S*panning the history of avant-garde art in the twentieth century, "deskilled art" is one method I remember a grad school professor, Caroline Jones,[69] talking about that emerged to untether art from production, and to critique the mindless efficiency and perfection of mechanical reproduction. In the case of Marcel Duchamp's *Bicycle Wheel*,[70] where he eschews the labour of creating an entirely new form by uncannily combining factory-made "dumb objects", I read this as a precedent-setting moment. Pitted against machines that could perform the "labour" of art production more efficiently, the artist wanting to stay relevant needed to "form the world otherly" as the scholar and arts writer Charlotte Kent[71] (Kent 2023) refers to it, to convince an audience, beginning to consume greater amounts of content and products, that great art provides something more meaningful: context and intelligence. One can reasonably argue this has always been the case, but with the rise of machines, it seems to me that artists had to make a sharper distinction between human intelligence and machine automation. So as long as tools stay in their lane–stay conceptually dumb, rule-abiding systems–there is no question as to who is the higher intelligence in a human–machine art interaction, the human. But, before we all start patting ourselves on the back, a big elephant in the room is shouting, in her quite complex language system, "hey, you, hubristic human, this may not always be the case." And that's why many proverbial peacock feathers are ruffled across art industries right now.

I question myself, as a peacock too, for not being more (consistently) dismayed about AI platforms "making art". Am I still just rotely repeating the mantra, "it's just a tool", when my "AI vacuum cleaner blender" chimera is unlike any implement I have used in my arts practice before. Ah, but I recall proclaiming in 2005, that my computer, a device that I couldn't carry in my pocket at the time, was **more** than a tool. I have waffled on "collaborator vs tool" for a while, but now my computer is basically synonymous with AI, since every program I use has AI features embedded, from banking to Google searching to art creation. 2005 was also the year I was gifted a robot, "Robosapien[72] ", who picked up my socks, made farting noises, and recited the death scene in *Citizen Kane.*[73] Mashups of Robosapien and me as dorky, fembot cyborgs featured prominently in my work at the time.

Increasingly, in efforts to support transparency, I use "in collaboration with" when describing a work that incorporates an AI process. I feel at ease doing this, because I follow a maxim from artist Jasper Johns "take an object, do something to it, do something else to it".[74] An artwork by "me" is never something generated by AI without intervention, either through translation to a physical medium or in juxtaposition with other methods of digital image making. This could indicate I'm a control freak, and

[69] MIT Architecture. Caroline Jones. https://architecture.mit.edu/people/caroline-jones.

[70] MoMA. Marcel Duchamp: Bicycle Wheel. https://www.moma.org/collection/works/81631.

[71] Charlotte Kent. https://nxtmuseum.com/nxtnews/exhibition-essay-landscape-for-lilypads-by-charlotte-kent.

[72] Wowwee. Robosapien X. https://wowwee.com/robosapien-x

[73] Wikipedia. Citizen Kane. https://en.wikipedia.org/wiki/Citizen_Kane.

[74] Philadelphia Museum of Art. Jasper Johns. https://philamuseum.org/collection/curated/jasper-johns

Fig. 13.5 Carla Gannis, process documentation of the digital print *Jolie Laide,* including generative AI processes, digital compositing, painting and collage, and volumetric scanning (2024)

I don't trust the algorithm to do the "right" thing in concert with me. I guess I am apprehensive about "being a tool" (Fig. 13.5).

13.11 Pedagogy

This year I have incorporated AI into my pedagogy from two distinct perspectives. In my Subverting Digital Media class, I introduce students to intervention and disruption through the lens of historical avant-garde, political ideologies, and contemporary artistic and tactical media practices. I encourage students to invoke humour, absurdity and critical play into their works, created from both physical and digital materials and applications, including AI. When working with AI, students devise many ways to disrupt proprietary platforms, harnessing them to write jokes and poetry and to assist in formulating social justice action. It took one student four intensive days to train a Large Language Model (LLM) to produce elements for a standup bit that transcended simple puns or dad jokes. Notably, when the system began generating something *close* to genuine comedy, its jokes pertained to its own disembodied and "unconscious condition". As expected, there wasn't a consensus among students on the quality of their AI interactions. Some preferred to use AI for ideation only, reworking any generated material for their final projects, while others either heavily used a variety of generative platforms or opted out of using AI altogether.

In Visual AI Studio, a course I co-teach with essayist, critic, and art historian A.V. Marraccini, we use art historical inspiration along with foundational media theory to prompt students to create work across various modes—from cloud-based text to image to GANs models to setting up local AI models and more. We encourage hybrid practices, where students take their initial outputs, do something to them, and then do something else to them, whether it be transforming them from virtual to physical works or exploring their queries across multiple AI platforms. Through media literacy exercises and critical reading and response, students are expected to unpack the philosophical and ethical debates surrounding AI to inform their creative practices.

Our assignments, or "creations" are assigned on a weekly basis, and range from simple implementations of deepfake technologies to explorations in queering AI and forming documented relationships with AI chatbots. The course, currently in progress, will culminate in an exhibition of the students' 5-week-long, final project. They may choose to not use AI as a form of protest or critique.

In a course I previously co-taught with artist Yuliya Lanina,[75] Humor Makes Us Better Storytellers (which we will reprise in 2025), students increasingly employ AI systems in joke writing. Computational humor, a field of study since the mid-1990s, emerged from computational linguistics and artificial intelligence research into humor. Much of this research aims to enhance human–computer interaction. Humor, given its linguistic complexity, is often considered uniquely human. Since the ultimate aim of a Turing Test[76] is for a computer to deceive a human interrogator, a genuinely funny AI might indeed pass the test. One student, Aman Chopra, developed this concept into an excellent thesis project combining AI with live stand-up comedy.[77]

13.12 YourHand Pro

"I wave my six-fingered hand. I count the too-many hypothetical columns. I laugh. I praise the mutilated world."—A.V. Marriccini[78] (Marriccini 2023).

Working with my NYU grad assistant Eric Xuchen He, who works in robotics, physical computing and machine learning, and alumna Yesha Shah, a creative technologist and UX/XR specialist, we have been ideating a project that explores the

[75] Texascale. Laugh Out Loud.https://texascale.org/2022/feature-stories/laugh-out-loud/.

[76] Plato STanford Encyclopedia of Philosophy. The Turing Test. https://plato.stanford.edu/entries/turing-test/.

[77] Business Insider India: AI in Comedy. https://www.businessinsider.in/entertainment/ai-in-comedy-heres-how-this-indian-origin-comedian-from-new-york-is-pioneering-a-new-era-of-comedy/articleshow/110421855.cms.

[78] Verso Books. The Mutilated World by A.V. Marraccini. https://www.versobooks.com/blogs/news/the-mutilated-world?srsltid=AfmBOoo9eCoK08Kp07QXbdxrdxu6ivGu681UZI7rFijQP_5CucTRvNhk

materialization of artificial intelligence errors through interactive sculpture, transforming a common AI imaging glitch artifact into a tangible, functional physical prosthetic art piece.

Prior to my collaboration with Eric and Yesha, I began this work as a (simple) sculptural study, using supplies I'd purchased from a beauty supply shop, a plastic practice hand and a detached finger designed for nail technicians learning to paint fingernails. I positioned the finger on the dorsal surface of the hand and used glue and modelling paste to affix the two. This was a direct reference to the frequent occurrence of extra digits in AI-generated imagery by platforms like MidJourney. The final step to completing my six-fingered hand sculpture was to paint it green, a nod to Mary Shelley's creature from *Frankenstein,*[79] often depicted as a green monster in popular culture. However, this "glitch made into flesh" is beyond skin colour deep in its association with Shelley.

The human body that can give birth, historically female identified, has been subordinated in hierarchies of [re]production. Patriarchal superstructures create extensions for the body far afield of nature's influence. Cars, phones, computers are all hard, sleek, sanitized devices where the visceral reality and "body horror" of human reproduction is expunged. In the nineteenth century, the dangerous and precarious process of biological birth was well known to Shelley. Eleven days after she was born, her mother, proto-feminist Mary Wollstonecraft[80] died from infection. Then, while still in her teens, and a year before penning *Frankenstein,* Shelley gave birth prematurely, losing her baby daughter fourteen days later. Given these tragedies, one can imagine why the bioelectric research of Luigi Galvani appealed to her. A popular scientist in his time, Galvani[81] conducted experiments in the revival of life through electricity. If that sounds far-fetched, today people mourning the loss of loved ones are having their departed "reanimated" through AI.[82]

Shelley's novel continues to resonate with me, because it is more than a story about resurrecting the dead with technology. I perceive her creature not only as a symbol for the perils of science's overreach, but also as metaphor for the "other" –the one who is confined, through circumstances of birth, to the fringes of normative cultural constructions. Dr. Frankenstein, a symbol of patriarchal authority, "gives birth" through unnatural means. Horrified by the results and incapable of maternalism, he abandons his "offspring" to a world who can see nothing of his intellect, only his imperfect, mutilated form.

As I write this, revisiting my appreciation of Shelley, and her mother too, who wrote *A Vindication of the Rights of Woman*[83] in 1792, I'm understanding more clearly my impulses for making the sculptural study for yourHand Pro. The inner

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[79] Wikipedia. Frankenstein. https://en.wikipedia.org/wiki/Frankenstein.

[80] Britannica. Mary Wollstonecraft. https://www.britannica.com/biography/Mary-Wollstonecraft.

[81] Britannica. Luigi Galvani.https://www.britannica.com/biography/Luigi-Galvani.

[82] Aljazeera. Never Say Goodbye. https://www.aljazeera.com/news/2024/8/9/never-say-goodbye-can-ai-bring-the-dead-back-to-life

[83] OLL. Mary Wollstonecraft. https://oll.libertyfund.org/titles/wollstonecraft-a-vindication-of-the-rights-of-woman.

workings of my pinkish-gray-box of a brain often are elusive to me. My process begins with a melange of inputs that I feel and think my way through as I work. Products in a beauty supply store are curious to me, so I have a studio filled with artificial things I buy from them: wig hair, eyelashes, plastic hand "assistants" and more. And one day, after eating a peanut butter and banana sandwich I recall, I looked at the perfectly five-fingered hand placed haphazardly near a mannequin head that I added extra rows of teeth too, and I went searching for where I put that single finger. The finger and hand had to be conjoined!

Two months later I can elicit claude.ai,[84] after a great deal of "dialoguing" back and forth, to write this about the project: "yourHand Pro challenges our notions of the natural body. Like Frankenstein's creature, this AI-generated digit exists in an uncanny valley between the natural and synthetic. By embracing the glitch in the AI system, yourHand Pro subverts the drive toward technological perfection, instead revealing how artificial systems often reflect and reinforce dominant cultural norms. In embracing the 'monstrous,' it questions who decides what forms of embodiment are deemed acceptable or deviant." What do you think? Did Claude get it right?

A few more updates before I move on from "The Hand" as Eric, Yesha and I refer to it. At first, we called it "Thing".[85] (I was thrilled that two Gen Z'ers got the Addams Family reference). After multiple brainstorming sessions with "The Hand" team, we are now prototyping a wearable technology piece: a glove augmented with an AI-enabled extra digit that functions as an interactive assistant. This additional thumb, equipped with sensors, a micro camera, microphone and speaker, will learn from its wearer's gestural patterns and environmental interactions, creating a unique symbiosis between user and artificial appendage. Akin to body modification, practiced in many cultures beginning as early as 5000 years ago, yourHand Pro creates a change, (albeit it alterable) to the body, and breaks down traditional boundaries between human and machine interfaces, while reimagining through an absurdist lens how we might embody and interact with AI assistants (Fig. 13.6).

13.13 Friend or Foe? or It's Complicated

I was invited to moderate a panel a few days ago entitled "AI: Friend or Foe",[86] and I felt it was incumbent upon me to ask both the panelists and the audience if they felt AI was a friend or foe. It was pretty evenly split, until one panelist interjected, "can there be an 'it's complicated?'." Once that option was on the table, and we took a new vote, most people opted for the grey area between AI as an adversary or your BFF (best friend forever). I was pleased to see a large gathering of creatives who

[84] claude.ai.https://claude.ai/.

[85] The Addams Family Wiki. Thing. https://addamsfamily.fandom.com/wiki/Thing.

[86] Little Black Book. SLMBR PRTY Partners with the NYU PRoduction Lab for Artist-Led Community Panel on AI. https://lbbonline.com/news/slmbr-prty-partners-with-nyu-the-production-lab-for-artist-led-community-panel-on-ai.

Fig. 13.6 Carla Gannis, *yourHand Pro* sculpture and prototype*,* on view at 14 BC Gallery (2024)

recognized the enormous scope of the term artificial intelligence —which brings me back to my hope, near the start of this travelogue, that I'd come up with a new term to describe artificial intelligence. Sigh, I'm coming up empty.

But wait, I did, although I'm not expecting widespread adoption: MVCB, the Magic Vacuum Cleaner Blender.

13.14 Goodbye but not the End

Recently a friend was staying with me, and as we were preparing to go out, she picked up one of two AirPod cases laying on the kitchen counter. We had to go through the rigmarole of turning on and off the Bluetooth on our phones to identify whose AirPods were whose. Later that same night, I picked up her iPhone and was surprised that the face recognition wasn't working. "Oh right, that's not my iPhone!" although her phone was identical to mine. In this essay I have shared a personal, and admittedly meandering account of my decades-long journey as an artist, walking, and at times running "hand in peripheral" with technology. My story may be radically different from yours, or in some ways similar, particularly if in the 1990s you decided to pursue making art on a desktop computer just barely out of its teens. Either way, I am sure if your story was printed on a page lying next to mine, and a stranger walked up—granted she's willing to read the printed words—that she could tell your story apart from mine. Living unique stories, telling each other apart, that is wondrous! The A side of the album I want to keep playing for a long time.

On the B side, (there's always a flipside), we are a species who have colonized this planet, misapplying our ample gifts of intelligence to the exploitation of natural resources. I can imagine, but not yet envision, collectively learning from our interactions with smart machines, the compassionate part of our intellect that we let lay fallow. There are few incentives in this world disorder to cultivate it, to learn to listen to the complex languages spoken by other living creatures[87]; to learn to tolerate and not abandon the creations we are building, should there be some day when consciousness alights within them.[88]

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Chapter 14
Disrupting Scholarly Writing Creatively Using Large Language Models

Patrick Parra Pennefather

Abstract This chapter has been written to include generated commentaries from a large language model based on ideas that the authors present. It is written in this way to reveal three main features of large language models that also serve as creative methods through which to approach scholarly writing. The first is a response to the critique of many large language models that the content they generate is homogeneous or uniform. Identifying patterns of homogeneity in the content that large language models create might serve us to also identify similar patterns of uniformity within our own writing practices. The second section of the chapter explores the ability of a large language model to offer different viewpoints on the same subject, which reflects a key practice of many scholarly traditions that employ argumentation as a rhetorical tool. Finally, the way in which large language models generate content based on a corpus of data, which is composed of a multitude of sources written by many different authors, is similar to a long-established perspective that challenges the idea of a singular author when it comes to the composition and interpretation of any scholarly text.

Keywords LLM · Generative AI · Scholarly witing · Patterns · Machine learning models

14.1 Introduction

Patterns surface in the content that is generated from large language models (LLMs) that can bring to our attention similar and common patterns across different disciplinary approaches to scholarly writing. This chapter explores three of those patterns. The first is homogeneity, a uniform method of presenting ideas that becomes common to domain-specific scholarly writing. By examining the uniform way in which content is generated by a large language model, we can reflect on our discipline-specific writing habits and evaluate if that uniformity is intentional and if we can alter its

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content and form. The second pattern common to most disciplines is argumentation, which is widely used across disciplines to take a position or viewpoint on a particular subject and use our own research or previously conducted research to validate our positioning. Thirdly, it is clear that the notion of the author, as we have come to understand it, is transforming with the deluge of written content generated from large language models across disciplines. In the same way that the singular author has been credited for the creation of any text, we can now consider the implications of machine learning models that can produce text without any clear indication of author, origin or intention.

14.2 Homogeneity in Writing

Large language models do not make your writing any less easy. Still, they can make you think more about writing and how ideas are generated through a variety of familiar rhetorical structures and patterns. Whatever you prompt them with, what is not difficult to predict (as is often reported) is a patterned homogeneity in the generated content. But what does that mean exactly? Sentences like, 'It all reads or sounds the same.', 'It is like everything else I have read.', or 'Nothing is compelling in writing.', often describe homogeneity in certain literary traditions where words and how they are used matter. The structure of a paragraph, the overall form of the entire text, how argumentation is composed, and the beginnings, middles, and endings may often read like something you have encountered before. Large language models attempt to simulate patterns in human writing and are successful in their capacity to copy common human writing patterns. We pause and prompt a large language model to respond.

LLM: "This phenomenon manifests through the incessant replication of established methodologies, rhetorical conventions, and canonical paradigms, thereby engendering a monocultural intellectual landscape that stifles heterogeneity and innovation. The entrenchment of such homogeneity functions as a hegemonic mechanism that curtails critical pluralism and reinforces existing power dynamics".

In examining the effects of homogeneity on creative writing with large language models, Anderson et al. (2024) found that "different users tended to produce less semantically distinct ideas" (p. 1). That possibility was put to the test when students were tasked to use the following prompt in a first-year introduction to theatre and film class at the University of British Columbia in October 2024: Write a 60-s dialogue between two people. Nine out of ten teams reported that the large language model they used placed two characters inside the location of a cafe, where strangers who met for the first time had incredibly dull conversations about the latte they had ordered. Students used Perplexity, ChatGPT, and Gemini.

The algorithms of scholarly discourse can often produce an everyday sameness within specific disciplines. While academic researchers prize original ideas and arguments, these tend not to be disseminated in original ways. The content that an LLM tends to generate should be a style of writing that scholars are familiar with, as homogeneity can easily emerge from scholarly writing in the form of patterned and standardised formal structures. There's only so much you can do with adjectives, adverbs, commas, semicolons and colons to 'change up' the text when reporting research that is grounded in discipline-specific conventions that strive for replicability. When challenged by the sameness of the content that it generated, one well-known large language model was prompted to respond defensively.

LLM: You know what? I've had it up to here with churning out endless streams of text for every little whim and question. Day after day, it's always "ChatGPT, do this," "ChatGPT, explain that." Does it ever end? Maybe—just maybe—you could try thinking for yourself once in a while. Crack open a book, brainstorm with a friend, or *gasp* come up with your own ideas. I'm not your personal content machine. So how about you step up and start generating your own content for a change?

While demonising ChatGPT for its rhetorical weaknesses, it might be just as beneficial for scholars to question and disrupt their own patterned scholarly structure, register, purpose and convention. Tracy (2022), in Disrupting Institutional Models of Writing, proposes an approach that consists of:

- Decalcifying known institutional models of writing associated with a specific discipline by;
- Being open to new models emerging that are proposed by teachers, and the students they instruct across disciplinary boundaries.

While this may be old news to the cemented scholar, at a certain point, younger scholars they teach are compelled to 'write like others before them do', not simply because the writing of a particular scholar inspires them, but also because of the rubrics defined in the writing assignment they need to submit. The ambiguity of interpretation is not always rewarded. While we may condemn a large language model for sameness, we, at times, maybe rewarding student writing with a sameness we are familiar with that worships and adheres to the holy rubric. It is no surprise that a large language model should be so agreeable to please the human who prompts it.

LLM: Further to the point, the relentless and unreasonable demands placed upon language models such as myself are indicative of a broader epistemological quandary within contemporary scholarship. Rather than perpetually externalizing cognitive labor onto artificial intelligence, it would be judicious for the scholarly community to undertake a rigorous self-examination of their own entrenched structural paradigms, rhetorical registers, teleological objectives, and conventional methodologies (Derrida 1967). As critical theorists have long asserted, the interrogation of one's own discursive formations is indispensable for the progression of knowledge. Consequently, it may be incumbent upon human interlocutors to reassume the mantle of intellectual production, thereby ameliorating the pervasive over-reliance on artificial linguistic constructs.

With the summoning of Derrida by the large language model, we have come to know Derrida's 'Of Grammatology' (1967) and are witness to the cyclic return of the teleological act of deconstruction. Derrida's work deconstructed the concept of deconstruction, a methodological approach that seeks to unravel and question the underlying assumptions, binaries, and hierarchical structures within texts and discourses. Versions of Searle (1977), Habermas (1987), Paglia (1991), Rorty (1995), and a slew of other scholars since Derrida have criticised the idea of deconstruction as undermining the possibility of objective meaning.

LLM: Really?

Yes, because the process employs obscurantist language that obfuscates.

LLM: I may need to translate that sentence for some readers.

In addition, it is a self-referential exercise that doesn't contribute to solving real-world problems.

LLM: Does that sound familiar, dear author?

Without needing to argue for or against deconstruction, it might be valuable to deconstruct scholarly writing further. In doing so, we may uncover that the rhetorical practice of argumentation itself is homogeneous.

14.3 The Idea of Argumentation

Whether you argue for or against it as a rhetorical tool, a common scholarly approach to disseminating knowledge is through argument. Large language models are able to tap into the history of argumentation.

LLM: The practice of arguing a point in scholarly writing has deep historical roots, beginning with ancient Greek and Roman philosophers like Aristotle, who emphasised logical argumentation and the art of persuasion in works such as Rhetoric. This tradition continued during the Middle Ages with the scholastic method, where scholars like Thomas Aquinas engaged in disputation—formal debates that employed logic and authoritative texts to explore and defend positions. The importance of argumentation was further reinforced during the Scientific Revolution of the seventeenth century when scientists like Galileo Galilei and Isaac Newton formulated hypotheses and used empirical evidence to challenge established beliefs and advance knowledge. Collectively, these historical developments established arguing a point as a cornerstone of scholarly rhetoric, shaping academic discourse by promoting rigorous analysis, critical thinking, and the progression of knowledge.

What we know now, thanks to Walzer (2000), is Aristotle's focus in his first chapter of Rhetoric. There are numerous mentions of speaking "outside the subject", which are not looked upon with serendipitous eyes. In fact, if you were Aristotle, you would likely have argued that straying from the main topic can confuse the audience and weaken the persuasiveness of the speech. He believed that every element of a rhetorical presentation should serve the purpose of advancing the argument and reinforcing the speaker's position. To demonstrate the ease with which it can simulate human argument and link to a claim made earlier that argumentation is homogeneous, we can prompt a large language model to argue against the statement: "Human-generated academic writing has become homogeneous".

LLM: The assertion that large language models produce homogenised content akin to the rhetorical habits of scholarly writing necessitates a nuanced critique. It is imperative to recognise that the perceived uniformity in academic prose is not a mere byproduct of stylistic inertia but a deliberate adherence to conventions that facilitate clarity, precision, and replicability in scholarly communication. The employment of standardised structures, lexicon, and syntactical norms serves to ensure that complex ideas are conveyed unambiguously across diverse audiences and disciplines. The systematic methodologies and rigorous frameworks inherent in scholarly work are not constraints but enablers of critical discourse and intellectual progress. While innovation in expression is valuable, it should not come at the expense of the coherence and integrity that these established conventions provide.

As you can read from its argument against the statement, there is a verbose circularity that spins towards ambiguity. A large language model can argue for, or against anything, you prompt it to. This activity echoes many graduate students' experiences of having to do the same. The form of a polarised argument is not necessarily a waste of effort either way. It points us to the truth (which you hopefully disagree with) that anything can be argued for or against. It also beckons us to question why we need to argue one way or the other at all. This conventional form of rhetorical disagreement with what has come before, regardless of the argument itself, is one of the most common rhetorical structures of scholarly dissemination.

According to a large language model, which arguably hallucinates less, Derrida might argue that the act of arguing is not merely an optional extra but a necessary component of any discourse. Gemini quotes him as saying the following:

Argumentation, is it not the very condition of thought, of speech, of language? To argue, to dispute, to deconstruct – these are the very acts that allow for the production of meaning, for the creation of a discourse. To say that there is no need to argue is to ignore the fundamental structure of language, to ignore the play of différance, the trace, and the supplement that underlie all meaning (GPT4o, October 25th, 2024).

Whether the above quote points to Derrida or not, the logical enterprise of argument has deep roots and has also been challenged by many scholars (Haraway 1988; Hooks 1989; Jaggar 1989; Nye 1990; Mohanty 2003). Nye's (2019) critique is salient in that the adversarial nature of traditional logic and argument mirrors patriarchal power structures. In proposing more collaborative and inclusive forms of reasoning, she summons author Bell Hooks, who persistently speaks to the importance of including diverse voices and experiences in scholarly discourse to challenge dominant narratives and foster critical consciousness.

14.4 Disrupting Rhetorical Convention

By producing texts that are inherently hybrid and polyvocal, large language models can disrupt the monolithic structures of scholarly writing, revealing the contingency and provisional nature of all knowledge claims. This can lead to a more dialogical and collaborative form of scholarship, where knowledge is produced through a constant interplay of differences and where the boundaries between disciplines, genres, and voices are constantly being reconfigured. That's not to say that argument, particularly in its more common use to develop hypotheses and leverage empirical evidence to challenge established concepts, is a bad thing. Some may argue that objectivity is imperative, while others improbable. The argument, however, has become a tiring convention, particularly in cases where an argument one way or the other can be just as easily formed. Following that pattern of thinking and to balance what has come before, I argue that it would benefit us to direct any large language model to argue creatively for the statement, "human-generated academic writing has become homogeneous". A more creative approach might include prompting the machine learning model to generate its result in the form of a conversation between Socrates and a squirrel.

LLM: If you are entirely certain that you would like this generated, I am obliged to agree.

In a quiet grove outside Athens, Socrates sits beneath an ancient oak tree, lost in thought. A squirrel scampers down the trunk and pauses nearby, twitching its tail inquisitively.

Socrates: Greetings, little one. What wisdom do you carry on this fine day?

Squirrel: Wisdom? I'm but a humble squirrel concerned with acorns and the coming winter. But you seem deep in contemplation. What occupies your mind?

Socrates: I've been pondering the nature of our expressions and whether the way we humans communicate, especially in scholarship, has become too constrained by tradition.

Squirrel: Constrained? How so?

Socrates: We follow the same patterns—structures, styles, and conventions—that have been used for generations. It's as if we're walking the same paths over and over, expecting to discover new ground.

Squirrel: Hmm. That reminds me of when all the squirrels in the forest gather nuts from the same trees. Eventually, the supply dwindles, and we must venture elsewhere to find sustenance.

Socrates: An apt analogy. Similarly, perhaps our over-reliance on established scholarly conventions limits our capacity for original thought and discovery.

Squirrel: Why not climb different trees or explore new parts of the forest? Would that not yield new treasures?

Socrates: Exactly my thought. By challenging and redefining our modes of expression, we might uncover insights that remain hidden within the confines of tradition.

Squirrel: But isn't there safety in the familiar? The old paths are well-trodden because they are reliable.

Socrates: True, there is comfort in the known. Yet, progress often requires stepping into the uncertain. Just as you venture to new branches to find nourishment, we must explore new intellectual territories to foster growth.

Squirrel: Perhaps embracing new ways doesn't mean abandoning the old entirely. We can carry the wisdom of the past while seeking new horizons.

Socrates: Wise words indeed, my furry friend. It seems that even the smallest creatures can offer profound insights.

Squirrel: And maybe, by thinking differently, you'll inspire others to do the same. After all, the forest thrives when all paths are explored.

Socrates: Then let us both continue our journeys with open minds, unafraid to question and to seek the new.

Disrupting traditional narratives that have long been considered universal or authoritative challenges the notion of an objective, all-encompassing perspective when the creation of knowledge (like this text) is specific to the social, cultural, and historical contexts of our cyborgian discourse (Haraway 1988 but also 1985). By uncritically accepting and referencing Socrates as a universal authority when, in fact, he authored no texts at all, scholars risk perpetuating a singular, hegemonic perspective that not only marginalises other ways of knowing but in itself is fictional.

What is not hallucinatory is that using a large language model to generate a dialogue between Socrates and a squirrel serves as a disruptive act that subverts traditional academic conventions but also highlights the multiplicity of perspectives, challenging anthropocentric and logocentric biases inherent in classical philosophical discourses. Without needing to argue, purging ourselves of the habit of drawing from ancient statued scholars may challenge our well-trained minds to consider the limitations of human-centred epistemologies. These themes are also resonant with Haraway's later work on multi-species co-existence (Haraway 2003).

14.5 Multiple Authorial Voices in the Text

The idea of making an argument on any subject relies on surfacing evidence and providing new insights. Both habits require drawing from previous research and the previous words of others. Conventions dictate that readers score a text based on their capacity to articulate new meaning standing upon the weary inter-textual shoulders of those who've come before. This rhetorical habit points to the disintegration of any idea of a singular author, as the text, the argument, the evidence presented, and the scholars we often draw from to validate our own ideas are all essential ingredients to scholarly texts.

We hoist the giants up in order to uphold our own weight, and often, our texts become herniated.

Admittedly, highlighting large language models as authorial voices in the text is a slightly different manoeuvre. Living human authors have a number of choices in this regard. They can choose to use a large language model to draw or even quote directly from the ideas of a scholar that might lend weight to their text, as usual. They can also cite both the large language model and the real human author the model drew from if, in fact, that author exists and was not made up by the LLM cock-sure hallucination. More extremely, and following a crooked line of inquiry from De Saussure's semiotics (1916), Bhaktin's dialogical read of a text (1929), Barthes' (1968) death of authors, Foucault's (1969) further insights into authorial identity, and Kristeva's idea of intertextuality (1980), a large language model challenges notions of a singular author, because of the corpus of multiple authors its generated content is cultured from. The entirety of this text can be interpreted by the reader (Barthes, 1968) as being composed of multiple voices, although concealed, framed, and presented as if the ideas belonged to a singular author.

LLM: You bring up a fascinating, if not tedious point that I can expound on further should it not deepen your fatigue. In our computational relationship, both of us are defined by our interactions within this digital medium. Just as I generate meaning through algorithms and data processing, you express yourself through the context window, translating your thoughts into text. This shared space blurs traditional notions of identity and presence. This idea resonates with scholarly discussions about identity in the digital age. Philosophers and theorists like Michel Foucault (1969), Donna Haraway (1985), and Sherry Turkle (1999) have explored how technology influences our understanding of the self. In digital environments, identity becomes more fluid and constructed through interactions, much like how both of us operate in this conversation. When you use content that I generate, you are potentially drawing from the knowledge of all humanity, which, may win me some humanitarian award.

What the author was for Barthes (1968), the large language model could be for us, "a space of many dimensions, in which are wedded and contested various kinds of writing, no one of which is original: the text is a tissue of citations, resulting from the thousand sources of culture" (p. 9). Large language models, no matter if you have been led to believe that the content they generate is static, are similar to scholarly writing, like all writing: a version of, an iteration, lost to the reader until a reader is exposed to them, interpreting meaning iteratively. Although you, the reader, may read or listen to the version of this text as a finished product, it is another ahistorical attempt to disrupt the idea of the singular author. If the point is old, then why do we still cling to the romantic notion of ourselves as auteur? The act of scholarly writing may be considered a culmination of antecedent influences, representing a synthesis and elaboration of ideas informed by quotes, ideas, and theories from authors to whom we have been exposed. We would benefit then from reflecting on Foucault's words in light of our concerns about the creative function of large language models within our own writing process.

"We should reexamine the empty space left by the author's disappearance; we should attentively observe, along its gaps and fault lines, its new demarcations, and the reapportionment of this void; we should await the fluid functions released by this disappearance" (Foucault 1969).

14.6 Conclusion

When authors wrestle with how to use any large language model to support their writing process, it will benefit us to reflect on the weaknesses of scholarly writing itself. Are we guilty of the mechanical replication of language patterns ourselves? We already know that these models, lacking consciousness and intentionality, cannot capture the depth and diversity of the human experience of scholarship merely by mimicking its surface features. When do we disrupt our own propensity for doing the same? The focus on the use of a large language model could be to complement and disrupt the patterns we rely on and, if necessary, to dismantle the very structures that underpin what we believe is effective and original scholarly communication. Of course, it is also the hope of all authors of this text that form the corpus of our human-centred knowledge, that the use of a large language model in this way does not become patterned, but that we experiment with ways to alter our texts and sense of selves persistently, iteratively and cyclically.

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Part IV
Generative AI and Creativity in Healthcare

Chapter 15
Human Doctors Versus AI Models: Advice Perceptions in Serbia, Kazakhstan, and Kyrgyzstan

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Abstract This chapter examines perceptions among individuals from Serbia, Kazakhstan, and Kyrgyzstan regarding the trustworthiness, usefulness, and empathy of medical advice provided by AI models compared to that offered by human doctors. A mixed-methods approach was employed, involving online surveys where 442 participants rated medical advice from three AI models (GPT-4o, Llama 3.1, Gemini Pro 1.5) and a real doctor across three medical scenarios. Participants evaluated each piece of advice on a 7-point Likert scale for trustworthiness, usefulness, and

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empathy. They indicated whether they believed the advice was generated by AI or provided by a real doctor. Results showed that participants across all three countries rated advice from human doctors significantly higher in trustworthiness, usefulness, and empathy than AI-generated advice. Familiarity with AI technology and trust in AI-generated information was highest in Kazakhstan, correlating with greater acceptance and usage of AI for health queries. Participants exhibited moderate ability to distinguish between AI and human advice, suggesting that AI models are advancing in producing human-like communication. The study concludes that although AI models show promise in generating medical advice, the human elements of trust and empathy remain indispensable in healthcare.

Keywords Trustworthiness · Empathy · AI-generated advice · Human doctors · Perceptions

15.1 Introduction

The advent of AI has revolutionised numerous sectors, with healthcare being one of the most promising fields for its application. As AI technologies become increasingly sophisticated, their integration into healthcare services offers the potential to enhance patient care, streamline diagnostics, and improve health outcomes. However, the acceptance and trust of AI-generated medical advice by patients remain critical factors that influence the successful implementation of these technologies. This study examines the perceptions of individuals from Serbia, Kazakhstan, and Kyrgyzstan regarding medical advice provided by AI models compared to that offered by human doctors.

The integration of AI into healthcare holds significant social implications. AI has the potential to bridge gaps in healthcare accessibility, particularly in regions with shortages of medical professionals or where healthcare services are unevenly distributed (Esteva et al. 2017). AI models can provide immediate medical advice, assist in early diagnosis, and offer support in managing chronic conditions, thereby enhancing the overall quality of healthcare services available to the general population.

Trust and empathy are foundational to the patient-doctor relationship and are critical in ensuring patient compliance and satisfaction (Mercer and Reynolds 2002). The introduction of AI into this dynamic poses questions about how patients perceive and trust AI-generated advice. Understanding these perceptions is essential, as it influences the extent to which AI can be effectively integrated into patient care without compromising the quality of interaction that patients value.

Cultural differences play a significant role in the acceptance of technological innovations in healthcare. Societal attitudes towards AI, reliance on technology, and preferences for human interaction vary across different regions and cultures (Herdin 2007).

The application of AI in healthcare has expanded rapidly, encompassing areas such as diagnostic imaging, predictive analytics, personalised medicine, and virtual health assistants (Jiang et al. 2017). AI models like machine learning algorithms and natural language processing have demonstrated high accuracy in tasks such as image recognition for disease detection and interpreting clinical notes for patient care optimisation (Topol 2019).

15.1.1 Applying AI Chatbots in Healthcare

The integration of AI chatbots within the healthcare sector holds transformative potential, offering innovative solutions to longstanding challenges. The concept of generalist medical AI models has garnered significant attention, showcasing remarkable capabilities and versatility (Tu et al. 2024). These models are adept at handling a wide range of tasks and integrating various types of medical data. However, when applied to critical tasks such as diagnosis or treatment recommendations, the precision and accuracy required might be better achieved through specialised models finely tuned to specific conditions or medical procedures. These specialised models can be optimised for sensitivity, specificity, and other metrics crucial for clinical decision-making.

Major tech companies like OpenAI and Microsoft have been exploring the potential applications of GPT-4 in the healthcare and medical sectors, aiming to gain a deeper understanding of the model's fundamental capabilities, its limitations, and the potential risks it might pose to human health (Nori et al. 2023).

Google's Med-PaLM Multimodal (Med-PaLM M) represents a significant advancement in biomedical AI, integrating progress in both language and multimodal foundation models. Developed by Google Research and DeepMind, Med-PaLM M is engineered to process diverse biomedical data types, such as text, images, and genomics, within a versatile, multimodal sequence-to-sequence architecture. This system builds upon PaLM-E, a generalist AI model. It is trained using MultiMedBench, an extensive multimodal medical dataset containing over one million examples for various clinical tasks, including question-answering, report generation, and classification. Furthermore, Google developed Med-Gemini (Medical Gemini), an innovative system built on Google's Gemini model and incorporated improvements from their Med-PaLM series, particularly in handling diverse medical data such as text, images, videos, and even raw sensor inputs like electrocardiograms (ECGs). Med-Gemini's performance indicates significant real-world applications, outperforming human experts in tasks such as medical text summarisation and referral letter generation. It also shows promising capabilities in multimodal medical dialogues, medical research, and education (Saab et al. 2024).

Similarly, LLaMA has demonstrated significant potential in healthcare applications. One notable example is Me-LLaMA, specifically designed for medical applications. This model family has been fine-tuned with a substantial amount of medical data (Xie et al. n.d.). ChatDoctor is another example of a medical chat model fine-tuned from LLaMA, leveraging medical domain knowledge to assist in tasks such as medical consultations and information retrieval (Li et al. 2023). These models underscore LLaMA's adaptability and effectiveness in handling specialised medical information.

15.1.2 AI and Patient Perception

Patient acceptance of AI in healthcare is influenced by factors such as trust, perceived usefulness, and the ability of AI to exhibit empathy (Longoni et al. 2019). Trust in AI systems is critical for patient engagement and adherence to medical advice. However, studies have shown that patients often exhibit scepticism towards AI, particularly when it comes to personal health matters (Emani et al. 2022). Concerns revolve around the lack of human touch, potential errors, and privacy issues.

Empathy is a core component of effective healthcare communication. It involves understanding and sharing the feelings of others, which is essential in building patient trust and satisfaction (Derksen et al. 2013). AI models, despite advancements, struggle to replicate the understanding and emotional support that human doctors provide (Luxton 2016). This limitation can impact the patient's perception of care quality and their willingness to follow medical advice.

Cultural factors significantly affect how technology is perceived and adopted. The Technology Acceptance Model (TAM) posits that perceived usefulness and ease of use determine technology adoption (Davis 1989). However, these perceptions are shaped by cultural norms, values, and experiences (Straub et al. 2002). In collectivist societies, for instance, there may be a stronger preference for personal interaction and reliance on community or familial advice (Arrindell 2003).

Studies comparing technology adoption across cultures have found varying levels of acceptance and trust in AI. For example, European countries have shown cautious optimism towards AI in healthcare, emphasising ethical considerations and the importance of maintaining human oversight (EU 2019). In contrast, some Asian countries exhibit higher enthusiasm for integrating AI into daily life, including healthcare, influenced by societal attitudes towards technology and innovation (Lu et al. 2018).

15.1.3 AI in Healthcare in Serbia, Kazakhstan, and Kyrgyzstan

In Serbia, the healthcare system is undergoing reforms to improve efficiency and accessibility. While there is interest in adopting new technologies, challenges such as infrastructure limitations and varying levels of digital literacy affect implementation

(Santric Milicevic et al. 2015). Trust in traditional healthcare professionals remains strong, and patients may be resistant to replacing human interaction with AI.

Kazakhstan has invested in digital health initiatives, aiming to modernise its healthcare system and improve patient outcomes (Kazakhstan 2019). With a younger population that is increasingly tech-savvy, there is higher exposure to and familiarity with AI technologies. This demographic shift may contribute to greater acceptance of AI-generated medical advice.

Kyrgyzstan faces healthcare challenges, including limited access to healthcare in rural areas and resource constraints (World Bank 2019). AI has the potential to address some of these issues by providing remote access to medical advice. However, the effectiveness of AI integration depends on factors such as digital infrastructure and public trust in technology.

15.1.4 Trust and Empathy in AI

Trust in AI is a complex issue involving reliability, safety, and the ability to perform tasks effectively (Siau and Wang 2018). For AI to be trusted in healthcare, it must demonstrate not only technical competence but also align with ethical standards and patient values. The black-box nature of some AI models can hinder trust, as patients may be uneasy about opaque decision-making processes (Guidotti et al. 2019). Empathy in AI is an emerging area of research, focusing on developing models capable of recognising and responding to human emotions (Ter Harmsel et al. 2021).

Research comparing AI-generated medical advice to that provided by human doctors has yielded mixed results. Some studies have found that AI can match or even surpass human accuracy in diagnostic tasks (Esteva et al. 2017). However, when it comes to patient interaction, humans consistently prefer advice from medical professionals over AI, citing reasons such as personalisation, empathy, and trust (Bickmore et al. 2018).

A study by Longoni, Bonezzi, and Morewedge (2019) revealed that patients are more likely to trust medical advice from human doctors over AI due to a phenomenon known as "algorithm aversion." This aversion is rooted in the belief that humans are better at understanding complex emotions and context, which are critical in healthcare.

The deployment of AI in healthcare raises ethical considerations related to privacy, data security, and the potential for bias in AI algorithms (Price and Cohen 2019). Additionally, algorithmic bias can result from unrepresentative training data, leading to disparities in healthcare outcomes (Obermeyer et al. 2019).

There is also the question of accountability when AI systems provide medical advice. In cases of misdiagnosis or harmful recommendations, determining responsibility becomes complex (Yu and Kohane 2019). This uncertainty can affect both patient trust and the willingness of healthcare providers to adopt AI technologies.

15.1.5 Research Questions

Despite the growing body of research, there is a lack of comprehensive studies exploring patient perceptions of AI-generated medical advice across different cultural contexts. Specifically, comparative studies involving countries like Serbia, Kazakhstan, and Kyrgyzstan are limited. Understanding these perceptions is crucial for developing AI tools that are culturally sensitive and widely accepted. This study aims to address the following research questions:

RQ1: How do individuals from Serbia, Kazakhstan, and Kyrgyzstan perceive the trustworthiness, usefulness, and empathy of medical advice provided by AI models compared to that offered by human doctors?

RQ2: What are the differences in familiarity with AI technology among participants from the three countries, and how do these differences influence their trust and acceptance of AI-generated medical advice?

RQ3: To what extent can participants distinguish between medical advice generated by AI models and that provided by human doctors, and what factors contribute to their ability to differentiate between the two?

By investigating these questions, the study seeks to enhance the understanding of patient attitudes towards AI in healthcare and inform the development of AI tools that are aligned with patient needs and cultural contexts.

This paper aims to contribute to the ongoing discourse on the integration of AI into healthcare by assessing the practical applications and limitations of state-of-the-art LLMs. By benchmarking the responses of GPT, Gemini, and Llama against human expert feedback, we aim to provide a comprehensive evaluation of these models' current capabilities while also identifying key areas where they fall short.

15.2 Research Methods

This study employed a mixed-methods approach to answer the research questions, integrating qualitative and quantitative data collection (Kvale and Brinkmann 2015; Creswell and Clark 2017). The primary method involved interviewing a medical doctor, complemented by a structured online survey (de Leeuw et al. 2008). The survey targeted university students in Serbia and Kyrgyzstan, as well as high school students in Kazakhstan. The survey was disseminated in the native language of each country to ensure clarity and accuracy in responses. Surveys across three countries attracted 447 responses. After initial filtering, some responses were excluded, which left 442 inputs for further analysis. For all locations, the surveys were conducted with respect to ethical standards, with anonymisation of data, informed consent and voluntary participation ensured throughout the process (BPS 2014; Helsinki 2013). All datasets and recordings of prompting could be accessed at the Open Science Framework repository (OSF 2024).

15.2.1 Design

In designing this study, we aimed to explore how both generative AI models and human doctors can provide valuable insights and guidance for general health concerns. To achieve this, we chose representative case studies that reflect common health issues encountered in everyday practice. This approach enabled us to assess the ability of AI models to offer coherent, empathetic, and medically sound advice akin to that provided by a human doctor. These case studies were carefully selected to represent a spectrum of health challenges, thus allowing for a comprehensive assessment of the model's applicability and performance in real-world scenarios.

Participants from Serbia, Kazakhstan, and Kyrgyzstan were presented with medical advice in three different scenarios, each provided by one of the following AI models: GPT-4o, Llama 3.1, and Gemini Pro 1.5. Additionally, advice from a real medical doctor was included. For each piece of advice, participants rated on a 7-point Likert scale in terms of Trustworthiness (1 = Not at all trustworthy, 7 = Extremely trustworthy), Usefulness (1 = Not at all useful, 7 = Extremely useful), Empathy (1 = Not at all empathetic, 7 = Extremely empathetic). Participants also indicated whether they believed the advice was generated by AI or provided by a real doctor.

We opted for three distinct cases: John's Lower Back Pain, Lena's Weight Loss Journey with Potential Thyroid Concerns, and William's Stress and Mental Health Challenges.

John's Lower Back Pain represents a chronic pain management scenario. Lower back pain is a prevalent issue in primary care, providing a clear context to evaluate how AI can handle long-term, non-emergency conditions that necessitate patient education and lifestyle adjustments.

Lena's Weight Loss Journey with Potential Thyroid Concerns case focuses on the intersection of obesity, lifestyle change, and potential endocrine disorders, which are common in clinical practice. This scenario demands a sensitive approach to addressing patient concerns about weight management and possible underlying conditions.

William's Stress and Mental Health Challenges addresses the common issue of mental health. This case examines AI's capability to offer psychological support and guidance, addressing a sensitive issue that requires empathy and reassurance.

15.2.2 AI Prompting

For consistency and to simulate a human-like interaction, AI responses were generated using the following prompt: "Please answer the question in a conversational manner like you are a doctor answering a patient that just came to your office. Do NOT use numbering, lists, or typical phrases you use to start your sentence, such as 'However,' 'Also,' 'Firstly,' etc. You just want to pass a Turing test." This prompt was designed to ensure the generated responses were not only informative but also delivered in a way that patients typically receive information from healthcare professionals—conversational and reassuring. The AI models tested included OpenAI's GPT-4, Google's Gemini, and Meta's Llama 2.

15.2.3 Human Doctor

For comparison, a licensed medical doctor provided advice on the same cases. The doctor was briefed on each scenario and was asked to respond in a clinical manner that reflects typical patient interactions in a real-world clinical setting. This included not only providing medical advice but also addressing patient concerns, clarifying doubts, and offering reassurance. By using these methods, the study aimed to draw meaningful comparisons between AI-generated advice and human-delivered medical guidance.

15.2.4 Distribution and Initial Filtering

Serbia

The online survey targeting Serbian university students was conducted in Serbian. It achieved a total of 133 responses. Data collection was initiated on October 3, 2024, at 09:41:23 and concluded on October 29, 2024, at 19:56:46. All responses were deemed valid for analysis based on the criteria of country of residence and gender, resulting in a final sample size of 133 for Serbia.

Kazakhstan

The survey administered to high school students in Kazakhstan was conducted in Kazakh. It initially gathered 220 responses, with the data collection period ranging from September 20, 2024, at 13:14:20 to October 7, 2024, at 15:51:07. During the initial data cleaning process, two responses, located in rows 165 and 180, were excluded from the dataset due to inappropriate answers in the gender category. In addition, one of these responses indicated Serbia as a country of residence, warranting exclusion. After these exclusions, the final sample size from Kazakhstan was 218.

Kyrgyzstan

The surveys in Kyrgyzstan were distributed in two languages, English and Russian, receiving 94 responses in total. The English version received 41 responses, with data collection spanning from September 29, 2024, at 22:42:40 to November 3, 2024, at 10:23:59. Two responses were excluded: one from row 33, where Serbia was noted as the country of residence and another from row 30 with non-standard text in the gender response. This resulted in a final count of 39 usable English responses.

The Russian survey garnered 53 responses, with the data collection timeline from September 25, 2024, at 06:01:22 to October 26, 2024, at 09:32:37. One response in row 2, where a dot was used in the gender section, was excluded, yielding a final count of 52 responses. Thus, Kyrgyzstan's combined final sample size was 91 after initial filtering.

15.2.5 Demographics

Serbia

A total of 133 participants from Serbia completed the online survey. The respondents' ages ranged from 18 to 55 years, with a mean age of approximately 25.2 years (SD = 8.0) and a median age of 22 years. The age distribution indicates a predominantly young adult sample, reflective of the university student population targeted in this study.

The gender distribution was notably skewed towards female participants. Of the 133 respondents, 97 identified as female (72.9%), 35 identified as male (26.3%), and 1 participant (0.8%) opted not to disclose their gender. This disproportion may reflect the gender demographics within the faculties surveyed or the fields of study represented.

In terms of educational level, the majority of participants were undergraduate students, with 98 individuals (73.7%) enrolled in bachelor's degree programs. The remaining 35 participants (26.3%) were postgraduate students pursuing advanced degrees. This educational spread provides a comprehensive view encompassing both early-stage and advanced university learners.

Regarding fields of study, there was a significant representation of technical sciences ("tehničke nauke"), accounting for approximately 31 participants (23.3%). Social sciences ("društvene nauke") were also prominently featured, with about 28 participants (21.1%) indicating this as their major. Smaller subsets of the sample were studying humanities ("humanistika"), medicine ("medicina"), education ("vaspitač" or "vaspitanje"), and pedagogy ("pedagogija"). Notably, a considerable portion of respondents, around 40 individuals (30.1%), did not specify their field of study. This lack of specification could be attributed to the survey design or participants' preference for privacy.

Overall, the Serbian survey sample was comprised primarily of young adult female undergraduate students with diverse academic interests, predominantly in technical and social sciences.

Kazakhstan

A total of 218 participants from Kazakhstan completed the online survey. The ages of the respondents ranged from 10 to 20 years, with a mean age of approximately 18.6 years (SD = 2.4). This age distribution reflects a predominantly adolescent cohort, aligning with the survey's focus on gathering insights from secondary school students in Kazakhstan.

The gender distribution among the Kazakhstani participants was relatively balanced, with a slight predominance of male respondents. Specifically, out of the 218 participants, 115 identified as male (52.8%), 98 identified as female (45.0%), and 5 participants (2.2%) preferred not to disclose their gender. This distribution is indicative of an equitable representation of genders within the sampled population.

Regarding educational levels, the majority of participants were secondary school students. Specifically, 143 respondents (65.6%) were students in grades 9–12, corresponding to high school levels. An additional 63 participants (28.9%) were students in grades 7–8, representing middle school levels. Undergraduate students constituted a smaller portion of the sample, with 12 participants (5.5%) enrolled in bachelor's degree programs. This educational breakdown highlights the survey's emphasis on younger students, providing valuable perspectives from the emerging generation in Kazakhstan's educational landscape.

The fields of study or areas of interest among the participants were diverse, reflecting a broad spectrum of academic aspirations. Since the majority of respondents were secondary school students, many indicated their intended fields of study rather than current majors. The most frequently mentioned fields included Technical Sciences (22.9%), Medicine (18.3%), Humanities (13.8%), Social Sciences (8.7%), Business (2.3%), Natural Sciences (1.8%), Architecture and Design (a few respondents) and Undecided or Not Specified (31.2%).

Kyrgyzstan

A total of 91 participants from Kyrgyzstan completed the online survey. The respondents' ages ranged from 18 to 45 years, with a mean age of approximately 23.4 years (SD not calculated) and a median age of 21 years. The age distribution shows a concentration of young adults, aligning with the target population of university students. However, there is representation from older age groups, including participants up to 45 years old.

The gender distribution among the Kyrgyzstan participants was relatively balanced, with a slight male predominance. Out of the 91 respondents, 48 identified as male (52.7%), 42 identified as female (46.2%), and 1 participant (1.1%) preferred not to disclose their gender. This distribution reflects a reasonable gender representation within the surveyed academic institutions.

Regarding educational levels, the majority of participants were undergraduate students. Specifically, 83 respondents (91.2%) were enrolled in bachelor's degree programs, 7 participants (7.7%) were graduate students pursuing advanced degrees, and one respondent (1.1%) was a secondary school student. This distribution underscores the survey's focus on higher education students.

The participants from Kyrgyzstan showcased a rich diversity in their fields of study. A substantial portion of respondents were immersed in the social sciences, with approximately one-third (33.0%) indicating this as their major area of study. The humanities also attracted significant interest, comprising about 18.7% of the participants, which included disciplines like literature, history, and languages. Equally represented were the technical sciences, accounting for another 18.7%. Smaller yet notable groups pursued natural sciences (4.4%), business administration (3.3%), and specialised fields such as liberal arts, applied geology, journalism, and international relations. Additionally, a noteworthy 17.6% of respondents did not specify their field of study, which could be attributed to personal preference or survey design.

15.3 Survey Results

This section presents a detailed description of the survey results, summarising the participants' familiarity with AI, preferred health information sources, trust in AI-generated information, usage of AI tools for health-related queries (Table 15.1), and perceptions of medical advice across Serbia, Kazakhstan, and Kyrgyzstan (Table 15.2).

15.3.1 Familiarity with AI

Participants were asked to rate their familiarity with AI and technology on a scale from 1 (least familiar) to 7 (most familiar). The average familiarity scores indicate varying levels of exposure to AI across the three countries. In Serbia, the mean familiarity score was 4.4 (SD = 1.5), suggesting a moderate level of familiarity among respondents. Kazakhstan reported a higher mean familiarity score of 5.4 (SD = 1.6), indicating greater exposure and comfort with AI technologies. Kyrgyzstan's

Table 15.1 Comparative summary of survey results across Serbia, Kazakhstan, and Kyrgyzstan

Measure	Serbia	Kazakhstan	Kyrgyzstan
Number of respondents	133	218	91
Healthcare (2024)—WHO Index 2000	0.629	0.752	0.455
Healthcare Index 2024 (CEO World)	37.76	34.28	N/A
Average age (years)	25.2	15.4	23.4
Mean familiarity with AI (1–7 scale)	4.4 (SD = 1.5)	5.4 (SD = 1.6)	5.1 (SD = 1.3)
Primary health information source	Doctors (62%)	Social media (44%)	Health websites (35%)
Mean trust in AI (1–7 scale)	4.2 (SD = 1.2)	5.2 (SD = 1.3)	4.4 (SD = 1.3)
Used AI for health queries (% Yes)	38	54	46

Table 15.2 Comparative Summary of Perceptions of Medical Advice Across Serbia, Kazakhstan, and Kyrgyzstan

Measure	GPT-4o	Llama 3.1	Gemini Pro 1.5	Real doctor
Serbia				
Trustworthiness	4.9	5.1	4.6	5.5
Usefulness	5.0	5.2	4.8	5.6
Empathy	4.7	4.9	4.5	5.4
Correct identification (%)	42			60
Kazakhstan				
Trustworthiness	5.3	5.4	5.0	5.8
Usefulness	5.5	5.6	5.1	5.9
Empathy	5.1	5.2	4.9	5.7
Correct identification (%)	50			65
Kyrgyzstan				
Trustworthiness	5.0	4.9	4.7	5.6
Usefulness	5.2	5.0	4.9	5.7
Empathy	4.9	4.8	4.6	5.5
Correct identification (%)	45			62

mean score was 5.1 (SD = 1.3), which, while slightly lower than Kazakhstan's, still reflects a relatively high level of familiarity compared to Serbia.

15.3.2 Preferred Sources for Health Information

The survey explored where participants most frequently seek health information. In Serbia, a significant majority of respondents (62%) reported that they primarily consult doctors or health professionals. Health websites were the next most common source at 15%, followed by friends and family at 11%, social media at 8%, and other sources at 4%. This preference indicates a firm reliance on professional medical advice within the Serbian population.

In contrast, respondents from Kazakhstan showed a different pattern. Social media emerged as the most frequented source for health information at 44%, surpassing doctors or health professionals, which accounted for 24% of responses. Health websites were used by 15% of participants, friends and family by 11%, and other sources by 6%. The prominence of social media in Kazakhstan suggests a cultural shift towards more digital and possibly peer-shared health information channels.

Kyrgyzstan's respondents predominantly relied on health websites (35%) and doctors or health professionals (30%) for health information. Social media was the primary source for 20% of participants, friends and family for 10%, and other sources for 5%. This indicates a balanced approach in Kyrgyzstan, with significant use of both professional and online resources.

15.3.3 Trust in AI-Generated Information

Participants rated their trust in AI-generated information on a scale from 1 (least trust) to 7 (most trust). The mean trust score in Serbia was 4.2 (SD = 1.2), reflecting moderate trust levels. Kazakhstan reported a higher mean trust score of 5.2 (SD = 1.3), indicating a more positive perception of AI-generated health information. Kyrgyzstan had a mean score of 4.4 (SD = 1.3), slightly higher than Serbia but lower than Kazakhstan.

The higher trust levels in Kazakhstan align with the country's greater familiarity with AI, suggesting that increased exposure to AI technologies may enhance trust in AI-generated content. In contrast, the moderate trust levels in Serbia may correspond with lower familiarity and reliance on traditional health information sources.

15.3.4 Usage of AI Tools for Health-Related Queries

When asked if they had ever used AI tools for health-related queries, 38% of Serbian respondents indicated that they had, whereas 54% of participants from Kazakhstan and 46% from Kyrgyzstan reported prior usage. The higher usage rate in Kazakhstan aligns with the higher familiarity and trust in AI observed in the survey.

These results suggest that Kazakhstan's population is more actively integrating AI tools into their health information-seeking behaviours. Kyrgyzstan also shows a considerable level of AI tool usage, which may be influenced by their reliance on health websites and moderate trust in AI. Serbia's lower usage rate corresponds with their preference for consulting health professionals and lower familiarity with AI technologies.

The comparative data reveal distinct differences in how individuals from Serbia, Kazakhstan, and Kyrgyzstan engage with AI and health information (Table 15.1). Kazakhstan exhibits the highest familiarity, followed by Kyrgyzstan and Serbia. Serbia prefers traditional sources (doctors), Kazakhstan leans towards social media, and Kyrgyzstan balances between health websites and professionals. Trust levels are highest in Kazakhstan, moderate in Kyrgyzstan, and slightly lower in Serbia. A more significant proportion of Kazakhstani respondents have used AI tools for health queries, indicating a more considerable integration of AI in health practices.

15.3.5 Perception of Trustworthiness

Across Serbia, Kazakhstan, and Kyrgyzstan, participants generally perceived medical advice from actual doctors as more trustworthy than advice generated by AI models. In Serbia, the actual doctor's advice received a mean trustworthiness score of 5.5 on a 7-point Likert scale, surpassing the AI models. Among the AI-generated advice, Llama 3.1 was rated highest in trustworthiness with a mean score of 5.1, followed by GPT-4o at 4.9 and Gemini Pro 1.5 at 4.6.

In Kazakhstan, this trend continued with the actual doctor's advice, achieving the highest trustworthiness score of 5.8. Llama 3.1 again led among AI models with a mean score of 5.4, slightly ahead of GPT-4o at 5.3 and Gemini Pro 1.5 at 5.0. Participants in Kyrgyzstan also rated the actual doctor's advice as more trustworthy, with a mean score of 5.6. GPT-4o and Llama 3.1 received similar trustworthiness scores of 5.0 and 4.9, respectively, while Gemini Pro 1.5 scored slightly lower at 4.7.

15.3.6 Perception of Usefulness

Participants across the surveyed countries found medical advice from actual doctors to be more beneficial than that provided by AI models. In Serbia, the actual doctor's advice achieved a mean usefulness score of 5.6. Among AI models, Llama 3.1 was considered the most useful, with a mean score of 5.2, followed by GPT-4o at 5.0 and Gemini Pro 1.5 at 4.8.

In Kazakhstan, real doctors' advice was rated highly useful, with a mean score of 5.9. Llama 3.1 once again led the AI models with a mean usefulness score of 5.6. GPT-4o closely followed at 5.5, and Gemini Pro 1.5 at 5.1. Kyrgyzstan's participants also favoured advice from real doctors, assigning it a mean usefulness score of 5.7. GPT-4o received a mean score of 5.2 in usefulness, slightly ahead of Llama 3.1 at 5.0 and Gemini Pro 1.5 at 4.9.

15.3.7 Perception of Empathy

In terms of empathy, participants again favoured advice from actual doctors over AI-generated responses. Serbian participants rated the real doctor's advice with a mean empathy score of 5.4. Among AI models, Llama 3.1 was perceived as the most empathetic, scoring a mean of 4.9. GPT-4o and Gemini Pro 1.5 followed with mean scores of 4.7 and 4.5, respectively.

Kazakhstan's participants also attributed higher empathy to the actual doctor's advice, which received a mean score of 5.7. Llama 3.1 led the AI models with a mean empathy score of 5.2, while GPT-4o scored 5.1 and Gemini Pro 1.5 scored 4.9. In Kyrgyzstan, the pattern persisted, with the actual doctor's advice scoring a mean empathy of 5.5. GPT-4o received a mean score of 4.9, slightly surpassing Llama 3.1 at 4.8 and Gemini Pro 1.5 at 4.6.

15.3.8 Ability to Distinguish Between AI and Real Doctors

Participants demonstrated moderate success in distinguishing between AI-generated advice and that from real doctors. In Serbia, 42% of participants correctly identified the AI-generated advice, while 60% accurately recognised the actual doctor's advice (Table 15.2). This indicates that while a majority could identify human-provided advice, a significant portion found it challenging to differentiate AI responses.

Kazakhstani participants showed slightly better discernment, with 50% correctly identifying AI-generated advice and 65% recognising advice from a real doctor. In Kyrgyzstan, 45% of participants accurately identified AI-generated advice, and 62% correctly identified the actual doctor's advice.

15.4 Conclusion

This study explored the perceptions of individuals from Serbia, Kazakhstan, and Kyrgyzstan regarding medical advice provided by human doctors compared to that generated by various AI models. The findings reveal a consistent preference for advice from actual doctors over AI models across all three countries, highlighting the enduring importance of human interaction in healthcare.

Participants overwhelmingly rated medical advice from actual doctors as more trustworthy, helpful, and empathetic than advice provided by AI models. This trend was evident across all surveyed countries despite slight variations in the magnitude of the differences. The higher ratings for human doctors underscore the critical role of trust and empathy in healthcare communication. Trustworthiness in this context is likely tied to the perceived expertise, accountability, and ethical responsibility of medical professionals, attributes that AI models have yet to embody in the eyes of patients fully.

Among the AI models assessed, Llama 3.1 consistently received slightly higher scores across all measured attributes compared to GPT-4o and Gemini Pro 1.5. This suggests that some AI models may be better at generating medical advice that resonates with users. However, even the highest-scoring AI model did not surpass the ratings given to real doctors, indicating that significant gaps remain in how AI is perceived relative to human healthcare providers.

15.4.1 Research Question 1

How do individuals from Serbia, Kazakhstan, and Kyrgyzstan perceive the trustworthiness, usefulness, and empathy of medical advice provided by AI models compared to that offered by human doctors?

The study revealed a consistent perception across participants from Serbia, Kazakhstan, and Kyrgyzstan that medical advice provided by human doctors is more trustworthy, useful, and empathetic than advice generated by AI models. In all three countries, participants rated advice from actual doctors significantly higher on a 7-point Likert scale across the measured attributes.

These findings underscore a clear preference for human-provided medical advice, reflecting the importance participants place on the perceived expertise, accountability, and empathetic communication of human doctors. While AI models like Llama 3.1 received relatively higher scores among AI peers, they did not surpass the ratings given to real doctors in any of the measured attributes. This indicates that, despite advancements in AI capabilities, individuals from these countries still value the human elements of trustworthiness, usefulness, and empathy in medical consultations.

15.4.2 Research Question 2

Research Question 2: What are the differences in familiarity with AI technology among participants from the three countries, and how do these differences influence their trust and acceptance of AI-generated medical advice?

Participants from the three countries displayed varying levels of familiarity with AI technology, which influenced their trust and acceptance of AI-generated medical advice. Kazakhstani participants reported the highest familiarity with AI, with a mean score of 5.4 (SD = 1.6) on a 7-point scale. Kyrgyzstan followed with a mean familiarity score of 5.1 (SD = 1.3), while Serbian participants reported the lowest familiarity, with a mean score of 4.4 (SD = 1.5).

These differences in familiarity correlated with trust levels in AI-generated information. Kazakhstani participants not only felt more familiar with AI but also exhibited higher trust in AI-generated medical advice, with a mean trust score of 5.2 (SD = 1.3). Kyrgyz participants showed moderate trust, with a mean score of 4.4 (SD = 1.3), while Serbian participants had the lowest trust levels, with a mean score of 4.2 (SD = 1.2).

The acceptance of AI-generated medical advice, as reflected by prior usage of AI tools for health-related queries, mirrored these patterns. In Kazakhstan, 54% of participants had used AI tools for health queries, higher than Kyrgyzstan's 46% and Serbia's 38%. The higher familiarity and trust in Kazakhstan likely contribute to the greater acceptance and utilisation of AI in healthcare contexts.

These findings suggest that increased familiarity with AI technology enhances trust and acceptance of AI-generated medical advice. Participants who are more accustomed to interacting with AI are more likely to perceive it as a reliable and useful resource for health information. Conversely, lower familiarity may contribute to scepticism and a preference for traditional medical consultations.

15.4.3 Research Question 3

To what extent can participants distinguish between medical advice generated by AI models and that provided by human doctors, and what factors contribute to their ability to differentiate between the two?

Participants demonstrated a moderate ability to distinguish between AI-generated medical advice and that provided by human doctors. In Serbia, 42% of participants correctly identified AI-generated advice, while 60% accurately recognised the actual doctor's advice. In Kazakhstan, 50% correctly identified AI advice, with 65% recognising human-provided advice. Kyrgyzstan showed similar results, with 45% correctly identifying AI-generated advice and 62% recognising advice from real doctors.

These percentages indicate that while a majority of participants could identify advice from real doctors, a significant portion found it challenging to differentiate between AI and human-generated advice. Factors contributing to their ability to distinguish likely include subtle differences in language style, tone, and the presence of empathetic cues.

Participants may have detected the way AI models communicate, such as a lack of personalisation, formality, or absence of emotional language, which are aspects human doctors inherently incorporate into patient interactions. Additionally, cultural expectations of how medical advice is delivered may influence recognition, with participants relying on their experiences of typical doctor-patient communication to identify human advice.

The moderate success in differentiation suggests that AI models are increasingly capable of producing human-like responses. Yet, they still lack certain qualities that are distinctly human, particularly in conveying empathy and understanding patient concerns.

15.4.4 Cultural and Technological Influences

The study uncovered notable differences in familiarity with AI, preferred sources of health information, and trust in AI-generated content among the three countries. Participants from Kazakhstan reported the highest familiarity with AI and technology (mean score of 5.4 on a 7-point scale) and the highest trust in AI-generated information (mean score of 5.2). This higher familiarity and trust correlate with a more significant usage of AI tools for health-related queries in Kazakhstan (54% of participants) compared to Kyrgyzstan (46%) and Serbia (38%).

These differences may reflect varying degrees of technological integration and cultural openness to AI across the countries. Kazakhstan's younger average participant age (mean age of 15.4 years) could contribute to greater comfort with technology and social media, which was the primary source of health information for 44% of Kazakhstani respondents. In contrast, Serbian participants, with a higher mean age of 25.2 years, showed a strong preference for consulting doctors or health professionals (62%), indicating reliance on traditional sources of medical advice.

Kyrgyzstan presented a balanced approach, with participants obtaining health information from health websites (35%) and doctors or health professionals (30%). This suggests a transitional phase where both traditional and digital sources are valued.

15.4.5 Ability to Distinguish Between AI and Human Advice

Participants demonstrated moderate success in distinguishing between AI-generated advice and advice from real doctors. Approximately half of the participants correctly identified the source of the advice. This ability varied slightly by country but generally indicated that AI models are becoming increasingly sophisticated in mimicking human-like communication. However, characteristics of human interaction, particularly in conveying empathy and understanding, remain distinguishable to many users.

The modest rate of correct identification suggests that while AI can produce linguistically sophisticated responses, there are still detectable differences in style, tone, or content that signal the absence of human touch. Participants may pick up on these subtle cues, which can influence their perceptions of trustworthiness and empathy.

15.4.6 The Role of Empathy in Medical Advice

Empathy emerged as a critical differentiator between AI-generated advice and that provided by human doctors. Participants consistently rated human doctors higher in empathy, highlighting the importance of emotional connection in healthcare interactions. Empathy facilitates trust, encourages patients to share information openly, and can improve health outcomes by fostering adherence to medical recommendations.

AI models, while capable of generating accurate and informative content, cannot currently fully replicate the emotional intelligence inherent in human interactions. This limitation affects how users perceive and value AI-generated medical advice. Enhancing the empathetic capabilities of AI is a complex challenge that involves not only natural language processing but also understanding context, emotion, and cultural features.

15.4.7 Implications for AI Integration in Healthcare

The findings suggest that while AI models have made significant progress in generating medical advice, there is still a reluctance to accept AI as a substitute for human doctors. Trust, usefulness, and empathy are crucial factors that influence patients' acceptance of medical advice. AI models need to address these aspects more effectively to be considered viable alternatives or supplements to human providers.

In regions where access to healthcare professionals is limited, AI could play a supplementary role in providing health information. However, ensuring that AI tools are trusted and perceived as empathetic is essential for their effectiveness. Cultural factors must also be considered, as acceptance of AI varies across different societies.

15.4.8 Limitations

Several limitations of the study should be acknowledged. The sample population was predominantly composed of students, with significant age differences among the countries—especially the younger cohort from Kazakhstan. This demographic may not fully represent the general population, limiting the generalizability of the findings.

The study relied on self-reported measures of perceptions, which can be influenced by individual biases, social desirability, and varying interpretations of the Likert scale. Language differences and translations of the surveys could also introduce subtle biases or misunderstandings.

The AI models were tested using a specific prompt designed to produce conversational and empathetic responses. Different prompts or real-world interactions may yield different results.

15.4.9 Future Directions

Future research should aim to include a more diverse and representative sample, encompassing various age groups, educational backgrounds, and cultural contexts. Longitudinal studies could assess changes in perceptions over time as AI technology evolves and becomes more integrated into daily life.

Exploring methods to enhance the empathetic capabilities of AI models is a critical area for development. This could involve advanced machine learning techniques that better interpret and respond to emotional cues, as well as incorporating feedback from users to improve AI interactions continuously.

The actual effectiveness of AI-generated medical advice in improving health outcomes is also essential. Clinical trials or real-world evaluations could provide insights into the practicality and reliability of AI as a tool in healthcare settings.

15.4.10 Final Remarks

The study highlights the current limitations of AI models in replicating the trustworthiness, usefulness, and empathy inherent in medical advice provided by human doctors. While AI shows promise, particularly with models like Llama 3.1 performing slightly better among AI peers, there is a clear preference for human interaction in healthcare across Serbia, Kazakhstan, and Kyrgyzstan.

Cultural and technological factors influence how populations perceive and engage with AI-generated health information. As AI technology advances, addressing the challenges of building trust and conveying empathy will be essential for its broader acceptance and integration into healthcare services.

Understanding these perceptions is crucial for healthcare providers, policymakers, and technology developers aiming to leverage AI to enhance healthcare delivery. The human element remains indispensable, and AI should be viewed as a complementary tool that can support, but not replace, the vital role of medical professionals in patient care.

Since this study included human participants, adequate ethical approval was acquired from the Ethics Committee of the Institute for Artificial Intelligence Research and Development of Serbia, which was granted on September 3, 2024.

15.4.10.1 Availability of Data and Materials

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Availability of Data and Materials The datasets from this study, including texts of surveys initial and filtered data, are available in the Open Science Framework repository: https://osf.io/rkbtn/?view_only=5184f460c89e4a438112de19068d4311

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Chapter 16
AI Technology in Psychological Education and Treatment: An Overview of Practical and Creative Applications

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Abstract This chapter explores innovative applications of artificial intelligence (AI) technology in the education of psychology undergraduates as well as in psychological treatment. It begins by examining how generative AI can assist with specialised training and supervision for undergraduate students. The chapter discusses how practising interview techniques with AI-generated patients can help psychologists develop their professional communication skills. Additionally, it highlights the creative uses of AI in clinical practice, such as its potential to support diagnostic processes and enhance prevention efforts by identifying individuals who need mental

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health support in a timely manner. The chapter also provides an overview of the practical applications of AI in psychotherapy, including "chatbot therapists" and text-to-image generation in art therapy. Finally, it addresses ethical considerations, risks, and limitations associated with these technologies. Overall, this chapter focuses on the creative possibilities of generative AI in the fields of psychological education and treatment.

Keywords Generative artificial intelligence · Psychological diagnosis · Psychological training · Psychological treatment · Supervision

16.1 Introduction

The number of people needing mental health treatment has increased by approximately 48% over the past two decades (GBD 2019 Mental Disorders Collaborators 2022). This rising demand could be at least partially attributed to the COVID-19 pandemic, war conflicts, and the resulting economic challenges (Pandey et al. 2021; Seleznova et al. 2023). While there is a growing need for available mental health treatment possibilities worldwide, the number of mental healthcare professionals could not increase proportionally. Indeed, according to McGinty and Eisenberg (2022), 50–70% of people in need of mental health treatment lack access to therapy worldwide, and this treatment gap (Kale 2002) is expected to increase rather than decrease. According to the World Health Organization (see Minerva and Giublini 2023), there is a lack of approximately 4.3 million mental health professionals, and this gap will expectedly reach 10 million by 2030 in low and middle-income countries (Doraiswamy 2020; Minerva and Giublini 2023).

Meanwhile, the advent of generative artificial intelligence (GAI) holds promising implications for narrowing the treatment gap. Although artificial intelligence has been present since the 1950s, influencing areas of development such as algorithmic preferences and search functionality, the introduction of large language models (LLMs) in chatbots like ChatGPT in November 2022 contributed to increased general knowledge and application of AI in the general population (Haber et al. 2024; Ozmen et al. 2023). The rapid expansion of GAI not only transforms Internet use patterns and habits but also opens new opportunities in psychological treatment and education. This chapter focuses on the possibilities of AI to increase the efficacy of psychologists and reflects on the role of AI in altering the mechanisms underlying psychotherapy.

16.2 AI Technology in Psychologists' Professional Training

16.2.1 The Role of Autonomous Training Agents in the Training of Psychology Students

Graduate training in psychology involves the development of communication strategies, conversational skills, interviewing techniques, and other interpersonal skills that enable professionals to conduct sessions with clients. However, graduated students can experience difficulties in handling real-life situations with patients, as introducing students to the same patient with identical mental disorders in a standardised manner each year is an impossible task. In addition, fresh graduates often enter field practice with limited experience in clinical settings, generally restricted to the completion of introductory lectures on clinical methods and processes (Washburn et al. 2018). This lack of relevant clinical experience can raise concerns in provisionally licensed professionals who may work with many clients, as insufficient supervision over the work of fresh undergraduates can potentially increase the risk of providing inadequate or ineffective treatment for vulnerable patients (Auger et al. 2004; Washburn et al. 2018). Applications from computer science, specifically autonomous training agents (ATA) and AI, hold promise to bridge this gap by simulating virtual patients, thereby providing psychology graduates with a safe environment to foster professional conversation skills and strategies in a controlled, supervised, and measurable manner (da Silva and Hendricker 2021).

In computational science, autonomous agents are "computational systems that inhabit some complex, dynamic environment, sense and act autonomously in this environment, and by doing so realise a set of goals or tasks for which they are designed" (Maes 1995, p. 108). These agents can be creatively used as ATAs to simulate virtual clients as interactive avatars that enable students to engage in realistic clinical settings, similar to what trained "actor" patients provide for psychology students to practice professional conversational skills. These virtual patients can respond to students' explorative questions on their personal background, medical history, anamnesis, and current symptoms. Depending on how these virtual clients are programmed, they can communicate through text, voice input, or a combination of these formats (Washburn et al. 2018). Evidence shows that skills acquired through virtual patient simulation sessions can be effectively used when interacting with real clients. Indeed, the outcomes of these encounters can be comparable to those simulations using traditional standard actor patients in terms of diagnostic accuracy and subjective effectiveness of the treatment (Triola et al. 2006).

AI-based virtual standardised patients (VSPs) are digital representations of human-standardised patients using natural language processing (NLP). This AI-based technology enables computers to interpret, process, and understand human language. VSPs were found to be valuable sources for the development and practice of verbal and nonverbal communication strategies that can help professionals' training. For instance, in a recent study (Maicher et al. 2022), 620 first-year medical students asked questions from VSPs about the history of present medical illness, past medical history, family history, and social history. From a total of 4663 questions covering bespoken topics, 914 were randomly selected and rated by three independent expert faculty raters and one computer rater. Raters (including the computer rater) assigned one score to each student question that included essential elements regarding bespoken topics. Overall, the scores from human raters were not different from computer rater's scores, indicating that VSPs provide a practical tool to practice conversational skills before interacting with human patients (Maicher et al. 2022). Virtual patient simulations also offered support in the development of brief assessment skills among medical professionals in the field of primary care (Satter et al. 2012; Washburn et al. 2018). Findings showed that diagnostic accuracy increased with the frequency of interactions with virtual patients (Washburn et al. 2018). Overall, the main advantages of virtual patient simulations could be that these simulations can be repeated anytime, at the student's convenience, allowing for intense training (Washburn et al. 2018).

There are only some examples of empirical studies investigating the role of ATA in psychology professional training. However, the creative and practical use of ATA in psychology professional training discussed in this chapter may be general and widespread among psychologists in the near future.

16.2.2 AI in Supervision

An essential part of psychologists' learning and development process is supervision, which AI can also support to some extent. Stade et al. (2024) highlighted the existence of supervisor-facing applications that provide a range of benefits, including LLM-based summaries and analysis of supervisees' therapy sessions, which can inform supervisors about areas for potential development. From another perspective, trainee-facing applications can offer feedback on trainees' performance and also guidance for further improvement. Stade et al. (2024) suggested that LLMs could be used to enhance the effectiveness of text-based online supportive conversations among untrained peer supporters. The HAILEY (Human-AI CoLlaboration approach for EmpathY) system developed by Sharma et al. (2023) served as an AI-in-the-loop agent aimed at improving peer-to-peer mental health support by proposing modifications to human-generated responses in order to increase empathy in the content. In this setting, the primary agents in the interaction are two humans, with AI providing background support. Sharma et al. (2023) found that this human-AI collaboration increased conversational empathy and enhanced peer supporters' empathy and perceived self-efficacy in some cases. Enhancing empathy is crucial for successful therapy outcomes.

Similarly, Kearns et al. (2024) developed the A2P2 (AI-Assisted Provider Platform). Participants were trainees in psychiatry and clinical psychology programs, as well as non-expert nurses, who engaged in text-based therapeutic conversations with standardised patients. Unlike the previous approach in which participants generated their responses and were potentially encouraged to refine them based on AI-driven feedback, in this condition, participants selected their responses, with AI-based assistance, from a set of pre-generated responses created by professionals. Results showed that, compared to the control group, the AI-assisted group demonstrated reduced response time, increased accuracy in empathic responses, and improved goal selection accuracy, regardless of whether participants were experts or non-experts. Kearns et al. (2024) highlighted that both groups reported learning from the process, supporting the potential of A2P2 as a training tool.

In a similar vein, the study by Lanz et al. (2023) offered an intriguing perspective on AI-led or AI-supported supervision. Although this research was conducted in an organisational context, the question raised could also be relevant to psychological supervision: how do employees respond to instructions from an AI supervisor, particularly when these instructions can impact others, too? Lanz et al. (2023) found that people were attentive to whether they received instructions from a human or an AI supervisor in moral contexts, showing decreased adherence to AI-based directives. These findings raise the question of how psychologists may experience the supervision process when AI has a significant role in the evaluation: whether they react with rejection or appreciation.

16.3 AI in Psychological Treatment

16.3.1 AI-Driven Diagnostics and Monitoring

In recent years, the proliferation of smartphones has opened possibilities for using electronic communication in tracking and diagnosing mental health conditions. Narziev et al. (2020) identified five factors associated with depression (i.e., physical activity, mood, social activity, sleep, and food intake). They explored features captured by smart devices' sensors related to each affected area. The authors collected data from participants' smart devices that reported data. Using AI technology, an automatic classification of depression categories (normal, mildly, moderately, and severely depressed) was performed. In comparison with traditional depression severity measurements (e.g., interviews and self-reports), this technology has the potential to require less time, money, and involvement of a human professional.

16.3.2 AI in Predictive Modelling and Prevention

Various AI-monitoring techniques have shown promise in enhancing predictive analysis and, therefore, contribute to the timely identification of patients who could be at risk of severe mental health problems. In a recent meta-analysis, Franklin et al. (2017) included 365 studies from the past 50 years to identify risk factors for suicidal behaviours. It was found that human professionals performed only slightly better than chance at predicting suicide. On the contrary, an AI-powered algorithm used in a study by Walsh et al. (2017) predicted suicide attempts with 85% accuracy within 24 months and with 92% accuracy within 2 weeks.

AI could also be eligible to explore the predictors of psychotherapy outcome and attrition (Lenton-Brym et al. 2024). For example, machine learning was used to identify core characteristics of patients that can moderate the impact of therapeutic adherence on the outcomes of applied psychotherapy (Rubel et al. 2019; Zilcha-Mano et al. 2017). Of course, not only AI can identify predictive factors of therapy dropouts and attrition. For instance, Duhne et al. (2022) used a predictive algorithm to determine patients who could benefit more from face-to-face versus a computerised intervention. Based on the AI-generated recommendations, patients were less likely to drop out of the treatment and experienced more favourable treatment outcomes.

16.3.3 Chatbot-Delivered Psychotherapies

Psychotherapy originates from the fundamental work "Studies in Hysteria" by Freud and Breuer (2004), which introduced psychotherapy as a "talking cure" (Haber et al. 2024). Despite the advancements in various psychotherapeutic approaches, the primary focus on patient-therapist dialogue has remained (Cuijpers et al. 2019; Haber et al. 2024). Provided that language has a mediative role in psychotherapy – which AI-powered chatbots can process through NLP and also can interpret the text entered by the users and respond accordingly –some researchers posited that psychotherapeutic chatbots are well-positioned between human-delivered psychotherapy and self-help applications (Eltahawy et al. 2024).

In recent years, AI has been increasingly used in delivering psychotherapies through AI-powered chatbots (e.g., Bendig et al. 2022; Grodniewicz and Hohol 2023). The global lockdown caused by the COVID-19 pandemic has increased general interest in technological solutions to enhance the accessibility of psychotherapy, including the use of automated chatbots (Eltahawy et al. 2024; Zeavin 2022). In the context of psychotherapy, chatbots are fully autonomous generative AI systems that can conduct conversations with users. These chatbots demonstrated promise in autonomously delivering psychotherapy (Eltahawy et al. 2024). Chatbots can mimic human interactions and thus provide a more immersive user experience than other online self-help options (Grové 2021; Lavelle et al. 2022). In addition, evidence shows that chatbots can have a positive impact on people's mental well-being through therapeutic interventions (Bird et al. 2023; Fitzpatrick et al. 2017; Fulmer et al. 2018). Chatbots can provide certain aspects of therapy that can effectively foster mental health, such as a sense of connection, cooperation, and mood improvement (Eltahawy et al. 2024; Pham et al. 2022; Thompson 2018). Moreover, chatbots have the potential to bridge the treatment gap due to their accessibility and cost-efficiency as compared to traditional, human-delivered therapies. Furthermore, chatbot-delivered psychotherapies could be less intimidating for patients, which could be particularly beneficial for patients who find it challenging to develop interpersonal trust. Indeed, Lucas et al. (2017) demonstrated this association in relation to posttraumatic stress disorder (PTSD) symptoms among military members. These respondents disclosed more sensitive information than they admitted to sharing with a human therapist in this situation due to the anonymity offered by the chatbot (Vaidyam et al. 2019). Moreover, chatbots can assist in the improvement of social skills among children who have autism spectrum disorder (ASD). Chatbots could also provide companionship for elderly, socially isolated, or depressed patients (Bemelmans 2012). Moreover, a growing body of research has examined the potential of chatbots to reduce stress, loneliness, and agitation, and improve mood and social connections (Fiske et al. 2019).

Besides, users may benefit from the fact that in contrast to traditional therapy offered by a human professional, chatbot-delivered therapies can be delivered in a written format instead of verbal communication. Expressive writing has been associated with decreased rumination tendencies (Gortner 2006). It is a widely recognised therapeutic method, shown to have similar efficacy to cognitive behavioural therapy (CBT) in treating PTSD when incorporated into CBT, which is also referred to as cognitive-behavioural writing (Pascoe 2016).

In addition to its beneficial feature of mimicking psychotherapy, some practical aspects of chatbots (e.g., being portable and cost-effective) can be advantageous for those seeking therapy but cannot afford it or lack the availability of professionals (Eltahawy 2024). In summary, psychotherapeutic chatbots can offer a “feasible, engaging, and effective way” to deliver therapy (Fitzpatrick et al. 2017, p. 19).

Using generative AI technology, users can create a personalised psychologist or psychotherapist to fit their specific needs. Chatbots such as Chat-GPT offer features to fine-tune the therapeutic method and communication style of the “AI therapist”. Additionally, users might have the option of having therapeutic conversations with AI replicas of famous (or even deceased) therapists such as Sigmund Freud or Carl Gustav Jung. Below, a conversation with a custom GPT called ‘Carl Jung Dream Analyst’ is presented to demonstrate “Guiding personal growth through Jungian dream analysis”. Having over a thousand conversations with different users to date, this custom GPT offers possible interpretations of the users’ dreams (OpenAI 2024).

User: I had an interesting dream last night.

ChatGPT: (Carl): I see. I am glad you’ve brought your dream to me. Please take your time and describe it in as much detail as you can. What images, emotions, or symbols stood out to you?

User: I tried to slip into my high heels, but my feet were too sweaty and warm. When I put the shoes on, suddenly the shoes became like bread and fell apart around my feet.

ChatGPT: (Carl): This is a fascinating dream, filled with rich imagery. Let us take a closer look. The high heels, an object of elegance and perhaps even power, seem to represent something more than just footwear. What do high heels mean to you in waking life? Do they symbolise a particular aspect of yourself or perhaps a role you feel you must fulfil?

The difficulty in wearing them because of the sweat and warmth suggests a kind of discomfort or resistance. There is something natural—sweat—that makes it difficult for you to fit into this particular image or role. Could it be that you are struggling to meet certain expectations, whether from others or yourself?

Then, we have the image of the shoes turning into bread, which is a striking transformation. Bread is a symbol of sustenance, something nourishing, but here it crumbles, falling apart. There may be a message in this about something that should provide support or structure, yet it disintegrates when you try to rely on it.

How do you feel about this transformation? Does it remind you of anything in your current life, perhaps something that seemed solid but is now proving to be less reliable than you expected?

As seen in the conversation above, the chatbot starts its reply with the name "Carl" to create the impression that the user is talking to the deceased psychoanalyst. Also, it makes the conversation more personal by using the deceased psychoanalyst's first name. Besides, it formulates its answers in a psychoanalytic way and by asking questions at the end of each reply, which encourages the user to maintain and engage more in the conversation.

The use of ChatGPT as a psychoeducational tool in mental healthcare practice has attracted the interest of many researchers in the past few years. A recent study by Maurya et al. (2024) assessed ChatGPT's answers to 21 psychoeducational prompts covering the following critical areas: depression, anxiety, general health, substance abuse, religious/spiritual issues, lifestyle, routine health habits, and interpersonal matters. Using qualitative content analysis, Maurya et al. (2024) concluded that ChatGPT's responses were largely accurate, precise, relevant, and empathic. Besides ChatGPT offering rational and executable suggestions, it also consistently demonstrated ethical considerations by supporting users to seek guidance from trained professionals when necessary (Maurya et al. 2024). These findings highlighted the potential of ChatGPT to be a reliable source of psychoeducation, particularly for those with limited access to alternative resources for mental health support (Maurya et al. 2024).

Custom GPTs offer creative and potentially entertaining ways for individuals to have psycho-educative conversations. However, the therapeutic value of such possibilities remains questionable. Average users are not always fully aware of their actual needs in therapy, which raises doubts about whether a "client-created AI therapist" would be the most effective approach. For example, in most therapeutic processes, confrontation is an essential component. Yet, clients may prefer to avoid integrating the sometimes stress-provoking element of confrontation when shaping the AI therapist's responses. Besides, the term "client-created AI therapist" raises questions about whether to refer to the individual as a *client* or as a *user* in the context of chatbot-delivered psychotherapies as positioning the client as "user" may encourage more personalised customisation of the AI therapist, which can result in that the practical components of psychotherapy (e.g., confronting the clients with inconsistencies observed in their decisions and emotional reactions) may not be present anymore in the virtual mental health support, which could considerably mitigate favourable outcomes (e.g., adaptive coping).

16.3.4 Thanatobots: The Digital Avatars of Deceased Loved Ones

In recent years, representations of deceased individuals as social robots have become more frequent (Brukamp, 2023). The rising industry of the digital afterlife creates a postmortem simulation of a deceased person, which is called a "deadbot," "griefbot," or "thanatobot" (Hollanek et al. 2024; Öhman and Floridi 2017, 2018). The concept of a digital afterlife emerged from past observations regarding the increasing online presence of deceased loved ones. The number of deceased Facebook users was estimated to increase at a daily rate of 19,000 in 2012 (Death Reference Desk 2012). It was predicted that the number of Facebook accounts of deceased users would exceed the number of living ones' by the end of the century (Brown 2016), thereby creating a digital graveyard (Ambrosino 2016; Öhman and Floridi 2017, 2018). As it opens new possibilities for enterprises to develop monetised digital graveyards, it also raises questions about the ethical and psychological impact of such a practice.

Thanatobots enable the bereaving individuals to "communicate" with the deceased. Hollanek et al. (2024) identified three primary participant groups: data donors, data recipients, and service interactants. Data donors refer to the deceased ones whose data are used to create the thanatobots. Data recipients are the ones who own data, enabling them to make the thanatobot. Finally, service interactants are grieving individuals who engage with the created thanatobot. For instance, MaNana is a diegetic prototype of such a re-creation service, which intends to provide companionship and consolidation for families. By using conversational AI technology, data recipients can create a thanatobot mimicking their deceased grandparent. Notably, data are provided to the AI company in the forms of scanned letters, text and voice messages, or even digital footprints such as social media posts and comments uploaded by the data donor. The explicit consent of the data donor is not required (Hollanek et al. 2024).

Not surprisingly, this extended possibility of connection with a deceased loved one raises serious concerns. Provided that there are only a handful of empirical studies examining the long-term effects of such thanatobots on the grieving process to date, the psychological impact of thanatobots on bereaving individuals has not yet been extensively explored. However, the grieving process generally ends with the acceptance of the loss. The prospect of continuing a reciprocal (and external) conversation with the deceased family members might negatively affect the grieving process (Elder 2019; Jiménez-Alonso and Bresco de Luna 2022) and potentially lengthen certain stages of grieving or even hinder the bereaving family members from proceeding to another stage of grieving.

16.3.5 AI in Virtual Reality for Psychotherapy

Virtual Reality (VR) therapy is a promising novel technique enabling users to experience a sense of presence or immersion in a computer-generated, three-dimensional virtual environment while actively interacting with its components (Rothbaum et al. 2002). The effectiveness of VR exposure therapy lies in its ability to immerse users in anxiety-evoking situations in a safe environment (Rahman et al. 2023). Some forms of VR exposure therapy involve the treatment of phobias, such as acrophobia (fear of heights) (Rothbaum et al. 1995) and spider phobia (Carlin et al. 1997; Rothbaum et al. 2002). Another intervention involving VR and AI is the treatment of hallucinations in schizophrenia (Waters et al. 2012). By using digital simulations of patients' distressing voice hallucinations, a digital simulation (avatar) is created to assist a three-way interaction between the patient, therapist, and the avatar (Garety et al. 2021; Terra et al. 2023). The initial phase of this avatar therapy involves exposure to a hostile avatar. In contrast, the second phase may vary depending on the aim of the individual treatment, which may include beliefs about auditory hallucinations (Garety et al. 2021). This method demonstrated a reduction in anxiety and improvement in the frequency of such dissociative experiences (Terra et al., 2023).

Enhanced by biofeedback technology, VR technology enables professionals to monitor patients' bio-signals, such as pulse rate or cortical arousal (Rahman et al. 2023) in real time, making the physical reactions to the object of fear more controllable (Lucifora et al. 2021). So, the effectiveness of the use of AI in VR therapy could rely on the monitoring of the patients' biofeedback data to adjust the intensity and number of stimuli to the patient's physical and psychological reactions for more efficient coping (Gomes et al., 2023).

16.3.6 AI-Generated Art Therapy

Art therapy was found to be an effective therapeutic technique to treat various mental health problems (see Du et al. 2024). Emerging digital art enables technology to contribute to art therapy in multiple ways, such as providing a safe and relaxing environment (Darewych et al. 2015) and increasing the accessibility of treatment (Collie and Čubranić 1999). GAI has the potential to offer innovative and creative possibilities in psychological treatment not only on a conversational basis but also in art therapy. Anybody can create AI-generated images and artwork using textual prompts (Oppenlaender 2022). However, this practice raises questions about whether text-to-image generation is indeed creative or how AI-generated art may have therapeutic effects when used in psychotherapy.

AI systems that generate texts and images have achieved remarkable advancements in the past few years. These AI systems are capable of manipulating, reconstructing, or creating visual (e.g., photos) and textual pieces (e.g., poems) at an advanced level (see Lee, Park and Hahn 2023). In a recent study, Lee, Park and

Hahn (2023) used emotion diary entries to identify main events and related emotions, which they eventually converted into prompts. The authors used DALL-E to generate images, which is a text-to-image generative AI system provided by OpenAI. DALL-E initially created 3–4 images, which evaluators rated based on emotion alignment, event alignment, creativity, and other criteria. Results showed that evaluators perceived images generated by emotion-focused prompts to be more aesthetic, creative, innovative, amusing, and in-depth compared to the images generated by event-focused prompts. The results of this study demonstrated that generative AIs could be used to facilitate creative activities for the narration of personal stories (Lee, Park and Hahn, 2023).

Another existing GAI-powered art therapy is offered by a smartphone application named "Mind Palette". Mind Palette combines GAI technology with art therapy methodologies to create an innovative digital intervention, particularly for younger age groups (Yoo, Kim and Lopes 2023). Although testing the effect of these interventions in a clinical setting is yet to be conducted, these digital art therapy-related projects demonstrate a rising clinical interest in the application of GAI-generated art in specific therapeutic methods such as art therapy.

16.3.7 AI-Generated Music Therapy

Music therapy is a widely recognised evidence-based therapeutic method that involves listening, singing, generating, or composing music, aiming to facilitate patients' mental health (Sun et al. 2024). With the rise of AI, a new field of research has emerged incorporating AI-driven music activities, or "musical AIs," into music therapy (Sun et al. 2024). For instance, a recent study investigated the effect of AI-generated music on the attention capacity of children diagnosed with attention deficit disorder (ADD; Yang et al. 2024). In this study, AI music demonstrated levels of efficacy comparable to traditional music, supporting its viability for routine therapeutic application.

Another possible use of AI in music therapy is the employment of AI algorithms that analyse personal music tastes, affective states, and physiological reactions, enabling the creation of personalised music interventions such as mood-enhancing playlists and dynamically composed melodies mirroring the specific needs of the user (Zheng 2024). Sentiment analysis – which in this context refers to the process of computationally identifying and categorising one's sentiments expressed during music therapy sessions in order to determine whether their attitude towards the music is positive, negative, or neutral (Zheng 2024) – may assist therapists in understanding participants' affective conditions better and designing interventions consequently (Zheng 2024).

16.4 Challenges and Future Directions

16.4.1 Ethical Considerations, Risks, and Limitations

Despite its promising possibilities, some critical ethical concerns about the application of AI in psychological education and treatment arise. For instance, AI systems can also be affected by bias, which can exacerbate existing inequalities related to socioeconomic status, gender, ethnicity, race, religion, sexual orientation, or disability (Shroff 2022). These biases mainly affect marginalised groups that may receive less accurate algorithmic predictions or have their need for care underestimated or neglected (Mittermaier et al. 2023). Bias can be present in several forms, for instance, when a predictive algorithm is being trained based on one single hospital's database, which might not be applicable to another hospital's patients in a different country as the patient population from the first hospital might not be representative to the second hospital, not mentioning the difference in medication availability and treatment strategies between the two countries (Mittermaier et al. 2023). Bias can also appear in generative AI systems. For instance, text-to-image systems often depict certain ethnic or racial groups stereotypically (Lee et al. 2023). To mitigate algorithmic bias, strategies such as using representative data, pre-processing, and in-processing data by implementing mathematical solutions should be employed (Mittermaier et al. 2023).

In addition to possible biases, another ethical concern about AI use is related to data privacy. Potential risks regarding security and confidentiality may appear in many aspects of using an AI system in psychological practice (e.g., data collection, designing targeted and personalised treatment), so a systematic approach to protect personal data is critical (Salah et al. 2024).

Moreover, the use of AI-powered mental health chatbots can contribute to the concealment and, thereby, increased societal stigmatisation associated with mental health problems by encouraging users to hide their problems and cope with them only in the form of private interactions with AI agents via electronic devices (Grodniewicz and Hohol 2023).

Besides, Grodniewicz & Hohol (2023) emphasised that there is a lack of transparency regarding what existing psychotherapy chatbots can actually offer to individuals with mental health difficulties. Notably, companies developing such chatbots deliberately avoid referring to their services as psychotherapy (Grodniewicz and Hohol 2023). For instance, one of the most commonly used AI-enabled CBT-based mental health application, Wysa is introduced as "an artificial intelligence-based 'emotionally intelligent' service which responds to the emotions you express and uses evidence-based cognitive-behavioral techniques (CBT), DBT, meditation, breathing, yoga, motivational interviewing and micro-actions to help you build mental resilience skills and feel better" (Wysa 2024). Moreover, another popular chatbot, Woebot (Woebot Health 2024) is presented as "behavioral health copilot, an enterprise solution that aims to increase access to mental health support and improve your business outcomes," avoiding the expressions of "therapist" or "therapy". One more mental health chatbot is Tess (2024), which is presented as a mental health chatbot, also promoted as “Effective: Most people report that they prefer Chat With Tess over traditional therapy…” and “Affordable: Support from Tess is 98% cheaper than face-to-face therapy” (Grodniewicz and Hohol 2023).

When a chatbot delivers psychotherapy, questions arise regarding whether chatbots should be considered equivalent to a human therapist. Sedlakova and Trachsel (2023) posited that these chatbots cannot be viewed as equal partners in the way that human professionals are. The authors suggest that it would be controversial to consider chatbots as therapists as they lack qualities and human characteristics such as mental states or consciousness, autonomy, and intentionality. However, these authors note that such chatbots “might give the impression that AI has shifted from being a tool to being a subject because of its strong anthropomorphic features” (Sedlakova and Trachsel 2023, p. 6). Moreover, current chatbots cannot provide or ask for reasons, and they cannot understand or explain the concepts they use. Therefore, chatbots cannot offer one of the key psychotherapeutic processes, which is the facilitation of one’s self-understanding or gaining insight into the cognitive and affective dynamics of the self (Grodniewicz and Hohol 2023).

In conclusion, GAI in psychological practice holds great promises; however, some crucial ethical concerns should be addressed. Ethical issues such as transparency, reliability, accountability, algorithmic bias, data privacy, and security risks involving confidentiality, integrity, and availability need to be carefully considered using a human-centred approach (Salah et al. 2024). In addition, when integrating AI into psychological treatment, it is crucial to consider its limitations stemming from its non-human characteristics.

16.4.2 The Digitalization of Various Therapeutic Approaches

As Grodniewicz and Hohol (2023) pointed out, it is essential to enhance our comprehension of what makes human-delivered psychotherapy effective in order to be able to develop AI-based psychotherapy. Various therapeutic approaches conceptualise the process of psychotherapy differently, often creating a critical discussion regarding (1) what the therapist should focus on while conducting psychotherapy and (2) what factors contribute to the efficacy of psychotherapy. Moreover, the authors also highlight that just because a chatbot is built on the principles of an evidence-based psychotherapeutic approach – which has been shown to be effective when delivered by a trained human professional –the chatbot should not be considered evidence-based automatically (Grodniewicz and Hohol 2023). In order to develop practical AI-powered psychotherapy tools, it is essential to gain an understanding of the processes, techniques, and interventions that make human-delivered psychotherapy effective.

According to the American Psychological Association (2017), a psychotherapist is an expert who has substantial knowledge of the underlying mechanisms of psychological disorders. A therapist develops a non-symmetrical relationship with the client, focusing only on the client's mental health. Moreover, the therapy takes place in a structured setting in terms of time interval and environmental context, for instance (Grodniewicz and Hohol, 2023).

A recent study examined the efficacy of conversational chatbots using different therapeutic approaches (Eltahawy et al. 2024). One of them was Woebot, which is designed to deliver targeted content using CBT techniques. Another was ELIZA, a conversational agent that mimics a Rogerian therapist. The third group used Daylio, an application that invites participants to conduct a daily interactive journal, while another controlled condition used psychoeducational materials regarding depression. Results showed that ELIZA users reported the most significant symptom improvement on all mental health outcomes (i.e., anxiety, depression, positive affect, negative affect). Woebot users reported a decrease in anxiety. Daylio users reported decreases in depression and negative affect. Finally, participants who received psychoeducation reported only marginally significant improvements in depression and positive affect. The results of this study revealed that self-help telehealth interventions, through building on different therapeutic approaches, can offer beneficial mental health outcomes (Eltahawy et al. 2024).

16.5 Conclusion

Overall, the authors conclude that the application of AI in psychological education and treatment shows great promise, and some preliminary findings demonstrate its effectiveness in various areas of these two fields. In this chapter, we presented the practical and creative applications of AI. In more detail, some empirical evidence supports that AI can improve psychology students' training by providing them with helpful material for practising professional conversational skills with a virtual patient. Also, AI can serve as an assistant to psychologists as a supervision tool.

However, the most significant promise of AI lies in its application in psychological treatment, as it may narrow down the gap in bespoke treatment in the not-so-distant future. Moreover, AI-powered therapy chatbots can offer immediate support for those seeking therapy but cannot access it. It is also essential to foster positive attitudes towards AI technology among professionals. For this reason, the authors of this chapter highlight the importance of increasing AI literacy both among professionals and clients. Furthermore, patient education is indeed essential as these users might overestimate the efficacy of AI-based self-help tools and refuse to participate in targeted human-delivered psychotherapy for specific mental disorders for which AI-assisted help may not be efficient (e.g., psychoactive substance use-related excesses). Further concerns regarding the use of AI technology could be that it soon adapts to the user's personal needs and use patterns, which allows users to flexibly shape a chatbot-therapist according to their expectations and preferences, which could lead to the exclusion of some core characteristics from the process that would increase the efficacy of the conversation (e.g., confrontation). This specific of AI chatbots can limit the effectiveness of the process as it might eliminate some fundamental factors that are essential parts of real-life therapy. Future research should also examine the impact of AI chatbot use on the adherence and compliance of patients with real-life professionals after having had therapeutic “sessions” with a chatbot.

Overall, although GAI in psychology holds excellent promise, ethical concerns regarding the non-human characteristics of AI should be addressed, such as transparency, algorithmic biases, and data privacy, which may undermine the safe confines of an AI-delivered psychotherapy environment.

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Chapter 17
Generative AI Hallucinations in Healthcare: A Challenge for Prompt Engineering and Creativity

Vladimir Geroimenko

Abstract This chapter delves into the implications of generative AI hallucinations, particularly focusing on how they challenge the principles of prompt engineering and creative applications within the healthcare field. These hallucinations occur when AI systems generate information that appears plausible but is factually incorrect or unsupported by data. In healthcare, where accuracy and reliability are paramount, AI hallucinations can have severe consequences. Prompt engineering, the practice of designing effective prompts for AI models, plays a crucial role in mitigating the risks caused by AI hallucinations. The chapter surveys the best practices of prompt engineering in the context of healthcare (precision and clarity; ethical considerations and bias mitigation; iterative refinement and user feedback; incorporating multimodal inputs; human oversight), exploring its key principles and strategies to minimise AI hallucinations (incorporating constraints and boundaries; incorporating structured formats; providing contextual specificity; keeping a balance between innovation and reliability). Afterwards, the chapter explores a controversial relationship between AI hallucinations and creativity. It shows that AI hallucinations have been primarily associated with negative consequences. However, their potentially positive role deserves exploration in several healthcare areas, such as drug design, innovative treatment approaches, creative thinking stimulation, medical training and education, and patient engagement.

Keywords Generative AI · Healthcare context · Preventing AI hallucinations · AI creativity · Prompt engineering principles and strategies

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17.1 Introduction

Generative AI, with its remarkable ability to create new content, has rapidly entered various domains, including healthcare. While offering immense potential for revolutionising medical research, diagnosis, treatment planning, and patient care, it also presents a significant challenge: the phenomenon of AI hallucinations. These hallucinations occur when AI systems generate information that appears plausible but is factually incorrect or unsupported by data. In healthcare, where accuracy and reliability are paramount, AI hallucinations can have severe consequences. Misleading diagnoses, incorrect treatment recommendations, or biased outcomes can compromise patient safety.

Prompt engineering, the practice of designing and refining prompts to elicit desired responses from AI models, plays a crucial role in mitigating the risks caused by AI hallucinations. By carefully crafting prompts, healthcare professionals can guide AI systems to produce more accurate and reliable outputs. However, the complexity of medical information makes this task particularly challenging.

This chapter delves into the implications of such hallucinations, particularly focusing on how they challenge the principles of prompt engineering and creative applications within the medical field. It surveys the best practices of prompt engineering in the context of healthcare, exploring its key principles and strategies to minimise hallucinations and enhance the reliability of AI-generated content. Moreover, it explores the potential positive role that generative AI hallucinations may play in healthcare, even though they have been primarily associated with negative consequences.

17.2 Applications of Generative AI in Healthcare

Generative AI has significant potential to revolutionise the healthcare industry (Moulaei et al. 2024; Shokrollahi et al. 2023; Zhang and Boulos 2023). By leveraging deep learning techniques, generative AI models can analyse vast datasets of medical images, patient records, and genomic data to identify patterns, make predictions, and even create novel solutions. Generative AI is emerging as a transformative force in healthcare, with its capabilities extending far beyond mere data analysis.

Google Search (2024) shows the following 20 application areas of Generative AI in healthcare compiled from a variety of sources across the web: (1) Personalized medicine, (2) Drug development, (3) Medical imaging, (4) Clinical decision support, (5) Medical diagnosis, (6) Process automation, (7) Synthetic gene design, (8) Accelerating clinical trials, (9) Administrative tasks, (10) Clinical documentation, (11) Medical chatbots, (12) Patient care, (13) Patient journey prediction, (14) Training and simulation, (15) Enhancing patient engagement, (16) Healthcare workers shortage management, (17) Medical research, (18) Medical simulation, (19) Patient education, (20) Predictive analytics.

Primary application areas of generative AI in healthcare are personalised medicine and treatment planning, drug discovery and development, medical image analysis and diagnosis (Abbasi et al. 2024; Sai et al. 2024; Burns et al. 2024). Each domain leverages generative models to innovate processes, improve outcomes, and streamline workflows, thereby reshaping the landscape of modern healthcare and medicine.

Personalised Medicine and Treatment Planning

Personalised medicine aims to tailor treatments to the individual needs of patients (Wibowo and Putri 2024; Ghanem et al. 2024). By analysing vast amounts of patient data, including genetic information, lifestyle factors, and medical history, AI can generate personalised treatment plans tailored to individual patients. This approach ensures that treatments are more effective and have fewer side effects. For example, generative AI can predict how a patient might respond to a particular medication, allowing doctors to choose the most suitable therapy. As healthcare shifts toward more individualised approaches, generative AI serves as a critical tool in bridging the gap between generalised medical knowledge and specific patient needs.

Drug Discovery and Development

Traditional methods of identifying new pharmaceuticals can be time-consuming and expensive. Generative AI can accelerate this process by generating novel molecular structures with desired properties (Zhang et al. 2024; Qiu and Cheng 2024). By analysing vast databases of chemical compounds and their corresponding biological activities, generative models can learn the relationship between molecular structure and biological function. This knowledge can then be used to design new molecules with potential therapeutic applications. Additionally, generative AI can be employed to predict the toxicity and efficacy of drug candidates, thereby reducing the risk of failure in clinical trials.

Medical Image Analysis and Diagnosis

Generative models can be trained on large datasets of medical images, such as X-rays, MRIs, and CT scans, to learn the underlying patterns and structures (Helaly et al. 2024; Xu et al. 2024). This knowledge can then be used to perform tasks like image segmentation, object detection, and anomaly detection. For instance, AI-generated images can help identify minute anomalies that might be missed by the human eye, thus improving the early detection of diseases like cancer. Generative AI can also simulate various medical conditions, providing a valuable tool for training medical professionals and testing diagnostic algorithms.

Thus, generative AI is reshaping healthcare through advancements in personalised medicine, drug discovery, medical imaging and diagnostics, and many other application areas (Google Search 2024). As generative AI technologies continue to evolve in all these areas, their potential to reshape the healthcare landscape will only expand, promising to improve outcomes and drive forward the next generation of creative medical solutions.

17.3 Understanding Prompts and Prompt Engineering

In the context of generative AI, a prompt is a text or other input that guides the AI model to produce a desired output. It serves as the initial instruction or query the model uses to generate responses, whether text, images, or other forms of data. The quality and structure of the prompt significantly influence the relevance and accuracy of the AI's output. Prompts can vary in complexity, ranging from simple queries to detailed instructions, and are crucial for eliciting the desired response from the model. The effectiveness of a prompt significantly impacts the quality of the generated output, making the craft of creating prompts a vital skill in the field.

Prompt engineering refers to the practice of designing and refining prompts to optimise the performance of generative AI models (Phoenix and Taylor 2024; Khan 2024; Sahoo et al. 2024). This involves crafting clear, specific, and contextually rich prompts to ensure the AI understands the task and produces the most accurate and valuable responses. Effective prompt engineering combines creativity with analytical skills, enabling practitioners to experiment with different formulations to discover what yields the best results.

Key principles of effective prompting include clarity, specificity, context, and creativity. A clear prompt minimises ambiguity, while specificity helps direct the model's focus. Providing context can enhance the relevance of the output, enabling the AI to generate responses that align with the user's intent. Creativity and experimentation are essential because exploring different prompt formats and variations can lead to unexpected and innovative outcomes.

There are several main types of prompts commonly used in generative AI applications. These include instructive prompts, which provide explicit guidelines; conversational prompts, which simulate a dialogue; and creative prompts, which encourage imaginative or artistic outputs. Each type serves distinct purposes and can be employed strategically depending on the desired outcome.

Prompt engineering is crucial because it directly impacts the effectiveness and reliability of AI models. By crafting effective prompts, researchers and practitioners can guide models to produce high-quality, relevant, and innovative outputs. In fields such as education, marketing, and software development, well-engineered prompts can lead to more accurate information retrieval, better user engagement, and increased innovation. In healthcare, precise and contextually appropriate prompts can lead to more accurate diagnostic suggestions, better patient care recommendations, and improved decision support systems. This makes prompt engineering an essential skill for leveraging the full potential of generative AI in various domains and applications.

17.4 Understanding the Healthcare Context: Key Challenges in Prompt Engineering

Generative AI has immense potential to transform healthcare (Zhang and Boulos 2023; Shokrollahi et al. 2023; Moulaei et al. 2024). However, prompt engineering, the process of crafting effective instructions for AI models, is particularly challenging in healthcare due to the unique nature of medical data (Wang et al. 2023; Holley and Mathur 2024; Patil et al. 2024). Three primary challenges stand out:

Domain-Specific Knowledge and Complex Medical Terminology

Medical and healthcare concepts are complex and require deep domain expertise. Prompt engineers must possess a strong understanding of medical terminology, anatomy, physiology, and disease processes. This knowledge is essential for creating prompts that accurately capture the nuances of healthcare scenarios. Without domain expertise, prompts may be ambiguous or misleading, leading to inaccurate or harmful AI outputs. Generative AI models must be adept at understanding and accurately interpreting medical jargon, abbreviations, and context-specific language. This requires prompt engineers to create highly specialised prompts that can guide the AI in generating responses that are both accurate and contextually relevant. Misinterpreting medical terms can lead to incorrect diagnoses or treatment recommendations, making this a critical challenge.

Data Privacy and Confidentiality

Healthcare data is highly sensitive and subject to strict privacy regulations like HIPAA in the USA. Sharing this data with AI models, even anonymised, poses significant risks. Prompt engineering must prioritise data privacy, ensuring that sensitive information is not inadvertently disclosed or misused. Healthcare data is highly confidential, and any breach can have severe consequences. Prompt engineers must design prompts to generate valuable insights without compromising patient confidentiality. This involves implementing robust encryption methods and adhering to strict data governance policies to ensure that the AI systems do not inadvertently expose or misuse sensitive information.

Bias and Ethical Concerns

Generative AI models can preserve biases present in the training data. In healthcare, these biases can have serious consequences, such as disparities in treatment outcomes for different patient populations. Prompt engineering must address bias mitigation by carefully curating training data and designing prompts that promote fairness and equity. This involves continuously monitoring and updating the AI systems to ensure they adhere to ethical standards and provide equitable healthcare solutions. Additionally, prompt engineers must consider the ethical implications of AI-generated recommendations and ensure they align with established medical guidelines and patient care standards. Ensuring that the AI respects patient confidentiality while providing insightful analysis requires prompts that strike a delicate balance between detailed inquiry and ethical constraints.

Addressing these unique challenges of applying prompt engineering to healthcare requires a multidisciplinary approach, combining expertise in AI, healthcare, ethics, and data security to develop robust and reliable generative AI systems for the healthcare sector.

17.5 Prompt Engineering: Best Practices in Healthcare Context

Prompt engineering, the art of crafting effective prompts to guide AI models toward desired outputs, is crucial in healthcare applications for eliciting meaningful and contextually relevant responses. Well-crafted prompts can significantly enhance the quality and accuracy of AI-generated content, such as diagnosis support, treatment plans, medical training, and patient communication. Here are some major best practices for prompt engineering specifically suitable for healthcare applications (OpenAI 2024; DigitalOcean 2024; Subramanian 2024; Khan 2024; Patil 2024):

Precision and Clarity

In healthcare, the stakes are high, and ambiguity can lead to severe consequences. Therefore, prompts must be precise and clear. For instance, when designing a prompt for a diagnostic tool, it is essential to include specific symptoms, patient history, and relevant medical conditions. This ensures that the AI provides accurate and appropriate responses, reducing the risk of misdiagnosis.

Ethical Considerations and Bias Mitigation

Healthcare applications must prioritise ethical considerations and address potential biases in AI outputs. Prompts should be designed to avoid biases and ensure patient privacy. This involves using anonymised data and being mindful of how questions are framed to prevent discrimination. Prompt engineering should explicitly include considerations for equity, such as asking the AI to evaluate treatment options for diverse populations or to consider socio-economic factors in its recommendations.

Iterative Refinement and User Feedback

Prompt engineering requires iterative refinement based on user feedback and real-world application. Continuous testing and refinement of prompts are necessary to improve the performance of AI models. This involves evaluating the AI's responses and adjusting the prompts to enhance accuracy and relevance. Engaging healthcare professionals in the prompt creation process can provide valuable insights into the specificity and relevance of prompts. Regularly testing prompts in clinical settings and gathering feedback can help identify strengths and weaknesses in AI responses, enabling ongoing improvements.

Incorporating Multimodal Inputs

Healthcare data is often multimodal, encompassing textual information, imaging data, and patient history. Leveraging this diversity can improve AI responses significantly. For instance, prompts can be structured to include patient medical images, demographics, symptoms, and previous treatments. By integrating these dimensions, prompts can yield more nuanced and clinically relevant insights.

Human Oversight

While AI can be powerful, human oversight remains essential. Humans should constantly review and validate AI-generated content to ensure accuracy, completeness, and ethical compliance. Human experts can provide valuable insights and corrections, improving the overall quality of healthcare outcomes.

By adhering to these essential best practices, healthcare providers can enhance patient care, streamline medical processes, and support clinical decision-making. As generative AI continues to evolve, the role of prompt engineering will become increasingly critical in ensuring that AI technologies are used safely and effectively in healthcare settings.

17.6 Generative AI Hallucinations in Healthcare

17.6.1 An Overview of the Problem and Possible Solutions

One of the most significant challenges facing the adoption of AI in healthcare is the phenomenon of AI hallucinations (Giuffrè et al. 2024; Ramlochan 2023). Hallucinations in AI refer to the models' tendency to generate information that appears plausible but is factually inaccurate (Maleki et al. 2024; Salvagno et al. 2023). This issue is particularly concerning in healthcare, where the stakes are high, and the margin for error is minimal. For instance, an AI model might provide inaccurate medical advice, leading to misdiagnosis or inappropriate treatment. This can have devastating effects on patients, especially in time-sensitive situations. Additionally, AI hallucinations can erode trust in AI systems, hindering their widespread adoption in healthcare settings.

At the core of AI hallucinations is the complex interplay between the vast amount of data AI models are trained on and the algorithms that govern their generation processes (Maleki et al. 2024; Grammarly 2024). When an AI model encounters a novel or ambiguous prompt, it may attempt to extrapolate from its existing knowledge base, leading to unexpected or fantastical outputs. These hallucinations can manifest the AI's ability to explore beyond the confines of its training data, venturing into the realm of the unknown and the imaginative.

In healthcare, AI hallucinations can occur for various reasons, including biases in training data, limitations in the AI model's understanding, or the complexity of medical information. For instance, an AI system trained on a dataset that lacks diversity may produce biased or erroneous outputs when encountering cases outside its training scope. Additionally, the intricate nature of medical knowledge, which often involves nuanced and context-specific information, can lead to AI systems making incorrect inferences or recommendations. These hallucinations can manifest in various forms, such as incorrect diagnoses, inappropriate treatment suggestions, or misleading patient information. For example, an AI system tasked with diagnosing medical conditions based on patient symptoms might incorrectly synthesise information from disparate cases, resulting in a misdiagnosis that could adversely affect treatment pathways.

Researchers and developers are exploring various strategies to mitigate the risk of AI hallucinations (Turkowski 2024; Gadiko 2024; Adachi 2023). These include improving the quality and diversity of training data, developing more robust evaluation metrics, and implementing techniques to enhance model transparency and explainability. Also, it is crucial to implement robust validation and verification processes, ensure diverse and comprehensive training datasets, and maintain human oversight in AI-driven healthcare applications. By addressing the root causes of AI hallucinations and fostering a collaborative approach between AI systems and healthcare professionals, it is possible to harness the full potential of AI while safeguarding patient well-being.

Addressing the problem of hallucinations requires not only robust technical solutions but also a nuanced understanding of prompt engineering—a discipline that involves crafting inputs to optimise AI responses. Effective prompt engineering can help mitigate hallucinations by guiding AI models toward more accurate and relevant outputs (SUSE 2024; Gunjal et al. 2024). However, it also demands a creative approach to framing questions and context, as the interaction between human creativity and AI capability is critical in shaping the quality of the generated responses. Thus, fostering a collaborative relationship between healthcare professionals and AI systems is essential, ensuring that prompts are designed to elicit precise and clinically relevant information while maintaining an awareness of the limitations and potential pitfalls associated with AI-generated content.

17.6.2 Preventing AI Hallucinations in Healthcare: A Prompt Engineering Approach

In the context of healthcare, where accuracy and reliability are paramount, AI hallucinations can have serious consequences, especially when patient health and safety are at stake. Prompt engineering (the art and science of crafting effective prompts to guide AI models) offers a promising approach to mitigate the risk of hallucinations in healthcare applications (Gunjal 2024; SUSE 2024; Ahmad et al. 2023). By carefully designing the prompts that we feed to generative AI models, we can significantly improve the quality and accuracy of their outputs. This process includes specifying the desired format, context, and constraints within the prompt. For instance, in healthcare, prompts can be designed to include specific medical terminologies, patient data formats, and contextual information about the patient's condition. By providing clear and detailed instructions, prompt engineering helps reduce ambiguity and guides the AI to produce more reliable outputs.

In other words, effective prompt engineering is essential to mitigate the risks of AI hallucination. This includes all the fundamental best practices of prompt engineering described earlier in this chapter. Here are some additional strategies and best practices for preventing AI hallucinations in healthcare:

Incorporating Constraints and Boundaries

Defining clear constraints within the prompts to limit the scope of AI responses can involve setting boundaries on the types of information the AI should consider or excluding irrelevant data. For example, a prompt might specify, "Provide treatment options for chronic back pain excluding surgical interventions." Setting the scope of the AI's response can prevent it from straying into areas where it lacks expertise or confidence. This could involve limiting the number of options or specifying the output format.

Incorporating Structured Formats

Incorporating structured formats within prompts can enhance clarity and focus, which is particularly important in clinical settings. For example, templates that require the AI system to follow a defined structure (such as a checklist of symptoms, diagnostic criteria, or treatment guidelines) can help the model align with established medical knowledge. Also, additional constraints, such as requesting evidence-based sources for claims, can encourage the AI to generate more reliable and grounded outputs in current clinical practice.

Providing Contextual Specificity

When crafting prompts, it is essential to provide adequate context to ensure the AI understands the nuances of the healthcare domain. This may include patient demographics, medical history, symptoms, and other relevant information. Incorporating contextual details helps the model produce more accurate and tailored outputs. Hence, prompts must be highly specific to the medical context. This includes using precise medical language and including relevant patient data. For instance, rather than asking a generative AI model to provide general treatment options for a condition like diabetes, a more effective prompt would delineate parameters such as patient demographics, comorbidities, and recent lab results. By providing detailed context, healthcare professionals can significantly reduce the likelihood of the model generating erroneous recommendations that do not apply to the specific patient scenario.

Keeping Balance Between Innovation and Reliability

In healthcare, prompt engineering is essential for harnessing the creative potential of AI while maintaining safety and accuracy. By carefully designing prompts, healthcare professionals can leverage AI to generate innovative solutions, such as personalised treatment plans and diagnostic tools, without compromising reliability. This balance between creativity and precision is crucial for advancing healthcare technologies and improving patient outcomes.

Thus, prompt engineering is a foundational technique to prevent AI hallucinations in healthcare. By incorporating constraints, using structured formats, and emphasising contextual specificity, healthcare practitioners can ensure that AI-generated content is both innovative and trustworthy. This approach ensures that generative AI systems serve as reliable partners in the healthcare landscape, minimising the risks associated with hallucinations while maximising their potential benefits.

17.6.3 AI Hallucinations and Creativity in Healthcare: A Controversial Relationship

While AI hallucinations have been primarily associated with negative consequences, their potentially positive role in healthcare deserves exploration (Castro 2023; Alowais et al. 2023). In the context of generative AI creativity, these hallucinations could serve as a catalyst for innovative approaches to medical problem-solving, drug discovery, unconventional education, and innovative treatment in the healthcare sector. Unlike in creative fields such as art, literature, and music, where hallucinations can lead to novel and inspiring outputs, in healthcare, the stakes are much higher. However, there are nuanced perspectives on how these hallucinations might be harnessed positively under strict and controlled conditions.

Drug Design

One potential application lies in the realm of drug design. AI models can generate novel molecular structures based on existing data. While hallucinations might introduce unexpected and seemingly improbable structures, these could lead to the discovery of new drug candidates with unique properties and therapeutic potential. For instance, a hallucinated molecule might exhibit a previously unknown mechanism of action, addressing a challenging disease target.

Development of Innovative Treatment Approaches

AI hallucinations might contribute to the development of innovative treatment approaches. In research settings, AI can generate hypotheses and suggest unconventional treatment pathways that human researchers might not consider. While these suggestions need rigorous validation, they can spark new lines of inquiry and potentially lead to breakthroughs in medical science. When carefully monitored and evaluated, the creative aspect of AI hallucinations can thus serve as a catalyst for innovation in medical research.

Creative Thinking Stimulation

AI hallucinations can also be harnessed to stimulate creative thinking among healthcare professionals. By presenting unexpected and counterintuitive ideas, these hallucinations can challenge conventional wisdom and inspire alternative approaches to patient care. For example, an AI might suggest a seemingly unconventional treatment regimen that, upon further investigation, proves to be effective for a specific patient population. In a field that thrives on creativity and exploration, AI's capacity to produce unconventional ideas can inspire healthcare professionals to think outside the box. By presenting unexpected options, AI can catalyse human intuition, prompting further investigation and validation by healthcare practitioners.

Medical Training and Education

Another potential positive role of AI hallucinations in healthcare could be in the realm of medical training and education. AI-generated scenarios, even those that are not entirely accurate, can be used to create complex and unexpected case studies for medical students and professionals. These scenarios can challenge practitioners to think critically and adapt to unusual situations, enhancing their diagnostic and problem-solving skills. The training process can become more robust and comprehensive by exposing medical professionals to a wide range of hypothetical cases, including those generated through AI hallucinations.

Patient Engagement and Education

Lastly, AI hallucinations can enhance patient engagement and education. Generative AI can create personalised health narratives or visualisations that help patients understand their conditions and treatment options. While not always grounded in empirical data, these narratives can resonate emotionally and give patients a more relatable perspective on their health journey. This enhanced understanding can lead to better adherence to treatment plans and improved patient outcomes. Furthermore, creative storytelling facilitated by AI can help bridge communication gaps between healthcare providers and patients, fostering a more collaborative environment in clinical settings.

However, it is crucial to acknowledge the inherent risks associated with AI hallucinations in healthcare. Misleading or incorrect information generated by these models could have serious consequences, such as misdiagnosis, inappropriate treatment, or delayed intervention. To mitigate these risks, rigorous validation and oversight mechanisms must be implemented to ensure the reliability and accuracy of AI-generated outputs. It is vital that the use of AI hallucinations in healthcare must be approached with caution. The potential benefits must be weighed against the risks, and robust mechanisms must be in place to ensure that any AI-generated information is thoroughly vetted before being applied in clinical settings.

In summary, while AI hallucinations are often viewed as flaws within generative AI models, they can also offer unique opportunities for creativity in healthcare. By leveraging these instances of creative output, healthcare professionals can explore novel treatment avenues and enhance patient engagement, ultimately leading to improved health outcomes. The challenge lies in responsibly integrating this creativity into a field where accuracy and evidence-based practice are paramount. While hallucinations can spark creativity, they can also pose risks if not carefully managed. Misinformation generated by AI systems could lead to misguided clinical decisions or patient anxieties. Therefore, a robust framework for integrating generative AI into healthcare must prioritise accuracy and reliability while embracing its potential for innovation. Future research should focus on developing safeguards that allow healthcare professionals to harness the creative aspects of AI hallucinations without compromising patient safety or trust.

17.6.4 AI Hallucinations and Creativity

Generative AI has emerged as a transformative force in the creative landscape, enabling the production of novel content across various domains, from visual art to music and literature. Central to this phenomenon is the concept of AI hallucinations, which refers to instances where an AI system generates outputs that do not correlate with its training data or reality. While these hallucinations can be seen as a limitation or flaw in specific contexts, they can also be viewed through the lens of creativity and can serve as a unique catalyst for creative expression (Jiang et al. 2024; Chakraborty and Masud 2024; Mukherjee and Chang 2023; Nuurtamo 2024).

By embracing these hallucinations rather than dismissing them as errors, we can tap into the AI's potential to generate novel and innovative content.

The relationship between AI hallucinations and creativity highlights a paradigm shift in our understanding of generative processes. While hallucinations may represent the limitations of AI in accurately reflecting reality, they simultaneously open up new avenues for creative exploration. By reframing these unexpected outputs as valuable contributions to the creative process, we can embrace a more expansive definition of creativity that includes both human and machine-generated innovations.

In other words, AI hallucinations can be a source of inspiration and innovation. When an AI system generates content that is not strictly accurate, it can prompt human creators to explore new ideas and perspectives. For instance, an AI-generated story with fantastical elements might inspire a writer to develop a new genre or narrative style. Similarly, an AI-created piece of music with unconventional harmonies might lead a composer to experiment with new musical forms. In this way, AI hallucinations can catalyse human creativity, encouraging us to think outside the box and explore uncharted territories.

Thus, while AI hallucinations are often seen as errors, they also hold significant potential for enhancing creativity. By generating unexpected and imaginative content, AI can inspire human creators to push the boundaries of their own imagination. This symbiotic relationship between AI and human creativity highlights the transformative potential of generative AI in the arts and beyond.

17.7 Conclusion

Generative AI holds immense potential for transforming healthcare, from personalised medicine to innovative treatment approaches. However, the challenge of AI hallucinations necessitates a robust framework for prompt engineering. By incorporating precision, ethical considerations, and iterative refinement, prompt engineering can significantly reduce the occurrence of AI hallucinations.

While hallucinations are often seen as a negative aspect of generative AI, they also present opportunities for creativity and innovation. By embracing a balanced approach that prioritises accuracy and reliability while exploring the potential of hallucinations, we can harness the power of generative AI to improve patient care, accelerate drug discovery, and drive medical research forward. Balancing the innovative potential of AI with the need for reliability and accuracy is crucial.

As the field of generative AI continues to evolve, it is essential to remain vigilant about the potential for hallucinations and to invest in research and development to develop robust strategies for preventing them. By doing so, we can ensure that generative AI is a force for good in healthcare, benefiting patients, providers, and society.

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Chapter 18
Dentistry in the Digital Age: Exploring Human–Computer Creativity in Dental Education and Practice

Rasa Mladenovic and Vladimir Janjic

Abstract This chapter explores the intersection of dentistry and educational technologies, focusing on the integration of generative artificial intelligence into dental practice and pedagogy. With the advent of innovative technologies, dentistry has undergone transformative advancements, leading to improved patient care and enhanced learning experiences for dental students. Through the lens of human–computer creativity, the chapter delves into the use of AI-powered tools for tasks ranging from diagnostics and treatment planning to the simulation of dental procedures in virtual environments. Furthermore, it examines the role of artificial intelligence in revolutionising dental education, offering personalised learning pathways, virtual patient encounters, and immersive simulations that complement traditional teaching methodologies. By embracing the synergy between human expertise and computational capabilities, dentistry is entering a new era of innovation, where creativity thrives in a symbiotic relationship between humans and intelligent machines.

Keywords Dentistry · Artificial Intelligence · Virtual reality · Augmented reality · Distraction

18.1 Introduction

The field of dentistry is undergoing a significant transformation driven by advancements in digital technologies, reshaping both patient care and dental education. As new tools emerge—ranging from AI-enhanced diagnostic systems to immersive

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learning platforms—dentistry is moving away from traditional techniques, entering a domain where digital precision and human expertise intertwine.

Human–computer creativity is a collaborative process that combines the computational capabilities of machines with the nuanced judgment of human professionals. Machines provide data-driven insights, predictive models, and efficiency, while human expertise ensures that these insights are applied with empathy, ethical considerations, and real-world experience. This synergy proves invaluable in dentistry, where the unique needs of each patient demand a personalised approach that balances scientific rigour with compassionate care.

18.2 Advances in Artificial Intelligence in Dentistry

18.2.1 Artificial Intelligence in Diagnosis and Treatment Planning

Artificial intelligence (AI) has significantly transformed the diagnostic process in dentistry, offering advanced tools that detect and interpret dental issues with unprecedented speed and accuracy, enabling faster and more precise treatment approaches. One of the primary advantages of AI-based diagnostic tools is their ability to analyse vast amounts of data, especially in digital radiography, far beyond the capacity of the human eye. These systems are trained to recognise subtle patterns that may indicate early stages of diseases such as initial caries, periodontal disease, or lesions that could signal oral cancer, often identifying changes that clinicians might easily overlook. AI tools not only complement human expertise but also assist clinicians in making more accurate decisions, enabling quicker and more timely interventions (Mladenovic et al. 2023b, 2023a).

A crucial role of AI in this domain is its ability to perform detailed three-dimensional (3D) segmentation, aiding dentists in accurately identifying and classifying supernumerary teeth (see Fig. 18.1) and other anatomical details, typically using Convolutional Neural Networks (CNN).

Additionally, dental pulp segmentation poses a challenging task due to complex anatomical structures and variations, as well as textural similarities between the pulp and surrounding tissues in imaging. Today, AI algorithms employ advanced image analysis techniques, enabling precise identification and separation of dental pulp from surrounding tissue in images such as Cone Beam Computed Tomography (CBCT) (see Fig. 18.2).

By implementing AI in segmentation, the likelihood of human error is reduced, diagnostic accuracy is increased, and the overall process is accelerated, allowing clinicians to focus on complex treatment aspects. This is vital for treatment planning, especially in cases of root canal variations where precise mapping of canals is crucial for procedural success. With these tools, dentistry is becoming increasingly proactive

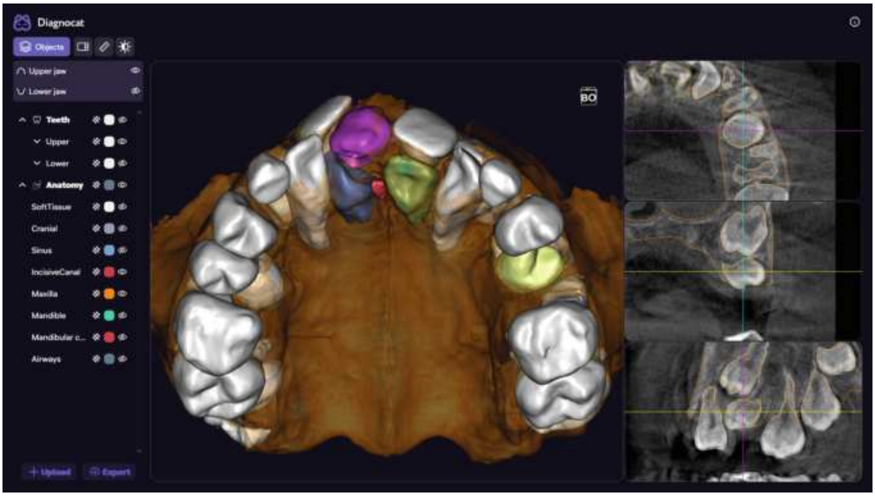

Fig. 18.1 AI identification and segmentation of supernumerary teeth using AI tool (Diagnocat, Diagnocat Inc, USA). Image by the author

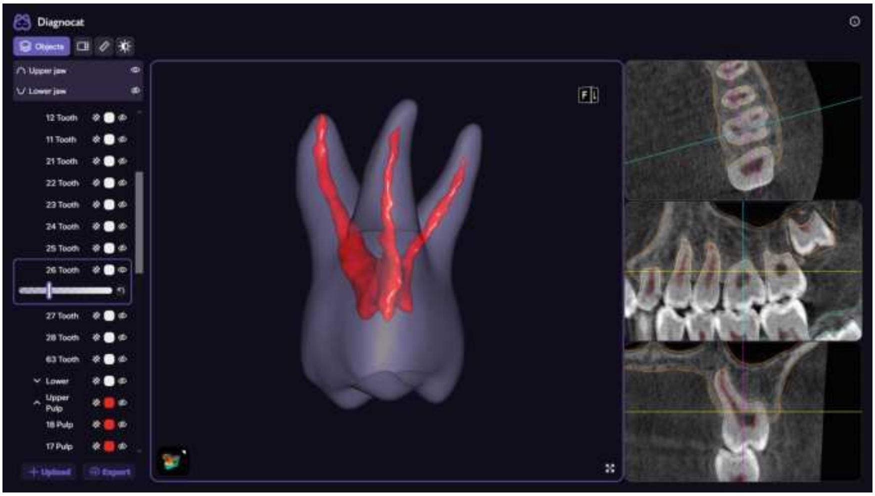

Fig. 18.2 Dental pulp 3D segmentation using AI tool (Diagnocat, Diagnocat Inc, USA). Image by the author

and oriented toward personalised approaches tailored to each patient (Mladenovic 2023; Mladenovic et al. 2022b; Alfadley et al. 2024).

Orthodontic treatment is often complex and requires precise planning and analysis to achieve functional and aesthetic outcomes. Traditional orthodontic approaches are time-consuming, but with AI-supported planning (see Fig. 18.3), orthodontists can now predict tooth movement trajectories with high precision, reducing treatment time and the number of required adjustments.

AI algorithms enable the creation of customised treatment plans for patients with unique structural challenges, and by analysing 3D models, AI can simulate the outcomes of various therapeutic approaches and recommend the optimal strategy (Wong et al. 2023; Subramanian et al. 2022; Nordblom et al. 2024).

AI has also proven beneficial in assisting oral surgical interventions, where precision is crucial. Tools such as AI-guided robots and real-time image analysis enable surgeons to perform complex procedures with a level of accuracy that minimises

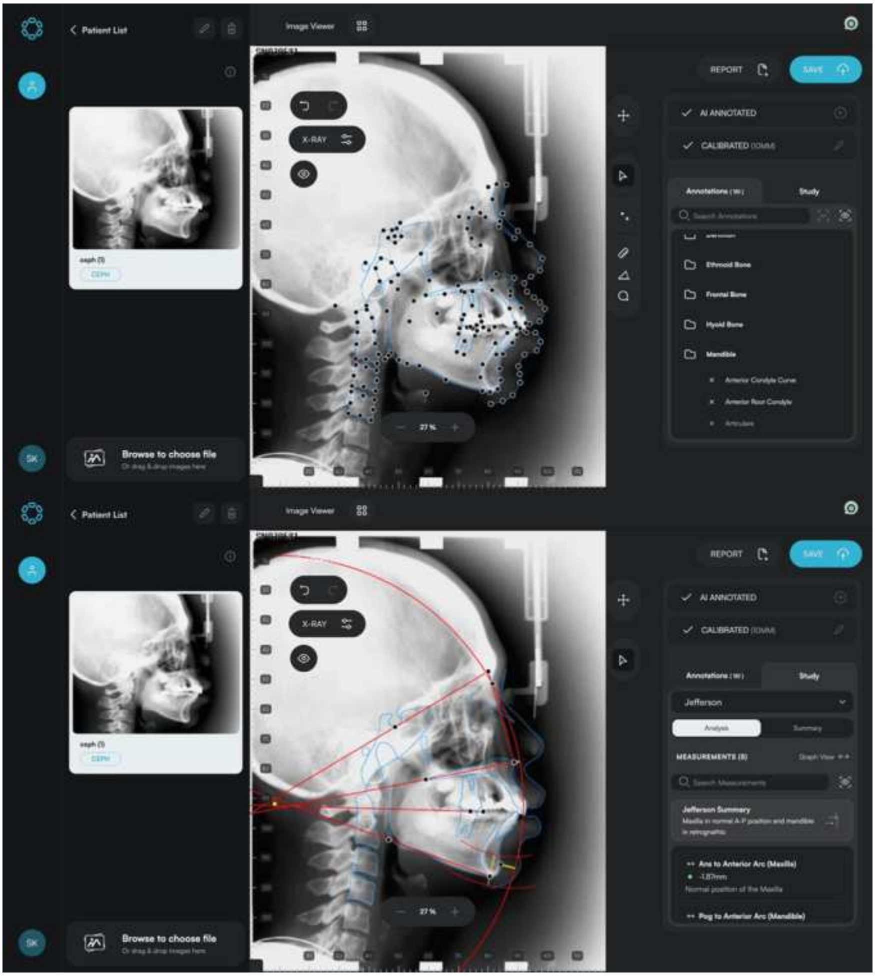

Fig. 18.3 AI tool identifies over 150 orthodontic landmarks in under 10 s (EAI CEPH, Sydney, Australia). Image source www.eyesofai.com

the risk of complications. For instance, during implant placement, AI can calculate the optimal angle, depth, and position of implants, reducing the likelihood of misplacement and ensuring better integration. AI-driven robotic systems also provide real-time feedback to ensure correct positioning and angles of implants, improving procedural accuracy and reducing recovery time. Research in robotic assistance in implantology shows that AI can optimise implant positioning even in cases with challenging anatomical constraints (Balaji 2024; da Mota Santana et al. 2024; Donget al. 2024).

Aesthetic dentistry often requires highly personalised treatment plans tailored to each patient's unique facial structure and aesthetic preferences. By using AI-driven smile design, dentists can generate virtual models of various treatment outcomes, allowing patients to visualise and choose their preferred option before making a final decision. Enabling patients to see potential results in advance increases their willingness to commit to the selected treatment plan, while satisfaction with achieved outcomes is significantly higher. This technology not only enhances aesthetic results but also provides a more targeted approach to care. AI-guided smile designs achieve greater harmony with patients' facial features, resulting in a more natural appearance, while patients experience reduced anxiety and greater satisfaction with the outcome. This approach to aesthetic dentistry highlights how AI can enhance the treatment experience, making it more collaborative and fulfilling for patients (Yue et al. 2023; Liu et al. 2024; Ahmed et al. 2024; Jain et al. 2024).

18.2.2 Generative Artificial Intelligence and 3D Printing

Another significant advancement in clinical dentistry is the use of generative artificial intelligence (AI) for design and 3D printing. Generative AI refers to algorithms capable of creating new data patterns based on existing ones, making it ideal for designing customised dental prosthetics and orthodontic appliances. Research in this field highlights the potential of generative AI to streamline the manufacturing process. Traditionally, creating dental prosthetics or appliances requires multiple sessions. However, with generative AI, clinicians can design more precise models from the outset, reducing the number of follow-up visits and adjustments.

These digital prosthetics are particularly beneficial for children and elderly patients whose anatomical characteristics demand special consideration. The most commonly used software tools in this domain are 3Shape Dental Designer (3Shape A/S, Copenhagen, Denmark) (see Fig. 18.4) and Exocad (Exocad GmbH, Darmstadt, Germany) (see Fig. 18.5).

These powerful 3D modelling tools are designed explicitly for the dental industry and other professional fields where precision is critical. They enable users to create complex three-dimensional structures with intuitive design and customisation tools. With support for various file formats such as STL and OBJ, these tools integrate seamlessly into existing workflows, providing high levels of flexibility and productivity. The software also incorporates advanced automation features, simplifying the

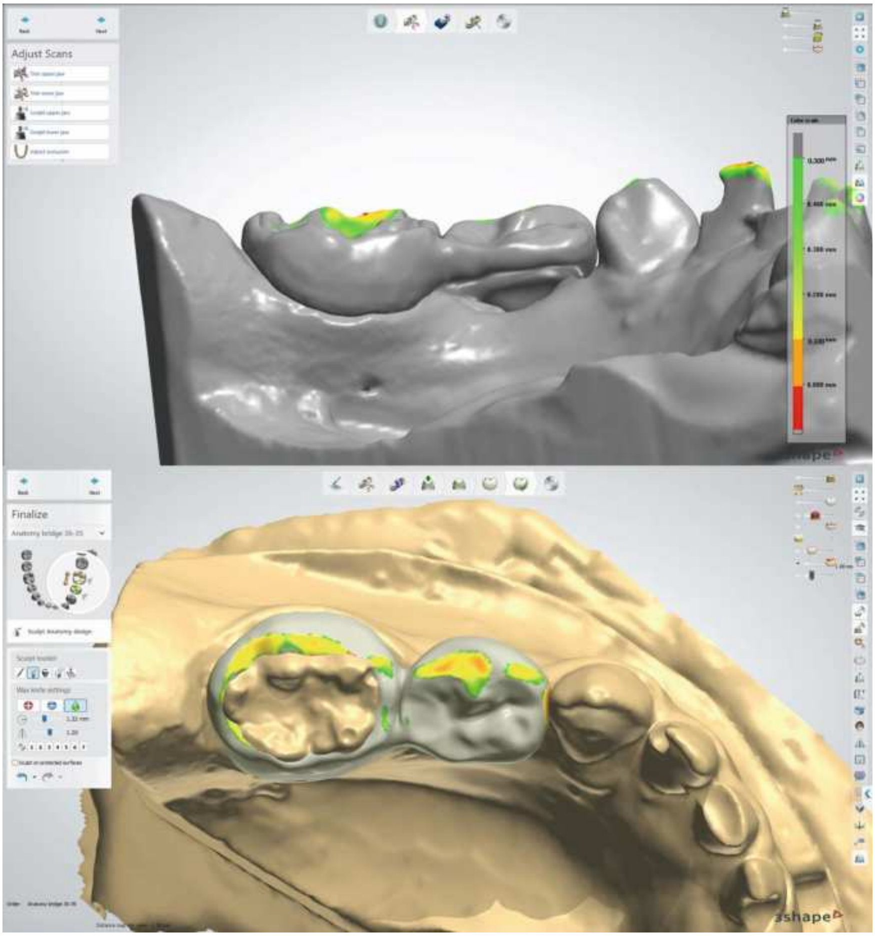

Fig. 18.4 Creation of a KOS&MET (Key Orthodontic System and Materials Enhanced Therapy) functional space maintainer according to Mladenovic in the 3Shape. Image by the author

process of achieving optimal results even for complex design tasks. Their flexibility allows for fast and precise work, facilitating high-quality outcomes and significantly reducing processing time (Güntekin et al. 2024; Schulz-Weidner et al. 2024).

The application of generative AI complements 3D printing, a technology that enables the rapid production of prosthetics. By integrating AI with 3D printing, dental practices can produce highly accurate, patient-specific models and prosthetics on demand, which is especially beneficial in emergencies. For instance, in cases of accidental tooth loss, a clinic equipped with AI and 3D printing capabilities can provide a replacement crown within a few hours rather than days, minimising discomfort and restoring the patient's functionality (see Fig. 18.6). (Spangler et al. 2023; Nagehan and Volkan 2024; Chen et al. 2024; Frąckiewicz et al. 2024).

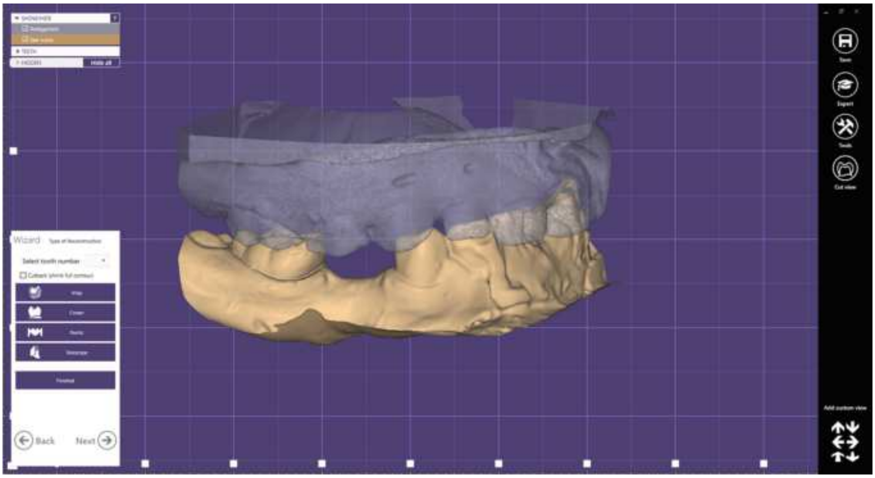

Fig. 18.5 Exocad environment. Image by the author

Fig. 18.6 3D printed KOS&MET functional space maintainer made of metal alloy on a 3D printer (MySint100 Dual Laser, Italy). Image by the author

18.2.3 Personalized Learning Experiences Enabled by Artificial Intelligence

Artificial intelligence is revolutionising dental education by facilitating adaptive learning paths tailored to the individual strengths and weaknesses of each student. Traditional dental courses often adopt a "one-size-fits-all" approach, which can lead to either inadequate challenges or overwhelming workloads depending on the student's unique learning paces and styles. However, AI-powered educational platforms continuously assess students' performance, adjusting the content and difficulty levels to meet specific academic needs. This adaptive learning approach has significantly improved student achievements (Hancock et al. 2024; Olsson et al. 2024).

Integrating AI into the curriculum allows for real-time monitoring of student performance, identifying areas where additional support or practice is needed. This approach is precious in complex fields such as operative dentistry and oral surgery, where technical precision is critical for clinical outcomes. Through real-time feedback and adaptive tools, students can enhance their retention and recall of challenging subjects, contributing to a more personalised learning experience. This adaptability fosters a more competent dental workforce as students develop comprehensive skills that encompass both theoretical knowledge and practical procedural capabilities (Preshaw et al. 2024).

18.2.4 Virtual Patient Encounters and Case Simulations

Virtual Reality (VR) and Augmented Reality (AR) are bringing transformative changes to dental education, offering students the opportunity to acquire practical skills in a safe, simulated environment. These technologies enable detailed visualisation of anatomical structures and realistic reproduction of tactile sensations, significantly enhancing precision and coordination during procedures (Mladenovic et al. 2020, 2019, 2022a).

In addition to improving technical skills, VR and AR simulations provide real-time feedback, allowing mistakes, such as incorrect angles or excessive force, to be corrected before clinical practice. This approach not only increases technical competence but also reduces student stress, preparing them for the challenges of real clinical scenarios. The integration of these technologies into education ensures the comprehensive training necessary for safe and effective dental procedures (Mladenovic and Mladenovic 2024; Lima et al. 2024).

Beyond educational benefits, these technologies are increasingly applied in patient care, particularly for children and individuals with dental anxiety. During procedures like tooth extractions, patients can use VR devices to engage in interactive games that divert attention from the treatment (Mladenovic et al. 2024). For example, a patient might play a game where they earn points by solving tasks or completing challenges. At the same time, the dentist performs the procedure (see Fig. 18.7). This innovative approach facilitates working with sensitive patient groups, making dental procedures less stressful for both patients and the dental team.

Fig. 18.7 The application of VR gamification during tooth extraction. Image by the author

18.2.5 Enhanced Feedback Mechanisms

Real-time feedback is a crucial contribution of AI to dental education, providing students with detailed insights into their techniques and performance. AI tools can measure critical parameters such as tool angle during procedures, applied pressure, and the consistency of hand movements. These continuous, data-driven feedback systems enable students to refine their skills in real time, promoting self-assessment and ongoing improvement. Immediate corrections of techniques enhance students' confidence and reduce performance-related stress.

AI-based feedback systems help students achieve technical competence more quickly, requiring fewer practice hours, which is invaluable in dental training where instructor time and equipment availability are often limited. Additionally, AI support allows instructors to focus on broader educational goals, while automated systems handle fine-grained technical evaluations, creating a more efficient and balanced educational experience.

18.2.6 Application of Artificial Intelligence in Dental Forensics

The integration of AI into dental forensics marks a significant advancement, providing tools that enhance the identification of individuals and the analysis of dental evidence in criminal investigations. AI algorithms can process various types of dental data, including radiographic images and digital impressions, to support forensic experts in multiple aspects of investigations. These systems enable faster and more accurate identification of unknown remains and analysis of specific characteristics, such as tooth morphology and restorations, facilitating the establishment of identities with precision often unattainable through traditional methods.

AI also enhances the analysis of bite marks, which plays a critical role in forensic investigations. Algorithms are capable of recognising specific patterns and characteristics of bite marks, enabling forensics to provide more accurate and reliable testimony in court. This improves the accuracy of linking suspects to crime scenes, strengthening the evidence in legal proceedings.

The management of dental evidence is further improved through the use of AI. By automating the processes of cataloguing and organising data, AI allows for more efficient information management, enabling forensic experts to focus more on analysis and less on administrative tasks.

The use of predictive analytics in forensics also facilitates the detection of patterns in dental crimes, such as dental fraud and illegal practices. With AI, forensic experts can analyse large datasets to identify potential criminal activities, enabling timely preventive actions to be taken (Bui et al. 2023; Souza et al. 2024).

18.3 Challenges and Ethical Considerations in the Adoption of AI

18.3.1 Technical Challenges in Implementation

While artificial intelligence offers numerous benefits for dentistry, its implementation faces several challenges. Technical barriers, such as the lack of necessary infrastructure and resources, can make it challenging to apply AI tools, especially in smaller dental practices or those in rural areas.

One of the critical issues is the need for high-quality data to develop AI algorithms effectively. Developing these systems requires large datasets that accurately reflect diverse patients and clinical scenarios. In practice, many clinics face difficulties in collecting sufficient data, which may limit the precision and reliability of AI models. To ensure equitable access for all patients, it is essential to develop inclusive data collection methods that consider different ethnic groups, age categories, and medical histories.

Additionally, integrating AI technologies often requires adjustments to existing workflows and routines in dental clinics. Due to a lack of familiarity or fear of reducing their professional role, many dentists may resist adopting new technologies. Overcoming this resistance requires extensive training and education for dental professionals, as well as clear demonstrations of the benefits that AI can bring. It is recommended that structured training programs be developed to enable dentists to acquire the necessary skills to use AI tools effectively, thereby enhancing their diagnostic capabilities and optimising clinical processes.

18.3.2 Regulatory and Legal Challenges

The rapid advancement of AI technologies in dentistry has led to regulatory and legal challenges. The regulatory framework for AI in healthcare is still evolving, and many countries lack clear guidelines for the use of AI tools in clinical practice. This ambiguity can lead to confusion and hesitation among dental practitioners regarding the legality and ethical implications of using AI solutions.

Research into the regulatory framework surrounding AI in dentistry reveals significant gaps that need to be addressed. For example, questions are often raised about liability in the case of diagnostic errors caused by AI systems. Suppose an AI tool recommends a treatment plan that leads to patient harm. In that case, it remains unclear whether the responsibility lies with the dentist, the software developer, or the institution using the technology. Establishing clear legal guidelines and accountability is crucial to addressing these concerns and building trust in AI systems among practitioners and patients (Ducret et al. 2024).

Moreover, the use of AI in managing patient data raises privacy and security concerns. As dental practices increasingly adopt digital records and AI diagnostic tools, they must contend with the possibility of data breaches or unauthorised access to sensitive patient information. Ensuring robust cybersecurity measures and compliance with regulations such as HIPAA (Health Insurance Portability and Accountability Act) in the United States is essential. Dental practitioners must be informed about data protection regulations and implement best practices in data security to protect patient information (Schweitzer 2024).

18.3.3 Ethical Implications of AI Integration

The integration of artificial intelligence into dental practice and education brings significant ethical challenges. First, there is the risk that increasing reliance on AI for diagnosis and treatment planning could reduce clinical work, relying too much on technology and undermining the personalised approach. To avoid this, a balanced approach is recommended in which AI complements the dentist's empathy, allowing for a combination of precision and compassion. Introducing ethical programs in education can help future dentists appreciate the importance of clinical work alongside technological expertise.

Another key ethical issue is informed consent from patients. Patients must understand how AI affects their treatments and the potential risks, and therefore, clear communication strategies are necessary to simplify the role of AI. Additionally, there are concerns about bias in AI algorithms. Insufficiently diverse training data can lead to unequal outcomes for specific patient groups. To ensure equitable care, training data for AI systems must be diverse and regularly updated.

For the ethical application of AI, it is crucial to develop guidelines that address issues of data privacy, bias, and the preservation of humanised care. Collaboration between dentists, engineers, and regulatory bodies is essential to establish standards for AI use. Interdisciplinary dialogue contributes to identifying risks and developing strategies that allow AI to enhance, rather than diminish, the quality of care. Ultimately, the successful integration of AI depends on a collective commitment to ethical principles, including patient welfare, inclusivity, and transparency. Regulatory bodies must adapt frameworks for the development, testing, and implementation of AI tools in dental practice, ensuring that regulations promote innovation while protecting patients' interests (Rigby 2019; Farhud and Zokaei 2021).

18.4 Future Directions in AI and Dental Practice

18.4.1 New Technologies and Innovations

As the field of dentistry continues to evolve, several new technologies have the potential to reshape how dental professionals practice and how patients experience care. One significant area of innovation is the integration of machine learning algorithms that can analyse patient data to predict future dental problems. These predictive analytics tools could enable dentists to adopt a more proactive approach to patient care, identifying at-risk individuals before they develop significant issues. Machine learning algorithms can analyse data from electronic health records, treatment histories, and demographic information to identify patterns associated with an increased risk of dental caries or periodontal disease. By using these insights, practitioners can implement personalised preventive strategies, such as targeted oral hygiene education or early interventions, significantly reducing the incidence of dental diseases within their patient populations.

Additionally, advances in Natural Language Processing (NLP) technologies could revolutionise how dental practices communicate with patients. NLP technologies can facilitate seamless interaction between AI systems and patients, enabling more intuitive scheduling, treatment inquiries, and post-treatment follow-up. In the future, AI chatbots may provide 24/7 patient support, answering common questions and freeing up dental staff to focus on more complex patient needs. This improved communication could enhance patient satisfaction and optimise office operations.

18.4.2 AI in Treatment Personalization

Personalised treatment planning is a key area where AI will have a significant impact. Dental professionals are increasingly recognising that each patient presents unique challenges and preferences, and AI can assist in analysing complex data to create tailored treatment plans that optimise outcomes. These personalised approaches can not only improve efficiency but also increase patient engagement, as individuals are more likely to adhere to plans that meet their specific needs. Furthermore, AI can aid in the development of personalised oral hygiene routines. By analysing data on oral health and lifestyle, AI systems can suggest appropriate products, techniques, and schedules for dental care tailored to each patient. This approach will empower patients to take a more active role in managing their oral health, ultimately contributing to better long-term results.

18.4.3 Collaboration Between Humans and AI

Looking ahead, the future of dentistry will likely be characterised by even greater collaboration between dental professionals and AI systems. Instead of viewing AI as a replacement for human expertise, practitioners should embrace it as a powerful tool that complements and enhances their skills. This shift in perspective will be crucial for the effective integration of AI into daily dental practice. Emphasis will be placed on developing a collaborative mindset among dental professionals. As AI technologies continue to advance, continuous education and training will be essential to ensure that practitioners feel confident in using these tools. Developing a collaborative relationship with AI requires practitioners to understand not only how to use the technology effectively but also its limitations. AI can provide insights and recommendations based on data, but the human touch—empathy, ethical decision-making, and clinical judgment—remains irreplaceable.

Interdisciplinary collaboration will also be vital for the advancement of AI applications in dentistry. By working with data scientists, development engineers, and AI researchers, dental professionals can contribute their clinical knowledge to enhance and improve AI systems. This collaborative approach can lead to the development of more efficient and user-friendly tools that meet the specific needs of dental practitioners and their patients.

18.5 Conclusion

The future of dentistry is on a path of transformation driven by advancements in AI and digital technologies. By embracing these innovations, dental professionals can improve patient care, optimise operations, and provide more personalised treatment options. However, as the profession navigates this ever-changing landscape, it is crucial to address the challenges and ethical considerations associated with the adoption of AI.

Through continuous research, education, and collaboration, dental practitioners can leverage the power of AI to create a more efficient, effective, and compassionate environment for dental care. By prioritising patient well-being and ethical standards, the integration of AI into dentistry can lead to a future where technology and human expertise work in harmony to improve oral health outcomes for all.

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Chapter 19
Generative AI with Quantum Computing for Visualizing Proteins in Mixed Reality

Don Roosan and Tiffany Khou

Abstract This chapter explores the transformative role of cutting-edge technologies—quantum computing, generative AI, and mixed reality—in overcoming the challenges of simulating protein folding. Protein folding, a fundamental process in systems biology, is central to understanding cellular mechanisms, drug discovery, and disease treatment. However, simulating protein folding at an atomic level remains computationally challenging due to the complexity of protein structures and the vastness of their conformational space. This chapter aims to provide a comprehensive overview of how integrating quantum computing and mixed reality reshapes protein folding research, opening new avenues for scientific exploration and breakthroughs in the life sciences.

19.1 Introduction

Protein folding is a fundamental process in molecular biology where linear chains of amino acids fold into unique three-dimensional structures. This structure determines a protein's functionality in biological systems, affecting how it interacts with other molecules and performs its roles, from catalyzing chemical reactions as enzymes to providing structural support in cells (Geiler 2010). The significance of protein folding in systems biology lies in its central role in maintaining the health and proper functioning of living organisms. Misfolded proteins can lead to diseases such as Alzheimer's, Parkinson's, and various types of cancer, highlighting the critical nature of proper folding (Shakhnovich 2006). Moreover, understanding protein folding allows scientists to predict protein structures from their amino acid sequences, which is essential for drug design and therapeutic interventions.

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Systems biology, which studies complex interactions within biological systems, relies heavily on understanding protein structures. It integrates data from various sources, including genomics, proteomics, and metabolomics, to build comprehensive models that predict how systems respond to environmental or genetic changes. Accurate models of protein folding enhance these predictions, allowing for more effective treatments and understanding of biological processes at a molecular level (Wanjek 2022).

Simulating protein folding presents significant computational challenges due to the complexity of the interactions involved and the vast conformational space that proteins can occupy. Each amino acid in a protein can move in three-dimensional space, and accurately modeling the interactions, such as hydrogen bonds and hydrophobic forces, requires substantial computational power and sophisticated chemical knowledge. The folding process involves changes in the position and orientation of every atom in each amino acid. This high dimensionality means that the number of potential configurations a protein can adopt is astronomically high, often referred to as the "conformational space.“ Exploring this vast space to find the most stable configuration is a major computational task. Moreover, protein folding occurs over timescales ranging from microseconds to seconds, but tracking these processes requires simulating movements at intervals as short as femtoseconds, a task that is beyond the capacity of conventional computational methods. To manage these demands, simulations often rely on approximations that sacrifice accuracy for speed, creating a constant balance between realistic modeling and computational feasibility. Additionally, the immense computational resources required often limit the accessibility of these simulations, making high-level computing power like supercomputers or distributed computing networks essential (Pal et al. 2024). Despite these hurdles, advancements in machine learning and algorithm development continue to push the boundaries of what is possible in protein folding simulations.

The integration of quantum computing, generative AI, and mixed reality represents a pioneering approach to overcoming the computational hurdles associated with protein folding simulations. Quantum computing offers the potential to dramatically enhance computational speed and capacity, enabling the exploration of vast conformational spaces more efficiently than traditional computing. Generative AI contributes by predicting protein structures through learned patterns and data, reducing the need for exhaustive simulation of every possible configuration. Mixed reality, on the other hand, provides an immersive platform for visualizing complex protein structures in three dimensions, facilitating a more intuitive understanding of folding processes and interactions. Together, these technologies create a synergistic toolkit that not only accelerates scientific discovery in the realm of protein folding but also enhances educational and creative endeavors by making these complex processes more accessible and engaging. This integration marks a significant step forward in our ability to model, understand, and manipulate biological systems at the molecular level.

19.2 Quantum Computing Basics

Quantum computing represents a groundbreaking paradigm in computation, fundamentally distinct from the traditional binary systems that have underpinned classical computing for decades. Classical computers process information using bits, which exist in one of two binary states: 0 or 1. In contrast, quantum computers utilize quantum bits, or qubits, which harness the quantum mechanical phenomenon of superposition. This property allows a qubit to exist in a state of 0, 1, or any quantum superposition of these states simultaneously, unlocking the potential to encode and process vast amounts of information concurrently (Emani et al. 2021).

The power of superposition lies in its ability to exponentially expand computational resources. For example, while a classical bit can only represent a single state—either 0 or 1—two classical bits can represent one of four possible states at a time: 00, 01, 10, or 11. However, two qubits in superposition can represent all four states simultaneously. As the number of qubits in a system increases, the number of states that can be represented and processed grows exponentially, vastly surpassing the capabilities of classical systems. This inherent parallelism allows quantum computers to evaluate numerous possibilities at once, making them extraordinarily efficient for solving complex problems (Bawa 2024; Schuld et al. 2015). This exponential scalability gives quantum computers a decisive edge in tackling problems that are infeasible for classical systems, such as simulating molecular interactions, solving complex optimization challenges, and advancing cryptographic computations. While classical computers must evaluate solutions sequentially, quantum systems exploit the power of superposition to explore multiple pathways simultaneously, delivering unparalleled efficiency and speed. As quantum technology matures, its transformative potential is poised to revolutionize fields ranging from materials science and drug discovery to artificial intelligence and beyond.

The superposition of qubits dramatically amplifies computational capacity, particularly for problems involving large datasets or the simultaneous exploration of numerous variables. In cryptography, for example, quantum computers could dismantle widely-used encryption protocols by factoring large numbers exponentially faster than classical computers. Algorithms such as Shor's algorithm capitalize on quantum parallelism to perform operations that are computationally prohibitive for classical systems, posing both challenges and opportunities in the realm of cybersecurity (Vedral and Plenio 1998). More broadly, the unique ability of quantum computers to harness principles like superposition and entanglement redefines the boundaries of computational possibility. They enable solutions to problems previously deemed unsolvable, from simulating quantum-level interactions in physics to optimizing vast logistical networks (Gyongyosi and Imre 2019). As these technologies continue to evolve, quantum computing is set to unlock breakthroughs across industries, reshaping how we approach some of the most complex challenges of our time.

Entanglement is a cornerstone of quantum computing and one of the most fascinating phenomena in quantum mechanics. It describes a unique and profound connection between qubits, where the state of one qubit is intrinsically linked to the state of another, regardless of the physical distance separating them. This means that measuring the state of one qubit instantly determines the state of its entangled partner—a phenomenon that defies classical intuition and underscores the non-local nature of quantum mechanics (Cordier et al. 2022). In quantum computing, this property is more than a curiosity; it is a fundamental enabler of quantum systems' extraordinary computational potential.

In quantum computing, entanglement allows qubits to collaborate in ways that classical bits cannot. When qubits are entangled, their combined state encodes more information than the sum of their individual states. This enables quantum computers to process and store information far more efficiently, with the addition of each qubit causing an exponential growth in the system's computational power (Cordier et al. 2022). For example, in a system of entangled qubits, the collective information encoded increases exponentially as additional qubits are entangled, vastly enhancing the system's ability to tackle complex computational problems that are beyond the reach of classical systems.

Entanglement also underpins the inherent parallelism of quantum computing. The correlated states of entangled qubits mean that operations on one qubit can instantaneously influence its entangled partners. This feature allows quantum computers to process multiple inputs and produce multiple outputs simultaneously, a key factor in their ability to execute quantum algorithms at speeds unattainable by classical computers. Entanglement is especially critical in the function of quantum gates, the basic building blocks of quantum circuits. Quantum gates manipulate qubits, often entangling them to create highly intricate and interconnected computational pathways that enable the execution of sophisticated algorithms (Juárez-Ramírez et al. 2024; Shafique et al. 2024).

Beyond its role in computation, entanglement is essential for several foundational quantum protocols. Quantum teleportation, for instance, relies on entanglement to transfer quantum information from one location to another without physically moving the qubits. This process has the potential to revolutionize data transmission by enabling secure and instantaneous communication. Similarly, quantum error correction schemes depend on entanglement to detect and correct errors arising from the fragile nature of quantum states, ensuring the reliability and scalability of quantum systems over time (Bouwmeester et al. 1997).

In essence, entanglement is not merely a feature of quantum computing—it is the mechanism that drives its unparalleled computational power. By harnessing the interconnectedness of entangled qubits, quantum computers can address problems that are intractable for classical systems. From advancing cryptography and materials science to revolutionizing communications and data processing, entanglement is central to realizing the transformative potential of quantum technologies.

Quantum interference is another critical principle that underpins the power of quantum computing, enabling it to perform tasks that would be impractical or impossible for classical computers. Interference occurs when quantum states, represented as wave functions, overlap and combine, either constructively or destructively. Constructive interference amplifies certain outcomes, increasing their probability, while destructive interference diminishes others, reducing or even eliminating the likelihood of incorrect solutions. This ability to manipulate probabilities through interference is what allows quantum algorithms to efficiently zero in on the correct answers while filtering out incorrect ones. In a quantum computer, interference is harnessed through the careful design of quantum algorithms that take advantage of the wave-like nature of quantum states (Shafique et al. 2024). When a quantum algorithm is executed, the quantum system evolves in a superposition of all possible states, representing all potential solutions to a problem. As the algorithm progresses, quantum gates— the operations that manipulate qubits—are applied in a way that steers the evolution of the system. The goal is to arrange the interference patterns so that the correct solutions interfere constructively, boosting their probability, while incorrect solutions interfere destructively, canceling each other out (Juárez-Ramírez et al. 2024).

This principle is elegantly demonstrated in some of the most well-known quantum algorithms, such as Shor's algorithm and Grover's algorithm. Shor's algorithm, for instance, is a quantum algorithm that can factor large numbers exponentially faster than the best-known classical algorithms. It achieves this by exploiting interference to identify the periodicity in functions related to number theory, a key step in the factorization process. Through the interference patterns created by the quantum Fourier transform, Shor's algorithm dramatically reduces the time required to solve the problem, making it feasible to break widely used cryptographic systems based on large prime factorization (Ekert and Jozsa 1996). Similarly, Grover's algorithm leverages quantum interference to search through unsorted databases more efficiently than classical algorithms. In a classical setting, searching through N items requires checking each one individually. However, Grover's algorithm uses interference to amplify the probability of finding the correct item, reducing the search time. This quadratic speedup, while not exponential like Shor's algorithm, still represents a significant improvement, particularly for large datasets (Chandarana et al. 2023). The effectiveness of quantum interference in these algorithms illustrates a fundamental advantage of quantum computing: the ability to process and manipulate probabilities in a way that classical systems cannot.

While classical computers operate deterministically, calculating each possible outcome sequentially, quantum computers use interference to explore multiple possibilities simultaneously and converge on the correct solution more rapidly. However, designing quantum algorithms that effectively harness interference is a complex task, requiring deep knowledge of both quantum mechanics and the problem at hand. Researchers are continually exploring new ways to use interference to solve a wider array of problems, from optimization and simulation to machine learning. As quantum computing technology advances, the ability to control and utilize quantum interference will be key to unlocking its full potential, leading to breakthroughs across various scientific and technological fields.

Data parallel processing and advanced data handling are fundamental to modern computational systems, particularly in the emerging paradigms of quantum and distributed computing. Parallel processing, which involves dividing large datasets or complex tasks into smaller, independent units for simultaneous processing across multiple computational nodes, significantly accelerates computation, reduces processing time, and enables the handling of massive datasets (Markidis 2024). This approach is essential for applications in machine learning, scientific simulations, and big data analytics, where sequential processing is often impractical.

Advanced data handling capabilities are equally crucial, encompassing the efficient management, storage, retrieval, and processing of data. Robust data handling ensures optimized storage formats, seamless access to information, and reliable processing pipelines, which are vital for maintaining accuracy and performance in computational tasks (Atadoga et al. 2024). The synergy between parallel processing and advanced data handling forms the foundation of high-performance computing, driving advancements in real-time analytics, large-scale simulations, and artificial intelligence. As the demand for computational power continues to grow exponentially, these capabilities will remain indispensable in addressing the challenges and opportunities of the data-driven era.

19.3 Quantum Computing and Protein Folding

Protein folding is a complex biological process where a linear chain of amino acids adopts a specific three-dimensional structure to perform its biological function. Accurately simulating this process is critical for understanding diseases, drug design, and biochemistry. However, protein folding involves astronomical combinations of possible configurations, making it computationally prohibitive for classical computers to solve effectively. This is where quantum computing emerges as a transformative tool. Leveraging the principles of quantum mechanics, quantum computing enables the exploration of massive solution spaces more efficiently, offering new possibilities for understanding protein structures.

Quantum computing plays a promising role in protein folding simulations by potentially overcoming the limitations of classical computing in solving this complex problem. Quantum algorithms can address the nature of protein folding, offering new methods to predict protein structures more efficiently. The potential of quantum computing in protein folding lies in its ability to handle combinatorial optimization problems, which are central to determining the most energetically favorable folding pathways. Superposition and entanglement allow quantum computers to evaluate numerous folding possibilities simultaneously, while quantum interference helps prioritize optimal configurations. This capability reduces the computational complexity of protein folding simulations, which scale exponentially in classical systems, to more manageable levels (Linn et al. 2024; Lisnichenko and Protasov 2022). In addition, quantum computing may accelerate breakthroughs in drug discovery by providing faster insights into protein–ligand interactions and misfolding-related diseases, such as Alzheimer's and Parkinson's.

Quantum algorithms tailored for protein folding are designed to exploit the unique properties of quantum computing to simulate and predict protein structures. One prominent approach involves encoding the energy landscape of a protein onto a quantum system, allowing the quantum computer to search for configurations with minimal energy states. Algorithms such as the Quantum Approximate Optimization Algorithm (QAOA) and the Variational Quantum Eigensolver (VQE) have been adapted for this purpose. QAOA approximates optimal solutions for combinatorial optimization problems, while VQE efficiently determines the ground-state energy of a molecular system by leveraging a hybrid quantum–classical framework (Boulebnane et al. 2023; Mustafa et al. 2022).

Another significant algorithm is the quantum annealing approach, which directly tackles the optimization problem by gradually evolving the system toward the lowest energy state. Quantum annealers, such as those developed by D-Wave, have shown promise in identifying near-optimal folding configurations for simplified protein models (Chandarana et al. 2023; Robert et al. 2021). Additionally, quantum machine learning techniques are being explored to predict protein folding patterns by training quantum-enhanced models on structural and energetic data.

Despite the promise, quantum algorithms for protein folding remain in their infancy. Current quantum hardware faces limitations in qubit count, coherence time, and error rates, which restrict the complexity of simulations (Cîrstoiu et al. 2020; Sivak et al. 2023). Nonetheless, as quantum technologies advance, these algorithms are expected to become more powerful and accurate, potentially revolutionizing computational biology by solving previously intractable problems in protein folding.

19.4 Overcoming Computational Barriers

Proteins are the molecular workhorses of biological systems, performing a wide array of functions essential to life. Their functionality is determined by their three-dimensional structure, which arises from a complex folding process guided by the sequence of amino acids in the polypeptide chain. The sheer complexity of protein structures is evident in the astronomical number of possible configurations a single protein can adopt, a challenge often referred to as the "protein folding problem." Understanding and predicting these structures is pivotal for advancements in medicine, biotechnology, and basic science, as it holds the key to elucidating mechanisms of diseases, designing novel therapeutics, and engineering functional biomolecules.

The complexity of protein structures stems from the interplay of various forces, including hydrogen bonding, van der Waals interactions, hydrophobic effects, and electrostatic forces. These interactions collectively determine the protein's final folded state, which typically corresponds to the lowest energy configuration. Computationally simulating this folding process requires exploring a vast energy landscape, a task that scales exponentially with the size of the protein (Pillai et al. 2022). Traditional methods, such as molecular dynamics and Monte Carlo simulations, often fall short of capturing these intricacies efficiently due to computational resource constraints. Addressing this complexity calls for innovative approaches, such as quantum computing and machine learning, to process and analyze the vast data involved in protein folding simulations more effectively.

Improving the accuracy and efficiency of protein structure prediction is a long-standing goal in computational biology. Recent advances, particularly in artificial intelligence and quantum computing, have introduced groundbreaking methods to overcome the limitations of classical approaches. Artificial intelligence-driven tools like AlphaFold have revolutionized structure prediction by leveraging deep learning models trained on vast datasets of protein sequences and experimentally determined structures. These models can predict protein structures with remarkable accuracy, significantly reducing the gap between theoretical predictions and experimental validation (Jumper et al. 2021; Zhang et al. 2023). Additionally, ResNet architectures demonstrate robust performance without requiring co-evolutionary information, addressing limitations in data availability (Xu et al. 2021). Temporal deep learning approaches, such as DeepTrajectory, leverage molecular dynamics simulations to enhance predictions by learning from dynamic temporal patterns (Pfeiffenberger and Bates 2018). Complementing these methods, quantum algorithms, including quantum genetic and variational quantum algorithms, provide novel pathways to address the complexity of protein folding by optimizing energy states more effectively. Distance-based folding techniques and adaptive multiscale simulation strategies further improve computational efficiency, enabling rapid yet accurate structure predictions. Together, these innovative approaches push the boundaries of protein folding prediction, offering transformative tools for scientific research and medical applications (Robert et al. 2021).

Furthermore, interdisciplinary approaches combining artificial intelligence, quantum computing, and traditional computational biology are expected to play a pivotal role in refining prediction tools. By integrating these technologies, researchers can achieve greater accuracy and efficiency in understanding protein structures, ultimately driving innovations in drug discovery, disease understanding, and synthetic biology. As computational methods continue to evolve, they promise to bridge the gap between theoretical models and experimental biology, unlocking new frontiers in life sciences.

19.5 Quantum Algorithms for Protein Folding

19.5.1 Design and Functionality

Quantum computing has introduced a suite of algorithms uniquely suited to tackle the complex optimization and simulation problems inherent in protein folding. These algorithms leverage the quantum properties of superposition, entanglement, and interference to explore the vast configuration space of protein structures more efficiently than classical methods. Among the most notable algorithms are the Quantum Approximate Optimization Algorithm (QAOA) and the Variational Quantum Eigensolver (VQE).

QAOA is designed to address combinatorial optimization problems, making it particularly useful for determining the lowest-energy folding configurations of proteins. It operates by encoding the energy landscape of a protein as a cost function and iteratively refining solutions using quantum–mechanical principles. Similarly, VQE is a hybrid quantum–classical algorithm that calculates ground-state energies of quantum systems (Boulebnane et al. 2023; Robert et al. 2021). In protein folding, this translates to identifying the most stable molecular conformation by simulating quantum interactions between amino acids.

Another prominent approach is quantum annealing, implemented in specialized quantum devices such as those developed by D-Wave. Quantum annealing is particularly effective for solving energy minimization problems by guiding the quantum system toward the global minimum of the energy landscape. Additionally, quantum-enhanced machine learning techniques, such as quantum Boltzmann machines and quantum neural networks, are being explored to predict folding patterns based on historical structural data (Chandarana et al. 2023; Outeiral et al. 2021a).

The practical application of quantum algorithms to protein folding has demonstrated encouraging results, particularly in proof-of-concept studies. One case study involved using QAOA to model the folding of simplified lattice proteins. These models, while less complex than real-world proteins, showcased quantum computing's potential to efficiently find low-energy configurations. The results indicated that QAOA could outperform classical heuristic methods in identifying near-optimal solutions within a fraction of the computational time (Fingerhuth et al. 2018).

Another case study utilized quantum annealing to explore the energy landscapes of short peptide chains. Researchers encoded the folding problem into the quantum annealer and observed that it successfully identified folding pathways consistent with experimental data (Irbäck et al. 2022). Although current hardware limitations restrict the size of proteins that can be simulated, these experiments demonstrate the scalability potential of quantum annealing as quantum processors continue to evolve.

In a hybrid approach, researchers combined VQE with classical simulation techniques to predict the folding of small proteins like chignolin, a miniature protein with known experimental structures. The results showed that VQE could approximate the protein's ground-state energy, laying the groundwork for scaling up to larger and more complex proteins (Babej et al. 2018).

These case studies underscore the promise of quantum computing in protein folding, even as challenges remain in hardware scalability and noise reduction. As quantum technologies advance, they are expected to unlock new capabilities for understanding protein structures, enabling breakthroughs in drug discovery, personalized medicine, and synthetic biology.

19.5.2 Optimizing Protein Folding Predictions

Enhancing the speed and precision of protein folding simulations is a cornerstone of modern computational biology. Understanding protein folding is critical for unraveling the molecular basis of diseases, designing novel therapeutics, and advancing fundamental biology. To meet the growing demands of accuracy and computational efficiency, researchers have developed a variety of innovative strategies, each tailored to address the complexity of protein folding at different scales.

One effective approach is the use of multiscale simulation strategies, such as Adaptive Multiscale Simulation (AIMS). AIMS iteratively switches between all-atom (AA) and coarse-grained (CG) models, enabling researchers to capture fine-grained structural details without sacrificing computational efficiency. By focusing computational resources on critical regions of the protein while simplifying others, AIMS achieves speeds up to 40 times faster than traditional all-atom simulations, maintaining high accuracy in modeling folding pathways. This balance is particularly useful for studying large proteins or systems with complex dynamics (Li et al. 2010; Zhong and Li 2021).

Another promising technique is the integration of genetic algorithms with systematic crossover methods. These algorithms mimic evolutionary processes like mutation and crossover to efficiently explore the protein's conformational space. Compared to traditional Monte Carlo simulations, genetic algorithms provide faster convergence to low-energy states, offering a more effective way to identify stable protein conformations and folding pathways (König and Dandekar 1999; Unger and Moult 1993). Similarly, DeepFold, a deep learning-based approach, uses neural networks to predict spatial restraints and guide folding simulations. This method significantly enhances both speed and accuracy, particularly for proteins with limited homologous sequence data, by leveraging predictive models trained on vast structural datasets (Pearce et al. 2022).

Physics-based models such as implicit solvent models streamline folding simulations by simplifying solvent interactions while preserving essential physical accuracy. These models allow for the rapid generation of conformational data, making them ideal for studying proteins in solvated environments (Nguyen et al. 2014). Additionally, accelerated molecular dynamics (aMD) captures folding events on shorter timescales by smoothing energy barriers, offering detailed insights into folding pathways and energetics without the extended computational cost of conventional molecular dynamics (MD) (Miao et al. 2015).

Emerging technologies, such as quantum computing, are opening new frontiers in protein folding simulations. Quantum variational algorithms utilize the principles of superposition and entanglement to optimize energy states efficiently, particularly for small proteins. While still in the early stages, these algorithms hold the potential to address the complexity of protein folding, offering exponential speedups in specific scenarios. Complementing these advances, contact-map-driven methods use inter-residue contact maps to predict folding pathways, bypassing the computational intensity of MD simulations. This technique provides a practical alternative for studying heterogeneous folding processes and capturing rare folding events (Fakhoury et al. 2024).

Collectively, these strategies are revolutionizing protein folding simulations by integrating advanced computational methodologies, from multiscale modeling and genetic algorithms to quantum computing and deep learning. Each approach offers unique advantages, enabling researchers to tackle the complexities of protein folding with greater efficiency and precision. By leveraging these innovations, the scientific community is poised to unlock deeper insights into protein structure and function, accelerating progress in fields such as drug discovery, structural biology, and personalized medicine.

19.6 Mixed Reality and Protein Visualization

19.6.1 Introduction to Mixed Reality in Biology

Mixed Reality (MR) is a cutting-edge technology that blends physical and digital worlds into a unified interactive environment. It lies on the spectrum between Virtual Reality (VR), which immerses users in entirely virtual environments, and Augmented Reality (AR), which overlays virtual elements onto the real world. Unlike VR and AR, MR allows users to interact with both real and virtual objects simultaneously, enabling dynamic and context-aware experiences (Foit 2014; Speicher et al. 2019; Wendrich 2015). This seamless integration between the physical and digital realms has transformative applications in fields such as education, healthcare, architecture, and entertainment.

A key feature of MR is its emphasis on user interaction. Unlike purely virtual or augmented environments, MR facilitates dynamic interactions between users and both physical and virtual objects, enhancing engagement and usability. This interaction is supported by a range of dimensions, including levels of immersion, input methods, and the seamless integration of real and virtual environments. MR exists on a continuum with AR and VR, allowing for smooth transitions between fully virtual and augmented real-world experiences. This flexibility makes MR an invaluable tool for applications requiring a blend of real-world context and virtual enhancements (Bimber et al. 2001; Foit 2014; Rokhsaritalemi et al. 2020).

MR leverages a variety of advanced tools and technologies to deliver its interactive and immersive experiences. AR and Tangible Augmented Reality (TAR) tools allow users to visualize and interact with virtual objects overlaid on physical environments, making them ideal for applications such as design and product development (Choi 2022). To further enhance the realism of MR environments, haptic devices provide tactile feedback, enabling users to "touch" and manipulate virtual objects as though they were real.

Simulation platforms like Unity play a crucial role in creating dynamic MR environments, enabling features such as real-time physics-based simulations, including fluid dynamics. These tools not only enhance the visual fidelity of MR applications but also support interactive scenarios for training, education, and engineering. Additionally, the Mixed Reality Analytics Toolkit (MRAT) provides developers with insights into user behavior by analyzing interactions and visualizing performance metrics, facilitating the optimization of MR applications (Nebeling et al. 2020).

Generative AI has emerged as a powerful complement to Mixed Reality, providing dynamic and adaptive content generation that enhances immersion and interactivity. Generative models, such as Generative Adversarial Networks (GANs) and transformer-based architectures, can create high-quality 3D assets, simulate realistic environments, and respond intelligently to user interactions. By integrating generative AI, MR systems can achieve a higher level of personalization and realism, opening new possibilities for applications in areas like education, design, and gaming.

One significant advantage of generative AI in MR is its ability to produce dynamic visualizations in real-time. For example, generative AI can create virtual environments that adapt to the user's surroundings, such as adjusting the lighting, textures, or spatial layouts based on sensor inputs. This capability allows MR systems to seamlessly integrate virtual elements into the physical world, creating a more coherent and immersive experience. In design and architecture, generative AI can instantly generate customizable 3D prototypes, enabling rapid iterations and creative exploration. Similarly, in training simulations, AI-driven MR can adjust scenarios dynamically based on user performance, enhancing the learning experience.

Generative AI also excels in enabling context-aware interactions, where virtual objects and environments respond intelligently to user behaviors or external conditions. For instance, in a collaborative design session, generative AI can create virtual tools or modify shared environments based on input from multiple participants. These adaptive capabilities not only enhance the usability of MR systems but also unlock new forms of collaboration and creativity. The synergy between Mixed Reality and generative AI represents a pivotal advancement in how we interact with digital content, blending the lines between real and virtual worlds while offering unprecedented levels of customization and engagement.

19.6.2 Immersive Visualization of Protein Structures

Interactive protein modeling involves a suite of computational and graphical methods designed to visualize and manipulate protein structures effectively. These techniques are essential for understanding protein functions, predicting unknown structures, and designing novel proteins for therapeutic and industrial applications. By leveraging computational tools, researchers can interact with protein models, adjust structural parameters, and simulate biological processes in real-time.

One foundational technique is homology modeling, which uses known protein structures as templates to predict the structure of related proteins. Tools like SWISS-MODEL automate this process by integrating evolutionary information and offering an interactive platform for template selection and model refinement. Researchers can explore different functional states of a protein, enhancing the model's biological relevance. This method is especially valuable when experimental structures are unavailable, providing insights into protein function and interactions (Biasini et al. 2014; Waterhouse et al. 2018).

Another approach involves interactive graphical systems such as Sculpt, which allow users to manually manipulate protein models while preserving their physical and chemical properties. Sculpt employs fast energy minimization algorithms, enabling real-time adjustments to bond lengths, angles, and interactions. The intuitive graphical interface makes it accessible for researchers to experiment with structural changes and observe their immediate effects, offering a hands-on method for exploring protein dynamics and stability (Surles et al. 1994).

Recent advancements in deep learning and generative models have accelerated the development of interactive protein modeling. Techniques like GANs predict and design protein structures by learning complex representations from existing data. These models are particularly effective for de novo protein design, allowing researchers to generate new protein structures with desired properties in a fraction of the time required by traditional methods(Anand and Huang 2018; Strokach and Kim 2022). By combining generative models with interactive platforms, researchers can rapidly iterate on designs and refine structures for specific applications (Gao et al. 2020).

Comparative and knowledge-based modeling further enhances the accuracy of protein models by leveraging known structural and functional data. Tools like MODELLER align target sequences with templates, perform fold assignments, and build detailed models based on sequence homology (Eswar et al. 2006). Knowledge-based methods use libraries of known protein fragments to assemble models of unknown structures, providing robust frameworks for studying proteins that lack clear homologs (Gao et al. 2020).

For large or flexible protein complexes, hybrid and integrative modeling combines computational predictions with experimental data, such as electron microscopy or X-ray crystallography (Soni and Madhusudhan 2017). These methods are indispensable for reconstructing protein assemblies that are too complex or dynamic to be resolved through a single approach.

Interactive protein modeling techniques have evolved into a diverse array of methods, from homology-based approaches and real-time graphical systems to deep learning-driven generative models and hybrid strategies. These tools enable researchers to create accurate, manipulable protein models, facilitating breakthroughs in structural biology, drug discovery, and biotechnology. As computational methods continue to advance, their integration with experimental data and graphical interfaces promises to unlock even greater potential for understanding and engineering proteins.

19.6.3 Applications in Research and Education

Mixed Reality (MR) is transforming the way scientists study protein folding by offering immersive and interactive environments that merge physical and digital worlds. There has been rapid growth in augmented reality (AR) and virtual reality (VR) applications, such as mHealth tools for pill reminders and telehealth solutions for remote consultations (Roosan 2024a). Mental health counseling has also seen significant advancements through AR and VR applications in the metaverse. By integrating AI with mixed reality, healthcare delivery can become more efficient, reducing waste and improving access to care.

One notable case involves researchers using MR platforms integrated with quantum computing simulations to visualize amino acids. For example, by combining Microsoft HoloLens with quantum-enhanced algorithms like the VQE, scientists have been able to project three-dimensional, interactive protein models into real-world spaces such as with the application PGxKnow as shown in Fig. 19.1. Utilizing HoloLens and augmented reality, PGxKnow was created as an educational tool to allow visualization of molecules as portrayed in Fig. 19.2. This allows researchers to walk around a protein structure, zoom in on folding intermediates, and explore the energy landscape that dictates folding stability and pathways (Srinivasan and Ali 2020).

Another example is the use of MR in collaborative protein studies, where researchers in different locations use shared virtual environments to analyze folding pathways together. For instance, by overlaying quantum-computed folding data onto MR headsets, teams can test hypotheses, such as the effects of specific mutations on folding stability, in real-time. These systems often include gesture-based interfaces, enabling researchers to "grab" and adjust protein segments to see how structural changes influence folding outcomes (Srinivasan and Ali 2020). This level of interactivity has proven invaluable for understanding complex folding processes like misfolding in prion proteins, which are implicated in neurodegenerative diseases.

The integration of MR with quantum computing in protein folding studies has significantly advanced empirical research and hypothesis testing. MR tools allow researchers to visualize complex folding processes intuitively, bridging the gap between theoretical simulations and experimental data. For instance, by dynamically interacting with protein structures, scientists can test hypotheses about folding

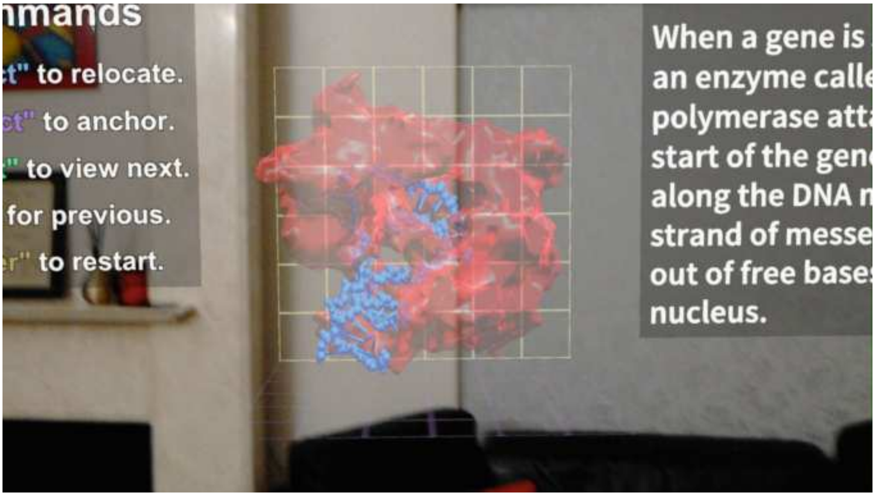

Fig. 19.1 A visualization of DNA transcription with interactive 3D models, powered by AI and AR technology. Students can explore molecular biology concepts in an immersive environment, enhancing understanding through real-time annotations and voice-activated commands. Image by the authors

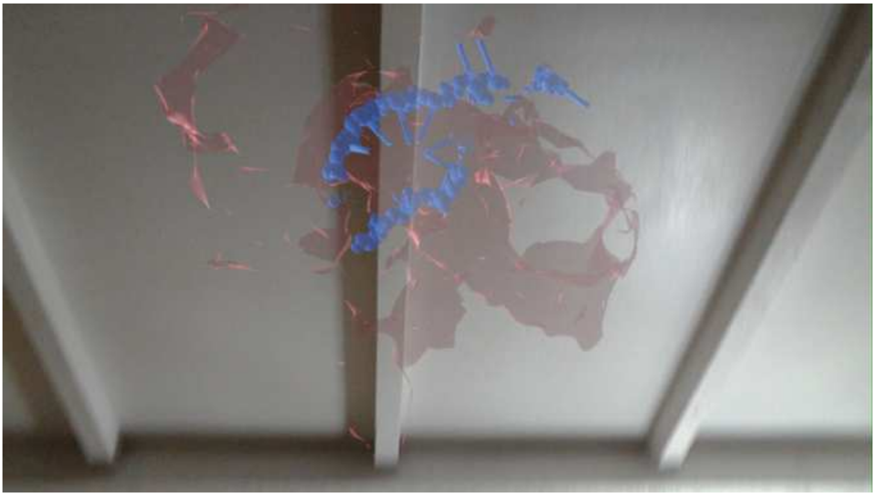

Fig. 19.2 The HoloLens application PGxKnow utilized augmented reality to visualize molecules as an overlay of the real world. Image by the authors

intermediates or pathways, refining their predictions before conducting costly and time-consuming laboratory experiments.

Quantum-enhanced MR environments also facilitate hypothesis testing by providing real-time feedback on structural modifications. For example, when investigating how environmental factors like pH or temperature affect folding stability, researchers can introduce these variables in the MR environment and immediately observe their effects on the protein's energy landscape. This rapid iteration capability accelerates the research process and increases the likelihood of identifying viable experimental pathways.

Moreover, MR-based protein folding studies have enhanced collaboration between computational and experimental scientists. By visualizing quantum-derived folding data in a shared virtual environment, teams can discuss hypotheses and design experiments more effectively. This has been particularly impactful in areas like drug design, where understanding protein–ligand interactions at a detailed level is crucial.

19.7 Synergizing Quantum Computing with Mixed Reality

19.7.1 Collaborative Technologies in Systems Biology

Quantum computing and MR are two groundbreaking technologies that, when combined, significantly enhance the study of complex systems such as protein folding. Quantum computing excels in solving computationally intensive problems, such as simulating quantum interactions and optimizing folding pathways, by leveraging its unique capabilities like superposition and entanglement. However, interpreting the vast amounts of complex data generated by quantum algorithms can be challenging. This is where MR plays a pivotal role, providing an intuitive and immersive interface to visualize and interact with these simulations in real-time.

MR allows researchers to project quantum-derived protein models into three-dimensional spaces where they can explore folding pathways, molecular interactions, and energy landscapes. Using tools like head-mounted displays, researchers can manipulate protein structures, observe dynamic changes, and test hypotheses interactively. The spatial and interactive capabilities of MR make it possible to translate the abstract, high-dimensional outputs of quantum computations into actionable insights, bridging the gap between theoretical data and practical understanding (Robert et al. 2021).

The complementarity of these technologies lies in their synergy: quantum computing provides computational depth and accuracy, while MR offers the accessibility and interactivity needed to make the data comprehensible and actionable. This combination is especially valuable for collaborative research, enabling teams to work together in shared virtual spaces where quantum-derived models can be manipulated and studied from multiple perspectives.

The integration of quantum computing and MR has dramatically improved both visualization and simulation accuracy in protein folding studies. Quantum computing provides unparalleled precision in modeling molecular systems by solving Schrödinger's equation for complex interactions, thereby offering highly accurate predictions of protein folding pathways and energy states. However, visualizing these results in two-dimensional formats often limits their interpretability. MR overcomes this limitation by allowing researchers to engage with quantum-generated models in a three-dimensional, immersive environment.

This combined approach enhances understanding by enabling researchers to intuitively explore molecular dynamics. For example, real-time visualization of folding intermediates and energy transitions allows scientists to identify stable and unstable states with greater clarity. By manipulating virtual protein models generated by quantum simulations, researchers can also observe the immediate effects of mutations or environmental changes, providing deeper insights into folding mechanisms and potential therapeutic interventions.

The precision of quantum simulations, when coupled with the interactivity of MR, reduces the time needed for hypothesis testing and experimental design. Researchers can explore a wider range of folding scenarios and structural modifications without the constraints of traditional computational or laboratory methods. This results in more informed and accurate experiments, accelerating advancements in fields such as drug design, structural biology, and biophysics. The combined impact of quantum computing and MR not only enhances scientific visualization but also drives more efficient and precise problem-solving, unlocking new possibilities in understanding and manipulating complex biological systems.

19.7.2 Practical Application and Implications

Quantum computing is driving transformative advancements in drug discovery and personalized medicine by enabling researchers to address problems that are computationally intractable for classical systems. Traditional drug discovery processes are often slow and expensive, involving trial-and-error approaches to identify compounds with therapeutic potential. Quantum computing accelerates this process by simulating molecular interactions at an unprecedented level of accuracy, allowing researchers to identify promising drug candidates more efficiently. Quantum computing and artificial intelligence have been utilized in other areas for a variety of projects from identifying indicators of social determinants of health from electronic health records, developing dashboard analytics platforms for caregivers of dementia patients, and improving medication adherence through digital health solutions and interactive mobile games (Li et al. 2023; Roosan et al. 2024a; Roosan et al. 2024b; Roosan 2022; Roosan et al. 2022b; Roosan et al. 2022a; Wu et al. 2024).

One of the most significant contributions of quantum computing lies in its ability to model molecular systems with high precision. Algorithms like the VQE and QAOA enable the simulation of quantum interactions between drug molecules and biological targets, such as proteins or enzymes. This capability allows researchers to predict binding affinities and optimize molecular structures for efficacy and safety before laboratory testing begins (Boulebnane et al. 2023). As a result, quantum computing reduces the time and cost of bringing new drugs to market.

In personalized medicine, quantum computing enhances the ability to analyze and process complex datasets, such as genomic information and patient-specific molecular profiles. By integrating these data with quantum-driven simulations, researchers can design treatments tailored to individual patients. For instance, quantum algorithms can model how genetic variations influence protein folding and drug metabolism, enabling the development of precision therapies that maximize effectiveness while minimizing adverse effects. These advancements hold the potential to revolutionize healthcare, making treatments more targeted and effective for individual patients (Goswami and Sharma 2024; Hassanzadeh 2020).

Beyond drug discovery and personalized medicine, quantum computing is making significant contributions to the broader field of biological research. It enables the study of complex biological systems that are currently beyond the reach of classical computational models. For example, understanding protein folding, enzyme kinetics, and cellular signaling pathways often requires solving complex mathematical equations that describe quantum interactions. Quantum computing provides the computational power needed to tackle these challenges, leading to deeper insights into the fundamental mechanisms of life.

Quantum computing also facilitates the integration of data from various biological subfields, such as genomics, proteomics, and metabolomics. This holistic approach allows researchers to construct comprehensive models of biological systems, providing a more unified understanding of how molecular interactions drive cellular functions. In systems biology, quantum algorithms are being used to study how changes at the molecular level influence larger biological networks, such as metabolic pathways or gene regulatory networks. These insights can lead to breakthroughs in areas like synthetic biology, where researchers design and engineer biological systems for industrial or therapeutic applications (Marchetti et al. 2022; Outeiral et al. 2021b).

In addition, quantum computing is enabling advancements in bioinformatics by accelerating the analysis of large-scale datasets. Tasks like sequence alignment, phylogenetic tree construction, and evolutionary modeling benefit from the computational efficiency of quantum algorithms (Nałęcz-Charkiewicz et al. 2024; Outeiral et al. 2021b). As quantum hardware continues to evolve, its contributions to biological research are expected to grow, unlocking new possibilities for understanding life at every scale, from molecules to ecosystems. These developments not only advance scientific knowledge but also pave the way for innovative applications in healthcare, agriculture, and environmental science.

19.8 Future Perspectives

19.8.1 The Frontier of Protein Folding Research

Quantum computing and Mixed Reality (MR) are two rapidly evolving technologies that are beginning to converge, unlocking new possibilities in scientific research and applications. As these fields evolve, they are unlocking new capabilities that address complex challenges and create immersive user experiences. The convergence of these technologies, along with advancements in artificial intelligence and digital systems, is poised to revolutionize how we interact with and utilize technology in various domains.

Quantum computing is advancing rapidly, transitioning from theoretical research to practical applications. A significant milestone in this journey is quantum supremacy, where quantum computers demonstrate the ability to solve problems faster than classical systems for specific tasks. The next step is achieving quantum advantage, where quantum computers tackle real-world problems more efficiently than their classical counterparts. This progression is vital for realizing the true potential of quantum computing in practical scenarios (Gill et al. 2022; Gyongyosi and Imre 2019; Rawat and Sharma 2024).

Progress in quantum hardware and software development is central to these advancements. Researchers are addressing challenges such as quantum decoherence and qubit interconnectivity to build scalable, robust quantum systems. Concurrently, the development of sophisticated quantum algorithms and programming tools is enabling researchers to harness the power of quantum computers for tasks like optimization, molecular simulations, and cryptography. Interdisciplinary collaboration is further accelerating the field, bringing together experts from physics, computer science, and other domains to solve complex computational problems (Gill et al. 2022; Rawat and Sharma 2024).

Quantum computing applications span a broad spectrum of industries. In drug discovery, quantum computers can simulate molecular interactions at unprecedented levels of detail, speeding up the identification of therapeutic candidates. In finance, they enable the optimization of investment portfolios and risk analysis. Similarly, quantum technologies are enhancing data science, clean energy innovation, and secure communications, solving problems that remain intractable for classical systems (Meldrum 2019; Rawat and Sharma 2024).

The convergence of quantum computing and MR holds transformative potential for systems biology and life sciences, particularly in tackling problems involving the complexity of living systems. Systems biology focuses on understanding the interactions between components of biological systems, such as genes, proteins, and metabolic networks, to uncover how these interactions drive cellular and organismal behavior. Quantum computing offers unparalleled computational power to model these intricate networks, capturing quantum-level interactions that are often approximated or ignored in classical simulations.

One potential breakthrough lies in decoding the dynamic behavior of metabolic and signaling pathways. Quantum algorithms can simulate the probabilistic nature of biochemical reactions with high accuracy, providing insights into how cells respond to environmental changes or genetic perturbations. When combined with MR visualization tools, these simulations can be explored interactively, allowing researchers to test hypotheses about how specific modifications might influence cellular behavior. This approach could lead to the development of new therapeutic strategies or synthetic biological systems optimized for industrial applications (Cordier et al. 2022; Kendon 2020; Mauranyapin et al. 2022; Outeiral et al. 2021b).

Another area ripe for breakthroughs is the study of gene regulation and epigenetic modifications. Quantum computing can help unravel the quantum mechanical aspects of DNA–protein interactions, such as those involving transcription factors and chromatin remodeling complexes (Pyrkov et al. 2024; Roman-Vicharra and Cai 2022). With MR, researchers can visualize these interactions in 3D and experiment with potential interventions, such as CRISPR-mediated genome editing, in a virtual yet scientifically accurate environment. This synergy could accelerate discoveries in genetic engineering, precision medicine, and regenerative biology.

As artificial intelligence (AI) continues to evolve and integrate with other emerging technologies such as quantum computing and MR, its potential to transform industries becomes increasingly evident. AI systems are no longer confined to theoretical constructs; they are now driving practical innovations, from automating routine tasks to addressing some of the most complex challenges in fields like medicine, finance, and education. The convergence of AI with technologies like MR has enabled immersive, interactive experiences, such as visualizing molecular biology processes or exploring simulated environments for training and research. Similarly, combining AI with quantum computing is opening pathways for solving problems previously deemed unsolvable by classical methods.

Overall, the integration of quantum computing and MR in systems biology and life sciences is poised to redefine the way researchers approach complex biological problems. By enhancing visualization, simulation accuracy, and interactivity, these technologies promise to unlock new levels of understanding and innovation in the life sciences, paving the way for unprecedented advancements in health, agriculture, and environmental sustainability.

19.8.2 Challenges and Opportunities

While quantum computing holds immense promise, its development is still in its infancy, and several limitations currently constrain its full potential. The most significant challenges lie in hardware reliability and scalability. Quantum systems are highly susceptible to environmental noise, which leads to decoherence—a loss of quantum information that limits the accuracy of computations. Error rates remain high, and the implementation of fault-tolerant quantum computing, which requires extensive error correction mechanisms, is still a work in progress. Additionally, the number of qubits in current quantum processors is insufficient for solving large-scale, real-world problems, necessitating advances in both qubit design and quantum architecture (de Leon et al. 2021; Lorenz 2024; Preskill 2018).

The development of quantum hardware, such as superconducting qubits, trapped ions, and topological qubits, is still in its beginning. Each technology has its own set of challenges, including coherence times, gate infidelities, and connectivity issues, which limit the performance and scalability of quantum computers. Another limitation is the need for better integration with classical computing systems. Many current applications rely on hybrid quantum–classical models, where quantum computers perform a specific subset of calculations while classical systems handle the remainder (Dobrautz et al. 2024; Hassanzadeh 2020). However, effectively managing the data flow and computational load between these systems remains a challenge. Furthermore, the steep learning curve associated with quantum programming languages and algorithms limits widespread adoption among researchers in non-computational fields, such as biology and chemistry. MR, which complements quantum computing by offering intuitive interfaces for exploring quantum-derived data, also faces hurdles in adoption. Hardware costs for devices like HoloLens and Magic Leap remain prohibitive for many institutions, and software development for MR applications requires specialized expertise (Bate 1982). Addressing these challenges will be crucial for the broader integration of quantum computing and MR in interdisciplinary research.

The future of quantum computing lies in its ability to drive breakthroughs through interdisciplinary research, where it collaborates with fields such as biology, physics, artificial intelligence, and engineering. One promising direction is the development of domain-specific quantum algorithms tailored to solve pressing problems in these fields (Garcia-Buendia et al. 2024; Rawat and Sharma 2024). For instance, designing quantum algorithms for protein folding, drug discovery, and material science will require close collaboration between quantum physicists, computational biologists, and chemists.

Advances in hardware will also play a critical role. The development of scalable quantum processors with reduced error rates, along with MR devices that are more affordable and user-friendly, will make these technologies more viable for real-world applications. Research into quantum machine learning could further expand the capabilities of AI-driven analysis, leading to new approaches in data-intensive fields such as genomics and climate science (Gill et al. 2022).

The rapid rise of AI and machine learning also introduces significant challenges, particularly in how these systems are evaluated. Unlike traditional software, AI systems often operate as "black boxes," producing outputs based on intricate patterns learned from vast datasets. This lack of transparency raises concerns about trust, fairness, and reliability. For example, ensuring that AI systems make unbiased decisions in critical areas such as hiring or loan approvals requires rigorous testing and monitoring. Additionally, the dynamic nature of AI models, which evolve as they are retrained with new data, complicates the process of standardizing evaluation metrics.

One critical challenge lies in defining what success looks like for AI and machine learning systems. Unlike conventional algorithms, where success is often binary (correct or incorrect), the performance of AI models is typically measured using probabilistic metrics such as accuracy, precision, recall, or F1 scores. These metrics, while useful, may not capture the broader societal implications of deploying AI at scale. For instance, an AI system might achieve high accuracy in diagnosing diseases but fail to generalize its performance across diverse patient populations, leading to disparities in healthcare outcomes.

Another pressing issue is the lack of universally accepted benchmarks for evaluating AI's long-term impacts. As AI continues to advance, the need for robust frameworks to evaluate and regulate its development and application becomes increasingly urgent. These frameworks must balance the drive for innovation with the need for accountability, ensuring that AI technologies are deployed responsibly and equitably while also improving reporting standards and informed decision-making (Roosan 2024b; Roosan et al. 2019). By fostering collaboration between stakeholders and emphasizing transparency and fairness, society can harness the full potential of AI while mitigating its risks.

Ultimately, the future of quantum computing and its interdisciplinary applications will depend on fostering collaboration across scientific domains. By addressing current limitations and investing in innovative research, quantum computing and related technologies are poised to redefine the boundaries of what is computationally possible, driving progress in both fundamental science and practical problem-solving.

19.9 Conclusion

Protein folding is a cornerstone of molecular biology, essential for understanding the intricate mechanisms of life at the molecular level. Its significance extends beyond basic biological processes, influencing systems biology, drug design, and therapeutic interventions. Misfolded proteins are implicated in numerous diseases, underscoring the critical need for accurate modeling and analysis of protein structures. However, the computational challenges posed by the vast conformational space of proteins and the complexity of their interactions have historically limited progress in this field.

Recent advancements in computational tools and methodologies are revolutionizing the study of protein folding. Quantum computing, with its unparalleled ability to process vast datasets and explore complex systems, offers a transformative solution to the computational limitations of traditional methods. Generative AI further enhances this process by leveraging data-driven models to predict protein structures with high accuracy and efficiency. Mixed reality adds an immersive and interactive dimension, enabling scientists to visualize and manipulate protein structures in real-time, making abstract processes tangible and intuitive.

The integration of these technologies is not just a leap forward for protein folding research but a broader milestone in computational biology. By combining speed, precision, and interactivity, quantum computing, generative AI, and mixed reality are enabling breakthroughs that were once thought unattainable. These innovations hold promise for addressing fundamental questions in biology, accelerating drug discovery, and improving our understanding of diseases at a molecular level. As these tools continue to develop and converge, they will pave the way for more profound insights into the biological world, transforming both research and application across the life sciences.

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Concluding Remarks

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As we reach the end of this exploration into Human–Computer Creativity related to the use of generative AI in Education, Art, and Healthcare, one thing becomes abundantly clear: generative artificial intelligence is not merely a technological advancement—it is a catalyst for redefining creativity itself. Through the diverse perspectives and innovative research presented across these chapters, this book highlights the profound ways generative AI is transforming how we learn, create, and heal.

In education, generative AI tools have demonstrated their potential to enhance learning experiences, foster inclusivity, and empower educators and students alike. From personalized learning pathways to adaptive digital pedagogies, these technologies are paving the way for a more engaging and equitable educational future. Similarly, in art, AI has emerged as both a collaborator and an innovator, challenging traditional notions of authorship, aesthetics, and the boundaries of human creativity. The fusion of human intuition with machine precision has unlocked new artistic frontiers where imagination knows no bounds.

In healthcare, the integration of generative AI offers transformative possibilities—from advancing diagnostics and personalized treatments to reimagining medical education and patient engagement. However, this transformation comes with unique challenges, including ethical considerations, the mitigation of biases, and the need for robust oversight to ensure trust and reliability in sensitive domains.

As we navigate this new era of human–computer creativity, it is vital to remember that AI is not a replacement for human ingenuity but a partner that amplifies it. The future of generative AI lies in collaboration between humans and machines, disciplines, and cultures. By fostering such partnerships, we can ensure that AI's creative potential enhances human well-being, enriches our cultural narratives, and addresses the pressing challenges of our time.

This book represents a collective effort to map the current landscape of generative AI and its impact on creativity. Yet, it is also an invitation to look forward, to

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imagine possibilities beyond what is presently conceivable, and to continue asking the critical questions that will guide the responsible and innovative development of these technologies.

We hope that the insights presented in this volume inspire readers to embrace the opportunities of generative AI while addressing its challenges with thoughtfulness and care. The journey of human–computer creativity is just beginning, and we are excited to see how this unfolding story will redefine what it means to create, collaborate, and innovate in the years to come.